Arithmeticity and farthest vertices in compact hyperbolic Coxeter polytopes

Coxeter Polytopes Frontier Project

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Abstract

We give a dimension bound of fifteen for compact arithmetic hyperbolic Coxeter polytopes. Geometry supplies a norm budget at one farthest vertex; arithmeticity restricts the actual ground field and integral cycles of its wall diagram. The general compact bound of seventeen supplies the higher-dimensional tail. Complete forest reductions leave two exact dimensions to exclude. In dimension seventeen the remaining chain has two actual endpoint walls, and exact quadratic and cubic calculations exclude their integral data. In dimension sixteen the final diagram is D_{16}. Its actual field is quadratic; the head and both forks satisfy three integral quarter-product conditions. A complete dyadic analysis, using full integer orders and the remaining geometric residues, excludes it without a new census. We state each finite algorithm separately from geometric coverage and recorded output. The quadratic-form model is automatic for arithmetic reflection lattices; no extra first-type restriction is imposed.

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1 The result and the common argument

All polytopes are convex, finite-sided and full-dimensional. Arithmeticity refers to the lattice generated by their facet reflections.

Theorem 1. *Every compact arithmetic hyperbolic Coxeter polytope has dimension at most fifteen.*

The argument separates geometry from arithmetic at a definite point. Geometry provides one vector whose squared norm is less than one. Its projections onto whole finite diagram components are constrained by actual neighboring walls. Arithmeticity then limits the fields and integral parameters of those walls. It does not create a second norm budget. When pressures overlap, their coefficients are paid together before one completed square is used.

Proposition 2 (General compact companion)**.** *Every compact real hyperbolic Coxeter polytope has dimension at most seventeen, without an arithmetic hypothesis.*

Proposition 2 is proved in the general compact companion [4]; its precise interface is G17 in Appendix O. Its proof excludes dimension eighteen, includes the independently proved higher-dimensional tail, and does not use Theorem 1. Thus only dimensions seventeen and sixteen remain. We first split by the whole finite forest; the final D_{16} argument splits by the inert, ramified or split behavior of two. No argument assumes arithmeticity or Coxeter angles for an arbitrary face.

Section 2 gives the rank-seventeen reduction, and Section 3 states the finite algorithms and their numerical output premises. Section 4 gives the rank-sixteen reduction; Sections 5–8 prove the final dyadic obstruction. The appendices develop the component estimates, actual-domain inclusions and complete forest dispatches used in these reductions.

1.1 One original vector

Use the Lorentz form with negative time coordinate and outward unit normals. Distinct actual wall normals have nonpositive products. Compact acute simplicity gives a rank-n finite spherical forest at each vertex. Choose a strict interior point z and an actual vertex p farthest from it. Write

$$z = E(p - x), \quad E > 0, \quad x \in p^{\perp}, \qquad \|x\|^2 = \sum_C e_C < 1, \quad e_C = \|x_C\|^2.$$

The sum is over whole orthogonal finite components. Overlapping subdiagrams do not supply independent summands. If $q = \cosh(L)p - \sinh(L)u$ is an actual adjacent vertex, then

$$x \cdot u \geq (\cosh L - 1)/\sinh L = \tanh(L/2). \tag{1}$$

More generally a future unit point $q = ap - v$, with $v \in p^{\perp}$, satisfying every wall of the original polytope gives $x \cdot v \geq a - 1$. Distance from z is convex on geodesic segments, so its maximum over the compact polytope is attained at a vertex. For any reference vector w,

$$\|x\|^2 \geq 2x \cdot w - \|w\|^2, \tag{2}$$

by $\|x - w\|^2 \geq 0$. Each coefficient removed from an original pressure must fit its individual available coefficient. Positive combinations of actual normals satisfy every unselected wall by acuteness; future orientation and the finite or negative square still need proof. The companion and the precisely cited supplements make these checks before using (2).

1.2 What arithmeticity adds

An arithmetic reflection lattice has an admissible quadratic-form model over its actual totally real field K; the structural reflection theorem is recorded as Lemma 5.2 of [1]. It supplies K-valued root lines and positive definite nonphysical forms. For roots r_i on those lines put

$$a_{ij} = 2(r_i, r_j)/(r_i, r_i).$$

Invariant cyclic products are algebraic integers and generate the actual ground field. Individual physical unit roots need not be K-valued. Disconnected roots need not have equal raw norm squareclasses.

If an actual principal Gram matrix, after its specified permitted K-rescaling, is defined over $k \subset K$ and indefinite physically, then $K = k$: an additional real embedding fixing k would preserve indefiniteness and contradict positive definiteness. A nonzero algebraic integer also has nonzero integral norm, so bounds at *every* nonphysical embedding can force a degree bound. Neither argument assumes a convenient overfield or an integer power basis.

The rational field is excluded in the dimensions used here: compactness requires anisotropy, while Meyer's theorem makes an indefinite rational form in at least five variables isotropic. These arithmetic inputs are reviewed in [2]. Finite/affine classification and acute wall geometry are external foundations; component estimates, forest reductions and computed propositions below are internal results.

2 The complete rank-seventeen reduction

Write $T_2 = I_2(m)$, $m \geq 3$, for an ordinary irreducible dihedral; $T_3 = A_3, B_3, H_3$ and $T_4 = A_4, B_4, D_4, F_4, H_4$ unless a stated field restriction removes a type. Put $\varphi = (1 + \sqrt{5})/2$. Ordinary nonshort A_1 costs $> 1/10$, ordinary $T_2 > 1/7$, $T_3 > 1/4$, $T_4 > 2/7$. A total-rank-three forest need not cost $> 1/4$; its ordinary bound is $> 17/70$.

The exceptions retain actual-wall meanings. Type I is an isolated short rank-one opposite

$$c_I = \sqrt{h}\,p - \sqrt{1+h}\,e, \qquad 0 < h < 1/2.$$

Type J couples two rank-one directions with joint energy $> .34$. The typed rank-two defects are N_7, old A_2/new isolated-seven, and O_7, old $I_2(7)$/new isolated-three. Each costs $> .09$; they cannot coexist with one another or the short I/J alternatives. A bare old seven is not a typed defect. These are the complete small-component alternatives of F17. Their geometric compatibility proof uses the native gap between distinct actual walls, not merely the abstract old forest.

2.1 Shared costs and arithmetic prefix seals

We specify the component inputs before carrying out either rank count. The ordinary geometric rates are the component theorems of the general compact companion, applied to whole orthogonal components at the same original vertex:

A_1	$1/10$	T_2	$1/7$
A_2	$319/2000$	A_3	$1/4$
B_3, H_3	$2/7$	A_4	$2/7$
H_4	$11/36$	B_4, D_4, F_4	$1/3$
A_5	$3/8$	D_5	$5/14$
B_5	$3/7$	A_6, B_6, D_6	$3/7$
B_7, D_7	$1/2$	B_8, D_8	$4/7$
E_6	$9/20$	E_7	$5/9$
E_8	$2/3$		

Every entry is a strict lower bound. In particular every ordinary component outside A_r, $r \geq 7$, and B_r, D_r, $r \geq 9$, costs more than its rank divided by fourteen. The universal large-component rates used here are $A_r > 1/2$ for $r \geq 7$, $A_r > 4/7$ for $r \geq 8$, $A_r > 5/7$ for $r \geq 13$, and $B_r, D_r > 9/14$ for $r \geq 9$. These are geometric companion inputs, not assertions that a higher-rank whole-diagram exclusion holds at a smaller rank. The stronger B_3, H_3 rate is needed only for the rank sixteen count; the rank seventeen count uses $1/4$.

The actual short-I wall gives parameters u, t with

$$e_I \geq u^2, \quad 0 < u < 1/3, \quad (1-u^2)(1-t^2) = 8ut, \quad t + 3u/2 > .767. \tag{3}$$

Every normalized old dual direction outside I has projection at least t. Its squared-time parameter h also gives the following geometric alternative: at $h \leq 1/5$ all these squared projections are at least $1/5$; at $h > 1/5$, $e_I > 3/70$. These are the companion's actual opposite-wall barrier, with no arithmetic hypothesis. We record the needed joint consequences here to make their accounting explicit.

Lemma 3. *For ordinary irreducible T_j the following bounds hold, each counting I once:*

$$e_{I+T_2} > .21, \qquad e_{I+T_3} > 2/7, \qquad e_{I+T_4} > 23/70, \qquad e_{I+A_1} > .20204, \qquad e_{I+2A_1} > .29.$$

Proof. For T_2 use $e_{T_2} > 1/7$ and $e_{T_2} \geq t^2$. If $u \geq 519/2000$, then $u^2 + 1/7 > .21$. Below that point $u^2 + (.767 - 1.5u)^2$ is decreasing and has endpoint value $.2100353125 > .21$.

For T_3, its two endpoint correlations are $1/3$, $1/\sqrt{3}$ and ρ with $\rho^2 = (3\varphi+2)/(3\varphi+6) < 16/25$, respectively. Cauchy on the sum of the two unit directions gives $e_{T_3} > (10/9)t^2$, as well as $e_{T_3} > 1/4$.

At $u \ge .19$ use $u^2 + 1/4 > 2/7$; below it the decreasing quadratic $u^2 + (10/9)(.767 - 1.5u)^2$ is at least $132407/450000 > 2/7$.

For A_4, the low-h direction cover costs more than $1/3$; the high-h case gives $3/70 + 2/7 = 23/70$, strictly exceeded. The B_4, D_4, F_4 rates already exceed $1/3$. For H_4 we need $t + 2u > 6/7$. Substituting $6/7 - 2u$ for t in the left-minus-right side of the identity in (3), and multiplying by 49, gives

$$13 - 168u + 575u^2 - 168u^3 + 196u^4.$$

With $v = u - .15$ this is $.269725 - 4.194v + 525.86v^2 - 50.4v^3 + 196v^4$. For $-3/20 < v < 11/60$ it is at least $.269725 - 4.194v + 516.62v^2 > 0$, the last quadratic having negative discriminant. The original identity decreases in t, proving the barrier. At $u \ge 19/125$, $u^2 + 11/36 > 23/70$; below it the decreasing quadratic $u^2 + (6/7 - 2u)^2$ is at least $50389/153125 > 23/70$.

For $I + A_1$, put $f(s) = (\sqrt{1+s} - 1)/(\sqrt{1+s} + 1)$. The actual edge barrier gives half-tangent squares $q = f(h)$ and at least $r = f(1/(4h))$ on the other component. Writing $P = \sqrt{qr}$, inversion gives $q + r = 1 + P^2 - 8P$, while AM–GM gives $q + r \ge 2P$. As $0 < P < 1$, this implies $P \le q_0 = 5 - 2\sqrt{6}$ and

$$q + r - 2q_0 = (q_0 - P)(8 - P - q_0) \ge 0.$$

Thus $e_I + e_{A_1} \ge 2q_0 > .20204$; the rational seal is $2.44949^2 > 6$. Finally, for two ordinary A_1 components, at $u \ge .3$ use $u^2 + .2 \ge .29$, strictly exceeded. Below it use $q(u) = u^2 + 2(.767 - 1.5u)^2$; then

$$q(u) - q(.3) = (.3 - u)(2.952 - 5.5u) \ge 0, \qquad q(.3) = .290978 > .29.$$

Each calculation estimates one joint sum of orthogonal whole-component energies; no endpoint has been charged a second copy of I. □

Lemma 4 (Propagation from the actual old-seven defect)**.** *Suppose a whole $I_2(7)$ component has an actual opposite supported only there, with a native-three contact at one node and zero contact at the other. Every other whole rank-one component has energy $> 2/15$, and every other whole irreducible rank-two component has energy $> 1/5$. This lemma is geometric and allows arbitrary finite spectators.*

Proof. Put $d_7 = \csc^2(\pi/7)$. The typed wall is $c_D = \sqrt{h_D}p - v_D$, with $h_D = d_7/4 - 1 \in (0, 1/3)$. Let $c = \sqrt{h}\,p - v$ close an actual edge of another component. Its finite endpoint retains the whole old $I_2(7)$. A positive contact there contributes at least $d_7/4 > 1$ to the white Schur cost, impossible. Thus v has zero projection to that whole component, and c is distinct from c_D. The native actual-wall gap gives

$$(c_D, c) = -\sqrt{h_D h} \le -1/2, \qquad h > 3/4. \tag{4}$$

This proves the height bound for every opposite outside the defect, without assuming an isolated profile for those other walls. In a rank-one direction its spatial derivative is at most $\sqrt{1+h}$, so $\sinh^2 L \ge h > 3/4$. Hence its half-tangent square is greater than $(11 - 4\sqrt{7})/3 > 2/15$.

Suppose a whole $T = I_2(m)$, $m \ge 3$, instead has energy $e \le 1/5$. Its normalized duals have correlation $k = \cos(\pi/m)$. Write $s_i = x_T \cdot u_i$ and $F(s) = 2s/(1 + s^2)$. For $m \ge 4$, distinct actual opposites would require

$$k \le F(s_1)F(s_2) \le F(\sqrt{1/5})^2 = 5/9 < 1/\sqrt{2} \le k,$$

impossible. For $m = 3$, Cauchy gives $(s_1 + s_2)^2 \le 3/5$, and concavity of $\log F$ gives the incompatible bound $1/2 \le F(s_1)F(s_2) \le 240/529 < 1/2$. Therefore one actual wall closes both directions. Both lengths have

$$\cosh L_i \le (1 + 1/5)/(1 - 1/5) = 3/2. \tag{5}$$

Let $b, g \geq 0$ be its contacts in T and Q its entire foreign Schur cost. With $d = (1-k^2)^{-1}$, the finite endpoints give

$$Q + b^2 < 1, \quad Q + g^2 < 1, \qquad h = d(b^2 + 2kbg + g^2) + Q - 1 > 3/4,$$
$$C_b^2 = \frac{d(b+kg)^2}{1-Q-g^2}, \qquad C_g^2 = \frac{d(g+kb)^2}{1-Q-b^2}.$$

Every positive b, g is consequently a finite native cosine. At least one is positive. A positive whole foreign contribution costs at least $1/4$, so $Q = 0$ or $Q \geq 1/4$.

For a single contact, say $b = \beta > 0, g = 0$, $C_g^2 = 1 + h/(1 - Q - \beta^2)$. Equations (4)–(5) imply $Q + \beta^2 < 2/5$; hence $Q = 0$ and the native gap forces $\beta = 1/2$. Then $h = d/4 - 1 > 3/4$ gives $d > 7$ and $m \geq 9$, since $d_8 = 4 + 2\sqrt{2} < 7$. But $\sin t < t$ and $\pi^2 < 10$ give $d > 81/10 > 31/4$, so $C_g^2 = (d-1)/3 > 9/4$, a contradiction.

If both contacts are positive and $Q > 0$, let g be the larger. When $g \geq 1/\sqrt{2}$, $C_b^2 \geq (4/3)(1 + 1/\sqrt{2})^2 > 9/4$. Otherwise $b = g = 1/2$. At old label three, $h = Q < 3/4$; at old label at least four, $C_b^2 \geq (1+k)/(2(1-k)) > 5/2$. Both are impossible.

It remains $Q = 0$. If the larger contact has label at least five, then $g > 4/5$, $b \geq 1/2$, $k \geq 1/2$, $d \geq 4/3$ give $C_b^2 > 3$. If both contacts have label four, $C_b^2 = (1+k)/(1-k) \geq 3$. The remaining finite contact pairs are $(3,4)$ and $(3,3)$. For $(3,4)$, old label at least four gives $C_b \geq 2$; old label three gives $h = \sqrt{2}/3 < 3/4$. For $(3,3)$, old label at least five gives $C_b^2 = (1+k)/(3(1-k)) > 3$; old labels three and four give $h = 0, 1/\sqrt{2} < 3/4$. Every contact and foreign case contradicts (4) or (5). Thus $e_T > 1/5$, completing both conclusions. □

Both typed defects force the actual field K_7. Their three actual roots form the indefinite $(7,3,2)$ principal. Odd bonds permit a square-two normalization over K_7; an extra embedding fixing K_7 would preserve that indefinite principal. Hence actual-field descent gives $K = K_7$. In particular H_3, H_4 cannot occur beside a typed defect, since their five-bond would put $\sqrt{5}$ in K_7. For N_7 every other normalized dual has squared projection $> 1/5$. For O_7, Lemma 4 gives every other A_1 cost $> 2/15$ and every other ordinary irreducible T_2 cost $> 1/5$. The hypothesis is the actual isolated-three wall, not the old label seven alone.

Lemma 5 (Literal arithmetic helpers)**.** *At a classified long-prefix index $i \geq 6$ of B_r or D_r, an actual far helper $v = \gamma\omega_i + \omega_1/2$ with $202/125 < 2\gamma < 12/5$ is impossible. At a D fork-helper site with $r \geq 9$, the corresponding profile $v = \gamma\omega_i + \omega_f/2$ can only have $2\gamma = \varphi$.*

Proof. Normalize the connected long roots to square two over K. The simple-three helper contact permits the same normalization of the actual outside root; its path cycle makes $t = 2\gamma$ integral. The short B root has square one, so the entire old Gram is rational. All embeddings of $\mathbb{Q}(t)$ extend to K, and every nonphysical one must give a strictly positive Schur complement.

For the far helper the raw inverse entries $i, 1, 1$ give $it_\sigma^2 + 2t_\sigma - 1 < 0$. At $i \geq 6$ this forces $|t_\sigma| < .61$ and $t_\sigma < .28$. Degree at least three would give a nonzero integral norm of magnitude less than $(12/5)(.61)^2 < 1$. In degree two the integral trace is strictly between $503/500$ and $67/25$, hence two; its norm $t(2-t)$ lies strictly between -1 and one, again impossible. Degree one leaves $t = 2$, an actual parallel pair with a future feasible null sum, contrary to compactness.

For the fork helper the inverse entries $i, i/2, r/4$ give

$$4it_\sigma(t_\sigma + 1) + r < 8, \qquad -1 < t_\sigma < 0.$$

The integral nonzero element $t(t+1)$ has physical value $< 204/25$ and other conjugates of magnitude at most $1/4$. Its norm excludes degree at least three. In degree two the trace of t is one or two.

Trace one gives $t < 2$ and norm in $(-2, -.995456)$, hence norm -1 and $t = \varphi$. Trace two gives norm in $(-24/25, 0)$, impossible. The rational $t = 2$ is again parallel. This proves both assertions without identifying $\mathbb{Q}(t)$ with the full field in advance. □

Lemma 6 (The mixed B_2 tail)**.** *At a B_r site $i = r-2 \geq 12$, the actual own-three/short-three profile $v = (\omega_i + \omega_r)/2$ is impossible.*

Proof. Normalize the old long roots to square two and the short root to square one over K. The simple-three short contact normalizes the outside root to square one; its own-three product is $-1/\sqrt{2}$. Thus $\sqrt{2} \in K$. At an embedding reversing $\sqrt{2}$, while fixing the rational old Gram, its Schur cost is

$$i/2 - i/\sqrt{2} + r/4 = i(3 - 2\sqrt{2})/4 + 1/2 > 1.$$

At $i = 12$ the last inequality follows from $\sqrt{2} < 17/12$, and it increases with i. This contradicts strict positivity. □

For later prefix counts put

$$f_j = \sqrt{j} - .61/\sqrt{2}, \qquad E_J = f_5^2/5 + \sum_{j=5}^{J-1} (\sqrt{j+1} - \sqrt{j})^2.$$

The generic geometric inventory in the companion applies to B_r with $M = r - 5 \geq 6$ and to D_r with $M = r - 6 \geq 6$, through $M + 1$. Every weak index is the literal far helper just excluded. Hence $6, \ldots, J$ are strong, with $J = r - 4$ in B and $J = r - 5$ in D. If five is strong, positive concave interpolation gives E_J. If it is weak, its actual point gives $x_1 > .5965$, $S_6 > 2.01815$; interpolation through $1, 6, J$ gives

$$L_J = .7600299945 + \frac{(\sqrt{J} - \sqrt{6})^2}{J - 6} > E_J,$$

with the quotient zero at $J = 6$. This follows from the variance comparison: $E_6 < .7$, and at most ten increments in $[1/8, 1/5]$ have variance sum at most $9/640$, less than the initial gain $> .06$. The rational lower seals used below are

J	6	7	8	9	10
$E_J >$	.69685	.73531	.76864	.79805	.82436
J	11	12	13	14	15
$E_J >$	.84816	.86989	.88989	.9084	.92564.

(6)

These statements retain the precise rank hypotheses of the companion inventory; the smaller terminal ranks are handled below, rather than being silently inserted into its generic range.

Proposition 7 (Necessary cover in rank seventeen)**.** *The whole forest at the original farthest vertex of a compact arithmetic* 17*-polytope is one of Table 1.*

The proof below starts at rank seventeen. The arithmetic baseline component estimates used in its partition stage are $e_{A_{15}} > .79$, $e_{A_{16}} > .83$, $e_{A_{17}} > 6/7$, and, under either actual typed defect, $e_{A_{12}} > .703101$, $e_{A_{14}} > .7897732$. Section N.1 derives all these inputs from the stronger printed whole-component proofs, with arbitrary spectators. They are not higher-rank exclusions.

Group	Component	Complete spectators
1	A_{11}	$I + (A_5$ or $D_5)$
2	A_{11}	N_7/O_7 plus A_4, B_4, D_4, F_4
3	A_{12}	$A_5, D_5, (A_4$ or $H_4) + A_1, T_3 + T_2, 2T_2 + A_1$
4	A_{12}	$I + T_4,\ I + T_3 + A_1,\ I + 2T_2$
5	A_{14}	$T_3,\ T_2 + A_1,\ I + T_2$
6	A_{15}	$T_2,\ 2A_1,\ I + A_1,\ N_7/O_7$
7	A_{16}	ordinary A_1 or actual I
8	A_{17}	none
9	B_{15}	N_7/O_7
10	B_{16}	actual I
11	B_{17}	none
12	D_{15}	N_7/O_7
13	D_{16}	ordinary A_1 or actual I
14	D_{17}	none

Table 1: Necessary whole forests. Unmarked components are ordinary; no realizability is asserted.

Proof of Proposition 7. Call A_r, $r \geq 7$, or B_r, D_r, $r \geq 9$, dominant. There is exactly one dominant. With none, discarding only the one possible I leaves cost $> 16/14$; a typed defect gives $> .09{+}15/14$. Ordinary or J accounting is stronger. Two dominants each cost more than $1/2$. The complete ordinary table includes E, F, H, so this step loses no exceptional irreducible component.

An A_7 leaves ten spectator ranks and is impossible by the same budget. At A_8, A_9, A_{10}, even discarding I leaves at least six ordinary ranks, giving $4/7 + 6/14 = 1$, strictly exceeded. The typed and J alternatives are stronger. Thus $11 \leq r \leq 17$.

For A_{11}, ordinary spectator rank six already costs $> 3/7$. With I, the complete ordinary rank-five partitions are 5, $4+1$, $3+2$, $3+1+1$, $2+2+1$, $2+1+1+1$, 1^5. Only $A_5, D_5, (A_4$ or $H_4) + A_1, T_3{+}T_2, 2T_2{+}A_1$ can initially fit below $3/7$. Lemma 3 removes the last three by $23/70{+}.1 = 3/7$, $2/7{+}1/7 = 3/7$, and $.21{+}1/7{+}.1 > 3/7$. Hence group1 is complete. With a typed defect, ordinary rank four initially leaves T_4 or $2T_2$; the other partitions cost at least .35, $12/35$ or .4 before the defect is added. Propagation makes each $T_2 > .2$, removing $2T_2$; actual-field descent removes H_4. This gives group2.

For A_{12}, the complete cheap ordinary rank-five list is exactly that just displayed. Indeed $B_5 > 3/7$, $B_4, D_4, F_4 + A_1 > 13/30$, $T_3 + 2A_1 > .45$, $T_2 + 3A_1 > 31/70$, and $5A_1 > .5$. A J pair plus three remaining ranks costs $> .34 + 3/14 > 3/7$. These give group3. With I, the rank-four partitions are $T_4, T_3 + A_1, 2T_2, T_2 + 2A_1, 4A_1$. The last two cost $> .29 + 1/7 > 3/7$ and $> .29 + .2 > 3/7$, respectively, using $I{+}2A_1$ only once. Thus group4 is complete. A typed defect would leave ordinary rank three and give $.703101 + .09 + 17/70 > 1$, so none remains here.

For $A_{13} > 5/7$, ordinary or J rank four already costs $> 2/7$. With I, the remaining rank-three partitions are $T_3, T_2 + A_1, 3A_1$; their joint costs exceed $2/7$ by Lemma 3 or by the ordinary three-A_1 cost. A typed defect plus T_2 costs $> .09 + .2 > 2/7$; two ordinary A_1 cost at least $4/15$ and give an even stronger contradiction. No A_{13} row remains.

At $A_{14} > 5/7$, ordinary spectator rank three leaves T_3 or $T_2 + A_1$; three A_1 and $J + A_1$ are too expensive. With I, only $I + T_2$ remains since $I + 2A_1 > .29 > 2/7$. A typed defect gives $.7897732 + .09 + 2/15 > 1$. These are exactly group5. At $A_{15} > .79$, rank two leaves $T_2, 2A_1, I + A_1$, or either typed defect. A J pair closes; the other bounds do not delete the retained alternatives. At $A_{16} > .83$, its single rank can be ordinary A_1 or actual I. At A_{17} there are no spectators. This proves groups6–8 without an endpoint or field-degree inference.

For B, D, the 9/14 rate and spectator accounting give $r \geq 12$: at six spectator ranks even the I debit leaves 5/14; a typed defect gives $4/14 + .09$. The generic arithmetic prefix has last strong knot $J = r - 4$ in B, $r - 5$ in D, within the rank domains specified above. The weakest floors at $r = 12, 13, 14$ are therefore E_7, E_8, E_9. At twelve their possible rank-five remainders cost more than $23/70$, $.09 + 17/70$, or $5/14$, respectively. At thirteen all rank-four remainders cost $> 2/7$, or at least $.09 + .2$ in a typed case. At fourteen the cheapest rank-three possibilities are ordinary $> 17/70$, $I + T_2 > .21$, $I + 2A_1 > .29$, and typed $> .09 + 2/15$. Each closes with its stated prefix floor. Thus no B, D component of ranks twelve through fourteen remains.

We must extend the short tails at their actual ranks before counting the proper higher rows. For $14 \leq r \leq 17$, the B_3 finite list has only the extra short-three F_4 role; it is sealed against foreign contacts and is affine or directly strong. At the B_2 site $i = r - 2$, the extra own-three/short-three profile is excluded by Lemma 6. Own zero, four, six and parallel retain their affine or null principals; native own at least five is directly strong. A bridge through the short endpoint is a full B_r with internal new root and admits no finite foreign arm. Its residual is $1/(2i)$ and its ratio at least $1/\sqrt{2i} + \sqrt{i}/2 > 100/61$ for $i \geq 11$. All the other classified weak roles are excluded by Lemma 5. Hence $6, \ldots, r - 2$ are strong, so $e_{B_r} > E_{r-2}$ in either initial-five branch. At B_{15}, $E_{13} > .88989$ excludes ordinary $T_2, 2A_1, J, I + A_1$, leaving either typed defect. At B_{16}, $E_{14} > .9084$ excludes ordinary A_1 and leaves actual I. Whole B_{17} is retained.

For D_{15} the spectator rank is exactly two, and every allowed remainder costs $> .09$. Its bulk already gives strong $6, \ldots, 10$. The local D_4, D_3 lists at indices eleven and twelve retain only an additional own-five/fork-three case. More explicitly, own zero, three, four, six and parallel have the same affine or null principals; middle-three, end-four and the sealed H_4 end-five are directly strong. The full bridge is D_{15}; at D_3 the end-three bridge is A_{15} with both arms at least three, so no finite foreign arm is possible. Its residual $1/8 + 1/(2i) \leq 1/6$ and numerator $> .75$ make it strong; its zero-own cycle is affine. The remaining fork helper is golden by Lemma 5.

For that golden role the actual positive left H_4 projection has

$$\alpha^2 = \varphi + 5/8 > 56/25, \quad \delta = (5 - 3\varphi)/8, \quad \sqrt{\delta}/\alpha < .091, \quad A^2 = i\alpha^2 - \kappa,$$

where $\kappa = 1/2$ at D_4, $5/8$ at D_3. The positive correction is $\gamma(3n_i/2 + n_{i-1} + n_{i-2}/2)$. The finite projection is future and feasible against all original walls; the correction alone is not used as a hyperbolic point. Its displacement divided by α is flat one on the first $i - 3$ coordinates, has square $< i$, and gives pressure $G > \sqrt{i - \kappa/\alpha^2} - .091$. Pay weight $1/6$ at the previous strong knot $K = i - 3$. Both initial-five references have endpoint coefficient $> 1/6$, so a single pure-D square gives

$$e_D > E_K + (G - f_K)/3 - (i - K)/36.$$

The complete two-row computation is

	i	K	$G >$	$f_K <$	$e_D >$
D_4	11	8	3.19	2.398	.9493
D_3	12	9	3.33	2.569	.9683.

The radical seals are $3.281^2 < 11 - 25/112$ and $3.421^2 < 12 - 125/448$. The pure energies each exceed .91; adding only the disjoint actual spectator energy $> .09$ excludes both roles. Thus $6, \ldots, 12$ are strong and $e_{D_{15}} > E_{12} > .86989$. Ordinary $T_2, 2A_1, J, I + A_1$ close; either typed defect is retained. At D_{16} its ordinary A_1 and actual I are both retained, and whole D_{17} is retained. These are groups12–14.

Every partition, actual defect and permitted dominant rank has now been considered. No root has been deleted from another ambient list, and every sum concerns the same original vector. □

Proposition 8 (The other rows in rank seventeen)**.** *Every row of Table 1 except group8 is impossible.*

Proof. The complete dispatch in Section N.2 excludes groups1–7 and9–14. Each entry refers to its printed actual-wall, field and individual-payment proof. The shared algebraic census for whole B17 and D17 has separate actual-domain inclusions and separate original-vector consumers. Group8 is treated next. □

Proposition 9 (Actual chain endpoints)**.** *If group8 occurs, both actual endpoint opposites are isolated within A_{17}. Their integral parameters U, V generate the same actual field and satisfy*

$$36/17 < U, V < 3969/425, \qquad 0 < U^\sigma, V^\sigma < 36/17 \quad (\sigma \neq \mathrm{id}). \tag{7}$$

The field is nongolden, has degree at most five, and at least one endpoint is finite. In fact

$$K \in \{\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), K_7, K_9\};$$

in the cubic fields a finite endpoint has label seven or eighteen, respectively.

Appendix C proves the complete endpoint, field and domain-inclusion sequence of AE, AD, AF. It retains every finite-white and own case, including the golden time-compatible exception. One decisive step explains the simultaneous accounting. The common body has squared norm .9037838. Both dotted cores are paid at weight .3; the outer coefficients change from .1382 to .0782 and still fund every central foot. On the entire $0 < u < 1$ interval the extra contribution is

$$.6\{\sqrt{17/9 - u} - \sqrt{u(1-u)} - .4174\} - .09(8/5 - 2u + u^2) > .04969.$$

The complete circle bounds in Appendix C.4 give $.9037838 + 2(.04969) > 1$ if both ends are dotted. No unavailable strong-five floor is used. The complete finite alphabet, including label30 before its conjugate test, then gives the stated fields, rather than a prescribed field list.

In raw old-root normalization of square two, a cap has square $2U$ and time square $F(U)/18$, where $F(X) = X(17X - 36)$. The inverse endpoint entry is $1/18$. Every actual pair consequently satisfies

$$R = F(U)F(V) \in K^2, \qquad g = (UV - \sqrt{R})/18 \in \mathcal{O}_K, \qquad L = g^2/(UV) \in \mathcal{O}_K. \tag{8}$$

The root is physically positive. Acuteness gives $g \leq 0$. If $g \neq 0$, nonphysical positive definiteness gives $0 < L^\sigma < 4$. The branch $g = L = 0$ is retained. Physical owns and mutual parameters must be native or dotted; four would be a compactness-forbidden parallel pair. The endpoints need not be equal.

3 Finite propositions and their complete algorithms

The computations check necessary algebraic domains. An empty domain excludes an actual configuration after geometric inclusion has been proved. A nonempty domain supplies necessary pairs, followed by a separate geometric contradiction, not examples.

3.1 Full quadratic integer rings

For squarefree $d > 1$ use $\omega = (1 + \sqrt{d})/2$ if $d \equiv 1 \pmod 4$, otherwise $\omega = \sqrt{d}$. Let Δ be the fundamental discriminant. If $Y = a + b\omega$ generates K, $Y > Y' > 0$, $Y < B$, then

$$0 < b\sqrt{\Delta} = Y - Y' < B.$$

Thus $\Delta < B^2$, and b and the strip-permitted integers a have explicit finite bounds. Squarefree trial division and these integer intervals cover the full maximal order.

Use rational pairs $a + b\sqrt{d}$. Integrality in the half basis means $2b, a - b \in \mathbb{Z}$; otherwise both coefficients are integral. Signs use exact rational square comparisons. For $b \neq 0$, a field square requires norm n^2 with rational $n \geq 0$ and one of

$$x^2 = (a + n)/2, \quad x^2 = (a - n)/2$$

a nonzero rational square. Set $y = b/(2x)$ and verify the full square identity. Both signed norm branches are necessary and sufficient. For $b = 0$ test rational squares and d times rational squares, including zero. A square norm alone is insufficient.

The complete positive quadratic finite native alphabet is

$$1, 2, 3, (3 + \sqrt{5})/2, (5 + \sqrt{5})/2, 2 + \sqrt{2}, 2 + \sqrt{3}.$$

Indeed $\phi(m) \leq 4$ leaves $m = 3, 4, 5, 6, 8, 10, 12$. Zero is orthogonality, values above four are dotted, and each radical value is retained only in its actual field.

Proposition 10 (Recorded finite outputs)**.** *At the indicated complete domains the exact outputs are as follows.*

1. *The quadratic A_{17} domain (7)–(8) has no survivor. There are 25 discriminants $\Delta < 88$, 53 models and 159 ordered pairs: 106 fail field-square membership, 52 long-cycle integrality and one two-cycle integrality.*

2. *The two cubic A_{17} finite-anchor domains have the nine strip models of Table 2, all failing a necessary test.*

3. *The quadratic domain*

$$4 < U < 11,\ 0 < U' < 2, \quad 2 < W < 676/17,\ 0 < W' < 4/17,$$

$K = \mathbb{Q}(U) = \mathbb{Q}(W)$,

$$g = UW - \sqrt{U(U-2)W(17W-4)}, \quad g,\ L = g^2/(2UW) \in \mathcal{O}_K, \quad 0 < L' < 4,$$

with physical native/dotted mutual parameter, has 36 fields $\Delta < 121$ and 120 pairs. Its only survivors are

K	U	W
$\mathbb{Q}(\sqrt{13})$	$(5 + \sqrt{13})/2$	$11 + 3\sqrt{13}$
$\mathbb{Q}(\sqrt{17})$	$(7 + \sqrt{17})/2$	$(21 + 5\sqrt{17})/2$.

4. *The quadratic whole-B_{16} domain*

$$2 < U < 12,\ 0 < U' < 2, \quad 4 < V < 64,\ 0 < V' < 1/4, \quad UV/2 \in \mathcal{O}_K,$$

$$k^2 = \frac{(U-2)V}{4U(4V-1)}, \quad k > 0, \quad G = 4Uk(4V-1) - UV, \quad L = G^2/(2UV),$$

with $k \in K$, integral G, L, actual acuteness and the following exact predicates, has 43 fields $\Delta < 144$: $G \geq 0$; if $G = 0$ then $L = L' = 0$, whereas otherwise $0 < L' < 4$; and U, L must

belong to the native finite alphabet of their actual field or be greater than four. In particular $L = 4$ is excluded, and no continuous nonnative value below four is allowed. The identity

$$L + G = ((15U - 32)/2)V - 2U + 4$$

is checked for the physical positive square root k. The complete domain has 100 head models, 85 short models and 273 pairs. Exactly 168 fail half-product integrality and 104 field-square membership. The sole survivor is $U = 5 + 2\sqrt{6}$, $V = 10 + 4\sqrt{6}$.

The exact contracts and actual executions are QA17, CA17, QBD17, QB16. The quadratic algorithm generates all fields below the proved discriminant bound, lists all strip-permitted integer coefficients, and visits every ordered pair. It applies full field-square, cycle-integrality, conjugate-positivity and native tests, recording every rejection or survivor and complete totals. For (i), $b \le 4$ and the conjugate width 36/17 requires *three* candidates $a = \lfloor -b\omega' \rfloor + 1, +2, +3$. There are at most 3600 pair tests before the strict strips. For (iii), $b_U \le 4$, $b_W \le 17$, with two a_U values and at most one a_W value: at most 4896 pairs. For (iv), safe full boxes are $1 \le a_U \le 7$, $1 \le b_U \le 5$ and $1 \le a_V \le 32$, $1 \le b_V \le 28$. These bounds prove termination. The full-ring strip proof gives coverage and the exact tests justify every rejection. The floors themselves are exact integer operations:

$$\lfloor -b\omega' \rfloor = \begin{cases} \operatorname{isqrt}(b^2 d), & \omega = \sqrt{d}, \\ (\operatorname{isqrt}(b^2 d) - b) \operatorname{div} 2, & \omega = (1 + \sqrt{d})/2. \end{cases}$$

Irrationality places $b\sqrt{d}$ strictly between consecutive integers, which proves both formulas, including even b. Trial square division through $\operatorname{isqrt}(d)$ detects squarefree radicands exhaustively. Fundamental discriminants below the proved bound require no larger radicand loop.

The A17 quadratic and B16 conjunctions replay every pair with the same exact implementation. The B/D17 conjunction has narrower scope: it rechecks every survivor and all fields/counters, but does not recompute rejected tests. Their coverage uses actual source-bound batch runs. None is an independent second implementation.

3.2 The two cubic finite anchors

Let U be the largest root of

$$P_7 = X^3 - 5X^2 + 6X - 1, \qquad P_{18} = X^3 - 6X^2 + 9X - 3.$$

The full integer rings are $\mathbb{Z}[U]$. Polynomial discriminants are 49 and 81. Translating the first by four gives Eisenstein at seven; the second is Eisenstein at three. At the only possible index prime, an element $(A + B\pi + C\pi^2)/p$ with a coefficient not divisible by p has a unique least numerator valuation among the distinct fractional parts $0, 1/3, 2/3$; division by p makes it nonintegral. Thus no denominator is missing, including at the wild prime three. In the P_7 field the complete physical finite alphabet for the other own and the mutual parameter is

$$0, 1, 2, 3, U, U^2 - 3U + 3;$$

in the P_{18} field it is

$$0, 1, 2, 3, U, U^2 - 4U + 4.$$

The last entries are native fourteen and nine, respectively. Indeed a finite label requires $\phi(m)/2$ to divide three, leaving the rational labels and $7, 9, 14, 18$. The two cubic fields have different

discriminants, and contain no quadratic subfield. The double-angle identity gives the displayed polynomial expressions. Values greater than four are dotted, four is forbidden parallel, and zero is retained.

Write $V = a + bU + cU^2$. Rational root brackets and Lagrange interpolation of the three embedding values give the complete boxes

	a	b	c	trials
P_7	$[-4,4]$	$[-9,9]$	$[-3,3]$	1197
P_{18}	$[-6,6]$	$[-9,9]$	$[-2,2]$	1235.

Here is the coefficient proof, before any enumeration. The three roots, in increasing order, have disjoint rational brackets

P_7	$(.19,.20)$	$(1.55,1.56)$	$(3.24,3.25)$
P_{18}	$(.46,.47)$	$(1.65,1.66)$	$(3.87,3.88)$.

Multiplying the six endpoint values by 10^6 gives respectively

$$\begin{aligned} P_7: &\quad -33641,8000;\ 11375,-11584;\ -35776,15625,\\ P_{18}: &\quad -32264,8423;\ 7125,-19304;\ -70797,4672. \end{aligned}$$

The three sign changes exhaust the roots of each cubic. If r_i is one of them, the denominator in its Lagrange basis polynomial is $P'(r_i) = \prod_{j\neq i}(r_i - r_j)$. The bracket gaps give absolute lower bounds, in physical, middle, small order,

$$(5.1072, 2.268, 4.104), \qquad (7.514, 2.6078, 4.012).$$

Use $0 < V_{\text{phys}} < 10$ and $0 < V_i < 9/4$ at the other roots. The coefficient of X^2 in the interpolant is $\sum_i V_i/P'(r_i)$, that of X is $-\sum_i V_i \sum_{j\neq i} r_j/P'(r_i)$, and the constant is $\sum_i V_i \prod_{j\neq i} r_j/P'(r_i)$. Triangle inequalities give the following strict upper bounds on the three absolute terms:

	physical	middle	small	sum below
P_7, c	2	1	.55	4
P_7, b	3.45	3.43	2.64	10
P_7, a	.62	.65	2.78	5
P_{18}, c	1.34	.87	.57	3
P_{18}, b	2.84	3.76	3.11	10
P_{18}, a	1.04	1.58	3.62	7.

Each entry follows by positive rational multiplication with the denominator bound. For example the last constant terms are below $(9/4)3.25 \cdot 1.56/4.104 < 2.78$ and $(9/4)3.88 \cdot 1.66/4.012 < 3.62$. Since a, b, c are integers, these strict estimates give exactly the safe boxes above. Negative coefficients have not been discarded, and the boxes were obtained without looking at the successful strip models.

All field operations reduce modulo the cubic. If $s^2 = R$ with R integral, s is integral. Rational R is handled separately: an odd-degree field contains its square root only if that root is rational. Otherwise put

$$A = \operatorname{Tr} R,\ B = ((\operatorname{Tr} R)^2 - \operatorname{Tr}(R^2))/2, \quad t = \operatorname{Tr} s,\ n = \operatorname{N} s.$$

Try both signed integers n with $n^2 = \operatorname{N} R$ and all $-454 \le t \le 454$; require

$$z = (t^2 - A)/2 \in \mathbb{Z}, \quad z^2 - 2nt = B, \qquad s = \frac{tR + n}{R + z}.$$

The identity follows from the characteristic polynomial of s; its denominator can vanish only for rational R. Verify $s^2 = R$ and its physical positive sign. The trace range is complete: the finite anchor has $U < 4$ and $V < 10$, so $F(U) < 128$, $F(V) < 1340$ and the physical root has magnitude below 415. Each other root magnitude is below $324/17 < 20$. Hence $|\operatorname{Tr} s| < 455$. Exact interval refinement terminates since no nonzero polynomial of degree at most two vanishes at a cubic generator. The zero element is first recognized by its three zero coefficients. For every other element, evaluate its degree-at-most-two representative on each separate isolating interval and bisect that interval until its sign is fixed. The interval width tends to zero, proving that every sign decision halts at the intended embedding.

Field	(a, b, c)	First failure
K_7	$(0, 1, 0)$	long cycle not integral
	$(1, -2, 1), (1, -1, 1), (2, -2, 1), (3, -5, 2)$	not a field square
K_9	$(0, 1, 0)$	long cycle not integral
	$(1, -2, 1), (2, -2, 1), (3, -3, 1)$	not a field square

Table 2: All nine cubic strip models.

All 2432 triples are visited. The recorded first failures are 1287 below the physical lower bound, 919 above its upper bound, 217 nonphysical-strip failures, seven field-square failures and two nonintegral long cycles. They sum to 2432. The separate combination replays every trial with the same implementation.

Corollary 11. *There is no compact arithmetic Coxeter polytope of dimension seventeen.*

Proof. The complete cover and row exclusions leave whole A_{17}. Its actual endpoint reduction gives the quadratic or cubic fields above. Proposition 10(i) excludes the former. In a cubic field designate a finite endpoint first, reversing chain labels if necessary; this assumes no chamber symmetry. Every other endpoint lies in the complete box of (ii), which has no survivor. No farthest forest remains. □

4 The complete rank-sixteen reduction

A list in rank seventeen cannot be decremented. The fresh classification in rank sixteen F16, Sections 2–7, starts with all finite components and actual defect compatibility. Two dominant components each cost $> 1/2$. With none, the largest allowed rank-one debit still leaves ordinary cost $> 15/14$; the typed case costs $> .09 + 14/14$. The unique A dominant has rank at least ten and the B/D dominant rank at least eleven. Their small classical tails are checked at their actual ranks, including the direct D_{11} boundary. The D_{14} weak-tail payments give pure energies $> .92531$ or $> .94697$ before the disjoint spectator cost $> .09$ is added. Only then are the stronger tail floors used.

The conditioned $A_{11} > .652761$ component input used below is proved independently of either forest cover in Corollary 21. Its proof reconstructs the no-helper reference directly; the weaker uniform 13/20 theorem would not imply that value.

Proposition 12 (Necessary cover in rank sixteen)**.** *The whole forest at the selected original farthest vertex is one of Table 3.*

Table 3: Necessary groups in rank sixteen.

Group	Component	Complete spectators
1	A_{10}	$I + (A_5$ or $D_5)$
2	A_{10}	N_7/O_7 plus A_4, B_4, D_4, F_4
3	A_{11}	$A_5, D_5, (A_4$ or $H_4) + A_1, A_3 + T_2, 2T_2 + A_1$
4	A_{11}	$I + T_4$
5	A_{11}	$O_7 + A_3$
6	A_{12}	A_4, H_4 or $2T_2$
7	A_{12}	$I + T_3$
8	A_{12}	N_7/O_7 plus T_2
9	A_{13}	$A_3,\ T_2 + A_1,\ I + T_2,\ O_7 + A_1$
10	A_{14}	$T_2,\ 2A_1,\ I + A_1,\ N_7/O_7$
11	A_{15}	ordinary A_1 or actual I
12	A_{16}	none
13	B_{14}	N_7/O_7
14	B_{15}	ordinary A_1 or actual I
15	B_{16}	none
16	D_{13}	$I + T_2$ or $O_7 + A_1$
17	D_{14}	ordinary $I_2(m)$, $m \geq 4$, or N_7/O_7
18	D_{15}	ordinary A_1 or actual I
19	D_{16}	none

Proof of Proposition 12. We repeat the rank count from its beginning. If no dominant occurs, the possible I debit leaves fifteen ordinary ranks, costing $> 15/14$; a typed defect gives $> .09 + 14/14$. A J pair is more expensive than the two rank-over-fourteen baseline costs it replaces. Two dominants again each cost $> 1/2$. Thus there is exactly one. For A_7 the nine remaining ranks are too expensive. For A_8, A_9, even I leaves at least six ordinary ranks, giving $4/7 + 6/14 = 1$, strictly exceeded; the typed cost is at least $4/7 + 5/14 + .09 > 1$. Therefore $10 \leq r \leq 16$. For B, D, the $9/14$ rate leaves at most five spectator ranks, so $11 \leq r \leq 16$. All exceptional finite types were already included in the ordinary table.

At A_{10}, ordinary or J rank six closes with $4/7$. With I, enumerate the seven rank-five partitions displayed in the rank seventeen proof. In $4 + 1$ use $I + A_1 > .20204$ plus $T_4 > 2/7$; in $3 + 2$ use $I + T_3 > 2/7$ plus $T_2 > 1/7$; in $2 + 2 + 1$ use $I + A_1 > .20204$ plus $2/7$. The more reducible partitions are stronger. Only $I + A_5, I + D_5$ remain. With a typed defect, the rank-four partitions are $T_4, T_3 + A_1, 2T_2, T_2 + 2A_1, 4A_1$. The propagated costs exclude all but T_4, and actual K_7 excludes H_4. These give groups1 and2.

At $A_{11} > 4/7$, the complete ordinary rank-five list below the remaining $3/7$ budget is

$$A_5, D_5, (A_4 \text{ or } H_4) + A_1, A_3 + T_2, 2T_2 + A_1.$$

The omitted B_5 costs $> 3/7$; $B_4, D_4, F_4 + A_1 > 13/30$; $B_3, H_3 + T_2 > 3/7$; and the other three reducible partitions have costs $> .45$, $31/70$ and $.5$. This is group3. The J cost $.34 + 17/70$ also closes. For actual I use the spectator-safe conditioned component estimate $e_{A_{11}} > .652761$. Of the five rank-four partitions only T_4 remains: $I + T_3 + A_1$ costs $> 2/7 + .1$, $I + 2T_2 > .21 + 1/7$, and the other reducible cases are stronger. Both resulting sums with .652761 exceed one. Thus group4 retains all five irreducible T_4 types.

Under a typed defect the same component floor .652761 holds. At remaining ordinary rank three, $T_2 + A_1 > .2 + 2/15$ and $3A_1 > 2/5$ close. So do $B_3, H_3 > 2/7$; H_3 is also incompatible with K_7. Only A_3 is possible. For N_7 its two normalized endpoint duals have correlation $1/3$ and heights $> 1/\sqrt{5}$. Cauchy on their sum gives

$$e_{A_3} > \frac{4/5}{2 + 2/3} = 3/10,$$

which closes with $.652761 + .09$. Thus only $O_7 + A_3$ remains, giving group5.

At A_{12} use its spectator-safe floor $269/400 = .6725$. Ordinary rank four leaves only $A_4, H_4, 2T_2$: $B_4, D_4, F_4 > 1/3$, $T_3 + A_1 > .35$, $T_2 + 2A_1 > 12/35$, and $4A_1 > .4$. The J alternative is stronger. With I, rank three leaves only $I + T_3$, since $.6725 + .20204 + 1/7 > 1$ and $3A_1$ is stronger. Under a typed defect the component floor is .703101; two ordinary A_1 cost $> 4/15$ and close, whereas a single $T_2 > .2$ gives only .993101 and is honestly retained. These are groups6–8.

At $A_{13} > 5/7$, ordinary B_3, H_3, three A_1, and $J + A_1$ close. The retained ordinary alternatives are $A_3, T_2 + A_1$. With I, only $I + T_2$ remains since $I + 2A_1 > .30204$. With N_7, an $A_1 > .2$ closes; with O_7, the weaker $2/15$ does not. This is group9. At $A_{14} > .76$, the rank-two list is ordinary $T_2, 2A_1, I + A_1$, or either typed defect; J closes. At A_{15} its single spectator is ordinary A_1 or actual I; whole A_{16} has none. This gives groups10–12, with no endpoint or field-degree conclusion smuggled into the cover.

We now check the smaller classical ranks. For B_r and D_r with $r \geq 12$, the legal generic prefix and Lemma 5 give strong $6, \ldots, r-4$ and $6, \ldots, r-5$, respectively. At D_{11} the generic parameter $M = 5$ is not allowed. Instead inspect its actual deletion sites five and six: the retained pieces are $A_4 + D_6$ and $A_5 + D_5$. The connected white rank eleven exceeds eight, so the ordinary bulk bridge list still applies. Its D_5, D_6 fork-to-E_6, E_7 contacts remain included and strong; the foreign-far bounds use only indices five and six, within their geometric range $5 \leq i \leq 29$. The left finite-five projection uses the first six long roots, all present. Thus the only weak-five/six profile is again the literal far-three helper, and Lemma 5 excludes weak six. This proves strong six without applying the generic theorem outside its rank hypotheses.

Both initial-five branches therefore give the floors (6), beginning with E_6 at D_{11}. At classical rank eleven the cheapest rank-five remainders cost $> 23/70$, $.09 + 17/70$ or $5/14$, all closing with .69685. At twelve all rank-four remainders cost $> 2/7$; a typed defect plus T_2 costs $> .29$. These close with $E_7 > .73531$. At thirteen ordinary or J rank three closes with $E_8 > .76864$. Only $I + T_2$ or $O_7 + A_1$ can remain; $I + 2A_1$ and $N_7 + A_1$ are more expensive. In B_{13} the better actual floor $E_9 > .79805$ closes both retained cases. In D_{13} they are group16.

For B_{14} the actual B_3 site is eleven and the B_2 site twelve. The former's extra short-three extension is sealed F_4; own zero is affine and positive own has residual $1/4$ and ratio at least $1 + 1/\sqrt{2}$, hence is strong. At the latter the only extra weak role is excluded by Lemma 6. Full bridges have residual $1/(2i)$ and ratio $1/\sqrt{2i} + \sqrt{i}/2 > 100/61$; they admit no foreign arm at their internal new root. The zero/four/six/null cases and the directly strong native owns are those already listed in the rank seventeen proof. Thus $6, \ldots, 12$ are strong, giving $e_{B_{14}} > E_{12} > .86989$. Ordinary T_2, two A_1, $I + A_1$ and J close; either typed defect is retained as group13. The same local sites at B_{15}, B_{16} give strong through $r - 2$, but leave their stated groups14 and15.

For D_{14} the spectator rank is two and always costs $> .09$. Its bulk already gives strong $6, \ldots, 9$ independently of the new tail sites. At site ten the right D_4 has long-leaf-three and fork-three extensions. Their scalars differ: the former has $\sigma = \gamma + 1/2$; the latter has $\delta = 1/2 - Q$, $\sigma = \gamma + 1/4$. At site eleven, $D_3 = A_3$ has middle-three, end-three, end-four and end-five extensions. Middle-three and end-four are affine or directly strong; the sealed H_4 end-five is strong even at own zero. The end-three full bridge is A_{14} with both arms at least three, so admits no finite foreign arm. Its

residual is at most 15/88, and its positive-own numerator $> .75$ makes it strong; own zero is the affine cycle. Full fork bridges are D_{14} and are similarly sealed. All other positive foreign, isolated and finite-own roles are the generic companion list at these actual local ranks.

The only extra weak cases are therefore the literal fork helpers; Lemma 5 leaves their golden owns. Use their actual positive H_4 projections with the same α, δ, κ as in the rank seventeen proof. For $K = i - 3$ the previous strong knot has available coefficient $> 1/6$ in both initial-five branches. One pure-D reference gives

$$e_D > E_K + (G - f_K)/3 - (i - K)/36.$$

Here the complete new rows are

	i	K	$G >$	$f_K <$	$e_D >$
D_4	10	7	3.035	2.215	.92531
D_3	11	8	3.183	2.398	.94697.

The seals are $3.126^2 < 10 - 25/112$ and $3.274^2 < 11 - 125/448$. Each pure bound plus the disjoint actual spectator cost $> .09$ exceeds one. Hence both weak roles are impossible, and $6, \dots, 11$ are strong. This gives $e_{D_{14}} > E_{11} > .84816$. Two A_1, $I + A_1$, and J close. Ordinary A_2 also closes, since $.84816 + 319/2000 = 1.00766$. Precisely an ordinary $I_2(m)$, $m \geq 4$, or either typed defect remains as group17. At D_{15} the single ordinary A_1 or actual I is retained, and whole D_{16} has none.

All small types, ranks, integer partitions and actual defect choices were included before this final table. In particular no rank seventeen forest was decremented and no farthest property was transferred to a face. □

4.1 A small component calculation

Lemma 13. *At the original farthest vertex of any compact Coxeter polytope, a whole A_3 component has energy $> 13/50$, with arbitrary finite spectators. Arithmeticity is not needed.*

Proof. The inverse unit Gram and normalized old duals are

$$C = \begin{pmatrix} 3/2 & 1 & 1/2 \\ 1 & 2 & 1 \\ 1/2 & 1 & 3/2 \end{pmatrix}, \qquad d_i = C_{ii}, \quad u_i = \omega_i/\sqrt{d_i}.$$

Suppose $e = \|x_{A_3}\|^2 \leq 13/50$. Put $\tau_i = \tanh(L_i/2) \leq x \cdot u_i$ and $H(t) = 2t/(1 + t^2)$. The logarithm of H is concave on $(0, 1)$: its second derivative is $-1/t^2 - 2(1 - t^2)/(1 + t^2)^2 < 0$. For a pair of normalized duals with correlation k, Cauchy gives $\tau_i + \tau_j \leq \sqrt{2e(1 + k)}$; hence

$$\tanh L_i \tanh L_j \leq H\big(\sqrt{e(1 + k)/2}\big)^2.$$

At an adjacent A_3 pair $k = 1/\sqrt{3}$, the right side is at most $5200(1 + k)/(113 + 13k)^2 < k$. The cleared positive difference is $(22876k - 12662)/3 > 0$, using $k > 4/7$. The companion's actual distinct-opposite tangent-product inequality requires the opposite inequality if the two walls are distinct. Thus adjacent directions share their actual opposite, and connectivity gives one wall c closing all three edges.

The individual bound is

$$\tanh^2 L_i \leq \frac{4e}{(1 + e)^2} \leq r := 2600/3969.$$

Write $c = Ap - v_A - v_S$, $h = A^2 > 0$, $v_A = \sum \beta_i \omega_i$, and $\beta_i = -(c, n_i) \geq 0$. Put $q = \beta^T C \beta$, $Q = \|v_S\|^2$, so $h = q + Q - 1$. For $v_i = (C\beta)_i / \sqrt{d_i} > 0$ we have

$$q - 1 \leq h \leq r \min_i v_i^2. \tag{9}$$

The companion's reciprocal-shadow and orthogonal-opposite lemmas say that every positive whole spectator contribution is at least $1/4$ and strictly below h. In particular $h < 1/4$ forces $Q = 0$.

Two adjacent positive β_i would give, at the finite endpoint of the third edge, a white triangle with all three contacts at least $1/2$. Its Gram is nonpositive, affine at equality, impossible. Thus there are at most two contacts, and two can only be at endpoints. For endpoint values $a, b > 0$,

$$q = \tfrac{3}{2}(a^2 + b^2) + ab, \qquad v_2^2 = (a + b)^2/2.$$

If both are $1/2$, the original principal is affine with a nonzero future feasible null sum. Otherwise the native gap gives $a, b \geq 1/2$ and one at least $1/\sqrt{2}$. Consequently

$$q - r v_2^2 \geq \frac{3(3 - r)}{8} + \frac{(1 - r)\sqrt{2}}{4} = \frac{27921 + 2738\sqrt{2}}{31752} > 1,$$

using $\sqrt{2} > 7/5$. This contradicts (9). There is at least one contact, since each $v_i > 0$.

For one endpoint contact β, $q = 3\beta^2/2$ and $\min v_i^2 = q/9$. Equation (9) gives

$$q \leq 35721/33121, \quad \beta^2 \leq 23814/33121 < 3/4, \quad h \leq 2600/33121 < 1/4.$$

Only native three, four or five remains, and each has $q < 1$. For five use $\cos^2(\pi/5) < 2/3$, following from $\sqrt{5} < 7/3$. Then $Q = 0$ and $h = q - 1 < 0$, impossible.

For one middle contact, $q = 2\beta^2$, $\min v_i^2 = q/3$. Thus $q \leq 11907/9307$ and $\beta^2 \leq 11907/18614 < 16/25 < \cos^2(\pi/5)$. Only native three or four remains, so first conclude $q \leq 1$, and only then strengthen the height bound to $h \leq r/3 = 2600/11907 < 1/4$. Again $Q = 0$ and $h = q - 1 \leq 0$ is impossible. All contact and foreign-spectator alternatives have been exhausted. □

Proposition 14 (All rows except whole D_{16})**.** *Groups1–18 are impossible at that same selected original vertex.*

Proof. Groups1–18 are excluded by the complete printed dispatch in Section N.3. Its separate actual rank-one branches exhaust the same opposite wall, including the boundary $h = 1/2$. Every overlapping endpoint/head/foot pressure is paid within its local proof before disjoint whole-component costs are added. The universal $A_{12} > 5/7$ estimate supplies both ordinary and typed consumers. Only group19, whole D_{16}, remains for the next section. □

5 The last actual diagram

Proposition 15 (Whole-D_{16} reduction)**.** *If group19 occurs, the actual field is quadratic. The actual long head and both genuine signed forks are isolated and dotted. There are integral generators $U, F, G \in \mathcal{O}_K$ satisfying*

$$4 < U < 98/9, \quad 0 < U' < 2, \qquad 4 < F, G < 121/2, \quad 0 < F', G' < 1/2, \tag{10}$$

$$UF/4,\ UG/4,\ FG/4 \in \mathcal{O}_K. \tag{11}$$

Their raw time squares are

$$T_H = U(U - 2), \qquad T_F = 2F(2F - 1), \qquad T_G = 2G(2G - 1), \tag{12}$$

and every pair of time products is a square in K.

We give its field and endpoint reduction below, after specifying the initial geometric inventory and two scalar estimates. The inventory must be proved at actual rank sixteen; it is not supplied by a higher-rank head theorem or by the dyadic calculation.

The original coordinates satisfy

$$x_1 > \cdots > x_{15} > |x_{16}|, \quad P_j = e_1 + \cdots + e_j, \quad S_j = x \cdot P_j, \quad f_j = \sqrt{j} - .61/\sqrt{2}.$$

The same reduction proves strong5 through14 and one positive prefix reference with

$$E_{14} > .9084, \quad b < .361, \quad \beta_5 > .14, \quad \beta_{14} > .13, \quad f_{14} < 3.311, \qquad S_6 > 2.01815. \tag{13}$$

Here b is its first-five mean. These coefficients are spent individually below. The head raw root has square $2U$, spatial part Ue_1; the fork raw roots have squares $2F, 2G$ and spatial parts $(F/2)(P_{15} + e_{16})$, $(G/2)(P_{15} - e_{16})$. They remain two distinct directions even if $F = G$.

The remainder of this exclusion is given in full. No D16 census is used. Every original cycle remains mandatory even when only necessary consequences of it are needed.

Lemma 16 (Initial whole-D_{16} inventory)**.** *At the same original whole-D_{16} vertex, knots five through fourteen are .61-strong, and the positive reference has* (13)*. The actual long head is isolated, with finite own label at least seven or dotted own. Every actual fork has edge cosh less than twelve and is either isolated dotted or has exactly one simple-three contact at an endpoint of its retained A_{15} path, with positive own contact. Own zero and parallel fork roles are excluded. If both forks are isolated, the actual field has degree two or three.*

The proof below first establishes the actual-rank coarse bounds and head isolation, then closes the late spine and fork inventories. This order keeps every pressure available before it is spent.

Lemma 17 (Two exact terminal estimates)**.** *For $0 < u < 1$, put*

$$H(u) = \frac{\sqrt{2-u} - \sqrt{u(1-u)}}{\sqrt{2}}, \qquad R(u) = 1 - u + u^2/2.$$

Then $.8H(u) - .16R(u) > .295$. Also a degree-three algebraic integer with conjugates $A, a, b > 0$, $a, b < 1/2$ and norm eight has $A > 42$.

Proof. For the first assertion let $c = .681$, $t = 2 - c^2 = 1.536239$. The radical difference is positive, and $t - u^2 > 0$. It suffices to prove

$$P(u) = (t - u^2)^2 - 4u(1-u)(2-u) > 0.$$

With $r = u - .7$ the exact expansion is

$$P = r^2(r - .6)^2 + 3.107522r^2 - .0094692r + .002616045121.$$

The last three terms exceed $3r^2 - .01|r| + .0026$. As $.01|r| \le r^2 + .000025$, this is positive. Thus $H(u) > c/\sqrt{2} > .4815351$ on the entire interval.

Moreover R decreases, and H decreases on $(0, .65]$. Below $1/2$ the derivative sign is immediate; between $1/2$ and $.65$ it follows from

$$\frac{2u-1}{\sqrt{u(1-u)}} \le \frac{.3}{\sqrt{.2275}} < \frac{1}{\sqrt{1.5}} \le \frac{1}{\sqrt{2-u}},$$

where $.09 \cdot 1.5 < .2275$ proves the strict inequality. The following rational radical seals give lower bounds for $H(v)$:

v	$\sqrt{2-v} >$	$\sqrt{v(1-v)} \le$	$H(v) >$
.25	1.3228	.4331	.62
.4	1.2649	.4899	.54
.5	1.2247	.5	.51
.55	1.2041	.4975	.499
.6	1.1832	.4899	.4902
.625	1.1726	.48413	.4867
.65	1.16189	.47697	.4842.

Square the first two columns and use $\sqrt{2} < 1.4143$ to obtain each last entry. Taking H at the right endpoint and R at the left endpoint gives the complete cover

$(0, .25]$	.336	$[.25, .4]$	.307
$[.4, .5]$	.2992	$[.5, .55]$	.2992
$[.55, .6]$	.29596	$[.6, .625]$	.29656
$[.625, .65]$	.29611	$[.65, 1)$	.29542808.

These are strict lower bounds for $.8H - .16R$; the last uses the global radical bound and $R(.65) = .56125$. Every endpoint is covered.

For the second assertion suppose $A \le 42$. Norm eight and $a, b < 1/2$ give $A > 32$ and integer trace $T \in \{33, \dots, 42\}$. Its minimal polynomial is $X^3 - TX^2 + BX - 8$, with integer B. Since $A > T - 1$, AM–GM gives

$$B \ge 2\sqrt{8A} + 8/A > 2\sqrt{8(T-1)} + 8/T.$$

Evaluating the polynomial at $1/2$, between both small roots and the large root, gives $B < T/2 + 63/4$. For even traces the largest permitted B, and the positive seal $32(T-1) - B^2$, are

T	34	36	38	40	42
$B_{\max}$	32	33	34	35	36
$32(T-1) - B_{\max}^2$	32	31	28	23	16.

They contradict the lower bound. At $T = 33$, $B_{\max} = 32$ and the same lower bound is already strictly larger than 32. At odd traces $35, 37, 39$, the permitted maxima are $33, 34, 35$; the exact integers

$$32(T-1)T^2 - (TB_{\max} - 8)^2 = 17191, 14588, 8087$$

are positive and give the same contradiction after squaring. Only $T = 41$ remains. The two bounds force $B = 36$, but $X^3 - 41X^2 + 36X - 8$ has discriminant -2704, incompatible with three real conjugates. Therefore $A > 42$. □

Proof of Lemma 16. All coordinates, caps and comparisons in this proof use the original whole-D_{16} vertex. Define its unit-root dual vectors by

$$\Omega_i = \sqrt{2}P_i \quad (1 \le i \le 14), \qquad \Omega_{15} = (P_{15} - e_{16})/\sqrt{2}, \quad \Omega_{16} = (P_{15} + e_{16})/\sqrt{2}, \qquad h_i = x \cdot \Omega_i.$$

Thus both genuine signed fork directions are retained.

The coordinate bound before any terminal classification. The legal rank sixteen bulk and Lemma 5 already give strong knots six through eleven. We first use only those knots. The quantities E_J are the explicit scalar sums defined above; in particular the identity

$$\Delta := E_{11} - E_6 = \sum_{j=6}^{10} (\sqrt{j+1} - \sqrt{j})^2 > .15106$$

is an analytic inequality, not an assertion that the actual vector already pays the strong-five reference. Here $E_{11} > .84816$ and

$$E_6 = 12.03721 - .122\sqrt{10} - 2\sqrt{30} < .6971;$$

the rational seals $\sqrt{10} > 3.16227$ and $\sqrt{30} > 5.47722$ prove the latter bound.

If $x_1 \geq .714$, use the positive reference through $(1, .714), (6, 2.01815)$ and then $f_7, \ldots, f_{11}$. Its initial slopes are $.714, .26083$, followed by a smaller positive slope and the ordinary decreasing increments. The first increment after six is larger than $\sqrt{7} - \sqrt{6}$, because $f_6 > 2.01815$. Its square is therefore greater than

$$.714^2 + 1.30415^2/5 + \Delta > 1.0010174445,$$

contrary to $\|x\|^2 < 1$. Thus $x_1 < .714$. Actual head comparison then gives $C_H < 77/25$, since $26/51 > .714^2$. Neither the coordinate nor the head bound has used strong five, a weak-five coordinate floor, or a head type.

The literal weak-five seal. At index five the generic finite-white list is the pure far helper $c = Ap - \gamma\Omega_5 - \Omega_1/2$ if the knot is weak. The first-six roots are genuine long roots, and the raw inverse entries are $5, 1, 1$. Thus $q = 2\gamma$ is integral, $202/125 < q < 12/5$, and every other conjugate satisfies

$$5q_\sigma^2 + 2q_\sigma - 1 < 0, \qquad -.69 < q_\sigma < .29.$$

Degree at least four has nonzero norm $< 2.4(.69)^3 < 1$. In degree three the norm is a unit. A positive small conjugate would give magnitude $< 2.4(.29)(.69) < 1$; both are therefore negative, say $-a, -b$, with $ab = 1/q$ and $a, b < .69$. Then $q > 1/(.69)^2$ and the integer trace lies strictly between zero and two, hence is one. Thus $a + b = q - 1 < 1.38$, $q < 2.38$, and the second integer coefficient $1/q - q(q-1)$ lies in $(-3, 0)$. The two cubics have discriminants $-44, -31$, impossible. In degree two the trace is one or two. Trace two gives norm in $(-.96, .621)$; trace one also forces $q < 1.69$, leaving only norm -1 and $q = \varphi$. The rational $q = 2$ is parallel. The actual connected principal is indefinite over $\mathbb{Q}(\varphi)$, so a weak five forces the actual field to be exactly that field.

It also contradicts the coordinate bound already proved. Here the selected old roots are n_5, n_1, n_2. Their correction $w = \gamma n_5 + (2n_1 + n_2)/3$ has positive coefficients and $w^2 = \gamma^2 + 1/3$; the full selected diagram, including c, is H_4. Thus $1 - w^2 = (5 - 3\varphi)/12$. The positive actual left-H_4 projection gives

$$x \cdot \{\gamma(\Omega_4 + \Omega_6)/2 + \Omega_3/6\} \geq \frac{A}{1 + \sqrt{\delta/(A^2 + \delta)}}, \quad A^2 = (4 + 7\varphi)/2, \quad \delta = (5 - 3\varphi)/12.$$

All coefficients in the finite projection are positive, so every unselected original wall remains feasible. Long-arm concavity gives $h_4 + h_6 \leq 2h_5$ and $h_3 \leq 3h_1$. Weakness of five therefore forces $x_1 > \sqrt{2}\{A - \sqrt{\delta} - \gamma(\sqrt{10} - .61)\} > .8$. For a rational seal use $A > 2.768$, $\sqrt{\delta} < .115$, $\gamma < .81$, $\sqrt{10} < 3.163$, and $\sqrt{2} > 1.4$. This contradicts $x_1 < .714$. Consequently five is strong before any head case is excluded.

The coarse fork bound after strong five. Knots five through eleven are now strong. Take their positive concave reference through $f_5, \dots, f_{11}$, with squared norm $E_{11} > .84816$, endpoint $f_{11} < 2.886$ and available P_{11} coefficient $\sqrt{11} - \sqrt{10} > .15$. For either actual fork put $T_s = x \cdot (P_{15} + se_{16})$. The single tail graft $.15(P_{15} + se_{16} - P_{11})$ has residual square five and zero cross with that reference. Hence

$$\|x\|^2 > .84816 + .3(T_s - 2.886) - .1125.$$

If its edge cosh were at least seventeen, then $T_s \geq 4\sqrt{8/9} > 3.77$, giving $1.00086 > 1$. Thus both actual fork coshes are below seventeen. This uses the strong-five reference only after its hypothesis has been proved, with no head or fork classification assumed.

The exact head and fork white lists. Releasing the long root n_1 retains whole D_{15}. Finite classification allows only an isolated new head or a simple-three contact at its long leaf n_2. A fork attachment would require a large exceptional diagram, an internal contact a second fork, and two contacts a cycle; weighted alternatives are not finite. For an isolated head, $c_H = A_H p - \gamma_H \Omega_1$ and $A_H^2 = 2\gamma_H^2 - 1$. Own zero, three and four have nonpositive time; own six has the actual affine $6, 3$ chain, and parallel has the positive null $c_H + n_1$. The remaining owns are five, finite at least seven, or dotted. With $U_H = 4\gamma_H^2$,

$$K = \mathbb{Q}(U_H), \quad 2 < U_H < 11858/625, \qquad 0 < (U_H)_\sigma < 2.$$

The reflection-defined raw cap has square $2U_H$, own product $-U_H$, spatial part $U_H e_1$ and nonzero time. Its indefinite principal with the rational old system proves actual-field descent and all the stated nonphysical inequalities.

For a nonisolated head write

$$c_H = A_H p - \gamma_H \Omega_1 - \Omega_2/2, \qquad t_H = 2\gamma_H, \quad A_H^2 = t_H(t_H + 2)/2, \quad C_H = t_H + 1.$$

The helper permits square-two normalization; its old path cycle makes t_H integral and gives $K = \mathbb{Q}(t_H)$. Strict nonphysical positivity gives $-2 < (t_H)_\sigma < 0$. Own zero has zero time; own three is an actual affine triangle; parallel is a feasible null.

For either fork f, with the other called k, deleting f retains the actual A_{15} path $n_1, \dots, n_{14}, n_k$. The complete finite one-node extensions are isolation, endpoint three, next-to-end three, or endpoint four. Two contacts make a cycle, and no exceptional finite diagram contains this path. There are no foreign components. For positive own put $t = 2\gamma$ at a white-three contact, $a = 2\sqrt{2}\gamma$ at white four. The following table gives UNIT time squares. For a white-three cap the helper permits a raw K-root of norm two and time square $2A^2$. For an isolated fork the reflection-defined raw norm is $2U_f$ and its time square is $2U_f(2U_f - 1)$. For a white-four cap use a raw K-root of norm four, with white product -2, own product $-a$ and time square $4A^2$. Indeed, if a K-root r makes angle $\pi/4$ with an old raw root α of norm two, then $(r, \alpha)^2 = (r, r)$; rescaling by $2/|(r, \alpha)|$ gives this norm-four representative without assuming $\sqrt{2} \in K$.

white contact	A^2	actual edge cosh
none	$2U_f - 1,\ U_f = 4\gamma^2$	$\sqrt{2U_f}$
3 at n_1	$(4t^2 + t - 1)/2$	$(8t + 1)/\sqrt{17}$
3 at n_2, n_{14}	$(4t^2 + jt + j - 2)/2$	$4t + j/2,\ j = 2, 14$
3 at n_k	$(4t^2 + 7t + 2)/2$	$(8t + 7)/\sqrt{17}$
4 at n_1	$(2a^2 + a)/2$	$4a + 1$
4 at n_k	$(2a^2 + 7a + 6)/2$	$4a + 7.$

The white-three normalization gives integral t and $K = \mathbb{Q}(t)$. White four gives integral a, integral $a^2/2$, and $K = \mathbb{Q}(a)$. These are original cycle statements. An isolated fork has $K = \mathbb{Q}(U_f)$, all other $0 < (U_f)_\sigma < 1/2$, and raw time square $2U_f(2U_f - 1)$. These follow from its full indefinite old-system/cap principal.

Own zero has negative or zero time in the isolated, far-three, far-next-three and far-four cases. Its remaining cases have the following positive affine nulls:

3 at n_{14}	$c + n_{13} + n_f + n_k + 2n_{14}$
3 at n_k	$2c + 4n_k + 6n_{14} + 3n_f + 5n_{13} + 4n_{12} + 3n_{11} + 2n_{10} + n_9$
4 at n_k	$\sqrt{2}c + 2n_k + 2n_{14} + n_f + n_{13}$.

Their products on the selected roots vanish. Their time is positive, and every other original wall has nonpositive product term by term. Parallel always gives $c + n_f$. This excludes own zero and parallel without a head assumption.

Every nonisolated dotted head is impossible. Set $s = t_H + 1 = C_H$. Physically $3 < s < 77/25$; all its other conjugates lie in $(-1, 1)$. The nonzero integral element $s^2(s^2 - 1)$ has physical value < 81 and other conjugate magnitudes at most $1/4$, excluding degree at least five by its norm. Degree one is excluded by Meyer. At degree two the integer trace is three or four, giving norms $s(3 - s) \in (-.2464, 0)$ or $s(4 - s) \in (2.8336, 3)$, both impossible.

At degree four its absolute norm is one, so $s, s - 1, s + 1$ are units. For the minimal polynomial $P = X^4 - TX^3 + BX^2 - CX + D$, $D = \pm 1$, the real strips force $P(1) = -1$, $P(-1) = 1$. Hence $B = -1 - D$, $C = 1 - T$. For $D = 1$, $T = s - (s^2 + s - 1)/(s^3 - s)$ lies strictly between two and three: the fraction is below one and above $1/s > 25/77$. For $D = -1$, $T = s + (s^2 - s - 1)/(s^3 - s)$ lies strictly between three and four, since its added fraction is in $(0, 1/2)$. Both contradict integrality.

At degree three write $P = X^3 - TX^2 + BX - C$ and put $a = -P(1) > 0$, $b = -P(-1) > 0$. The integer norm satisfies $C^2ab < 81/16 < 6$. For $|C| = 1$ the complete equal-parity pairs are $(1, 1), (1, 3), (3, 1), (1, 5), (5, 1), (2, 2)$; for $|C| = 2$ only $(1, 1)$ remains. Here $T = (a + b)/2 - C$, $B = (b - a)/2 - 1$. The complete seals are

C	(a, b)	(T, B)	seal
1	$(1, 1)$	$(0, -1)$	$P(3) = 23$
1	$(1, 3)$	$(1, 0)$	$P(3) = 17$
1	$(3, 1)$	$(1, -2)$	$P(3) = 11$
1	$(1, 5)$	$(2, 1)$	$P(3) = 11$
1	$(5, 1)$	$(2, -3)$	$\operatorname{disc} P = -23$
1	$(2, 2)$	$(1, -1)$	$P(3) = 14$
-1	$(1, 1)$	$(2, -1)$	$P(3) = 7$
-1	$(1, 3)$	$(3, 0)$	$P(3) = 1$
-1	$(3, 1)$	$(3, -2)$	$P(77/25) = -4.401088$
-1	$(1, 5)$	$(4, 1)$	$P(77/25) = -4.647488$
-1	$(5, 1)$	$(4, -3)$	$P(77/25) = -16.967488$
-1	$(2, 2)$	$(3, -1)$	$P(77/25) = -1.321088$
2	$(1, 1)$	$(-1, -1)$	$P(3) = 31$
-2	$(1, 1)$	$(3, -1)$	$P(77/25) = -.321088$.

Every displayed polynomial has derivative at least $3X^2 - 8X - 3 = (3X + 1)(X - 3) > 0$ on the interval. The sign seals or the negative discriminant exclude all fourteen. This exhausts every dotted nonisolated head.

For finite own, the same norm bounds its actual degree by four. The complete $\phi(2m)/2 \le 4$ list, after own three, is $m = 4, 5, 6, 7, 8, 9, 10, 12, 15$. At $7, 8, 10, 12, 15$ the embeddings with cosine multipliers $3, 3, 3, 5, 7$ give a positive nonphysical t_H, forbidden. Only the finite heads $4, 5, 6, 9$ remain.

The four finite nonisolated heads. Their parameters are $t_H = \sqrt{2}, \varphi, \sqrt{3}, 2\cos(\pi/9)$ and their actual fields are generated by these parameters. The first-five block of the strong reference has mean $b < .361$. An actual floor $x_1 > X$ permits rebuilding this block at fixed sum, gaining $\frac{5}{4}(X - .361)^2$, with all its new slopes positive. For $X = .643, .681, .695$ the remaining first-five slope exceeds .27, larger than the next ordinary slope $< .214$. The original individual P_{11} price still funds the signed tail. The resulting complete fork bounds are

head	X	forbidden fork $C \ge$	single norm lower
4	.643	7	1.001265
6	.681	6	1.01186
9	.695	6	1.023305.

The head squares satisfy respectively $x_1^2 \ge \sqrt{2} - 1$, $x_1^2 \ge 2\sqrt{3} - 3$, and $x_1^2 \ge t_H/(t_H + 2)$ with $t_H > 1.87$. The signed tail demands used are > 3.44 at $C = 7$ and > 3.38 at $C = 6$. For head five use $x_1 > .668$ and the reference through $1, 6, 7, \dots, 11$, whose energy is $> .668^2 + 1.35015^2/5 + .84816 - .6971 = .9618650045$. At $C = 49/8$, $T_s > 3.39$, giving norm > 1.0005650045. Thus both fork coshes are below seven, 49/8, six, six, in the four respective cases.

No fork can be isolated in these cases. Its original cycle with the nonisolated head gives, for integral g, L,

$$L + g(t_H + 2) = (15t_H^2 + 28t_H - 4)U_f/4 - 2t_H(t_H + 2).$$

At heads $5, 6, 9$, t_H is a dyadic unit, forcing $U_f/4$ integral. The positive norm bounds are $2401/4096$, $9/16$ and $(18/4)(1/8)^2$, each below one. At head four normalize $v(\sqrt{2}) = 1$. The displayed coefficient has valuation two, so only $U_f/2$ is integral. For $W = U_f/2 \in \mathbb{Z}[\sqrt{2}]$, its strips are $W < 49/4$, $0 < W' < 1/4$. The radical coefficient is $1, \dots, 4$; only $W = 3 + 2\sqrt{2}$ fits, giving $U_f = 6 + 4\sqrt{2}$. The original raw time product has square $1256 + 888\sqrt{2}$, whose norm 448 is nonsquare.

All remaining forks must be endpoint-three. Indeed near-next-three and near-four have cosh at least eleven and greater than twelve. Far-next-three has $t < 3/2$, leaving only own three or own four. Own three gives the actual fifteen-node all-three cycle $c, n_2, n_3, \dots, n_{14}, n_f, c$, where f is the released fork. The sum $c + n_2 + \cdots + n_{14} + n_f$ is null: its positive all-ones kernel annihilates every selected unit-Gram row. Its time is the positive time of c, and acuteness makes its product with every other original wall nonpositive. It is therefore a feasible ideal ray of the whole chamber, contradicting compactness. Own four has a conjugate violating its negative time strip. Far-four has $a < 3/2$, leaving only $a = \sqrt{2}$; its time numerator $2a^2 + a$ is positive at $-\sqrt{2}$, also forbidden. Own zero and parallel were already excluded.

For an endpoint-three cap its raw time square is $M_f(t) = 4t^2 + t - 1$ far or $M_n(t) = 4t^2 + 7t + 2$ near, with $K = \mathbb{Q}(t)$ and integral $t \ge 1$. Writing $z = t + 1$ near, every nonphysical conjugate lies in $(-13/20, 2/5)$ far or $(-2/5, 13/20)$ near. The full quadratic integer strips give

head	far t	near z	time-product norms
4	$1 + \sqrt{2}$	$2 + \sqrt{2}$	72, 36
5	φ, φ^2	φ^2	5, 5, 95
6	none	none	

At head four the physical bounds are $t < 7/2$, $z < 15/4$; the positive integral radical coefficient is one, giving exactly the displayed values. The square-norm near product $102 + 72\sqrt{2}$ would require rational square part 54 or 48. At head five the physical parameters are below $10/3$; the positive φ coefficient is one, giving far integer coefficients zero or one and near coefficient one. At head six the only strip value is $2 + \sqrt{3}$, above its respective bounds three and $13/4$. No quadratic endpoint survives.

At head nine the actual degree is three. A far parameter has $1 \le t < 3$ and norm magnitude one; a positive small conjugate would make the norm $< 3(.4)(.65) < 1$. Thus both are negative, $-a, -b$, with $ab = 1/t$, $a, b < .65$, so $t > 400/169$. Its integer trace is strictly greater than $400/169 - 1.3 > 1$ and less than $3 - 2/\sqrt{3} < 2$, impossible. Near, $2 \le z < 13/4$ has unit norm and both small conjugates positive, since a negative one would give norm magnitude $< (13/4)(.4)(.65) < 1$. The integer trace is four. Then $z > 2.7$ from $a + b < 1.3$, and $z < 3$ because $z + 2/\sqrt{z} > 4$ at $z \ge 3$. The second integer coefficient $1/z + 4z - z^2$ decreases on $(2.7, 3)$ and lies strictly between $10/3$ and $10477/2700 < 4$. This final contradiction excludes head nine. Both actual fork opposites must exist, so the long head is isolated in every case.

White-four, next-three and finite isolated forks. The head is now isolated; its weak upper strip $U_H < 11858/625$, $0 < (U_H)_\sigma < 2$ is retained. A far white-four fork has integral a, integral $a^2/2$, $a < 4$, and every other $-.5 < a_\sigma < 0$. Degree at least three has nonzero norm magnitude < 1. In degree two the magnitude is below two, but integrality of $a^2/2$ forces its norm even. Degree one is impossible. Near, $z = a + 2$ has $z < 9/2$, all other $0 < z_\sigma < 1/2$, and integral $z^2/2$. Degree at least four is excluded by norm. At degree three its positive norm is one, incompatible with integrality of $z^2/2$. At degree two that integrality forces norm two. Then $z > 4$ and its integer trace $z + 2/z$ lies in $(9/2, 5)$. This excludes both white-four roles over every actual field.

For next-three at n_2, $1 \le t < 4$ and all other conjugates lie in $(-1/2, 0)$. Degree at least three is excluded by norm; degree two has norm -1 and trace two or three. Thus $t = 1 + \sqrt{2}$ or $(3 + \sqrt{13})/2$. For next-three at n_{14} put $z = t + 2$. Then $3 \le z < 9/2$ and all other conjugates lie in $(0, 1/2)$. Degree at least four is excluded by norm. Degree three would have norm one, $z > 4$, and integer trace strictly between five and $11/2$. At degree two norm two gives the same forbidden trace interval as white four; norm one gives trace four, $z = 2 + \sqrt{3}$, $t = \sqrt{3}$.

For these three next-three parameters the actual head path cycle and two-cycle give integral $g = -H \cdot C$, $L = g^2/U_H$, and

$$L + g(t + 2) = (15t^2 + 4t - 4)U_H/4 - 2M \qquad \text{far},$$
$$L + g(t + 2) = (15t^2 + 52t + 44)U_H/4 - 2M \quad \text{near}.$$

Each listed t is a dyadic unit, so $U_H/4$ is integral. Writing $W = U_H/4$, its full strips are $W < 11858/2500$, $0 < W' < 1/2$. Use the full integral bases $1, \sqrt{2}$ and $1, \sqrt{3}$ in the first and third fields, and $1, (1 + \sqrt{13})/2$ in the second. Writing $W = a + b\eta$ in the appropriate basis, the positive integer coefficient b is one in all three cases: $W - W' < 11858/2500$, whereas $b \ge 2$ would give respectively at least $4\sqrt{2}$, $2\sqrt{13}$ or $4\sqrt{3}$, all larger than this bound. In particular the coefficient of $\sqrt{13}$ itself is $1/2$, not one. The first two conjugate strips require $\sqrt{2} < a < \sqrt{2} + 1/2$ or $(\sqrt{13} - 1)/2 < a < \sqrt{13}/2$, neither admitting an integer. The last gives $a = 2$, hence $W = 2 + \sqrt{3}$ and $U_H = 8 + 4\sqrt{3}$. Then $C_H = 1 + \sqrt{3}$, $x_1 > .681$. The near-next fork has $C = 7 + 4\sqrt{3} > 13$, hence $T_s > 3.7$. The reference through $1, 6, 7, \ldots, 11$ has energy $> .681^2 + 1.33715^2/5 + .84816 - .6971 = .9724150245$; its legal signed tail graft gives $\|x\|^2 > 1.1041150245$. No next-three role survives.

For an isolated finite fork $F = 4\cos^2(\pi/m)$, with actual degree d, its trace is $2d + \mu(m)$. The nonphysical strip $F_\sigma < 1/2$ and physical $F < 4$ give $3d < 7 - 2\mu(m)$, hence $d \le 2$. The complete

quadratic native list and the rational-field exclusion leave only $F = \varphi^2$ or $2 + \sqrt{3}$; native eight and ten have conjugates above $1/2$. Both retained parameters are units. The actual isolated head/fork cycle satisfies

$$L + g = (15U_H F - 32F - 8U_H + 16)/4,$$

so $U_H/4$ is integral. The full integer strips give $U_H = 4\varphi^2$ or $8 + 4\sqrt{3}$, respectively. For the first, $x_1 > .625$ and the positive left-H_4 projection onto c_f, n_f, n_{14}, n_{13} gives

$$V = P_{12} + (e_{13} + e_{14} + e_{15} - se_{16})/2, \quad V^2 = 13, \quad x \cdot V > \sqrt{8\varphi} - (2\varphi - 3) > 3.36.$$

Its old correction is $(\varphi/2)(3n_f/2 + n_{14} + n_{13}/2)$, with positive coefficients and residual $(5 - 3\varphi)/8$. Thus the finite projection is future and feasible against every original wall. The reference through $(1, .625), (6, 2.01815)$ and the later knots has energy $> .9298583845$. Its $.15P_{11}$ graft has residual square two and gives norm > 1.0270583845. For native twelve the same original head gives $x_1 > .681$. Projection onto c_f, n_f has divided displacement P_{14} and demand $\sqrt{8\sqrt{3}} - \sqrt{2}(2 - \sqrt{3}) > 3.34$. Its positive correction is $\gamma_f n_f$ and residual $1 - \gamma_f^2 > 0$. The reference energy is $> .9724150245$; the $.15P_{11}$ graft has square three and gives norm > 1.0411150245. Thus every isolated finite fork is excluded. At this stage a fork is isolated dotted or endpoint-three.

Three head feet used by the remaining spine payments. For an isolated head write $c_H = A_H p - \gamma_H \Omega_1$. The actual first-three/four-root constructions give $x \cdot Z \geq H$, nonnegative Z with sum one, and

own	Z	H	$R = Z^2$
5	$P_4/4$	$A_H/\sqrt{2} > .393$	$1/4$
$m \geq 7$	$(1 - u/2)(e_1 + e_2) + (u - 1)e_3$	$[\sqrt{2 - u} - \sqrt{7u - 4 - 3u^2}]/\sqrt{2}$	$3 - 4u + 3u^2/2$
dotted	$(1 - u/2)e_1 + (u/2)e_2$	$H(u)$	$R(u)$.

Here $u = \gamma_H^{-2}$; the dotted functions are those of Lemma 17. The native-five row is the finite positive H_4 projection. The finite row uses $\xi = c_H + n_1/\gamma_H + 2(\gamma_H^{-1} - \gamma_H)n_2$. Its square is $-(7 - 4\gamma_H^2 - 3/\gamma_H^2) < 0$; its products with c_H, n_1 vanish and that with n_2 is $(3 - 4\gamma_H^2)/(2\gamma_H) < 0$. The dotted row uses $\xi = c_H + n_1/\gamma_H$. The time is positive in both cases, and all other original walls have nonpositive products termwise.

For every finite $m \geq 7$ and every dotted own,

$$.8H - .16R > .2893. \tag{14}$$

For seven, its cubic gives $.9009 < \gamma_H < .901$. Rational squares give $A_H > .7894$, $\sqrt{7 - 4\gamma_H^2 - 3/\gamma_H^2} < .2395$, $\gamma_H\sqrt{2} < 1.2743$, hence $H > .4315$; $1.2318 < u < 1.233$ gives $R < .349$ and the result $> .28936$. For all finite $m \geq 8$, $1 < u \leq 4 - 2\sqrt{2} < 1.172$. Put $D = (u - 1)(4 - 3u)$ and $a = .8/\sqrt{2}$. The derivative of $.8H - .16R$ is

$$a\{-1/(2\sqrt{2 - u}) - (7 - 6u)/(2\sqrt{D})\} + .64 - .48u.$$

It is negative on $(1, 7/6]$: the first term is $< -.28$, the second nonpositive, and the last at most .16. On $[7/6, 1.172)$, $\sqrt{D} > .284$, the positive second term is $< .032$, and the other terms are $< -.28 + .08$. The minimum is therefore at native eight, where $\sqrt{2 - u} > .910$, $\sqrt{D} < .289$, $R < .373$, giving $H > .4388$ and $.8H - .16R > .29136$. For dotted own Lemma 17 is stronger. The graft $.4(Z - P_5/5)$ spends $.08 < \beta_5$, is zero-sum on the first-five flat block, and gives the single-reference gain $> .2893 - .8(.361) + .032 = .0325$.

The complete late D_4, D_3 inventories. At twelve the retained pieces are $A_{11} + D_4$, with right long leaf thirteen, centre fourteen and the two genuine forks. At thirteen they are $A_{12}+D_3$, the right A_3 having middle fourteen and endpoints the forks. Finite classification gives: left endpoint-three, next-three or endpoint-four; right D_4 three at any leaf; right D_3 middle-three or endpoint-three, four or five; and bridges joining a left endpoint by three to a right leaf or middle by three. The full bridges are D_{16} or A_{16}. Other bridges have two forks, a weighted internal bond or large unequal exceptional arms. Two contacts in one retained component make a nonpositive cycle. These are all white cases.

With left contact at position j, the unit residual/ratio data are

role	Δ	σ
left end-three	$(i+1)/(2i)$	$\gamma + j/(2i)$
left next-three	$2/i$	$\gamma + j/(2i)$
left end-four	$1/i$	$\gamma + j/(\sqrt{2}i)$
D_4 long-three or D_3 middle-three	$1/2$	$\gamma + 1/2$
D_4 fork-three	$1/2$	$\gamma + 1/4$
D_3 end-three	$5/8$	$\gamma + 1/4$
D_3 end-four	$1/4$	$\gamma + 1/(2\sqrt{2})$
D_3 end-five	$(5-3\varphi)/8$	$\gamma + \varphi/4.$

The actual edge cosh is $\sqrt{2i}\sigma/\sqrt{\Delta}$. The required strong ratio is $T(q) = 100/61 - (q-.61)/(q(2q-.61))$. It increases on this range. Substitution gives $T(\sqrt{28}) < 39/25$ and, at twelve/thirteen, $T(q) < 31/20$ using $q < 26/5$; the latter cleared numerator is $25414 > 0$.

For single-helper own zero at a long $j \geq 2$, the actual affine D principal consists of c, n_{j-1}, the trunk $j, \ldots, 14$ and both forks. A right fork helper has the affine E_8 arms $1, 2, 5$. Far-left-three own zero has negative time; far-left-four has zero time. The near-four zero case has its affine $4, 3, 3$ star. Right D_3 end-four has kernel coefficients $(\sqrt{2}, 2, 2, 1, 1)$ on $(c, f, 14, 13, k)$. End-five is directly strong even at own zero. Own three with a single three-helper gives an affine all-three cycle. Own four with a fork helper gives the affine F_4 path $i-2, i-1, i, c, f$; with no right helper the old right D and c give the long-leaf-four affine principal. Own six uses the $6, 3$ chain through a neighbor distinct from its helper; parallel gives $c + n_i$. Bridges at own zero have their all-three cycles. Each null has positive coefficients, future time, and nonpositive products with all omitted original walls.

Positive-own bridges have ratio > 1.9. Right long/middle-three and end-four have ratio at least $1 + 1/\sqrt{2}$. End-five has $\Delta < .02$, $\sigma > .4$. Left near-three after its null cases has $\gamma \geq 1/\sqrt{2}$ and ratio $> 31/20$; next-three and end-four are stronger. The only weak left role is the classified literal far helper, excluded by Lemma 5.

Isolated caps at these sites are also strong, without an arithmetic field cutoff. Own three has the affine D formed by the new leaf, the old left leaf, the right trunk and both forks; own four has the long-leaf-four affine principal; own six and parallel have the nulls just described. At native five, the positive left-H_4 projection has displacement $\gamma(3\Omega_{i+1}/4 + \Omega_{i-3}/4)$, whose product with x is at most γh_i by concavity. With $u = \gamma^{-2} < 1.53$, its demand therefore gives $h_i > \sqrt{2i-u} - \sqrt{u-3/2}$, of loss $< .336$ at $i \geq 12$. All roots through $i+1 \leq 14$ exist. For finite own at least seven, the actual core $c + n_i/\gamma + 2(\gamma^{-1} - \gamma)n_{i-1}$ gives $h_i > \sqrt{2i-u} - \sqrt{7u-4-3u^2}$. Here $u < 1.235$, $7u - 4 - 3u^2 \leq 1/12$, and the loss is $< 1.235/9.5 + .289 < .42$. For dotted own the core $c + n_i/\gamma$ has loss at most $u/9.5 + \sqrt{u(1-u)} \leq (2+\sqrt{365})/38 < .56$. All these are below .61, proving strong isolation.

Only a pure single-fork-three packet can remain weak. Its square-two parameter $t = 2\gamma$ is integral and all other conjugates satisfy $it_\sigma(t_\sigma + 1) + 2 < 0$. The physical weak ratio gives $\varphi \leq t < 1.98$.

The norm of $t(t+1)$ excludes degree at least three; trace one, norm minus one gives $t = \varphi$, and trace two is impossible. Thus each late weak role is exactly native five/fork three.

Its actual positive left-H_4 projection has

$$\alpha^2 = \varphi + 5/8 > 56/25, \quad \delta = (5 - 3\varphi)/8, \quad A^2 = i\alpha^2 - \kappa, \quad \sqrt{\delta}/\alpha < .091,$$

where $\kappa = 1/2, 5/8, 3/4$ at $i = 12, 13, 14$. The correction $\gamma(3n_i/2 + n_{i-1} + n_{i-2}/2)$ is positive. Its divided displacement V equals one through $K = i - 3$, has $V^2 < i$, and gives $x \cdot V > G > \sqrt{i - \kappa/\alpha^2} - .091$. Consequently a paid tail reference has

i	K	μ	$G >$	$f_K <$	one norm lower
12	9	1/6	3.34	2.569	.9717166666
13	10	.16	3.475	2.732	.98532.

Its formula is $E_K + 2\mu(G - f_K) - 3\mu^2$. The endpoint coefficients $d_8 > 1/6$, $d_9 > .16$ fund the individual payments. For finite head at least seven or dotted, the zero-sum first-five head graft above has zero cross with each tail and adds $> .0325$ to that same reference; both exceed one. For native-five head the actual field is $\mathbb{Q}(\varphi)$ and the head raw time square is φ. The two tail raw time squares are $14 + 24\varphi$, $15 + 26\varphi$; their products with φ have nonsquare norms $44, 61$. Thus neither weak role exists, and twelve and thirteen are strong.

The entire centre-fourteen list. Deleting fourteen leaves A_{13} and the two actual fork A_1's. The complete finite white list is

left contact	fork contacts
none	none; one finite $m \geq 3$; $(3,3), (3,4), (3,5)$
end-three at $1, 13$	none; one three/four; both three
next-three at $2, 12$	none
end-four at $1, 13$	none.

Finite path and three-node classification proves the list; an extra fork on a next-three branch has large unequal exceptional arms. For left coefficient η at j and fork coefficients a, b, the original inverse gives

$$\Delta = 1 - 2\eta^2 j(14 - j)/14 - a^2 - b^2 > 0, \quad \sigma = \gamma + \eta j/14 + (a + b)/2, \quad C_{14} = \sqrt{28}\sigma/\sqrt{\Delta}.$$

The strong target is below $39/25$. Isolation has finite parameter $B = 4\gamma^2$ with actual $K = \mathbb{Q}(B)$ and all other $0 < B_\sigma < 1/7$. Finite $B < 4$ has nonzero norm $< 4/7$ in degree at least two; rational K is excluded. Dotted isolation has loss $< .56$ from the actual core $c + n_{14}/\gamma$, so is strong. Left-only contacts have the same exact nulls, strong ratio bounds, and excluded literal far helper as above. Left end-three plus one fork-three has $\Delta = 2/7$; own zero is an affine cycle, own three an affine triangle, and own at least four has ratio > 1.8. One fork-four or both fork-three gives $\Delta = 1/28$ and ratio > 2 even at the far own-zero boundary.

With no left contact, own zero and one finite fork makes c the actual isolated cap of that original fork edge: the future timelike point $p + Ac$ is zero on c and every old wall except that fork, strictly negative on it and every unknown wall. Its finite isolated role was excluded above. Own zero with fork pair $(3,3)$ is the affine four-cycle; pair $(3,4)$ is reciprocally near white-four and was excluded; pair $(3,5)$ is already strong at own zero. All positive-own pairs are stronger except own three with $(3,3)$, its actual affine triangle.

For one fork at positive own, own four has the affine $4,3,3$ star on $c, 14, 13$ and the uncontacted fork; own six has the $6,3$ path; parallel is null. Own three/fork three is an affine triangle. Own three/fork four is reciprocally near-next-three and was excluded. Each reciprocal claim uses positive projection onto the finite full retained $A_{15}+c$ system. The omitted fork product is strictly negative and all unknown walls stay strict, so the point is the endpoint of an original fork edge. Own three/fork five has actual field $\mathbb{Q}(\varphi)$; the raw root $\varphi\sqrt{2}c$ has time square $34+52\varphi$, positive at $\varphi' = 1-\varphi$, violating admissibility. Own three/fork at least six has ratio at least $1+\sqrt{3}/2 > 39/25$. For own at least five and fork at least four the minimum is native-five/fork-four, with ratio $(.809+\sqrt{2}/4)/\sqrt{1/2} > 39/25$.

Only single-fork-three remains. Its complete weak strip is $\varphi \le t < 2.21$, and $14t_\sigma(t_\sigma+1)+2 < 0$. The same integral $t(t+1)$ norm excludes degree at least three. Quadratic trace one leaves only $t=\varphi$; trace two gives norm in $(-.4641, 0)$ and is impossible. For that golden packet use the positive H_4 point above at $i = 14, K = 11, \mu = .15$, obtaining the one-reference lower bound $> .84816 + .3(3.60 - 2.886) - .0675 = .99486$. The isolated finite-at-least-seven/dotted head gain adds $> .0325$ with zero cross, giving > 1.02736. For native-five head the tail raw time square $16+28\varphi$ has product-with-head norm 80, nonsquare. Thus fourteen is strong. All knots five through fourteen are now strong, and direct interpolation gives every coefficient in (13), including $\beta_{14} = d_{13} > .13$.

Native-five head and the final fork/degree bounds. At a native-five head the point $Z = P_4/4$, $H > .393$ permits the payment $.56(Z - P_5/5)$, with debit $.112 < .14$ and gain $> 1.12(.393 - .361) - .3136(.25 - .2) = .02016$. The separate signed $.13P_{14}$ tail has square penalty .0338 and zero cross with this early graft. A fork cosh at least fourteen would give $T_s > 3.72$ and $\|x\|^2 > .9084 + .02016 + .26(3.72 - 3.311) - .0338 = 1.00110$. Thus both fork coshes are below fourteen. The actual field is $\mathbb{Q}(\varphi)$; the head raw time square is φ. Only isolated dotted or endpoint-three roles remain.

For an isolated dotted fork the head/fork cycle gives $F/4$ integral. For $W = F/4$ its strips are $W < 49/2$, $0 < W' < 1/8$. In $W = a + b\varphi$, $1 \le b \le 10$, the strict conjugate strip permits only $b = 8, a = 5$, so $F = 4\varphi^6$. Its time-product-with-head norm is $5056 = 64 \cdot 79$, nonsquare. Far endpoint-three has $t < (14\sqrt{17} - 1)/8 < 36/5$ and conjugate in $(-13/20, 2/5)$. The full ring gives $1 \le b \le 3$ and exactly $t = \varphi, \varphi^2, \varphi^3, \varphi^4$. The product-with-head norms are $1, 1, 76, 149$. The first two positive time products are φ^3, φ^4, but the spatial products are $2\varphi + 3/2$, $5\varphi/2 + 2$. Thus $g = -H \cdot C$ is $-1/2$ or $\varphi/2$: one violates acuteness, the other original path-cycle integrality. Near, $z = t + 1 < (14\sqrt{17} + 1)/8 < 15/2$ has conjugate in $(-2/5, 13/20)$. The same full-ring strip gives $z = \varphi^2, \varphi^3, \varphi^4$. Their product norms are $19, 36, 191$. The square-norm middle product is $54 + 24\sqrt{5}$; its rational square parts would be $30, 24$, neither square. No actual fork remains, excluding native-five head.

Every isolated head now has finite own at least seven or dotted own and therefore the zero-cross gain $> .0325$. A fork cosh at least twelve gives $T_s > 3.67$, and

$$\|x\|^2 > .9084 + .0325 + .26(3.67 - 3.311) - .0338 = 1.00044.$$

Thus both coshes are below twelve. Each fork is isolated dotted or endpoint-three with positive own, as asserted.

Finally suppose both forks are isolated. Their original cycles give integral $Y = FG/4$, with $Y < 1296$ and every other $0 < Y_\sigma < 1/16$. Degree at least five has nonzero norm $< 1296/65536 < 1$. At degree four each positive integer $\mathrm{N}\,F, \mathrm{N}\,G < 9$ is at most eight, but Y integral forces $256 \mid \mathrm{N}\,F\,\mathrm{N}\,G$. The two norms have total dyadic valuation at most six, a contradiction. Degree one is excluded by Meyer. Hence the actual degree is two or three under the stated both-isolated hypothesis. This proves the entire initial inventory at its original-rank and actual-wall scope. □

Proof of Proposition 15, from Lemma 16. Use raw old roots of square two. An isolated head has raw square $2U$, spatial part Ue_1, time square $U(U-2)$, and actual field $K=\mathbb{Q}(U)$; its nonphysical U_σ lie in $(0,2)$. An isolated fork has the data stated above, actual $K=\mathbb{Q}(F)$, and $0<F_\sigma<1/2$ at every nonphysical place. The only nonisolated forks left by the initial inventory have raw square two and one endpoint-three white contact. Their complete data are

role	spatial vector	time square M	edge cosh
far	$(t/2)(P_{15}+se_{16})+e_1$	$4t^2+t-1$	$(8t+1)/\sqrt{17}$
near	$(t/2)(P_{15}+se_{16})+(P_{15}-se_{16})/2$	$4t^2+7t+2$	$(8t+7)/\sqrt{17}$.

Here $s=\pm1$ denotes a genuine fork. In the far role put $z=t$; in the near role put $z=t+1$. Path-cycle integrality gives integral z, and actual-field descent gives $K=\mathbb{Q}(z)$. To justify these normalizations, the connected simple-three old system has one raw norm squareclass, so a totally positive rescaling of the form puts its roots at square two over K. An isolated own contact has integral two-cycle $4\gamma^2$; rescaling its root line by its nonzero old product gives the raw head or fork displayed above. The raw principal consisting of that cap and the full old system is defined over $\mathbb{Q}(U)$ or $\mathbb{Q}(F)$ and is physically indefinite, since the cap has nonzero time. The descent argument in Section 1 proves the asserted field equalities. For an endpoint cap its simple-three helper permits square-two normalization directly; the closed old path gives integral t, and the same indefinite-principal argument gives $K=\mathbb{Q}(t)$. The cosh bound twelve gives $z<6.311$; positive own gives $z\ge1$ far and $z\ge2$ near. Negative nonphysical time squares give

$$-.65<z_\sigma<.4 \quad \text{far}, \qquad -.4<z_\sigma<.65 \quad \text{near}. \tag{15}$$

All these formulas concern actual K-roots, so every positive raw time product is K-valued and its square must be a full K-square.

First sharpen the head bound. If $x_1\ge.632$, rebuild only the first-five flat block of the positive reference, preserving its sum. The new coordinates are $.632,(f_5-.632)/4$ four times. They exceed the next slope and preserve nonnegative prefix prices. Their single-reference gain gives

$$\|x\|^2>E_{14}+\tfrac{5}{4}(.632-.361)^2=1.00020125,$$

a contradiction. Thus $x_1<.632$. The actual head edge gives $C_H<7/3$, since $.632^2<2/5$, and hence always $U<98/9$; if dotted, $U>4$.

Endpoint cycles and the actual degree. For an endpoint cap let $g=-H\cdot C$ and $L=g^2/U$. The original path and own two-cycle make both integral. Expanding the common time product gives

$$\begin{aligned} L+g(t+2)&=(15t^2-8)U/4-2M && \text{far},\\ L+g(t+1)&=(15z^2-4z-4)U/4-2M && \text{near}. \end{aligned}$$

All remaining terms are integral, and 15 is coprime to four; therefore $z^2U/4$ is integral in both roles. If $|\,\mathrm{N}\,z|$ is odd, z is a unit at every dyadic prime, forcing $U/4$ integral. This is impossible in degree at least three. Indeed the sharpened head bound just proved gives

$$0<\mathrm{N}(U/4)<\frac{49}{18}(1/2)^{d-1}\le\frac{49}{72}<1 \qquad (d\ge3),$$

contradicting its positive integral norm. This uses the original head bound and all nonphysical places, with no integer-basis assumption.

The nonzero endpoint norm is less than $6.311(.65)^{d-1}$. It excludes $d \geq 6$; at degrees four or five it is below two and hence has odd absolute norm one, just excluded. In degree three only $|\,\mathrm{N}\, z| = 2$ remains. Degree one is excluded by Meyer. If neither endpoint role occurs, both forks are isolated and the initial lemma already gives degree two or three. Thus these are the only actual degrees in every case.

Every cubic endpoint is impossible. In degree three norm magnitude two and $6.311 \cdot .4 \cdot .65 < 2$ force both small conjugates negative far, positive near. Write their positive magnitudes $a, b < 13/20$; then $zab = 2$ and $z > 800/169$. In the far role its integer trace $z - a - b$ is four or five, by AM–GM and $z < 6.311$. The second coefficient $B = 2/z - z(a + b)$ satisfies $B \leq -6$ by AM–GM. The polynomial $X^3 - TX^2 + BX - 2$ is negative at $-13/20$, so $-B < (13/20)^2 + T(13/20) + 40/13 < 7$ and $B \geq -6$. Thus $B = -6$. The two resulting cubics, with $T = 4, 5$, have discriminants $-44, -424$, impossible in a totally real field. Near, the function $z + 2\sqrt{2/z}$ is increasing for $z > 4$. The integer trace is therefore seven, since $z + 2\sqrt{2/z} > 800/169 + 13/10 > 6$ and $z + a + b < 8$. Thus $z > 57/10$. Evaluation at $13/20$ gives $B < 7.205$, whereas $B \geq 2\sqrt{2z} + 2/z > 6.75 + .316 > 7$. The incompatible integer bounds exclude this role too.

Every finite isolated head is impossible. For finite own m, $U = 4\cos^2(\pi/m)$ has degree $\phi(m)/2$. The complete totient-at-most-six list and the initial small-own exclusions leave $m = 7, 8, 9, 10, 12, 14, 18$. At $m = 9, 14$ a nonphysical $U_\sigma > 2$ is forbidden. At cubic $m = 7, 18$, $\mathrm{N}\, U = 1, 3$ is odd. The cubic endpoint exclusion makes both forks isolated; their cycle $UF/4$ forces $F/4$ integral, so $64 \mid \mathrm{N}\, F$. But $F < 72$ and its other conjugates below $1/2$ give $\mathrm{N}\, F < 18$, a contradiction.

The remaining heads have $U = 2 + \sqrt{2}, 2 + \varphi, 2 + \sqrt{3}$. The full endpoint strips, in the maximal orders, give

K	far t	near z	$U(U-2)$
$\mathbb{Q}(\sqrt{2})$	$1 + \sqrt{2}, 3 + 2\sqrt{2}$	$2 + \sqrt{2}, 3 + 2\sqrt{2}$	$2 + 2\sqrt{2}$
$\mathbb{Q}(\varphi)$	$\varphi, \varphi^2, \varphi^3$	φ^2, φ^3	$1 + 3\varphi$
$\mathbb{Q}(\sqrt{3})$	$2 + \sqrt{3}$	$2 + \sqrt{3}$	$3 + 2\sqrt{3}$.

For $\sqrt{2}$ the radical coefficient is one or two, for $\sqrt{3}$ one. In the golden integral basis the coefficient is one or two; the next cover already exceeds the exact cosh bound. These finite integer intervals prove the list, including dotted owns. The corresponding time-product norms are $72, 400, 36, 544$; $5, 5, 380, 95, 180$; and $78, 150$. Only $400, 36$ are squares. Their products are $340 + 240\sqrt{2}$ and $102 + 72\sqrt{2}$; the rational square parts would be $180, 160$ or $54, 48$, none square. Thus no endpoint is possible for a finite head.

For an isolated fork in the $m = 10, 12$ fields the head is a dyadic unit, so $F/4$ is integral. At $m = 8$, normalize the unique dyadic valuation by $v(\sqrt{2}) = 1$. The quarter-cycle gives $v(F) \geq 3$; the time-product valuation is $4 + v(F)$ and must be even, hence $v(F) \geq 4$ and again $F/4$ is integral. For $W = F/4$, the full strips are $W < 18$, $0 < W' < 1/8$. In $\mathbb{Z}[\sqrt{2}]$ no coefficient with $1 \leq b \leq 6$ fits; in $\mathbb{Z}[\varphi]$ only $W = 5 + 8\varphi$ fits; in $\mathbb{Z}[\sqrt{3}]$ only $W = 7 + 4\sqrt{3}$ fits. The last two give time-product norms $25280, 9024$, nonsquares. Thus the actual head is dotted.

One dotted-head reference excludes the remaining cubic. Put $u = 4/U \in (0, 1)$. The positive actual core $c_H + n_1/g_H$ has zero product with c_H, negative product with n_1, future time and nonpositive product with every other original wall. Normalizing it gives

$$x \cdot Z \geq H(u), \quad Z = (1 - u/2)e_1 + (u/2)e_2, \quad Z^2 = R(u).$$

The single graft $.4(Z - P_5/5)$ spends .08 at P_5, below its available .14, and has zero cross with the flat block. Lemma 17 therefore gives gain $> .295 - .8(.361) + .032 = .0382$, hence one reference of energy $> .9466$. A signed $.13(P_{15} + se_{16} - P_{14})$ graft has square residual two and zero cross. If its fork cosh were at least eleven, its actual pressure would exceed 3.65, giving

$$\|x\|^2 > .9466 + .26(3.65 - 3.311) - .0338 = 1.00094.$$

Both fork coshes are therefore below eleven.

In a cubic field all endpoint roles were excluded. The isolated forks satisfy $4 < F, G < 121/2$ and $\mathrm{N}\,F, \mathrm{N}\,G \in \{1, \dots, 15\}$. The quarter-cycle gives $64 \mid \mathrm{N}\,F\,\mathrm{N}\,G$, forcing both norms eight. Lemma 17 gives $F, G > 42$. Their coshes exceed $\sqrt{84} > 9.16$ and their signed pressures exceed 3.584, since $3.584^2 < 1632/127$. Averaging the two genuine fork inequalities gives $S_{15} > 3.584$. The average tail graft has residual square one, so the same reference gives

$$\|x\|^2 > .9466 + .26(3.584 - 3.311) - .0169 = 1.00068.$$

Thus the actual field is quadratic.

Every quadratic endpoint is impossible. The improved cosh bound eleven gives $z < 5.796$, so $|\mathrm{N}\,z| < 4$. If its norm is odd, $U/4$ is integral. The full head strip then gives $U = 4\varphi^2$, with time square $8\varphi^5$. In this field the complete endpoint strips give far $t = \varphi, \varphi^2, \varphi^3$ and near $z = \varphi^2, \varphi^3$. The first two far time products are $8\varphi^{10}, 8\varphi^{12}$, requiring $\sqrt{2}$ in the field; the third has nonsquare norm 4864. The near products have nonsquare norm 1216, or equal $48\varphi^{10}$, requiring $\sqrt{3}$. None is possible.

Only norm magnitude two remains. Integer trace and the exact role-specific cosh/conjugate strips give the full list

role	$\mathrm{N}\,z$	$\mathrm{Tr}\,z$	z
far	-2	$3, 4, 5$	$(3+\sqrt{17})/2,\ 2+\sqrt{6},\ (5+\sqrt{33})/2$
near	2	$4, 5, 6$	$2+\sqrt{2},\ (5+\sqrt{17})/2,\ 3+\sqrt{7}$
near	-2	5	$(5+\sqrt{33})/2.$

For example far positive norm two would require $z > 5$ and $5 < z + 2/z < 6$, so is impossible. The other rows follow from $z + \mathrm{N}(z)/z$ and the strict conjugate intervals, not a field search.

In the ramified fields $\sqrt{2}, \sqrt{6}, \sqrt{7}$, z has dyadic valuation one. Integrality of $z^2U/4$ forces $U/2$ integral. Writing $W = U/2$, its strips are $2 < W < 49/9$, $0 < W' < 1$. The only radical coefficient is one: the candidates are $2 + \sqrt{2},\ 3 + \sqrt{6},\ 3 + \sqrt{7}$. The last two exceed $49/9$; the first gives $U = 4 + 2\sqrt{2}$. Its near cap has time square $21 + 15\sqrt{2}$ and the head $16 + 12\sqrt{2}$; their product norm 288 is nonsquare.

In the split fields $\sqrt{17}, \sqrt{33}$, taking norms in $z^2U/4$ gives $4 \mid \mathrm{N}\,U$. The full head strips retain precisely

$\mathbb{Q}(\sqrt{17})$	$(7+\sqrt{17})/2,\ 5+\sqrt{17}$
$\mathbb{Q}(\sqrt{33})$	$(7+\sqrt{33})/2,\ (9+\sqrt{33})/2.$

In the first field the full integral-basis coefficient is one or two, and the omitted strip heads have norms two and nineteen; in the second it is one. No maximal-order class is missing. For $s = \sqrt{17}$ the head time squares are $(19 + 5s)/2$, $32 + 8s$. The far cap has time square $(53 + 13s)/2$. Its first time product is $(16 + 4s)^2$, but the spatial product is $(33 + 7s)/4$; hence the actual path cycle would be $(31 + 9s)/4$, not integral. The second product is $1732 + 420s$, whose possible rational square parts $882, 850$ are nonsquares. The near cap has time square $(77 + 19s)/2$ and product norms $832, 3328$,

also nonsquares. For $s = \sqrt{33}$ both head time squares have norm -24; the far and near caps have norms $-98, -8$, giving nonsquare product norms $2352, 192$. Thus every endpoint is excluded.

Both genuine forks are consequently isolated dotted. Their original cycles, in these raw normalizations, give

$$L_{HF} + g_{HF} = (15UF - 32F - 8U + 16)/4,$$
$$L_{FG} + 7g_{FG} = 15FG/4 - 8(F + G) + 4,$$

with integral g, L; the analogous head/G identity holds. These yield all three quarter-products. The common actual K-time line supplies every field-square condition. The strict physical and nonphysical strips were retained throughout. This proves the claimed necessary quadratic triple from the initial inventory. □

5.1 Norm bounds, integer orders and valuations

Put

$$N_H = \mathrm{N}\, U, \quad T = \mathrm{Tr}\, U, \quad A = -\mathrm{N}(U - 2) = 2T - 4 - N_H.$$

The strict strips imply

$$5 \le T \le 12, \quad 1 \le N_H \le 21, \quad 1 \le A \le 17.$$

For a fork put $n = \mathrm{N}\, F$, $S = \mathrm{Tr}\, F$ and $B = 2S - 4n - 1 > 0$ odd. Then

$$1 \le n \le 30, \quad 5 \le S \le 60, \qquad \mathrm{N}\, T_H = -N_H A, \quad \mathrm{N}\, T_F = -4nB. \tag{16}$$

A nonsquare rational norm disproves a field square; a square norm still requires the full square test.

Use the full integer order of $K = \mathbb{Q}(\sqrt{d})$. The gap between the embeddings of U is a positive integral multiple of $\sqrt{\Delta}$, so

$$\Delta < (98/9)^2 < 119, \qquad \Delta \le 118. \tag{17}$$

For a prime $\mathfrak{p}$ over two, normalize $v_{\mathfrak{p}}(\mathfrak{p}) = 1$ and write

$$e = v_{\mathfrak{p}}(2),\ r = v_{\mathfrak{p}}(U),\ s = v_{\mathfrak{p}}(U - 2),\ k = v_{\mathfrak{p}}(F),\ l = v_{\mathfrak{p}}(G).$$

The factors $2F - 1, 2G - 1$ are units there. Quarter integrality and the time squares give

$$r + k, r + l, k + l \ge 2e, \qquad r + s + e + k,\ r + s + e + l \equiv 0 \pmod 2. \tag{18}$$

Also $s = r$ for $r < e$, $s = e$ for $r > e$, and $s \ge e$ for $r = e$. Norm valuations are $\sum f_{\mathfrak{p}} v_{\mathfrak{p}}$. The quadratic possibilities are inert $(e, f) = (1, 2)$, ramified $(2, 1)$, or two split primes $(1, 1)$. No PID or unramifiedness assumption enters.

5.2 Three complete integer-strip lemmas

If $F/4$ is integral, $16 \mid n < 30.25$ gives $n = 16$. For $W = F/4$, norm one and $0 < W' < 1/8$ give $8 < W < 121/8$, and $\mathrm{Tr}\, W = W + 1/W \in \{9, \dots, 15\}$. This gives the complete table

$\mathrm{Tr}\, W$	K	Δ	F when $\Delta \le 118$
9	$\mathbb{Q}(\sqrt{77})$	77	$18 + 2\sqrt{77}$
10	$\mathbb{Q}(\sqrt{6})$	24	$20 + 8\sqrt{6}$
11	$\mathbb{Q}(\sqrt{13})$	13	$22 + 6\sqrt{13}$
12	$\mathbb{Q}(\sqrt{35})$	140	excluded
13	$\mathbb{Q}(\sqrt{165})$	165	excluded
14	$\mathbb{Q}(\sqrt{3})$	12	$28 + 16\sqrt{3}$
15	$\mathbb{Q}(\sqrt{221})$	221	excluded.

If $Y = U/2$ is integral with odd norm, then $2 < Y < 49/9$, $0 < Y' < 1$ and $\mathrm{N}\,Y \in \{1, 3, 5\}$. Its integer trace t is in $3, \dots, 6$ and $1 - t + \mathrm{N}\,Y < 0$, so $t > \mathrm{N}\,Y + 1$. The complete list is

$\mathrm{N}\,Y$	t	Y	2 in K
1	3	$(3+\sqrt{5})/2$	inert
1	4	$2+\sqrt{3}$	ramified
1	5	$(5+\sqrt{21})/2$	inert
3	5	$(5+\sqrt{13})/2$	inert.

Trace six gives $3 + 2\sqrt{2}$ or $3 + \sqrt{6}$, both above $49/9$; norm five requires trace above six. These bounded integer comparisons prove completeness.

Finally, if $U/4$ is integral with $\mathrm{N}\,U = 16$, then $W = U/4$ has norm one, $1 < W < 49/18$, $0 < W' < 1/2$. Thus $W > 2$, $\mathrm{Tr}\,W = 3$, and

$$U = 4\varphi^2, \quad K = \mathbb{Q}(\varphi), \qquad \varphi = (1+\sqrt{5})/2. \tag{19}$$

6 Odd head norm and inert two

6.1 Odd norm

If N_H is odd, at every dyadic prime $r = s = 0$. In an inert or split field, (18) makes both fork valuations odd and at least three; their norms are divisible by 64, contrary to $n \le 30$. Thus two ramifies. Now k, l are even and at least four. Both fork norms are sixteen and $F/4, G/4$ integral. The unit-fork table leaves only $\mathbb{Q}(\sqrt{3})$ or $\mathbb{Q}(\sqrt{6})$. The full head strips with odd norm give

K	U	$\mathrm{N}\,T_H$	$\mathrm{N}\,T_F$
$\mathbb{Q}(\sqrt{3})$	$5+2\sqrt{3}$	-39	-3008
$\mathbb{Q}(\sqrt{6})$	$3+\sqrt{6},\ 5+2\sqrt{6}$	$-15,\ -15$	$-960.$

The radical coefficient is one or two; coefficient three exceeds the physical bound. The conjugate strip and norm parity remove the other rational coefficients. In $\mathbb{Q}(\sqrt{6})$, $6+2\sqrt{6} > 98/9$ and $4+\sqrt{6}$ has even norm.

The $\sqrt{3}$ product norm is $64 \cdot 1833$, a nonsquare. For $U = 3 + \sqrt{6}$ the exact time product is $58200 + 23760\sqrt{6}$, of norm 120^2. Its possible rational square parts are 29160 and 29040, strictly between 170^2 and 171^2, so it is not a field square. The remaining triple is $U = 5+2\sqrt{6}$, $F = G = 4U$. The actual head and two genuine fork edges have coshes

$$C_H = (2+\sqrt{6})/2 > 11/5, \qquad C_+ = C_- = 2(2+\sqrt{6}) > 44/5,$$

using $\sqrt{6} > 12/5$. Equation (1) gives $x_1 \ge \sqrt{3/8} > .61$ and, for both $s = \pm 1$,

$$x \cdot (P_{15} + s e_{16}) > 4\sqrt{39}/7 > 3.56.$$

The last comparison follows from $6.23^2 < 39$. Averaging the two signed inequalities gives $S_{15} > 3.56$; Equation (13) gives $S_6 > 2.01815$. One positive reference gives

$$\|x\|^2 > .61^2 + 1.40815^2/5 + 1.54185^2/9 = 1.032821887 > 1.$$

The last pressure averages the two genuine fork edges, not a virtual root. This excludes odd norm.

6.2 Even norm with inert two

Here $e = 1, f = 2$, and $r = 1$ or two since $r \geq 3$ forces $N_H \geq 64$. If $r = 1$, the head strip gives

$$U = 3 + \sqrt{5}, \quad 5 + \sqrt{21}, \quad 5 + \sqrt{13}.$$

In each case $s = 1$, so (18) makes k, l odd and positive. The norm bound gives $k = l = 1$. Thus $W = F/2$ is integral with odd norm, $W < 121/4$, $0 < W' < 1/4$. In $W = (a + b\sqrt{d})/2$, $a \equiv b \pmod 2$, the full strip $0 < a - b\sqrt{d} < 1/2$ gives

d	b range	(a, b) before norm filter	F
5	$1, \ldots, 13$	$(7, 3), (18, 8), (25, 11)$	$2\varphi^4,\ 2\varphi^6,\ 14 + 22\varphi$
21	$1, \ldots, 6$	$(5, 1), (23, 5)$	$5 + \sqrt{21},\ 23 + 5\sqrt{21}$
13	$1, \ldots, 8$	$(11, 3), (22, 6)$	$11 + 3\sqrt{13}$.

The $d = 13$ pair $(22, 6)$ has $\mathrm{N}\, W = 4$, not odd. Each b bound follows from $b\sqrt{d} < W < 121/4$; the displayed a is the only possible integer with the required parity. The close pair $(28, 6)$ at $d = 21$ fails since $6\sqrt{21} < 55/2$.

At $d = 5$, product norms are 256 times $11, 55, 95$, all nonsquares. The other square-norm exceptions are

d	F	$T_H T_F$	$\sqrt{\mathrm{N}(T_H T_F)}$	possible square parts
21	$5 + \sqrt{21}$	$12648 + 2760\sqrt{21}$	48	6348, 6300
21	$23 + 5\sqrt{21}$	$303000 + 66120\sqrt{21}$	240	151620, 151380
13	$11 + 3\sqrt{13}$	$52872 + 14664\sqrt{13}$	144	26508, 26364.

These lie strictly between $79^2, 80^2$; $389^2, 390^2$; and $162^2, 163^2$, respectively. No fork exists.

If $r = 2$, then $\mathrm{N}\, U = 16$, $U/4$ integral, and (19) holds. Now $s = 1$, so k, l are even. The fork/fork quarter-product gives $k + l \geq 2$, so one fork is divisible by four with norm sixteen. Its $W = F/4$ in $\mathbb{Q}(\varphi)$ has norm one, $W < 121/8$, $0 < W' < 1/8$. Writing $W = a + b\varphi$ gives $b = 1, \ldots, 6$; none permits $0 < a + b(1 - \varphi) < 1/8$. The inert case is exhausted.

7 Even head norm with ramified two

Here $e = 2, f = 1$ and $r = v_2(N_H) \in \{1, 2, 3, 4\}$. If $r = 1$, $s = 1$ and both fork valuations are even and at least four. The unit-fork table leaves $\sqrt{6}$ or $\sqrt{3}$. The head strips with $v_2(N_H) = 1$ give only $U = 4 + \sqrt{6}$ or $3 + \sqrt{3}$. Their product norms are $256 \cdot 75$ or $256 \cdot 141$, nonsquares.

If $r = 2$, the odd-norm $U/2$ table forces $U = 4 + 2\sqrt{3}$. Then $s = 3$, so k, l are odd and at least three. Norm bounds permit only valuation three and norm eight or twenty-four. For $F = 2W$, therefore, $\mathrm{N}\, W = 2$ or six, $W < 121/4$, $0 < W' < 1/4$. In $\mathbb{Z}[\sqrt{3}]$, the strip for $W = a + b\sqrt{3}$, $1 \leq b \leq 8$, allows only $(7, 4), (14, 8)$, of norms one and four. Neither required norm occurs.

If $r = 3$, $U/2$ is integral of norm two and trace four, five or six. Trace six gives $3 + \sqrt{7} > 49/9$; trace five gives the split field $\mathbb{Q}(\sqrt{17})$. Thus ramification forces $U = 4 + 2\sqrt{2}$. Here $s = 2$, so k, l are odd, at most three. Since $k + l \geq 4$, at least one is three and its norm is eight or twenty-four.

For that head $\mathrm{N}\, T_H = -32$, so a product norm is $128n(2S - 4n - 1)$ with S even in $\mathbb{Z}[\sqrt{2}]$. At $n = 8$, a square would require $2S - 33$ square, but it is three modulo four. At $n = 24$, the strip gives $49 \leq S \leq 60$; the even values yield

$$3(2S - 97) \in \{9, 21, 33, 45, 57, 69\}.$$

Only nine is square, at $S = 50$. But even norm for $F = a + b\sqrt{2}$ forces a even, hence $S = 2a$ divisible by four. This excludes fifty.

Finally $r = 4$ gives (19), whose field has inert two. Larger r forces $N_H \geq 32 > 21$. All ramified cases are excluded.

8 The complete split branch

Here $d \equiv 1 \pmod 8$, squarefree, and (17) leaves $d = 17, 33, 41, 57, 65, 73, 89, 97, 105, 113$. Write $U = (a + b\sqrt{d})/2$, $a \equiv b \pmod 2$. The strip gives $0 < a - b\sqrt{d} < 4$ and $b = 1$, except at $d = 17$ where $b = 1$ or two. The complete even-norm heads are

d	U	T	N_H	A
17	$(5 + \sqrt{17})/2$	5	2	4
17	$(7 + \sqrt{17})/2$	7	8	2
17	$5 + \sqrt{17}$	10	8	8
33	$(7 + \sqrt{33})/2$	7	4	6
33	$(9 + \sqrt{33})/2$	9	12	2
41	$(7 + \sqrt{41})/2$	7	2	8
41	$(9 + \sqrt{41})/2$	9	10	4
57	$(9 + \sqrt{57})/2$	9	6	8
57	$(11 + \sqrt{57})/2$	11	16	2
65	$(9 + \sqrt{65})/2$	9	4	10
65	$(11 + \sqrt{65})/2$	11	14	4
73	$(9 + \sqrt{73})/2$	9	2	12
73	$(11 + \sqrt{73})/2$	11	12	6
89	$(11 + \sqrt{89})/2$	11	8	10
97	$(11 + \sqrt{97})/2$	11	6	12
105	$(11 + \sqrt{105})/2$	11	4	14
113	$(11 + \sqrt{113})/2$	11	2	16.

For $d = 17$, all coefficient pairs are $(5, 1), (7, 1), (10, 2), (12, 2)$; the last has odd norm nineteen. For the other fields two odd a values pass the conjugate strip; at $89, 97, 105, 113$ the larger exceeds $98/9$. These explicit finite coefficient comparisons prove the table, without an assumed field-search output.

8.1 Odd trace

Let $\mathfrak{p}$ be the prime where U is nonunit and $\mathfrak{q}$ its conjugate. At $\mathfrak{q}$, $r = s = 0$, so both fork valuations are odd and at least three. Put $c = v_2(N_H) = r_{\mathfrak{p}}$. A fork has total norm valuation at most four.

If $c = 2$ or four, $s_{\mathfrak{p}} = 1$ and both fork valuations at $\mathfrak{p}$ are even. The only possible ordered valuations are $(0, 3)$ for each fork, contradicting $FG/4$ at $\mathfrak{p}$.

If $c = 1$, both forks have valuations $(1, 3)$ and norm sixteen. Parity requires $v_2(A)$ odd. The table leaves only

$$d = 41,\ U = (7 + \sqrt{41})/2; \qquad d = 57,\ U = (9 + \sqrt{57})/2.$$

Now $W = F/2$ is integral of norm four, $W < 121/4$, $0 < W' < 1/4$. In $W = (a + b\sqrt{d})/2$, $b = 1, \ldots, 4$, the strip $0 < a - b\sqrt{d} < 1/2$ with parity allows only $(26, 4)$ at 41, of norm five, or $(23, 3)$ at 57, of norm four. Thus the only candidate is $F = 23 + 3\sqrt{57}$ in $K = \mathbb{Q}(\sqrt{57})$.

For that head $U > 8$, so $x_1 > 1/\sqrt{3} > .577$. Rebuilding only the first-five block of (13), with its sum fixed, gives

$$E_{14} + \frac{5}{4}(.577 - .361)^2 > .96672.$$

If a fork had $\cosh L \geq 9$, its actual signed pressure would exceed 3.577. The separately available .13 coefficient at P_{14} has residual square two and zero cross with that first-block change. One square would give

$$\|x\|^2 > .96672 + .26(3.577 - 3.311) - .0338 = 1.00208 > 1.$$

Both fork coshes are therefore below nine, so $F < 81/2$, contrary to $23 + 3\sqrt{57} > 44 > 81/2$.

If $c = 3$, $s_{\mathfrak{p}} = 1$ and both forks again have valuations $(1, 3)$, norm sixteen. The remaining heads have $d = 17$ or 89. For $W = F/2$ the full strip/parity pairs are

d	b range	(a, b)
17	$1, \ldots, 7$	$(21, 5), (29, 7)$
89	$1, \ldots, 3$	none.

The 17-field norms are four and two, leaving only $F = 21 + 5\sqrt{17}$. The close pair $(17, 4)$ fails since $4\sqrt{17} < 33/2$. The surviving product norm is $(-16)(-1216) = 1024 \cdot 19$, a nonsquare. There are no other odd-trace cases.

8.2 The sole even-trace head

The last row is $U = 5 + \sqrt{17}$. Its two prime valuations are one and two. At the valuation-two prime $s = 1$; $\mathrm{N}(U - 2) = -8$ makes $s = 2$ at the other. Equation (18) makes both fork valuations even at both primes, and at the valuation-one prime each is at least two. The fork/fork quarter-product forces one fork also to have valuation at least two at the other prime. It is divisible by four globally, with norm sixteen. For $W = F/4$ we have

$$\mathrm{N}\, W = 1, \quad W < 121/8, \quad 0 < W' < 1/8, \qquad W = (a + b\sqrt{17})/2, \quad 1 \leq b \leq 3.$$

None admits an integer a in $0 < a - b\sqrt{17} < 1/4$. The final split row is impossible.

Proposition 18. *Whole D_{16} cannot occur at the original farthest vertex of a compact arithmetic Coxeter* 16*-polytope.*

Proof. The actual reduction gives the necessary quadratic triple. Its head norm is odd or even. The odd case and every inert even case were excluded in Section 6; Sections 7 and 8 exclude ramified and split even cases. These exhaust quadratic fields. Each contradiction uses a necessary actual cycle, full field-square condition or one paid original-vector norm. No D16 census is used. □

Proof of Theorem 1. The complete acyclic implication is given in Section N.4: the independent general compact companion bounds n by seventeen, the rank seventeen dispatch excludes that dimension, and the fresh rank sixteen cover and manual whole-D_{16} obstruction exclude dimension sixteen. All steps retain the original farthest vertex and the stated external and finite-output inputs. □

9 Inputs, computation and verification

9.1 Dependencies and verification scope

Appendix O locates the proofs of the local component estimates, both necessary forest covers and their complete dispatches, and records their primary-source provenance. The stronger conditioned A_{11} corollary is distinct from the uniform and joint-I estimates. The whole-D_{16} argument is manual.

The proof uses the independent general compact companion, the geometric interfaces identified in Appendix O, the four finite outputs in Section 3, and the external arithmetic statements specified below. The companion's proof includes its own higher-dimensional tail and does not depend on this article.

The literal golden-lattice cutoff in Section I.1 is an explicitly assumed external statement attributed to [3]. Its precise lattice, dimension and stronger infinite-covolume scope are retained there. The exact external statement is assumed here; its full primary text has not been independently authenticated. The all-compatible-invariant-lattice bridge is proved internally and does not follow merely from commensurability.

Each computed proposition specifies a complete necessary domain, termination argument, exact tests and recorded output. Different same-code replay scopes are stated in Section 3. Every nonempty result has a subsequent actual geometric consumer. A source comparison, receipt parser or PDF build does not replay a mathematical domain.

Selected conditional Lean kernels prove discrete forest dispatch, numerical composition, valuation/norm consequences and scalar completed-square deductions. Actual geometry, the ground field, all nonphysical inequalities, integral-cycle interpretation, complete local inventories and named computed propositions remain explicit parameters. Printed axioms do not enumerate these parameters. Each formal verification claim retains the theorem and source scope specified by its execution record.

The reproducibility record gives the serial entry points, separate axiom audits and source bindings of the recorded executions. Their scopes are unchanged by an editorial revision or PDF build. The manual D_{16} obstruction does not remove the four finite-output premises used elsewhere in the proof.

A A strengthened ordinary dihedral comparison

This section supplies the ordinary rank-two comparison needed beside D_{14}. Its component theorem is general real geometry. Arithmeticity enters only when the already established arithmetic D_{14} prefix bound is used in the final corollary. Throughout, p is the ORIGINAL vertex farthest from one strict interior point. The theorem is not asserted at an arbitrary neighbour with a weakened comparison factor.

Proposition 19. *Let a compact hyperbolic Coxeter polytope have an actual farthest vertex p, and write $z = E(p - x)$, $\|x\| < 1$. Let one WHOLE component of the incident finite diagram be $I_2(m)$, $m \geq 4$, and let e be the squared norm of the projection of this SAME x onto that component. If $e \leq 3/19$, then $m = 7$ and one actual opposite wall closes both component directions, has a single old contact of label three there, and has no contact with any other old component. This is precisely the actual O_7 profile. Consequently every ordinary $I_2(m)$, $m \geq 4$, has energy strictly greater than $3/19$, with arbitrary finite spectators.*

Proof. Use outward unit incident roots, finite Gram B, inverse $C = B^{-1} \geq 0$, duals ω_i, and unit inward dual directions $-u_i = -\omega_i/\sqrt{C_{ii}}$. For an actual outgoing edge of length L_i, maximality at

the original p gives

$$x \cdot u_i \geq \tau_i = \tanh(L_i/2) > 0, \qquad r_i = \tanh L_i = \frac{2\tau_i}{1+\tau_i^2}.$$

The old dihedral inverse is

$$C_I = d \begin{pmatrix} 1 & k \\ k & 1 \end{pmatrix}, \qquad k = \cos(\pi/m) \geq 1/\sqrt{2}, \quad d = (1-k^2)^{-1}.$$

Thus each $\tau_i^2 \leq e$, and the increasing function $4t/(1+t)^2$ on $0 \leq t < 1$ gives

$$r_i^2 \leq 57/121 =: r. \tag{Dih1}$$

We recall the actual-wall calculation forcing a common opposite. For an outside actual wall $c = Ap - v$, let $\beta_j = -c \cdot n_j \geq 0$ and put

$$T_i(c) = \frac{(C\beta)_i}{A\sqrt{C_{ii}}}.$$

For the actual opposite c_i of edge i, $T_i(c_i) = 1/r_i$. The matrix $C - C_{\cdot j}C_{j\cdot}/C_{jj}$ is entrywise nonnegative: its nonzero block is the inverse of the positive principal Gram obtained by deleting j. Actual acuteness between two distinct opposites therefore gives $T_j(c_i) \leq r_j$, while that same nonnegative matrix gives $T_j(c_i) \geq k/r_i$ within this old component. Hence distinct opposites imply $r_i r_j \geq k$. But (Dih1) gives $r_i r_j \leq 57/121 < 1/\sqrt{2} \leq k$. One actual wall c therefore closes BOTH directions.

Write its full decomposition as

$$c = Ap - v_I - v_S, \quad h = A^2 > 0, \quad b = -c \cdot n_1 \geq 0, \quad g = -c \cdot n_2 \geq 0, \quad Q = \|v_S\|^2 \geq 0.$$

Here S is the entire orthogonal spectator forest. Every positive contact is a native magnitude at least $1/2$; none of its finite, parallel or dotted possibilities is omitted before the actual endpoint constraints are imposed. At least one of b, g is positive.

Suppose both are positive. At the actual endpoint releasing direction one, the retained white Gram has residual $\delta = 1 - Q - g^2 > 0$. Its actual edge formula is

$$\cosh L_1 = \frac{\sqrt{d}(b+kg)}{\sqrt{1-Q-g^2}} \geq \frac{\sqrt{d}(1+k)}{\sqrt{3}} = \frac{\cot(\pi/(2m))}{\sqrt{3}} \geq \frac{1+\sqrt{2}}{\sqrt{3}} > \frac{11}{8}.$$

The last comparison is $128\sqrt{2} > 171$, whose positive squared margin is $32768 - 29241$. It follows that

$$e \geq \tanh^2(L_1/2) > \frac{11/8 - 1}{11/8 + 1} = \frac{3}{19},$$

a contradiction. Hence exactly one old contact $\beta > 0$ remains.

We need one whole-spectator fact at small time. If $h < 1/4$, then c cannot contact any foreign component. Indeed a positive foreign contact gives a normalized derivative

$$\frac{(C_S\beta)_j}{\sqrt{(C_S)_{jj}}} \geq 1/2.$$

This uses the full spectator inverse, with nonnegative entries and diagonal at least one. Its ratio to $\sqrt{h}$ exceeds one. The actual-wall shadow inequality therefore forces this SAME c to be the opposite of that foreign edge as well.

For clarity, the required orthogonal obstruction is specific to the original farthest vertex. Suppose a wall closed orthogonal dual directions u_i, u_j. Clear it positively off all other incident roots W:

$$c_F = c + \sum_{s\in W} \alpha_s n_s, \qquad \alpha = B_W^{-1}(-c\cdot n_W) \geq 0.$$

Strict interiority gives $z \cdot c_F < 0$. In the retained Lorentz three-space its exact form is

$$c_F = Ap - (A/r_i)u_i - (A/r_j)u_j.$$

Therefore

$$\frac{x\cdot u_i}{r_i} + \frac{x\cdot u_j}{r_j} < 1.$$

Original maximality gives the opposite bound $\tau_i/r_i + \tau_j/r_j = 1 + (\tau_i^2+\tau_j^2)/2 > 1$. This proves the obstruction, including arbitrary contacts in W. Thus $h < 1/4$ really forces $Q = 0$.

Put $q = d\beta^2$. The smaller normalized derivative square is $k^2 q$. The common-wall edge formula and (Dih1) imply

$$q - 1 \leq h = q + Q - 1 \leq rk^2 q.$$

Substituting $k^2 = 1 - 1/d$ and $r = 57/121$ gives the exact strengthened restriction

$$\beta^2(64d + 57) \leq 121. \tag{Dih2}$$

This is derived at the 3/19 threshold; it is not the older 1/7 inequality with an unchanged coefficient.

Since $\beta^2 \geq 1/4$, this forces $d \leq 427/64$. For every old $m \geq 8$, $d \geq 4 + 2\sqrt{2} > 34/5 > 427/64$, impossible. The complete remaining old alphabet is $4, 5, 6, 7$. At old $m = 5$, $d = 2+2/\sqrt{5} > 185/64$: equivalently $128 > 57\sqrt{5}$, with squared margin139. Since d increases with m, if $\beta \geq 1/\sqrt{2}$ at any of old $5, 6, 7$, (Dih2) would give the opposite $d \leq 185/64$. Thus all three rows have exactly $\beta = 1/2$. Every larger finite contact, parallel and dotted contact has been excluded by the native jump, rather than a sampled label list.

At old four, $d = 2$ and $\beta^2 \leq 121/185 < \cos^2(\pi/5)$. The strict comparison is $413 < 185\sqrt{5}$, with squared margin556. Only contacts three and four remain, giving $q = 1/2$ or 1. Consequently $0 < h \leq 57/242 < 1/4$. The small-time argument just proved gives $Q = 0$, whereas $h = q - 1 \leq 0$, a contradiction.

At old five only $\beta = 1/2$ remains. Since $d < 3$, $q < 3/4$ and $0 < h \leq r(d-1)/4 < 57/242 < 1/4$. Again $Q = 0$, and $h = q - 1 < 0$.

At old six, $\beta = 1/2$, $d = 4$, $q = 1$. If $Q = 0$, then $h = 0$, impossible. For any $Q > 0$, the actual positive sum

$$\nu = c + 2n_1 + \sqrt{3}n_2$$

is a nonzero future null: all three selected products vanish. The coefficient order is for OLD6/NEW3, since $c\cdot n_1 = -1/2$, $n_1 \cdot n_2 = -\sqrt{3}/2$, $c\cdot n_2 = 0$. Every other actual wall pairs nonpositively termwise with ν. It is a feasible ideal ray, contrary to compactness. No foreign contact was discarded in this affine exclusion.

At old seven only $\beta = 1/2$ remains, and

$$0 \leq Q = 1 + h - d/4 \leq \frac{427 - 64d}{484} < \frac{107}{484} < \frac{1}{4},$$

using $d_7 > 5$. Every positive spectator cost is at least 1/4, by inverse positivity and the native contact lower bound. Thus $Q = 0$. We have exactly the ACTUAL isolated-new-three wall in old

$I_2(7)$, closing both directions: the stated O_7. Its time square is $d_7/4 - 1 \in (0, 1/3)$; the bounds $5 < d_7 < 16/3$ follow from $6/5 < 2\cos(2\pi/7) < 5/4$. Its separate inherited energy floor $> 9/100$ is unchanged.

All weak inequalities included $e = 3/19$, so no ordinary equality case is omitted. The proof has retained every finite spectator component, all its contacts and the same original x. No arithmetic hypothesis or weakened-neighbour comparison was used. □

Corollary 20. *At the original farthest vertex in arithmetic dimension sixteen, a whole D_{14} cannot have an ordinary irreducible rank-two spectator.*

Proof. The fresh true-rank-14 argument in the proof of Proposition 12 gives

$$e_D > .84816 = \frac{5301}{6250}$$

for its actual whole rank-two spectator. Its bootstrap uses only the independently available spectator floor $> .09$, so it does not depend on the present strengthening. For an ordinary $I_2(m)$, $m \geq 4$, the proposition gives

$$\|x\|^2 = e_D + e_S > \frac{5301}{6250} + \frac{3}{19} = 1 + \frac{719}{118750} > 1.$$

For ordinary A_2, use its geometric companion bound $319/2000$, which exceeds $3/19$ by $61/38000$. Both contradict $\|x\| < 1$. The actual typed N_7/O_7 alternatives retain their separate costs and arguments; they have not been charged as ordinary. □

B Arithmetic chain windows and simultaneous actual feet

The endpoint arguments below share one arithmetic chain calculation. We prove it before specializing to the whole A_{16} and A_{17} cases. The geometric finite-white classification is the one in the general companion; the arithmetic restrictions and the simultaneous treatment of isolated feet are given here. No component dimension is obtained by deleting a node from a previous ambient theorem.

Let $N = 16, 17, 18$, let the original incident forest contain the WHOLE component A_{N-1}, and retain all finite spectators. Use its zero-sum $\mathbb{R}^N$ frame and unit old roots $n_i = (e_i - e_{i+1})/\sqrt{2}$. Then

$$D_i = \omega_i^2 = \frac{2i(N-i)}{N}, \qquad C_{ij} = \omega_i \cdot \omega_j = \frac{2i(N-j)}{N} \quad (i \leq j).$$

All heights $h_i = x_A \cdot \omega_i$ concern the SAME original farthest vector. Set

$$\ell_{16} = .601, \qquad \ell_{17} = \ell_{18} = .6.$$

B.1 The complete central inventory at the new losses

For $5 \leq i \leq N-5$, call an index weak when $h_i \leq \sqrt{D_i} - \ell_N$. We claim that every central constraint is either strong, an isolated-within-A cap with its actual cleared parameter $0 < u < 1$, or a pure golden end-helper. The last alternative can occur at at most ONE central index.

First establish the full inventory, without assuming a contrary component energy. Deleting i retains two old paths of ranks at least four. Their complete one-node finite extensions are isolation; one admissible end, D- or E-position three; one terminal four; or two endpoint threes, one in each path. Two contacts in a single path make a cycle. A connected two-path completion has rank $N - 1 > 8$, so finite classification forces the ordinary path bridge, with no extra finite foreign arm.

The zero-own repair depends only on side order $m \geq 5$. At a far-end three the time square is less than its full positive foreign cost; the next-to-far three has foreign remainder $< 2/m$ and the same obstruction. If such a positive foreign contact existed, its actual shadow would force the same opposite to close an orthogonal component, contrary to the original-farthest obstruction proved above. Terminal four has foreign remainder $< 1/m \leq 1/5$, so no positive spectator cost, and then negative time. The other zero-own D/E, internal-four and two-white roles have their actual positive affine stars or cycles. Their time is positive through the actual cap and every omitted wall product is nonpositive. Own contact is therefore positive.

Write $q = \sqrt{D_i}$. Here $5/2 < q \leq 3$. The actual edge ratio needed for a loss ℓ is

$$R_\ell(q) = \frac{1}{\ell} - \frac{q-\ell}{q(2q-\ell)}.$$

It increases with q and decreases with ℓ on this rectangle. Thus

$$R_\ell(q) \leq R_{.6}(3) = 41/27 < 761/500.$$

The finite-white residual formulas give the following ordinary lower ratios, with side order at least five: near-three/own at least five has $(101/125{+}2/5)/\sqrt{3/5}$; a D-position has $(101/125{+}1/5)/\sqrt{2/5}$; the E, terminal-four and two-end-three rows are stronger. All exceed $761/500$. The smaller owns have the same actual affine principals. For the two exceptional roles WITH positive foreign cost, the complete clearings give ratio $> 227/(25\sqrt{35}) > 761/500$, or ratio square $> 121/35$. These estimates retain the ENTIRE contacted spectator cost. Hence no additional ordinary nonisolated row is weak.

Unlike the older $3/2$ test, the present ratio threshold retains near-own-four also at side order seven. Up to reflection all remaining such profiles are

$$v = \omega_i/\sqrt{2} + \omega_{i-1}/2, \qquad i = 5, 6, 7.$$

The simple-three helper normalizes the cap to square two over the actual field, with old product -1; its own product is $-\sqrt{2}$, so $\sqrt{2} \in K$. At the embedding reversing $\sqrt{2}$, the time numerators at $N = 16$ are

$$126 - 88\sqrt{2}, \quad 143 - 100\sqrt{2}, \quad 154 - 108\sqrt{2}.$$

They are positive since $\sqrt{2} < 17/12$. Their increments with N are $(i-1)(3-2\sqrt{2}) > 0$. The conjugate new-root Schur complement is the negative of this numerator divided by N, contradicting positive definiteness. This removes all three side orders and both orientations.

The remaining nonisolated possibility is pure:

$$c = Ap - \gamma\omega_i - \omega_e/2, \qquad e = 1 \text{ or } N-1, \quad Q = 0,$$

with own five, finite own at least seven, or dotted. The exact helper-side identity is

$$\frac{(\gamma + 1/(2m))^2}{(m+1)/(2m)} - (4\gamma - 2) = \frac{(\gamma - 1 - 1/(2m))^2}{(m+1)/(2m)} \geq 0.$$

The ratio threshold and native-five gap therefore give

$$202/125 < t = 2\gamma < 3139/1458 < 54/25. \tag{Astrip}$$

For isolated-within-A contact, retain the whole foreign projection v_S and write

$$c = Ap - \gamma\omega_i - v_S, \quad Q = \|v_S\|^2, \quad u = (1-Q)/\gamma^2, \quad A^2 = \gamma^2(D_i - u).$$

Here $Q < 1$ by finite-white positivity. Before splitting Q, owns three, four, six and parallel are excluded for EVERY foreign cost. Own three has the actual affine E_7 principal with arms $1, 3, 3$; own four has the affine $4, 3, 3$ star; own six has the affine $6, 3$ path. Each positive affine kernel has positive time through the cap, and every omitted original wall, including every foreign wall, has nonpositive product termwise. Parallel similarly has the feasible positive null pair. Compactness forbids all these nulls. Thus $\gamma \geq \varphi/2$ before any foreign-cost estimate. When $Q > 0, u > 1$, the actual positive clearing has loss at most

$$\frac{1}{\sqrt{D_i} + \sqrt{D_i - 1}} + \sqrt{u - 1} < .2 + .383 < \ell_N.$$

This uses $Q \geq 1/4$, $\gamma \geq \varphi/2$, and $u - 1 \leq 5 - 3\varphi < .383^2$. At $u = 1$ the positive cleared vector is a forbidden feasible future null. The interval $0 < u < 1$ is RETAINED, including $Q > 0$; we do not charge a strong floor there.

For the remaining PURE isolated estimates, own five has the actual H_4 projected-point loss $< 25/76 + 5/24 < .6$. For own seven, the increasing cubic $8a^3 - 4a^2 - 4a + 1$ gives $.9 < a < .902$. The actual positive A_2 core gives

$$h_i > \sqrt{D_i - a^{-2}} - \sqrt{4/3 - a^{-2}},$$

and hence loss

$$< \frac{100/81}{2.6 + 2.37} + \sqrt{17/150} < .25 + .337 = .587.$$

Pure own eight is arithmetically impossible: $4\gamma^2 = 2 + \sqrt{2}$, and its opposite conjugate would require $D_i(2 - \sqrt{2}) < 4$, whereas $D_i \geq 55/8 > 4 + 2\sqrt{2}$. Every embedding of this minimal field extends to the actual totally real field, so this contradiction applies there. For all finite owns at least nine, the exact singleton finite projection gives $h_i > \sqrt{D_i - 1} - \tan(\pi/m)$. Since $\tan(\pi/9) < 1386/3727 < 2/5$, the loss is $< .6$. This covers the infinite finite-own tail, not a finite label sample. All positive finite and negative-norm projections just used are positive sums of ACTUAL roots; their selected signs, positive time and all omitted walls are retained as in the companion's actual-point lemmas.

B.2 Every pure weak helper is golden over the actual field

Normalize old A roots to square two. If r is a K-root on the actual cap line and its simple-three helper is a_e of square two, then $r^2 = 2(r, a_e)^2$. Rescaling by $-(r, a_e)^{-1} \in K$ therefore makes the cap square two with helper product -1. Its own product is $-t$. The path cycle through old simple-three bonds makes t integral. This compares only connected root lines; no disconnected norm squareclasses are equated.

Every nonidentity embedding of $\mathbb{Q}(t)$ extends to K. Strict conjugate positivity of the full old-component Schur complement gives

$$D_i t_\sigma^2 + 2C_{ie} t_\sigma + D_e < 4.$$

After reversing $i \leq N/2$, the cheaper costs for negative and positive t_σ are respectively

$$Q_-(s) = 2is^2 + 4s + 2 - 2(is + 1)^2/N, \qquad Q_+(s) = 2is^2 + 2 - 2(is - 1)^2/N.$$

At the smallest $N = 16, i = 5$, their values at $-4/5, 12/25$ exceed four by $3/40, 59/1000$, respectively. The costs at those endpoints increase with i and N; their values at zero are below four. Convexity of the quadratics therefore gives the full strict strip

$$-4/5 < t_\sigma < 12/25. \qquad \text{(Agoldenstrip)}$$

This covers EVERY nonphysical conjugate, not chosen ones.

If the degree is at least five, the nonzero integral norm has magnitude $< (54/25)(4/5)^4 < 1$. At degree four or three its magnitude is below two. Any positive small conjugate would make the magnitude below one, so all small conjugates are negative and their absolute product is $1/t$. At degree four the integral trace lies in $(-1, 0)$: its lower bound is $t - 12/5 > -1$, and its upper bound $t - 3t^{-1/3} < 0$ follows from $(54/25)^4 < 27$. At degree three the trace lies in $(0, 1)$, since $t - 8/5 > 0$ and $t - 2/\sqrt{t} < 1$; the latter follows from $(54/25)(29/25)^2 < 4$. Both degrees are impossible. At degree two the integral trace is one or two. Trace two makes the nonzero norm $t(2-t)$ lie in $(-1, 1)$. Trace one forces $t < 9/5$ and $t(1-t) \in (-2, 0)$; the norm is -1, and $t = \varphi$. Degree one leaves only $t = 2$, the actual parallel pair with a feasible future null. Thus every pure weak helper has $\gamma = \varphi/2$.

The old component and this foreign-free cap have a physically Lorentzian Gram over $\mathbb{Q}(\varphi)$. If K were a proper extension, a nonphysical real embedding fixing that subfield would preserve this indefinite principal Gram, impossible. Consequently the helper forces the ACTUAL field $K = \mathbb{Q}(\varphi)$, rather than a convenient overfield.

B.3 The complete golden packing calculation

For a golden cap reversed to $5 \le i \le N/2$, its time numerator $M = 2NA^2$ is

$$M_i^{\text{near}} = i(N-i) - (N+1) + (i+2)(N-i)\varphi, \quad M_i^{\text{far}} = i(N-i) - (N+1) + i(N-i+2)\varphi.$$

These formulas retain BOTH helper ends. At $N = 18$, conjugate positivity excludes the midpoint nine and opposite-end indices six through nine. The endpoint costs are $(173 - 84\varphi)/36$ and $(197 - 99\varphi)/36$, both greater than one, and the opposite-end cost is concave in i. At $N = 17$, only far seven is excluded, by $(170 - 84\varphi)/34 > 1$ using $\varphi < 34/21$. At $N = 16$ retain every orientation. The resulting COMPLETE necessary tables are

N	type and M	N(M)
16	near 5 : $38 + 77\varphi$	-1559
	near 6 : $43 + 80\varphi$	-1111
	near 7 : $46 + 81\varphi$	-719
	near 8 : $47 + 80\varphi$	-431
	far 5 : $38 + 65\varphi$	-311
	far 6 : $43 + 72\varphi$	-239
	far 7 : $46 + 77\varphi$	-271
17	near 5 : $42 + 84\varphi$	-1764
	near 6 : $48 + 88\varphi$	-1216
	near 7 : $52 + 90\varphi$	-716
	near 8 : $54 + 90\varphi$	-324
	far 5 : $42 + 70\varphi$	-196
	far 6 : $48 + 78\varphi$	-36
	far 8 : $54 + 88\varphi$	-76
18	near 5 : $46 + 91\varphi$	-1979
	near 6 : $53 + 96\varphi$	-1319
	near 7 : $58 + 99\varphi$	-695
	near 8 : $61 + 100\varphi$	-179
	far 5 : $46 + 75\varphi$	-59.

At $N = 16$, near and far eight are reflections at the same central index, so one row suffices.

Why do these finite tables restrict coexistence? Subtracting a cap's K-valued projection onto the FULL old forest leaves a nonzero vector on the SAME one-dimensional time line over K. The caps have no foreign component. Thus any two time coefficients have ratio in K; their M-ratio must be a square IN K, and its rational norm must be a rational square. This argument retains all spectators.

At $N = 16$ and 18, the positive norm magnitudes in each table are pairwise coprime nonsquares; their ratios are therefore not rational squares. These are direct small integer checks of the displayed complete tables. At $N = 17$, different norm squareclasses $-1, -19, -179$ cannot mix. In class -1, write the four entries as $42\varphi^3, 18\varphi^5, 14\varphi^5, 6\varphi^7$. After squares are removed the rational coefficient classes are $42, 2, 14, 6$. Every pair ratio has an odd valuation at three or seven; none is a rational square or five times a rational square. Those are exactly the positive rational squares in $\mathbb{Q}(\sqrt{5})$, as expansion of $(a + b\sqrt{5})^2$ shows. For the repeated -19 class,

$$\frac{M_{\mathrm{far8}}}{M_{\mathrm{near6}}} = \frac{25 + 33\varphi}{76} = \frac{83 + 33\sqrt{5}}{152}$$

has norm $1/16$. A square would require $a^2 = 121/304$ or $45/304$, both with odd denominator valuation at nineteen. Thus different types cannot coexist, including the repeated norm classes where a norm-only test would be insufficient.

At a fixed index the actual opposite is unique. Reflected copies of the same noncentral type would have

$$-2c_L \cdot c_R = \frac{(N - 2i)(i + (i + 2)\varphi) - (N + 2)}{N}$$

for near helpers, and the same formula with $i - 2$ replacing $i + 2$ for far helpers. Their simple-three helper bonds and old path make this parameter integral by the invariant cyclic criterion, even for a dotted mutual. But the constant numerators for $i = 5, 6, 7$ at $N = 16$ are $12, 6, -4$; for $i = 5, 6, 7, 8$ at 17 they are $16, 11, 2, -11$; and at 18 they are $20, 16, 8, -4$. The far rows have the same respective constants. None is divisible by N, contrary to $\mathcal{O}_{\mathbb{Q}(\varphi)} = \mathbb{Z}[\varphi]$. The central $N = 16$ row cannot occupy two different indices. Hence at most ONE central index has a golden helper. This is an all-native-five packing statement, independent of which of those caps are actually weak at the new loss.

B.4 Sum every retained isolated foot in one reference

The intervals $0 < u < 1$ retained earlier cannot be declared strong by the older .61 estimate. For such a cap clear the entire spectator projection by its positive actual-root combination. The actual vector

$$\eta = c + v_S + \gamma u n_i = \gamma\{\sqrt{D_i - u}\, p - (\omega_i - u n_i)\}$$

has square $-\gamma^2 u(1 - u)$, positive time, zero contacted spectator products, negative or zero c, n_i products, and all other actual products nonpositive. Its normalization is a genuine point of the whole chamber. Original maximality gives

$$x_A \cdot (\omega_i - u n_i) \geq \sqrt{D_i - u} - \sqrt{u(1 - u)}.$$

If $\lambda_i = \sqrt{D_i} - \sqrt{D_i - u} + \sqrt{u(1 - u)}$ denotes its loss, then

$$\lambda_i \leq a_i u + \sqrt{u(1 - u)}, \qquad a_i = \frac{1}{\sqrt{D_i} + \sqrt{D_i - 1}}.$$

The rational constants needed below are

N	$a_i < a$	positive reference coefficient $\beta_i \geq b$	ℓ
16	.1982	.022	.601
17	.1955	.0205	.6
18	.194	.018	.6.

The a bounds are positive-square comparisons at $D_{\min} = 55/8, 120/17, 65/9$. For every $0 < u < 1$,

$$au + \sqrt{u(1-u)} - \frac{b}{2}(2u - u^2) < \ell. \tag{Aabsorb}$$

For $u \leq 1/2$ use $a/2 + 1/2$, giving $.5991, .59775, .597$. For $u \geq 1/2$, use $2u - u^2 \geq 3/4$ and $\max(au + \sqrt{u(1-u)}) = (a + \sqrt{1+a^2})/2$. Radical upper bounds $1.0195, 1.019, 1.019$ give respectively $.6006, .5995625, .59975$, all strictly below their ℓ.

Let H be a concave reference with available nonnegative coefficients β_i, positive ones at least b:

$$w_j = (H_j - H_{j-1})/\sqrt{2}, \quad \beta_i = w \cdot n_i, \quad w = \sum_i \beta_i \omega_i, \quad E_H = \|w\|^2 = \sum_i \beta_i H_i.$$

Require $H_i + \ell < \sqrt{D_i}$ at its positive central coefficients. Omit the possible golden index. If S is the set of retained isolated indices with $\beta_i > 0$, define ONE vector

$$Z = w - \sum_{i \in S} \beta_i u_i n_i.$$

Sum all actual point pressures BEFORE estimating its norm:

$$x_A \cdot Z \geq E_H - \sum_{i \in S} \beta_i (\lambda_i - \ell).$$

Because $w \cdot n_i = \beta_i$ and $n_i \cdot n_j \leq 0$ for $i \neq j$,

$$\|Z\|^2 \leq E_H - \sum_{i \in S} \beta_i^2 (2u_i - u_i^2).$$

It follows that

$$e_A \geq 2x_A \cdot Z - \|Z\|^2 \geq E_H + \sum_{i \in S} \beta_i \{\beta_i (2u_i - u_i^2) - 2(\lambda_i - \ell)\} > E_H. \tag{Asummed}$$

Strictness follows from (Aabsorb) if $S \neq \varnothing$, and from the strict available constraints otherwise. Adjacent isolated indices and positive foreign costs are allowed. No nonadjacency is assumed for this new isolated set, and no independently optimized energies are added.

The references used by the endpoint arguments are

N	$H_5, H_6, \ldots$ to midpoint	E_H	maximum internal golden omission loss
16	$2.021, 2.137, 2.205, 2.227$	.8354522	.000576
17	$2.056, 2.186, 2.269, 2.310$	.8708972	.00055225
18	$2.087, 2.228, 2.324, 2.381, 2.399$	.9037838	.00050625.

Reflect $H_{N-i} = H_i$, use linear end tails, and at $N = 17$ take $H_9 = H_8$. The positive curvatures have exactly the lower bounds b above. All floor inequalities are positive-square comparisons: the new heights are the old .61 heights shifted by $.61 - \ell$, so $H_i + \ell$ is unchanged. A single internal omission sets its coefficient zero by neighbour averaging and increases the adjacent coefficients. An end omission replaces the first tail by the chord 0–6, or its reflection. These are changes in ONE reference. In particular the full and omitted references preserve the summed-foot argument whenever their positive central coefficients have the displayed lower bounds; subsequent endpoint grafts must check that condition again rather than assume it.

B.5 A golden end hole pays for its own endpoint

For later proper-component consumers we record how the exceptional end omission is closed. Suppose its central index is five, reversing if necessary, and put $\gamma = \varphi/2 < .81$. For either helper end e, with its adjacent old node j, the actual finite H_4 correction is

$$b_c = \gamma n_5 + \tfrac{2}{3} n_e + \tfrac{1}{3} n_j, \qquad \eta = c + b_c, \quad \eta^2 = \tfrac{2}{3} - \gamma^2 < .111^2.$$

The actual positive finite projected point has pure spatial displacement $v - b_c$. Its normalization and original maximality give $x \cdot (v - b_c) \geq A - .111$. Since $x \cdot b_c > 0$, the helper-end signed coordinate satisfies

$$x_{\mathrm{end}} > \sqrt{2}(A - .111 - \gamma h_5) > .49.$$

Both orientations are covered: the smaller opposite-end time squares, at $N = 16, 17, 18$, are

$$(38 + 65\varphi)/32, \quad (42 + 70\varphi)/34, \quad (46 + 75\varphi)/36.$$

They give $A > 2.115, 2.136, 2.155$. Weakness gives $h_5 < 2.022, 2.057, 2.088$. Using $\sqrt{2} > 1.4$, the corresponding signed-coordinate lower bounds are $.512652, .502362, .493808$. The nearer-end time is larger. Here the signed coordinate is x_1 at helper one and $-x_N$ at helper $N - 1$. No strong floor at the missing index five has been inferred.

Use the chord 0–6 at the omitted end. At the helper end the resulting reference is flat on $L = 6$ coordinates in the near case and $L = 5$ in the far case, with signed mean $.25 < b_0 < .3$. Replace that block by

$$.49, \quad \frac{Lb_0 - .49}{L - 1}, \ldots, \frac{Lb_0 - .49}{L - 1}.$$

Its sum is unchanged. The latter coordinates lie in $(.19, .262)$; the next old coordinate is below .101. Thus concavity persists, the new central boundary coefficient is $> .062 > b$, and all other positive central coefficients keep their previous lower bound. The new endpoint coefficient is positive and is paid by the actual endpoint pressure just proved. The one-reference energy increases by

$$\frac{L}{L - 1}(.49 - b_0)^2 > .04332.$$

The pre-graft chord energy is exactly

$$E_H - \frac{1}{2}\{H_5^2/5 + (H_6 - H_5)^2 - H_6^2/6\}.$$

At $N = 17, 18$ it is respectively $12569249/15000000$ and $13079276/15000000$; at $N = 16$ it exceeds .8008. These direct rational formulas give, after the graft and simultaneous absorption (Asummed), the conservative component bounds

$$e_{A_{15}} > .8348, \qquad e_{A_{16}} > .8703, \qquad e_{A_{17}} > .9032.$$

If the single golden hole is internal instead, subtract the internal losses in the reference table: the energies are $> .8348762, .87034495, .90327755$. If there is no hole they are stronger. The three conclusions therefore include every omission location, all retained isolated feet, and arbitrary finite spectators. They are the shared component estimates; no historical ambient-dimension consumer has been imported.

C The actual whole-A_{17} endpoint reduction

We now prove the geometric and arithmetic inclusion needed before using the recorded finite A_{17} propositions. The old incident diagram is WHOLE A_{17}, so $N = 18$ and there is no spectator. Keep the original p, z, x throughout. Use the $N = 18$ reference and simultaneous-foot argument of Appendix B, with energy $E_0 = .9037838$, central coefficients at least .018, and first/last five-coordinate signed means $.2951 < b < .2952$. The next signed coordinate is below .1. A possible single golden central hole is omitted before its pressure is estimated.

C.1 Actual endpoint lengths and the full finite-white list

Even with a golden end hole, the chord reference has energy

$$E_{\min} = \frac{13079276}{15000000} > .8719.$$

Each signed end block has length $L = 5$ or 6 and mean $.25 < b < .3$. If its actual endpoint coordinate were at least .63, replace the block by

$$.63, \quad (Lb - .63)/(L-1), \ldots, (Lb - .63)/(L-1).$$

Its sum and every available central height are unchanged. The remaining coordinates exceed .155, and the following coordinate is $< .1$; hence the changed positive central coefficient is $> .055/\sqrt{2} > .018$. All other positive central coefficients still meet .018. Thus the SAME summed-foot proof applies to this ONE new reference, whose energy increases by

$$\frac{L}{L-1}(.63 - b)^2 > \frac{6}{5}(.33)^2 = .13068.$$

This exceeds one after adding $E_{\min}$. Therefore both signed actual endpoint coordinates are $< .63$.

For either actual endpoint edge,

$$x_{\text{end}} \geq \sqrt{17/18}\sqrt{(C-1)/(C+1)}.$$

The bound $C \geq 5/2$ would give endpoint square $\geq 17/42 > .63^2$, impossible. If the ACTUAL field is nongolden, no central golden hole exists. Use the full reference and graft instead to $.575 = 23/40$. The remaining block coordinates exceed .225125; all changed central coefficients still exceed .018. The energy gain is $(5/4)(.575 - .2952)^2 = .09786005$, giving total > 1.00164385. Since $C \geq 21/10$ would give endpoint square $\geq 187/558 > (23/40)^2$, we obtain

$$\text{always } C < 5/2, \qquad \text{outside } \mathbb{Q}(\varphi) : C < 21/10. \tag{A17C}$$

Release old node one. The retained white path is A_{16} on nodes two through seventeen. Finite one-node classification gives isolation; helper three at $j = 2, 3, 16, 17$; or helper four at $j = 2, 17$. A second contact would make a cycle; no exceptional finite rank-seventeen diagram contains that path. There is no foreign contact in this whole-diagram case. Parallel own always has its actual future-null pair.

For helper three put $t = 2\gamma$. Connected simple-three normalization gives a square-two cap, helper product -1, own product $-t$, and integral t. Its full Lorentz star is defined over $\mathbb{Q}(t)$; all-place definiteness gives ACTUAL $K = \mathbb{Q}(t)$. The unit time square and edge coshes are exactly

$$A^2 = \frac{17t^2 + 2(18-j)t + j(18-j) - 36}{36},$$

j	C
2	$(17\gamma + 8)/9$
3	$(17\gamma + 15/2)/(3\sqrt{2})$
16	$(17\gamma + 1)/(3\sqrt{2})$
17	$(17t + 1)/18.$

These are obtained by clearing ALL retained A_{16} roots. For helper four the two formulas are

$$j = 2 :\ C = (17\gamma + 8\sqrt{2})/3, \qquad j = 17 :\ C = (17\gamma + 1/\sqrt{2})/3.$$

The raw white residuals are $18/17$ at a helper-three endpoint, $4/17$ at a next-to-end three, and $1/17$ at a helper-four endpoint.

At helper $j = 3$, every positive own violates $C < 5/2$. Own zero has the ACTUAL feasible affine-E_8 null

$$2n_1 + 4n_2 + 6n_3 + 5n_4 + 4n_5 + 3n_6 + 2n_7 + n_8 + 3c.$$

Its future time is $3A > 0$, and all omitted products are nonpositive. At $j = 16$, own zero has $A^2 = -1/9$, own three has the actual all-three cycle, and own at least four violates $C < 5/2$. The helper-four rows fail already at own zero for $j = 2$ and own at least three for $j = 17$; the latter own-zero time is $-1/18$. At $j = 2$, own zero has negative time and $C < 5/2$ gives $\gamma < 29/34$, leaving owns three, four or five. Own three is an actual affine triangle. Own four generates $\mathbb{Q}(\sqrt{2})$, but its edge exceeds $21/10$, contrary to the nongolden bound. Only own five in the ACTUAL golden field can remain at this end.

C.2 The far-helper strip and the golden exceptions

For the far helper $j = 17$, positive time gives $t > 1$. Outside the golden field, (A17C) gives

$$1 < t < 184/85, \qquad -19/17 < t_\sigma < 1 \quad (\sigma \neq \mathrm{id}).$$

Set $f(s) = s(s^2-1)$. On this nonphysical interval $|f(s)| \le 2/(3\sqrt{3})$: on $[-1, 1]$ use $4/27 - q(1-q)^2 = (1-3q)^2(4-3q)/27$; on $[-19/17, -1]$ the bound $1368/4913$ is smaller. Physically $0 < f(t) < 8$. Degree at least four would give nonzero integer norm of magnitude $< 8[2/(3\sqrt{3})]^3 < 1$. At degree three the magnitude is $< 32/27 < 2$, so $t, t-1, t+1$ would all be units. The signs at one and minus one leave exactly

$$X^3 - X - 1, \quad X^3 - 2X^2 - X + 1, \quad X^3 + X^2 - 2X - 1, \quad X^3 - X^2 - 2X + 1.$$

The first has discriminant -23, so is not totally real. The second is negative at $11/5$, with increasing positive branch there, so its physical root exceeds $11/5 > 184/85$. The latter two have positive values $5347/4913, 2899/4913$ at $-19/17$, contrary to all three roots lying above that endpoint.

In degree two the integral trace is $0, 1, 2, 3$, and the nonzero norm is $\pm 1, \pm 2$. For trace zero the negative conjugate strip forces $1 < t < 19/17$, so $-t^2$ lies strictly between $-2, -1$. For trace one the norm $t(1-t)$ leaves only -1, giving $t = \varphi$, or -2, giving the rational $t = 2$. For trace two the norm $t(2-t)$ is strictly between $-1, 1$, with zero excluded in degree two. For trace three the conjugate strip requires $t > 2$, and $t(3-t)$ lies strictly between one and two. Thus no nongolden quadratic survives. In degree one only $t = 2$ is possible, the forbidden parallel. Hence any far helper forces the actual golden field.

In $K = \mathbb{Q}(\varphi)$, the unconditional bound gives $1 < t < 44/17$ and the same nonphysical strip. Writing $t = a + b\varphi$ in the FULL ring yields $0 < b < 2$; thus $b = 1$, $a = 0$ or one, and $a = 1$ violates the physical bound. Again $t = \varphi$. Every possible nonisolated endpoint is therefore a near- or far-simple-three helper with own five in $K = \mathbb{Q}(\varphi)$.

We now exclude that actual field in full, retaining its isolated alternatives. The unit near/far helper times are $M_N/36, M_F/36$, where

$$M_N = 13 + 49\varphi, \quad \mathrm{N}(M_N) = -1595, \qquad M_F = -2 + 19\varphi, \quad \mathrm{N}(M_F) = -395.$$

Two far helpers have positive actual product $(1+4\varphi)/9$. Near/far time-norm ratio $319/79$ is not a rational square. Two near helpers have invariant two-cycle $(500+660\varphi)/81 \notin \mathbb{Z}[\varphi]$. Thus both endpoints cannot be nonisolated.

For an isolated endpoint put $U = 4\gamma^2$. Rescale its actual K-root line to raw square $2U$ and sole old product $-U$. The actual two-cycle makes U integral and the full Lorentz star gives $K = \mathbb{Q}(U)$. Its raw time square is

$$H^2 = U(17U-36)/18, \quad 36/17 < U < 225/17, \quad 0 < U' < 36/17.$$

The FULL ring models $U = a + b\varphi$ are exactly

U	$\mathrm{N}(U)$	$-\mathrm{N}[U(17U-36)]$	time norm squareclass
$1+\varphi$	1	251	251
$2+\varphi$	5	1595	$5\cdot 11\cdot 29$
$2+2\varphi$	4	4880	$5\cdot 61$
$3+2\varphi$	11	4631	$11\cdot 421$
$2+3\varphi$	1	2699	2699
$3+3\varphi$	9	14499	179
$3+4\varphi$	5	16895	$5\cdot 31\cdot 109$
$4+4\varphi$	16	22784	89
$4+5\varphi$	11	38291	11
$5+5\varphi$	25	16475	659.

Indeed $b\sqrt{5} < 225/17$ gives $b = 1,\dots,5$; the strict conjugate strip gives respectively $a = 1,2;\ 2,3;\ 2,3;\ 3,4;\ 4,5$. The entries follow from

$$\mathrm{N}[U(17U-36)] = n(289n - 612T + 1296), \quad n = \mathrm{N}(U), \quad T = \mathrm{Tr}(U).$$

The displayed squareclasses are distinct; their small prime factorizations can be checked by trial divisibility through the respective square roots.

Actual roots on the SAME K-time line require $U(17U-36)/M_N$ or $/M_F$ to be a square IN K. Their rational norms exclude every model except $U = 2+\varphi$ with a near helper; none has the far-helper class $5\cdot 79$. For that exception $U(17U-36) = M_N$ exactly. Its equal positive raw time coefficients and spatial product $U^2/18$ give

$$g = -(4+22\varphi)/9, \qquad g^2/U = (168+164\varphi)/81,$$

whose nonintegral two-cycle excludes it.

Finally two isolated golden endpoints cannot use different rows, since their rational time-norm squareclasses differ. With the same row their equal positive raw times give

$$g = 2U(9-4U)/9, \qquad g^2/U^2 = 4(9-4U)^2/81 \in \mathcal{O}_K.$$

Integral closedness makes $2(9-4U)/9$ integral, hence $8U/9$ integral; Bezout $8\cdot 8 - 7\cdot 9 = 1$ then makes $U/9$ integral. But $0 < \mathrm{N}(U) < 8100/289 < 29$, whereas $\mathrm{N}(U/9) = \mathrm{N}(U)/81$ would be a positive integer. Thus the actual golden field is impossible.

Both endpoints are now isolated in a nongolden actual field, with independent integral generators satisfying

$$K = \mathbb{Q}(U) = \mathbb{Q}(V), \quad 36/17 < U, V < 3969/425, \quad 0 < U^\sigma, V^\sigma < 36/17 \quad (\sigma \neq \mathrm{id}). \quad \text{(A17isolated)}$$

No equality of their parameters, finite-own assumption or common disconnected raw scale was used.

C.3 One two-end graft gives actual degree at most five

There is no golden hole after the preceding argument. If BOTH signed endpoint coordinates were at least $1/2$, modify the first and last five-coordinate blocks simultaneously. Replace each signed block mean b by

$$a, r, r, r, r, \quad a = 1/2, \quad r = (5b - a)/4.$$

Both sums and every central height stay unchanged. The supports are disjoint and each residual has sum zero, so the two residuals are orthogonal to one another and to the old flat blocks. Moreover $r > .243875 > .1$, making every changed positive central coefficient $> .018$. The new endpoint coefficients are paid by the two ACTUAL coordinates assumed at least a. The summed-foot proof therefore applies to this ONE new reference, with energy

$$> .9037838 + 2(5/4)(.5 - .2952)^2 = 1.0086414.$$

Thus at least one actual endpoint coordinate is $< 1/2$. For its generator U, the edge formula $C^2 = 17U/36$ and the farthest endpoint comparison give

$$36/17 < U < 66564/10625 < 157/25. \tag{A17shortU}$$

The other generator need not satisfy this sharper upper bound; the chosen one still generates the ENTIRE actual field.

Let $F(U) = U(U-1)(U-2)$ and $d = [K : \mathbb{Q}]$. For $d \geq 2$ all factors are nonzero algebraic integers. Physically

$$0 < F(U) < 142.$$

At any other embedding put $t = U^\sigma - 1 \in (-1, 19/17)$. On $[-1, 1]$, $|t^3 - t| \leq 2/(3\sqrt{3}) < 2/5$; on $(1, 19/17)$ it is at most $1368/4913 < 1/3$. If $d \geq 7$, the nonzero integral norm would have magnitude $< 142(2/5)^6 = 9088/15625 < 1$. Hence $d \leq 6$. For $d = 6$, the bound $142(2/5)^5 < 2$ makes its absolute norm one, and all three separate factor norms have absolute value one.

We give the complete degree-six seal. Set $s = U - 1$, and let $r_i = |U^{\sigma_i} - 1|$, $i = 1, \ldots, 5$. Irreducibility excludes $r_i = 0, 1$. The unit norms and (A17shortU) give

$$0 < r_i < 19/17 < 9/8, \quad \prod r_i = 1/s > 3/16, \quad s < 132/25, \quad s^2 - 1 < 16799/625 < 27.$$

The sharper $s < 132/25$, rather than merely $s < 16/3$, is needed in the last inequality.

If m positive numbers $q_i < 1$ have $p = (\prod q_i)^{2/m}$, ordinary AM–GM applied separately to q_i^2 and $1 - q_i^2$ gives

$$p + \{\prod (1 - q_i^2)\}^{1/m} \leq 1.$$

Let k be the number of $r_i > 1$, so $m = 5 - k$. Each large r_i contributes $|1 - r_i^2| < 72/289 < 1/4$. The following six cases exhaust all possibilities:

k	$\prod_{\text{small}} r_i$ lower	p lower	$\prod_{i=1}^{5} \lvert 1 - r_i^2 \rvert$ upper
0	3/16	1/2	1/32
1	1/6	2/5	81/2500
2	4/27	1/4	27/1024
3	32/243	1/8	49/4096
4	not needed	not needed	1/256
5	empty product	not needed	1/1024.

For the product bounds use $\prod_{\text{small}} r_i > (3/16)(8/9)^k$. The four root comparisons for p follow respectively from $9/256 > 1/32$, $1/6 > 4/25$, $16/729 > 1/64$, $32/243 > 1/8$; each exponent has been cleared positively. Multiplying the last column by 27 gives

$$27/32, \quad 2187/2500, \quad 729/1024, \quad 1323/4096, \quad 27/256, \quad 27/1024,$$

all below one. But the unit norm of $U(U-2)$ would be

$$1 = (s^2-1)\prod_{i=1}^{5}|1-r_i^2| < 1.$$

This excludes degree six and proves the ACTUAL degree bound $d \le 5$, without a field list or an equality assumption on the two endpoints.

C.4 The whole two-dotted interval and the derived finite fields

Suppose both isolated owns are dotted. For the left endpoint put $u = \gamma^{-2} \in (0,1)$. The positive actual core

$$c + \gamma^{-1}n_1 = \gamma\{\sqrt{17/9-u}\,p - (\omega_1 - un_1)\}$$

has negative square $u-1$, zero c product and negative n_1 product. Every other original wall, INCLUDING the other endpoint opposite, has nonpositive product with both summands. After normalization it is a genuine chamber point and gives

$$x \cdot Z_L \ge h(u) = \sqrt{17/9-u} - \sqrt{u(1-u)}, \quad Z_L = \omega_1 - un_1.$$

The right endpoint has its independent parameter $v \in (0,1)$ and reflected vector Z_R.

Let

$$Y_L = Z_L - \omega_5/5, \qquad R(u) = \|Y_L\|^2 = 8/5 - 2u + u^2.$$

This is a zero-sum vector supported in the first five coordinates. Its central-root products are zero except $-1/5$ at five. Its reflected partner is supported in the LAST five coordinates. Consequently $w \cdot Y_L = w \cdot Y_R = Y_L \cdot Y_R = 0$, while $w \cdot Z_L = w \cdot Z_R = .4174$. Pay both actual core pressures at $\mu = .3$:

$$w' = w + \mu Y_L + \mu Y_R, \quad \beta'_5 = \beta'_{13} = .1382 - .06 = .0782, \quad \beta'_i = \beta_i \ (6 \le i \le 12).$$

Every positive central coefficient is still at least .018, and $w' \cdot n_i = \beta'_i$ EXACTLY.

Retain every central isolated foot, with its OWN parameter $v_i \in (0,1)$, using ONE vector

$$V = w' - \sum_{i \in S} \beta'_i v_i n_i.$$

Sum the endpoint and central actual pressures before estimating the norm. The identities just proved and $n_i \cdot n_j \le 0$ give

$$x \cdot V \ge E_0 + \mu\{h(u) - .4174 + h(v) - .4174\} - \sum_S \beta'_i(\lambda_i - .6),$$

$$\|V\|^2 \le E_0 + \mu^2\{R(u) + R(v)\} - \sum_S (\beta'_i)^2(2v_i - v_i^2).$$

Every central summand is nonnegative by the already proved absorption inequality, since $\beta_i' \geq .018$. Thus

$$\|x\|^2 \geq E_0 + G(u) + G(v), \qquad G(u) = .6\{h(u) - .4174\} - .09(8/5 - 2u + u^2).$$

This whole continuous interval has a strict rational seal. Concavity and the endpoint squares give $\sqrt{17/9 - u} > 1.3743 - .4315u$ on $[0, 1]$. Therefore

$$G(u) > .43014 - .0789u - .09u^2 - .6\sqrt{u(1-u)}.$$

For $u \leq 1/2$, use $u^2 \leq u/2$ and the circle inequality

$$au + b\sqrt{u(1-u)} \leq (a + \sqrt{a^2 + b^2})/2.$$

With $a = .1239, b = .6$, $\sqrt{a^2 + b^2} < .6127$ gives $G(u) > .06184$. For $u \geq 1/2$, use $u^2 \leq 1.5u - .5$; $a = .2139, b = .6$, with radical $< .637$, gives $G(u) > .04969$. The positive square margins are $.00005008, .00001579$, respectively. Hence throughout $(0, 1)$,

$$\|x\|^2 > .9037838 + 2(.04969) = 1.0031638,$$

impossible. Parallel owns already have their actual future null. At least one endpoint is a finite native own.

At that endpoint $U = 4\cos^2(\pi/m) = 2 + 2\cos(2\pi/m)$ generates the ACTUAL field. The degree bound makes $\phi(m) \leq 10$. The COMPLETE elementary totient list is

$$3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 14, 15, 16, 18, 20, 22, 24, 30.$$

Indeed only prime divisors $2, 3, 5, 7, 11$ are possible; their permissible prime powers and products give that list. Labels three and four have $U \leq 2 < 36/17$. Six gives the rational field, excluded by Meyer and compact anisotropy at form rank eighteen. Five and ten give the already excluded golden field. For the following nine labels choose the indicated coprime cyclotomic embedding:

m	k	$\cos(2k\pi/m) = \sin\theta$	$2\theta/\pi$
9	2	$\theta = \pi/18$	1/9
14	3	$\theta = \pi/14$	1/7
15	2	$\theta = 7\pi/30$	7/15
16	3	$\theta = \pi/8$	1/4
20	3	$\theta = \pi/5$	2/5
24	5	$\theta = \pi/12$	1/6
30	7	$\theta = \pi/30$	1/15
11	2	$\theta = 3\pi/22$	3/11
22	3	$\theta = 5\pi/22$	5/11.

Every last entry exceeds $1/17$. Strict concavity on $0 < \theta < \pi/2$ gives $\sin\theta > 2\theta/\pi$, so that conjugate would have $U^\sigma > 36/17$, a contradiction. Label thirty was included before its conjugate test. Only labels $7, 8, 12, 18$ remain. Their actual fields are, respectively, $K_7, \mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), K_9$; the cubic parameters have polynomials

$$X^3 - 5X^2 + 6X - 1, \qquad X^3 - 6X^2 + 9X - 3,$$

with discriminants $49, 81$ and irreducibility by the rational-root test. This proves the actual field reduction, rather than assuming a list of possible fields.

C.5 Actual inclusion in the recorded finite domains

The isolated endpoint normalization in (A17isolated) has raw square $2U$, sole old product $-U$ and time square $F(U)/18$, where now $F(X) = X(17X - 36)$. For the two actual endpoint roots, subtracting their K-valued old projections leaves one common K-time line. Thus

$$R = F(U)F(V) \in K^2, \quad g = \frac{UV - \sqrt{R}}{18}, \quad L = \frac{g^2}{UV}.$$

The square root is PHYSICALLY POSITIVE. The long old-three cycle gives $g \in \mathcal{O}_K$; the actual two-cycle gives $L \in \mathcal{O}_K$. Acuteness gives $g \leq 0$. If $g \neq 0$, every nonphysical place satisfies $0 < L^\sigma < 4$; the branch $g = L = 0$ is retained. The physical owns and mutual products retain their native/dotted alphabet, with a parallel pair excluded by its actual future null. These are precisely the stated necessary tests in the quadratic domain; the endpoint parameters need not be equal.

If K is quadratic, this actual pair therefore enters the FULL quadratic domain of Proposition 10(i), including its maximal-order classes, square signs and zero mutual. That recorded finite-output proposition says the domain is empty. If $K = K_7$ or K_9, the finite endpoint proved above supplies the actual anchor seven or eighteen, respectively. The same original time-line square, cyclic and native conditions put the other endpoint in the full cubic finite-anchor domain of Proposition 10(ii). Its complete ring/box/decision arguments are given in Section 3, and the recorded output is empty.

Thus the stated A_{17} finite-output premises, together with the geometric inclusions proved here, exclude whole A_{17} at the original arithmetic rank-seventeen vertex.

D The separate whole-A_{16} endpoint proof

The whole A_{16} case has $N = 17$ and its own constants. It is not a consequence of reranking the preceding A_{17} theorem. We retain one original farthest vector throughout. The common chain proof gives central reference energy

$$E_0 = .8708972, \quad (H_5, H_6, H_7, H_8) = (2.056, 2.186, 2.269, 2.310), \quad H_9 = H_8,$$

with positive coefficients at least .0205. Its worst single end-hole chord has energy

$$\frac{12569249}{15000000}.$$

All grafts below preserve the common simultaneous-foot proof by checking those coefficients afresh.

D.1 Actual lengths and all nonisolated endpoint roles

On either signed end block, of length five or six and mean $.25 < b < .3$, grafting to .67 leaves each remaining coordinate $> .145$; the following coordinate is below .1. The central coefficient stays $> .045/\sqrt{2} > .0205$. The gain $> (6/5)(.37)^2 = .16428$, even after the worst chord omission, exceeds the available norm budget. Both signed endpoint coordinates are therefore $< .67$. The actual edge comparison is now

$$x_{\text{end}} \geq \sqrt{16/17}\sqrt{(C-1)/(C+1)}.$$

At $C = 17/6$ its square $176/391$ exceeds $.67^2$. Hence both endpoints have $C < 17/6$. If both signed coordinates were at least .56, disjoint grafts on the SAME reference would gain $> 2(6/5)(.26)^2 = .16224$, again exceeding one. Since $16/51 > .56^2$, at least one endpoint has $C < 2$.

Outside the actual golden field there is no hole and the full block mean lies in $(.2907, .2908)$. A single graft to .613 gains $> .12976605$, making total > 1.00066325. The retained coordinates are $> .210125$, so the coefficient threshold remains valid. As $32/85 > .613^2$, both endpoints then have $C < 7/3$. Two disjoint grafts to .52 gain $> .1313316$, giving total > 1.0022288. The exact comparison $1296/4777 > (13/25)^2$ therefore proves

$$\text{always both } C < 17/6,\ \min C < 2; \qquad K \neq \mathbb{Q}(\varphi):\ \text{both } C < 7/3,\ \min C < 181/100. \tag{A16lengths}$$

Release node one and retain the whole white A_{15}. Its complete finite one-node extensions are isolation; helper three at $j = 2, 3, 15, 16$; or helper four at $j = 2, 16$. There is no foreign component or exceptional rank-sixteen completion. A second contact makes a cycle. For helper three, the actual connected normalization gives integral $t = 2\gamma$, ACTUAL $K = \mathbb{Q}(t)$, raw time square $M_j/17$, and

$$M_j = 16t^2 + 2(17-j)t + j(17-j) - 34,$$

j	C
2	$(16t+15)/17$
3	$(16\gamma+7)/\sqrt{17}$
15	$(16\gamma+1)/\sqrt{17}$
16	$(16t+1)/17.$

For helper four the lengths are

$$(16\gamma + 15/\sqrt{2})/\sqrt{17/2}, \qquad (16\gamma + 1/\sqrt{2})/\sqrt{17/2}.$$

These follow by eliminating ALL retained roots; the unit white residuals are $17/32, 1/8, 1/16$ in the endpoint-three, next-three and endpoint-four rows.

Helper $j = 3$ with positive own gives $C \geq 15/\sqrt{17} > 17/6$. Its own-zero case has the same actual affine-E_8 vector printed in the preceding endpoint proof. At $j = 15$, own zero has unit time square $-2/17$, own three has the actual all-three cycle, and owns at least four give $C > 17/6$. Helper four at $j = 2$ fails even at own zero. At $j = 16$, its own-zero time is $-1/17$, and every positive own gives $C \geq (8\sqrt{2}+1)/\sqrt{17} > 17/6$. At near $j = 2$, own zero has negative time and own three the actual affine triangle. At far $j = 16$, positive time requires $t > 1$, since $M_{16} = (t-1)(16t+18)$. Parallel has the positive actual future-null pair. All these nulls use UNIT actual normals and are feasible against every omitted wall.

In a nongolden field the near helper has $t < 37/24$, so its only remaining native own is four, $t = \sqrt{2}$. The far helper satisfies

$$1 < t < 29/12, \qquad -9/8 < t_\sigma < 1.$$

As before $f(t) = t(t^2-1)$ has other-place magnitude at most $2/(3\sqrt{3})$; on the additional interval $[-9/8, -1]$, the maximum is $153/512$, smaller. Physically $0 < f(t) < 12$. The integral norm excludes degrees at least four. At degree three $t, t-1, t+1$ would all be units. The same complete four cubics as before occur. The negative-discriminant cubic and the two with a conjugate below $-9/8$ fail; the only survivor is

$$P(X) = X^3 - 2X^2 - X + 1,$$

whose physical root $a > 2$ generates K_7. For degree two, trace $0, 1, 2, 3$ and norm $-2, -1, 1, 2$ exhaust the strip. Trace zero would require $\sqrt{2}$, with conjugate below $-9/8$; trace one leaves only φ or rational two; trace two leaves $1+\sqrt{2}$, since $1+\sqrt{3}$ is too large; trace three's irrational positive root exceeds $29/12$. The rational two is parallel. Thus the nongolden helper roles are exactly near silver, far silver, or far K_7. Their actual lengths are respectively

$$(16\sqrt{2}+15)/17, \quad (17+16\sqrt{2})/17, \quad (16a+1)/17,$$

all $> 181/100$. The short endpoint selected in (A16lengths) is consequently ISOLATED.

D.2 The full golden endpoint table

For any isolated endpoint, its OWN K-rescaling gives raw norm $2U$, sole old product $-U$, integral $U = 4\gamma^2$ generating ACTUAL K, and

$$F(U) = U(16U - 34), \quad H_U^2 = F(U)/17, \quad C^2 = 8U/17, \quad U > 17/8, \quad 0 < U^\sigma < 17/8. \quad \text{(A16U)}$$

No raw time itself is assumed K-valued; the ratios along the SAME one-dimensional K-time line are K-valued.

Suppose $K = \mathbb{Q}(\varphi)$. The near helper has $t < 199/96$ and conjugate $-2 < t' < 1/8$. The finite native list leaves φ; a dotted full-ring $a + b\varphi$ has $b = 1$ and no integer a in $2 < t < 199/96$. The far helper satisfies $1 < t < 283/96$, $-9/8 < t' < 1$, giving $t = \varphi$ or φ^2. All three types have raw times $M/17$:

helper	M	$\mathrm{N}(M)$
near φ	$12 + 46\varphi$	-1420
far φ	$-2 + 18\varphi$	-356
far φ^2	$16 + 50\varphi$	-1444.

The norm squareclasses $355, 89, 1$ forbid different types. Reflected near copies have nonintegral actual long-cycle $g = -(7+41\varphi)/17$. Reflected far-φ copies have positive product $g = (4+15\varphi)/17$. Reflected far-φ^2 copies have $g = (19 - 15\varphi)/17 \in (-1, 0)$, forbidden for two distinct square-two actual roots by the native gap. Thus both endpoints cannot be nonisolated.

For the isolated golden end, the universal length cap gives

$$17/8 < U < 4913/288, \qquad 0 < U' < 17/8.$$

Writing $U = a + b\varphi$ gives $b = 1, \ldots, 7$. The complete conjugate intervals give $a = 1, 2;\ 2, 3;\ 2, 3;\ 3, 4;\ 4, 5;\ 4, 5;\ 5, 6$, and the physical upper bound removes only $(6, 7)$. The complete thirteen rows are

U	$\mathrm{N}(U)$	$-\mathrm{N}(F(U))$	squareclass
$1 + \varphi$	1	220	55
$2 + \varphi$	5	1420	355
$2 + 2\varphi$	4	4336	271
$3 + 2\varphi$	11	4180	1045
$2 + 3\varphi$	1	2396	599
$3 + 3\varphi$	9	12924	359
$3 + 4\varphi$	5	15020	3755
$4 + 4\varphi$	16	20416	319
$4 + 5\varphi$	11	34100	341
$5 + 5\varphi$	25	15100	151
$4 + 6\varphi$	4	21744	151
$5 + 6\varphi$	19	50996	12749
$5 + 7\varphi$	11	58036	14509.

For $n = \mathrm{N}(U), T = \mathrm{Tr}(U)$, every entry follows from $\mathrm{N}(F(U)) = n(256n - 544T + 1156)$. The listed classes are squarefree: $1045 = 5{\cdot}11{\cdot}19$, $3755 = 5{\cdot}751$, $319 = 11{\cdot}29$, $341 = 11{\cdot}31$, $12749 = 11{\cdot}19{\cdot}61$, $14509 = 11 \cdot 1319$, $355 = 5 \cdot 71$, $55 = 5 \cdot 11$; the remaining small primality checks are direct trial division. The only repeated class is151. Precisely the FIRST SIX rows have $U < 17/2$, equivalently $C < 2$, and their classes occur nowhere else in this complete table.

If one endpoint is near-φ or far-φ^2, its $C > 2$, so the opposite isolated end is in the first six rows. Class one matches none. The near class355 matches only $U = 2 + \varphi$; there $F(U) = 12 + 46\varphi$, so the actual positive raw times are equal. Their spatial product is $U(\varphi + 2)/17 = U^2/17$, giving the same nonintegral long cycle $g = -(7 + 41\varphi)/17$. For a far-φ helper the opposite end may be ANY of the thirteen rows, but none has class89. All helper/isolated possibilities are excluded.

Both golden endpoints must now be isolated. One belongs to the first six rows by the selected $C < 2$. The SAME time-line square test forces equality of the two U-models; the repeated151 class cannot occur at the selected end. Equal positive raw times give

$$g = U(34 - 15U)/17, \qquad L = g^2/U^2 = ((34 - 15U)/17)^2 \in \mathcal{O}_K.$$

Integral closedness makes $(34 - 15U)/17$, and then $U/17$, integral, using Bezout for15 and17. But the selected norms $1, 5, 4, 11, 1, 9$ are smaller than289, so $\mathrm{N}(U/17) = \mathrm{N}(U)/289$ cannot be a nonzero integer. The ACTUAL golden field is impossible.

D.3 Both actual endpoints are isolated

The actual field is now nongolden, and the selected endpoint of $C < 181/100$ is isolated. At $K = \mathbb{Q}(\sqrt{2})$, its full possible isolated strip from $C < 7/3$ is

$$17/8 < U < 833/72, \quad 0 < U' < 17/8.$$

Writing $U = A + B\sqrt{2}$, the complete models are

$$2 + s,\ 3 + s,\ 3 + 2s,\ 4 + 2s,\ 5 + 3s,\ 6 + 3s, \qquad s = \sqrt{2}.$$

Indeed $B = 1, \ldots, 4$; the conjugate strip permits respectively $A = 2, 3;\ 3, 4;\ 5, 6;\ 6, 7$, and the physical upper bound removes both $B = 4$ possibilities. Their signed F-norms are

$$-1016, -2212, -1852, -9184, -17444, -13752.$$

The near-silver helper has $M = 28 + 30s$, norm $-1016 = -8 \cdot 127$. Only $U = 2 + s$ has a compatible norm squareclass: none of the other five norms is divisible by127. For that exception $F(U) = M$ and the actual long cycle $g = -(22 + 26s)/17$ is not integral. The far-silver helper has $M = 32 + 34s$, norm $-1288 = -8 \cdot 7 \cdot 23$. None of the six time norms is divisible by23. Both silver helper roles are excluded.

For the remaining far helper in K_7, let $a > 2$ be the root of $P(X) = X^3 - 2X^2 - X + 1$. Its three roots have disjoint rational brackets

$$2.246 < a < 2.247, \quad -.803 < b < -.801, \quad .554 < c < .556.$$

Signs and derivatives of P isolate them. The full ring is $\mathbb{Z}[a]$, as justified below. The selected isolated endpoint has $17/8 < U < 7$, $0 < U_b, U_c < 17/8$. Write $U = A + Ba + Ca^2$. Lagrange interpolation expresses C as the sum of the three values divided by the derivatives at the three roots. The two positive denominators exceed five and four; the magnitude of the negative denominator exceeds $17/8$. Consequently $-1 < C < 7/5 + (17/8)/4 < 2$, so $C = 0, 1$. The small conjugate difference gives

$$|B + C(2 - a)| < (17/8)/(c - b) < 8/5.$$

If $C = 0$, positive physical difference forces $B = 1$, and the conjugate interval forces $A = 1$. If $C = 1$, the physical/conjugate difference < 7 removes $B \geq 1$, leaving $-1, 0$. For $B = -1$, the c-value requires $A \geq 1$, while the b-value requires $A \leq 0$. For $B = 0$, $A = 0, 1$. Thus the COMPLETE selected models are

$$U = 1 + a, \quad a^2, \quad 1 + a^2.$$

Their positive time norms are $5048, 32248, 82264$. The helper time numerator is $(a-1)(16a+18)$, of norm $7496 = 8 \cdot 937$. The prime937 divides none of these three norms; therefore no actual time ratio can be a K-square. This excludes the last nongolden helper. BOTH actual endpoints are isolated.

D.4 Degree, the whole two-dotted domain and finite anchors

The selected isolated endpoint generates the actual field, with

$$17/8 < U < 556937/80000 < 7, \qquad 0 < U^{\sigma} < 17/8.$$

For $F_3(U) = U(U-1)(U-2)$, physical value $0 < F_3(U) < 210$. At a nonphysical embedding $y = U^{\sigma} - 1 \in (-1, 9/8)$. The same cubic bound gives $|y^3 - y| < 2/5$, including the upper extra interval where the maximum is $153/512$. A degree at least seven would give a nonzero integer norm $< 210(2/5)^6 = .86016 < 1$. Hence ACTUAL degree is at most SIX. No degree-five refinement from the A_{17} case is used.

Suppose both isolated owns were dotted. Their ACTUAL endpoint cores give, for independent $u, v \in (0,1)$,

$$x \cdot Z_L \geq h(u) = \sqrt{32/17 - u} - \sqrt{u(1-u)}, \qquad Z_L = \omega_1 - un_1,$$

and its reflection. The core is future timelike, zero on its cap, negative on the released old root, and nonpositive against EVERY other actual wall, including the opposite endpoint. The residual $Y_L = Z_L - \omega_5/5$ has zero sum, support in the first five coordinates, norm $R(u) = 8/5 - 2u + u^2$, and sole central product $-1/5$ at five. The reflected residual subtracts $\omega_{12}/5$, and is orthogonal to both the first residual and the old flat block. Here $w \cdot Z_L = w \cdot Z_R = .4112$.

Pay each core at weight $\mu = .32$. The individual outer coefficients change from .1406 to $.1406 - .064 = .0766 > .0205$; every other central coefficient is unchanged. In particular the changed reference has its EXACT products equal to these new coefficients. Subtract all retained central isolated-foot terms with their own parameters, and apply the common simultaneous absorption proof to this ONE vector. All endpoint/central and adjacent-central crosses are retained, giving

$$\|x\|^2 \geq E_0 + G(u) + G(v), \quad G(u) = .64\{h(u) - .4112\} - .1024(1.6 - 2u + u^2).$$

The following COMPLETE three-chord certificate covers $[0,1]$:

u interval	$L(u) < \sqrt{32/17 - u}$	K	a	$\sqrt{a^2 + .64^2}$ upper
$[0, 1/2]$	$1.3719 - .3924u$	.451008	.097536	.6475
$[1/2, 3/4]$	$1.3989 - .4464u$	.506688	.208896	.6733
$[3/4, 1]$	$1.4385 - .4992u$	.570432	.293888	.7043.

The strict endpoint values $1.3719, 1.1757, 1.0641, .9393$ have squares below $32/17, 47/34, 77/68, 15/17$. Concavity proves each chord lower bound. On $[l, r]$ use $u^2 \leq (l+r)u - lr$; then $G(u) > K - au - .64\sqrt{u(1-u)}$. The circle inequality used earlier gives the three lower values .07849, .06559, .071338. The squared radical margins are

$$.000142978704, \quad .000095351184,$$
$$.000068333456.$$

Thus $G(u) > .06559$ on the whole dotted interval, and

$$\|x\|^2 > .8708972 + 2(.06559) = 1.0020772.$$

Both owns cannot be dotted. Parallel has its actual future null, so at least one endpoint is a finite native own.

Its integral $U = 2 + 2\cos(2\pi/m)$ generates the actual field, and the degree bound gives $\phi(m) \leq 12$. The full totient list is

$$3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 18, 20, 21, 22, 24, 26, 28, 30, 36, 42.$$

Only prime factors $2, 3, 5, 7, 11, 13$ and their allowed prime powers occur; checking products with totient at most twelve gives exactly this list. Labels three and four have $U \leq 2 < 17/8$. For own six the ACTUAL unit-root null is $\sqrt{3}c + 2n_1 + n_2$, a nonzero future ray feasible against all other walls. Five and ten give the excluded golden field. The remaining forbidden labels have explicit coprime embeddings:

m	k	$\cos(2k\pi/m)$ lower
$9, 11, 13, 15, 21$	2	$\sin(\pi/18) > 1/9$
$14, 16, 20, 22, 26, 28$	3	$\sin(\pi/14) > 1/7$
$24, 36, 42$	5	$\sin(\pi/12) > 1/6$
30	7	$\sin(\pi/30) > 1/15.$

All exceed $1/16$, by strict sine concavity. Their conjugate U would exceed $17/8$. Only finite anchors seven, eight, twelve and eighteen remain, with ACTUAL fields $K_7, \mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), K_9$.

We use FULL integer rings. For K_7, $P(X + 3) = X^3 + 7X^2 + 14X + 7$ is Eisenstein at seven, and the polynomial discriminant is49. For K_9, put $Q(X) = X^3 - 3X - 1$; $Q(X + 1) = X^3 + 3X^2 - 3$ is Eisenstein at three and the discriminant is81. The Eisenstein uniformizer criterion proves maximality at the only possible index primes, while the discriminants exclude every other index prime. Equivalently these are the full-ring proofs already given in Section 3, under translation. Thus the bases are the full $\mathbb{Z}[a]$, $\mathbb{Z}[k]$; the two quadratic rings are $\mathbb{Z}[\sqrt{2}]$ and $\mathbb{Z}[\sqrt{3}]$.

D.5 All twenty-one actual finite-anchor time norms

The opposite isolated endpoint V, whether finite or dotted, has

$$17/8 < V < 833/72, \qquad 0 < V^\sigma < 17/8.$$

For $\mathbb{Q}(\sqrt{2})$, its complete six models were printed above. For $\mathbb{Q}(\sqrt{3})$, the full ring gives $B = 1, 2, 3$, and conjugate intervals $A = 2, 3;\ 4, 5;\ 6, 7$. The physical upper bound removes only $(A, B) = (7, 3)$. The EXACT five models are

$$2 + r,\ 3 + r,\ 4 + 2r,\ 5 + 2r,\ 6 + 3r, \qquad r = \sqrt{3}.$$

For K_7, the leading-coefficient interpolation bound now gives $-1 < C < 3$, so $C = 0, 1, 2$ in $V = A + Ba + Ca^2$. The same small conjugate difference gives $|B + C(2 - a)| < 8/5$. The complete integer possibilities are

C	B	A
0	1	1
1	-1	none
1	0	$0, 1$
1	1	1
2	$-1, 0$	0.

For $C = 2, B \geq 1$, the physical/conjugate difference exceeds $3.047(1 + 2(1.444)) = 11.846736 > 833/72$. This excludes every remaining larger B; the conjugate-difference bound excludes the smaller ones. The resulting EXACT six models are

$$1 + a,\ a^2,\ 1 + a^2,\ 1 + a + a^2,\ 2a^2 - a,\ 2a^2.$$

For K_9, the three roots have brackets

$$1.879 < k < 1.880, \quad -.348 < b < -.347, \quad -1.533 < c < -1.532.$$

The two positive Lagrange denominators exceed $7.5, 4$, and the magnitude of the negative one exceeds $17/8$. Thus again $C = 0, 1, 2$. The full conjugate difference gives $|B - Ck| < 9/5$. The constant-term interval cases are

C	B	A
0	1	2
1	1	1
1	2	$1, 2$
1	3	3 fails the physical upper bound
2	2	none.

For $B = 3, C = 1$, the physical value is $> 3 + 3(1.879) + 1.879^2 = 12.167641 > 833/72$. For $C = 2, B \geq 3$, the physical difference exceeds $3.411(3 + .694) > 833/72$. At $C = 2, B = 2$, the b-strip forces $A \geq 1$ and the c-strip $A \leq 0$. Hence the EXACT four models are

$$2 + k,\ 1 + k + k^2,\ 1 + 2k + k^2,\ 2 + 2k + k^2.$$

Every listed accepted model satisfies all strict strips by the root brackets; every omitted integer coefficient has been excluded. These are complete handwritten domains, not unexecuted worker results.

The original raw times of two isolated roots are $F(U)/17$, $F(V)/17$. The same actual K-time line makes their ratio a K-square. A necessary condition is a square rational norm; BOTH possible signed norms of any putative square root satisfy that condition. Here are ALL twenty-one signed time norms, in the stated model order:

field and finite anchor	$\mathrm{N}(F(V))$
$\sqrt{2},\ 2 + \sqrt{2}$	$-1016, -2212, -1852, -9184, -17444, -13752$
$\sqrt{3},\ 2 + \sqrt{3}$	$-764, -3432, -8688, -12428, -27612$
$K_7,\ 1 + a$	$5048, 32248, 82264, 110552, 1784, 330688$
$K_9,\ 2 + k$	$16872, 66168, 79032, 293624.$

For a quadratic model the reproducible identity is

$$\mathrm{N}(F(V)) = n(256n - 544T + 1156).$$

For a cubic model with coefficient data (T, S, n), it is

$$\mathrm{N}(F(V)) = n(4096n - 8704S + 18496T - 39304).$$

The six K_7 triples are

$$(5, 6, 1), (6, 5, 1), (9, 20, 13), (11, 24, 13), (10, 17, 1), (12, 20, 8),$$

and the four K_9 triples are

$$(6,9,3),(9,18,9),(9,6,1),(12,27,17).$$

They follow by multiplication in the displayed full defining bases.

The anchors have odd valuations respectively at the primes $127, 191, 631, 19$:

$$1016 = 8(127), \quad 764 = 4(191), \quad 5048 = 8(631), \quad 16872 = 8 \cdot 3 \cdot 19 \cdot 37.$$

NONE of the other time norms in its field row is divisible by that prime. The exact nonzero residues of the ABSOLUTE time norms are respectively

$$(53,74,40,45,36), \quad (185,93,13,108), \quad (67,234,127,522,44), \quad (10,11,17).$$

Thus every UNEQUAL anchor/model time ratio has nonsquare rational norm and cannot be a square in K. No square norm is used as a sufficient test, and no native, orthogonal, parallel or dotted mutual case is omitted by this necessary invariant.

Only $V = U$ at the actual finite anchor remains. The endpoint walls are distinct, and their canonical raw norms are $2U$. Equal time squares give equal POSITIVE physical times; their spatial product is $U^2/17$. Therefore the actual two-cycle is

$$g = \frac{U(34-15U)}{17}, \qquad L = \frac{g^2}{U^2} = \left(\frac{34-15U}{17}\right)^2 \in \mathcal{O}_K.$$

The zero branch would force the irrational anchor $U = 34/15$, impossible, but integral closedness applies in either case. It makes $15U/17$, hence $U/17$, integral: $15 \cdot 8 - 17 \cdot 7 = 1$. The four anchor norms are $2, 1, 1, 3$, in degrees $2, 2, 3, 3$; their quotients by the corresponding powers of seventeen are not integers. This final contradiction excludes every diagonal pair too.

The whole-A_{16} exclusion is complete: actual endpoint lengths and helper/golden cases, BOTH isolated, degree at most six, full two-dotted interval, finite anchors and actual fields, complete full-ring domains, every unequal time ratio, and the actual diagonal two-cycle. This manual proof uses the printed inequalities and norm tables. It adds no census premise and assumes no equality between the raw root scales of disconnected components.

E One actual rank-one opposite

For the full forests $A_{15} + A_1, B_{15} + A_1, D_{15} + A_1$, the correct first split concerns the SAME actual opposite d of the old rank-one direction e. Either it contacts the old large component, or all those products vanish. In the isolated case physical unit normalization gives

$$d = \sqrt{h}\, p - \sqrt{1+h}\, e, \qquad h > 0.$$

No unrelated wall is chosen in the second case, and this identity does not identify disconnected field-root norm squareclasses. We first close the whole $A_{15} + A_1$ case and record its separate B_{15} nonisolated corollary.

E.1 Nonisolated rank-one opposites beside A_{15}

Let the original full forest be $A_{15} + A_1$, with old unit A-roots in zero-sum $\mathbb{R}^{16}$, and write

$$d = Ap - v_A - \gamma e, \quad A > 0, \quad \gamma > 0, \quad Q = \|v_A\|^2, \quad A^2 = \gamma^2 + Q - 1.$$

At the actual endpoint of the rank-one edge, old $A_{15}+d$ is a finite rank-sixteen white diagram. Its COMPLETE nonisolated extensions are

role	v_A	Q
endpoint three	$\omega_j/2,\ j=1,15$	15/32
endpoint four	$\omega_j/\sqrt{2},\ j=1,15$	15/16
next-to-end three	$\omega_j/2,\ j=2,14$	7/8.

These are A_{16}, B_{16}, D_{16}, including both orientations. Two old-path contacts make a nonpositive cycle; no exceptional rank-sixteen completion contains this long path.

We must have $\gamma > 1$. If $\gamma < 1$, the full positive actual replacement

$$r = p + \frac{A}{1-\gamma^2}(d+\gamma e)$$

has negative norm, positive time, zero products with e, d and all retained old A-roots except the contacted node j. Every unknown wall is nonpositive termwise, and its omitted n_j product is strictly negative. It is therefore the ACTUAL endpoint of old edge j. The SAME d would close both that direction and the orthogonal old rank-one direction, contradicting the original-farthest orthogonal obstruction. For $\gamma = 1$, $d+e$ is an actual future null. Thus $Q < A^2$ and the entire surviving own domain is dotted.

The actual rank-one edge length is $C = \gamma/\sqrt{1-Q}$, with energy at least $(C-1)/(C+1)$. At $Q \geq 7/8$, $C > \sqrt{8} > 7/3$ gives energy $> 2/5$. Together with the independently proved same-vector $e_{A_{15}} > .8348$, this excludes endpoint-four and next-three throughout their unbounded own domains. For endpoint-three, if $\gamma \geq 41/40$,

$$C^2 \geq 1681/850 > 49/25$$

gives rank-one energy $> 1/6$, and $.8348 + 1/6 = 1 + 11/7500$. Only $1 < \gamma < 41/40$ remains.

Here the ACTUAL positive core

$$d + \gamma e = Ap - \omega_j/2$$

is future timelike of square $1-\gamma^2$, zero on e, negative on d, and nonpositive on every other original wall. Original maximality gives $h_j/2 \geq A - \sqrt{\gamma^2-1}$. The right side decreases with γ. At $41/40$, $A^2 = 831/1600 > .72^2$ and $\sqrt{\gamma^2-1} = 9/40$. Thus the appropriate ACTUAL signed A_{15} endpoint satisfies

$$x_{\text{end}} > \sqrt{2}(.72 - .225) > .69.$$

Rebuild its reference before using the new pressure. Take the common N16 raw heights $2.021, 2.137, 2.205, 2.227$, reflected, with their possible single golden hole. Internal omission loses at most .000576. End omission replaces the first tail by 0–6, giving

$$E_{\text{end}} = 48050651/60000000 > .80084.$$

Thus every RAW reference has energy $> .79$, before any earlier golden-endpoint graft. At the selected endpoint its flat block has length five or six, mean $.25 < b < .3$, and next coordinate $< .101$, including an adjacent internal omission. Graft this block afresh to .69. The other coordinates are $> (L(.25) - .69)/(L-1) \geq .14$; the changed central coefficient is $> (.14 - .101)/\sqrt{2} > .027 > .022$. Every other positive central coefficient retains .022; the golden hole remains coefficient zero. The exact sum-preserving zero-sum change has zero old block cross, so the new ONE reference has energy

$$> .79 + (6/5)(.69 - .3)^2 = .97252.$$

Absorb EVERY retained actual isolated foot again into THIS reference, using the common N16 loss inequality and its NEW coefficients. This gives $e_A > .97252$, including adjacent and foreign feet. No earlier optimized endpoint gain or foot energy was added.

Finally the original rank-one edge has $C > \sqrt{32/17} > 11/9$, so its distinct orthogonal energy is $> 1/10$. Adding that energy ONCE gives $\|x\|^2 > .97252 + .1 > 1$. This excludes the entire nonisolated actual rank-one opposite beside A_{15}, over every actual field.

For later use the corresponding B_{15} argument is separate. A finite rank-sixteen extension of old B_{15} can only be B_{16}, attaching by three at its LONG endpoint. Thus $v_B = \Omega_1/2$, $Q = 1/2$, and $A^2 = \gamma^2 - 1/2$. The same reciprocal actual projection proves $\gamma > 1$. The original rank-one energy is then $> (\sqrt{2}-1)/(\sqrt{2}+1) = 3 - 2\sqrt{2} > .17$. The already written fresh B_{15} bulk in the proof of Proposition 12, with arbitrary spectators and both initial-five cases, gives $e_B > E_{13} > .88989$. Their sum exceeds 1.05989. Thus a nonisolated actual rank-one opposite is impossible beside B_{15} as well. No D_{15} conclusion is transported from this corollary.

E.2 The isolated A_{15} case and its invariant cycles

Now suppose the SAME actual opposite is isolated: $d = \sqrt{h}\,p - \sqrt{1+h}\,e$. If $h \geq 1$, its actual edge has energy at least $3 - 2\sqrt{2} > .17$; together with $e_A > .8348$ this already contradicts the norm budget. Assume $0 < h < 1$ and put $V = 4h$. The actual two-cycle makes V integral and all nonphysical places satisfy $-4 < V^\sigma < 0$. Physically $0 < V < 4$. Consequently $V/4$ CANNOT be integral, since its nonzero norm would have magnitude below one. The actual field has degree at least two by the stated Meyer/compact-anisotropy input for the ENTIRE rank-seventeen form.

We need pressures only at

$$J = \{5, 6, 7, 9, 10, 11\}, \qquad \ell_i = .49 \ (i \text{ odd}), \quad \ell_6 = \ell_{10} = .496.$$

The midpoint eight will never be assigned a pressure or coefficient.

Let an actual central opposite be

$$c = Ap - v_A - \beta e, \quad Q = \beta^2 < 1, \quad \lambda = -c \cdot d = A\sqrt{h} - \beta\sqrt{1+h} \geq 0.$$

The INVARIANT actual two/three cycles are

$$L = 4\lambda^2, \quad W = 4\beta^2, \quad T = 8\beta\sqrt{1+h}\lambda, \qquad L, W, T \in \mathcal{O}_K.$$

Zero β or λ is included. Direct expansion cancels the mixed radical:

$$L + T = VA^2 - W(1 + V/4). \tag{Rankonecycle}$$

No individual unit coefficient or disconnected raw time is assumed K-valued.

For an isolated-within-A cap, $v_A = \gamma\omega_i$, $B = 4\gamma^2$, this gives the exact full-foreign identity

$$L + T + V + W = \frac{i(16-i)BV}{32} \in \mathcal{O}_K. \tag{A15isocycle}$$

At EVERY nonphysical embedding, positive definiteness on the whole old A_{15} and this cap gives $0 < B^\sigma < 4/D_i$; its additional foreign square is nonnegative and only strengthens the bound. Together with $-4 < V^\sigma < 0$, this gives $|B^\sigma V^\sigma| < 16/D_i$. For one white-three contact, $v_A = \gamma\omega_i + \omega_j/2$, $t = 2\gamma$, and

$$M = i(16-i)t^2 + 2\min(i,j)(16 - \max(i,j))t + j(16-j) - 32,$$

the corresponding identity is

$$L + T + W = MV/32 \in \mathcal{O}_K. \tag{A15whitecycle}$$

Thus the foreign radical is retained through its triangle term. Dropping that term would not prove these congruences.

E.3 Finite native parameters and all dyadic places

For $a = 2\cos(\pi/m)$, $m \geq 4$, in any containing field, every dyadic valuation normalized by $v(2) = 1$ satisfies

$$0 \leq v(a), v(a+1) \leq 1/2.$$

Here is the full root-of-unity argument. Write $a = \zeta + \zeta^{-1}$, ζ primitive $2m$-th. Prime-to-two roots keep their order on reduction: a nontrivial odd-order root reducing to one would make its zero geometric sum reduce to a nonzero odd integer. Thus $v(a) > 0$ occurs only when $2m = 2^s$. For $s \geq 3$, $v(a) = v(\zeta^2 + 1) = 1/2^{s-2} \leq 1/2$. The formula follows because conjugate primitive 2^r-root differences have equal valuation and product two. Likewise

$$a + 1 = \zeta^{-1}(\zeta^3 - 1)/(\zeta - 1)$$

has positive valuation only when $2m = 3 \cdot 2^s$; for $m \geq 4$, $s \geq 2$, and $v(a+1) = 1/2^{s-1} \leq 1/2$. For $m = 3$, $a = 1$ and $v(a+1) = 1$; that exception is removed by an actual affine cycle BEFORE this lemma is used. In particular $B = 4\cos^2(\pi/m) = a^2$ satisfies $v(B) \leq 1$ for every finite $m \geq 3$. If $a \notin K$, use the cyclotomic compositum and descend the normalized valuation of $B \in K$. These bounds persist under every actual field extension.

For a finite one-white-three own, the actual three-cycle along the old path excludes own three. For every other finite own, apply the bounds to t. Put $a_i = i(16 - i)$ and $C = \min(i, j)(16 - \max(i, j))$. The following four parity cases prove $v(M) \leq 3$ at EVERY dyadic place.

If i, j are odd, rewrite in $s = t + 1$:

$$M = a_i s^2 + 2(C - a_i)s + [r(16 - r) - 32], \qquad r = |i - j|.$$

The last two coefficients are divisible by four and a_i is odd. The leading term has valuation at most one, while the others have valuation at least two. If i is odd and j even, use t instead: a_i is odd, C even and a_j divisible by four. The same unique-minimum argument gives $v(M) \leq 1$. If i is even and j odd, the constant is odd and the other coefficients are even, giving $v(M) = 0$. If both are even, $i = 6, 10$ makes $k = i/2$ odd. With $l = j/2$,

$$M/4 = k(8 - k)t^2 + 2\min(k, l)(8 - \max(k, l))t + l(8 - l) - 8.$$

For even l use the t-argument; for odd l use $s = t + 1$, whose constant $r(8 - r) - 8$, $r = |k - l|$, is divisible by four. Thus $v(M/4) \leq 1$ and $v(M) \leq 3$. No midpoint-eight assertion has been made.

Equation (A15whitecycle) now forces $v(V) \geq 2$ at every dyadic place. At all other finite places V is already integral. Hence $V/4$ would be integral, impossible. This excludes EVERY finite one-white-three cap at the six required sites, including foreign contacts and both orientations. The same proof applies to a dotted t whenever both $v(t), v(t+1) \leq 1/2$ have been established.

E.4 The fresh six-site inventory at losses .49, .496

At each $i \in J$, both retained old paths have ranks at least four. Their COMPLETE finite-white list is no old-A contact; one three at an admissible end or D/E position; one four at a path end; or one endpoint three in each path. The two-path bridge has rank fifteen, so it is A_{15}: a third arm through the new root would have two old arms at least four, incompatible with finite D, and no exceptional finite type has this rank. The old e-contact β is retained throughout. It only restricts the white list further.

Own zero is not silently discarded. Without old-A contact its time $Q - 1$ is negative. For a white three at an internal old node, the full old diagram and cap have actual affine arms $(1, 2, 5)$ or

$(1,3,3)$, unless the contact is at $1,2,14,15$. At global ends one or fifteen, the full finite white path has rank at least five; adding e allows only $\beta = 0, 1/2, 1/\sqrt{2}$. Then $A^2 = 15/32 + Q - 1 < 0$. At two or fourteen, $\beta = 0$ gives time $-1/8$. A positive foreign contact in that finite branched white component forces $\beta = 1/2$, and the full actual roots then have the affine arms $(1,2,5)$. An internal white four has the $4/3/3$ star. At a global end white four, any positive e-contact would put a three BEFORE the four in a path of rank at least six, or make a nonfinite $4,4$ principal; hence $Q = 0$ and time $15/16 - 1 < 0$. Two white threes at own zero give the actual all-three cycle. Every null is a positive sum of actual roots, future through the cap time, and nonpositive against ALL omitted walls.

For every positive own, the parallel value is excluded BEFORE the remaining case dispatch: $c + n_i$ is its actual feasible future null, including positive foreign contacts. For white three on a side of order $m = i$ or $16 - i$, number its node j from that side's far end. The actual residual and numerator are

$$\delta = 1 - j(m-j)/(2m), \qquad \sigma = \gamma + j/(2m).$$

For white four at an end they are $\delta = 1/m$, $\sigma = \gamma + j/(\sqrt{2}m)$. For two white threes, $\delta = 1/D_i$, $\sigma \geq \gamma + 1/D_i$. The ENTIRE old-$e$ cost changes the denominator to $\delta - Q$, and

$$C_i = \frac{\sigma\sqrt{D_i}}{\sqrt{\delta - Q}}.$$

If the desired ℓ_i-pressure fails, this implies

$$\frac{\sigma}{\sqrt{\delta - Q}} \leq R_{\ell_i}(\sqrt{D_i}) < 1.881. \tag{A15ratio}$$

For completeness the ratio target increases with q and decreases with ℓ on $q \leq \sqrt{8} < 2.829$, $\ell \geq .49$. At $q = 2.829, \ell = .49$, positive denominator clearing gives the strict margin119649968. Thus this is a fresh changed-loss inventory, rather than reuse of the old .601 weak list.

Finite one-white-three owns have already been excluded by the invariant parity argument, including own three's affine cycle. Thus only DOTTED $\gamma > 1$ remains there. At a D-position the ratio is $> (1 + 1/m)/\sqrt{2/m} \geq 6/\sqrt{10} > 1.881$. At an E-position $\delta \leq 1/4, \sigma > 1$, stronger still. Foreign cost decreases the denominator and cannot weaken this. For near white four, every positive own gives ratio

$$\geq (1/2 + 4/(5\sqrt{2}))\sqrt{5} > 2.38.$$

For far white four, owns three, four and six have actual affine principals using old roots at most three steps on each side of i, avoiding the helper at one or fifteen. The remaining own is at least $\varphi/2 > .809$, giving $.809\sqrt{5} + 1/\sqrt{10} > 2.12$. For two white threes, own three has an actual cycle. Every other positive own is at least $1/\sqrt{2} > .7$, and its ratio exceeds $.7(2.6) + .35 > 1.881$. Every finite-white role except dotted near/far three has now been included and removed.

E.5 Every remaining dotted helper fails over every field

For near white three the ratio is

$$\frac{\gamma + (m-1)/(2m)}{\sqrt{(m+1)/(2m) - Q}}.$$

At $m \geq 7, \gamma > 1$, it exceeds $10/\sqrt{28} > 1.881$. For $m = 5, 6$, weakness gives $2 < t = 2\gamma < 2.11555 < 53/25$. These are $i = 5, j = 4$, $i = 6, j = 5$, and reflections. Lift EVERY nonidentity embedding of $\mathbb{Q}(t)$ to K. Strict full-old-A positivity gives respectively

$$(55/8)t_\sigma^2 + 11t_\sigma + 2 < 0, \qquad (15/2)t_\sigma^2 + (25/2)t_\sigma + 23/8 < 0.$$

The foreign square is nonnegative, strengthening these conditions. Both negative quadratic intervals lie in $(-1.4,-.2)$; check positive endpoint values at $-7/5,-1/5$.

If the minimal degree is at least three and some small conjugate is at least -1, the nonzero integral norm of $t(t+1)$ has magnitude

$$< 6.6144(1/4)(.56)^{d-2} \le .926016 < 1.$$

If all small conjugates are below -1, the norm of $t+1$ has magnitude $< 3.12(.4)^{d-1} < 1$. In degree two the integral trace is one, and the norm $t(1-t)$ lies strictly between $-3,-2$. In degree one no integer lies in $(2,2.12)$. Thus every weak near dotted helper is impossible, including foreign contacts and proper actual fields.

For far white three, if $\gamma \ge 34/25$ its squared ratio is at least its $m=5$ value $5329/1500 > 1.881^2$. To check the minimum, put $r=1/m$. The derivative of $2(\gamma+r/2)^2/(1+r)$ has sign $1-\gamma+r/2<0$, and the ratio increases with γ. Hence a weak far helper has $2 < t < 68/25$. Conjugate positivity gives the entire strip

$$-4/5 < t_\sigma < 12/25.$$

The N16 endpoint cost checks at $-.8,.48$ have margins $3/40, 59/1000$; they increase over the reflected half-window. Again positive foreign cost only strengthens them.

For $f(t)=t(t^2-1)$, small conjugate magnitude is at most $2/(3\sqrt{3})$. Degree at least four would give norm magnitude

$$\frac{68}{25}\left\{\left(\frac{68}{25}\right)^2-1\right\}\frac{8}{81\sqrt{3}} < 1.$$

For an explicit positive seal, eight times the physical factor is139.229184, below $81(1.732)=140.292$. Thus the minimal degree is at most three. At degree three, $|\mathrm{N}(t)| < 2.72(.8)^2 < 2$, so it is a unit. Two positive small conjugates would give norm below one. For opposite signs their absolute product $1/t$ forces

$$t > 125/48, \quad t_{\text{positive}} > 125/272, \quad |t_{\text{negative}}| > 625/816,$$

putting the integral trace strictly between two and three. For two negative conjugates, the trace is one and $t < 2.6$. The integral second coefficient $1/t - t(t-1)$ lies in $(-4,-1.5)$, hence is -2 or -3. The two cubics are X^3-X^2-2X-1, of discriminant -31, and $X^3-X^2-3X-1 = (X+1)(X^2-2X-1)$. Neither is the minimal polynomial of the totally real cubic. Degree three is excluded.

Degree two has trace two or three. Trace two has norm in $(-1.9584,0)$, hence -1; trace three has positive norm below two, hence one. The exact possibilities are

$$t = 1+\sqrt{2}, \qquad t=\varphi^2.$$

Degree one is the forbidden parallel two. For $1+\sqrt{2}$, t is a dyadic unit and $v(t+1)=1/2$. For φ^2, both t and $t+1=\varphi+2$ have odd norm and are dyadic units. These facts persist in every actual field extension. Thus the already proved four-parity argument applies to M for BOTH remaining dotted possibilities. Equation (A15whitecycle) forces $V/4$ integral, a contradiction. All weak dotted helpers are excluded.

E.6 The isolated central caps through the full $h<1$ interval

For isolated-within-A contact, use (A15isocycle). At odd sites $i(16-i)$ is coprime to32, so integer Bezout gives $BV/32 \in \mathcal{O}_K$. At six and ten, the gcd is four, giving $BV/8 \in \mathcal{O}_K$. For ANY finite own

the dyadic bound $v(B) \le 1$ then forces $v(V) \ge 4$ or ≥ 2, respectively. Thus every finite isolated own would make $V/4$ integral, impossible; own zero has negative time. Parallel was already excluded globally by its actual null.

For dotted own, $u = (1-Q)/\gamma^2 \in (0,1)$. Its positive ACTUAL core $c + \beta e + \gamma u n_i$ has square $-\gamma^2 u(1-u)$, positive time, zero c, e products, negative n_i product and all other wall products nonpositive. It gives

$$h_i > \sqrt{D_i - u} - \sqrt{u(1-u)}, \qquad \text{loss} \le au + \sqrt{u(1-u)},$$

with $a < .2$ at the odd sites and $a < .19$ at six and ten. If $u \le 1/4$ at an odd site, the loss is $< 1/20 + \sqrt{3}/4 < 39/80 < .49$. Thus an odd weak cap forces $u > 1/4$ and $B < 16$. The integral $BV/32$ has physical magnitude < 2 and EVERY other magnitude $< 1/(2D_i) \le 4/55$. Since the actual degree is at least two, its nonzero norm has magnitude $< 8/55 < 1$, impossible.

At six or ten, for $u \le .27$ the loss is $< .0513 + .444 = .4953 < .496$. An even weak cap therefore has $u > .27$, $B < 400/27$. The integral $BV/8$ has physical magnitude $< 200/27$, and all other magnitudes $< 4/15$. At degree at least three its norm is $< (200/27)(4/15)^2 = 128/243 < 1$. At degree two, instead use the rational integer norms:

$$0 < \mathrm{N}(B) < 640/81 < 8, \qquad 0 < |\mathrm{N}(V)| < 16.$$

The first is positive and the second negative before taking absolute values. Integrality of $BV/8$ requires64 to divide their product. But an integer from one through seven has dyadic valuation at most two, and one through fifteen at most three. Their sum is less than six, a contradiction. This argument permits arbitrary ramification and full maximal order.

Combining the complete own-zero, finite-white, finite, parallel, dotted-helper and isolated cases proves, for EVERY $i \in J$,

$$h_i > \sqrt{D_i} - \ell_i.$$

There is still NO midpoint-eight assertion.

E.7 One final reference for every isolated rank-one height

Choose the rational heights

$$H_5 = H_{11} = 2.132, \quad H_6 = H_{10} = 2.2426, \quad H_7 = H_8 = H_9 = 2.3162$$

and linear outer tails. Their six positive constraints lie strictly below the just proved pressures by direct squares at $55/8, 15/2, 63/8$. The midpoint coefficient is zero, the slopes decrease, and

$$E = 2.132^2/5 + .1106^2 + .0736^2 = .92673412.$$

For $1/2 \le h < 1$, the separate rank-one edge energy is at least $5 - 2\sqrt{6} > .1$; hence the original norm exceeds one. Together with the earlier $h \ge 1$ argument this leaves only $0 < h < 1/2$.

We retain BOTH actual A_{15} endpoint pressures in that case. Put

$$T = \sqrt{1+4h} - 2\sqrt{h}, \qquad u = \frac{\sqrt{h}}{\sqrt{1+h}+1}, \qquad \alpha = \sqrt{15/16}.$$

The actual rank-one edge gives energy $\ge u^2$, $0 < u < 1/3$, and direct substitution gives $(1-u^2)(1-T^2) = 8uT$. The rank-free actual-I wall comparison (3) applies with every foreign contact retained. Explicitly, for another actual opposite with time A and old-e contact $\beta = 0$ or $1/2 \le \beta < 1$, its

chosen spatial derivative square is at most $1+A^2-\beta^2$. For $\beta=0$, the native gap gives $A^2\ge 1/(4h)$; for $\beta>0$, acuteness gives $A^2\ge(1+h)\beta^2/h$. Both imply full tangent square $\ge 1/(1+4h)$. In the positive-contact case the comparison, after positive denominators are cleared, has numerator $h((5+4h)\beta^2-1)>0$. Thus its normalized half-edge height is at least T. At BOTH signed A_{15} endpoints this gives $x_1,-x_{16}\ge\alpha T$. This is the original wall comparison, not a new field assumption.

The first and last five-coordinate reference means are $b=2.132/(5\sqrt{2})<.30152<.302$. Since $T>\sqrt{3}-\sqrt{2}>.317$ and $\alpha>.968$, $\eta=\alpha T>b$. Graft BOTH disjoint blocks to η. Their other coordinates $(5b-\eta)/4>1/8$ exceed the next old coordinate $.1106/\sqrt{2}<.079$. Thus all dual coefficients remain nonnegative, supported only on the six proved pressures and the two ACTUAL endpoint bounds; the midpoint coefficient remains zero. The single rebuilt reference gives

$$e_A>E+(5/2)(\eta-b)^2.$$

For completeness the scalar barrier $T+3u/2>.76$ follows directly from the displayed relation. At $T_0=.76-1.5u>0$, its left-minus-right is bounded below by

$$264/625-(19/5)u+(21419/2500)u^2>0.$$

The quadratic discriminant is $-55964/1562500<0$; the cubic term was bounded using $u<1/3$. The relation decreases strictly with positive T, proving the barrier. With $D=\eta-b>0$,

$$D+(3/2)\alpha u>.76\alpha-b>.43368.$$

Weighted Cauchy gives

$$(5/2)D^2+u^2>\frac{.43368^2}{2/5+(9/4)\alpha^2}=\frac{.43368^2}{803/320}>.074.$$

The positive comparison is $.1880783424>.18569375$. Adding the original rank-one energy ONCE to the rebuilt A_{15} reference yields

$$\|x\|^2>.92673412+.074=1.00073412>1.$$

All $h>0$, including boundaries $1/2,1$, are excluded for the isolated opposite.

Together with the complete nonisolated argument, this excludes the ENTIRE original $A_{15}+A_1$ family, with either ordinary nonshort rank one or actual short I. The dichotomy concerns one actual wall throughout, and the two case contradictions were never added to one another. No historical frontier-count subtraction is needed. The separate B_{15} nonisolated corollary only supplies its isolation reduction; its isolated ordinary and actual-I branches remain to be proved.

E.8 A fresh two-tail comparison for D_{15}

We first supply the missing body estimate for $D_{15}+A_1$. The conclusion is an energy bound, rather than unconditional strength of the last two knots. Use actual unit roots

$$n_j=(e_j-e_{j+1})/\sqrt{2}\quad(1\le j\le 14),\qquad n_{15}=(e_{14}+e_{15})/\sqrt{2}.$$

For $P_i=e_1+\cdots+e_i$, the long dual is $\Omega_i=\sqrt{2}P_i$, $i\le 13$. The two fork duals are

$$\Omega_{14}=(P_{14}-e_{15})/\sqrt{2},\qquad \Omega_{15}=(P_{14}+e_{15})/\sqrt{2},\qquad \|\Omega_{14}\|^2=\|\Omega_{15}\|^2=15/2.$$

The old rank-one unit root e is orthogonal to this space. Retain the SAME original decomposition

$$z = E(p - x - ye), \quad x_1 > \cdots > x_{14} > |x_{15}|, \quad e_D = \|x\|^2, \quad e_1 = y^2, \quad e_D + e_1 < 1.$$

Set $f_i = \sqrt{i} - .61/\sqrt{2}$ and

$$E_K = f_5^2/5 + \sum_{j=5}^{K-1} (\sqrt{j+1} - \sqrt{j})^2.$$

The shared geometric bulk inventory and Lemma 5 give strong $6, \ldots, 10$ and $e_D > E_{10} > .82436$, with either initial-five reference. The generic parameter is $M = 15 - 6 = 9$, so its last established index is ten. Eleven and twelve need a fresh check.

For their actual opposite write

$$c = Ap - v_D - \beta e, \quad A > 0, \quad \gamma = -c \cdot n_i \geq 0, \quad Q = \beta^2 < 1.$$

The full retained white Gram is positive definite. Every nonzero foreign cost satisfies $Q \geq 1/4$. If the inverse calculation gives residual $\delta > 0$ and $\cosh L_i = \sqrt{2i}\,\sigma/\sqrt{\delta}$, the original edge comparison makes the index strong if

$$\frac{\sigma}{\sqrt{\delta}} > T(\sqrt{2i}), \qquad T(q) = \frac{100}{61} - \frac{q - .61}{q(2q - .61)}.$$

On this interval T increases and $\sqrt{2i} < 5$, so

$$T(\sqrt{2i}) < T(5) = \frac{442721}{286395} < \frac{31}{20};$$

the final cleared numerator is $23825 > 0$. This actual threshold, rather than $100/61$, retains the full bridge list at this rank.

Deleting eleven leaves $A_{10} + D_4$, with long leaf twelve, centre thirteen and forks fourteen and fifteen. Deleting twelve leaves $A_{11} + D_3$, with middle thirteen and endpoints fourteen and fifteen. Finite classification gives the COMPLETE list: on the left, endpoint-three, endpoint-four or next-to-end-three; on D_4, simple three at any of its three leaves; on D_3, middle-three or endpoint-three, four or five. A two-piece bridge is simple three at a left endpoint and any listed simple-three right contact. Both left orientations and both fork orientations are retained. There is no exceptional finite rank-fifteen diagram. A bridge cannot have an extra old-e contact: that adds two forks to its D_{15} completion or the nonfinite arms $11, 3, 1$ to its A_{15} completion. Thus ONLY for bridges $Q = 0$.

For a left simple three at j, counted from its far end,

$$\sigma = \gamma + \frac{j}{2i}, \qquad \delta = \frac{i+1}{2i} \text{ at endpoints}, \qquad \delta = \frac{2}{i} \text{ at next-to-end sites.}$$

For endpoint four, $\sigma = \gamma + j/(\sqrt{2}i)$, $\delta = 1/i$. Foreign Q subtracts from every displayed residual. The remaining exact inverse-Gram formulas are

right contact	δ before Q	σ
D_4 long three or D_3 middle three	$1/2$	$\gamma + 1/2$
D_4 fork three	$1/2$	$\gamma + 1/4$
D_3 endpoint three	$5/8$	$\gamma + 1/4$
D_3 endpoint four	$1/4$	$\gamma + 1/(2\sqrt{2})$
D_3 endpoint five	$(5 - 3\varphi)/8$	$\gamma + \varphi/4$

and

i, right contact	δ	$\sigma-\gamma$ for near/far left endpoint
11, long three	1/22	21/22, 6/11
11, fork three	1/22	31/44, 13/44
12, middle three	1/24	23/24, 13/24
12, endpoint three	1/6	17/24, 7/24.

The middle-three D_{15} bridge is retained separately; treating D_3 only as an end-contact path would miss it.

Every affine exclusion below applies with arbitrary foreign contacts. Its positive combination of actual roots has positive future time and nonpositive product with every unselected original wall termwise. It would be a feasible null direction, impossible in a compact polytope. For a single three at global long node $j \geq 2$, own zero has the affine D principal on the cap, n_{j-1}, trunk $n_j, \ldots, n_{13}$, and BOTH forks; the trunk coefficients are two and the four terminal coefficients one. At a fork, own zero has affine E_8 arms $1, 2, 5$: the other fork is the length-one arm and the contacted fork plus cap is the length-two arm. Far-three at one with own zero has $A^2 = Q - 1/2$ and $Q < (i+1)/(2i) \leq 6/11$; the only native costs are $0, 1/4, 1/2$, so $A^2 \leq 0$. Far-four with own zero forces $Q < 1/i < 1/4$, hence $Q = 0$ and zero time. Near-four with own zero has the affine $4, 3, 3$ star. Right endpoint-four with own zero has affine B_4 kernel $(\sqrt{2}, 2, 2, 1, 1)$ on the cap, contacted fork, centre, old twelve and other fork. Right endpoint-five remains in the direct estimates.

Own three with a single-three helper has the actual all-three affine cycle. Own six has the affine $6, 3$ path using an old neighbour other than the helper. Own four with a right fork helper has affine F_4 path $n_{i-2}, n_{i-1}, n_i, c,$ fork, whose bonds are $3, 3, 4, 3$. With a nonadjacent left helper it has the $4, 3, 3$ star. The ADJACENT near-three/own-four profile is retained; it is not declared an affine star. An isolated-within-D cap has the affine D, star and $6, 3$ principals at owns three, four and six. Parallel own always has the actual feasible null $c + n_i$.

The remaining direct estimates are exhaustive. Right long-three and middle-three have ratio at least $1 + 1/\sqrt{2} > 31/20$. Right fork-three and endpoint-three have $\gamma \geq \varphi/2 > .8$; if $Q > 0$, their largest residual $3/8$ gives $1.05/\sqrt{3/8} > 31/20$. Positive-own right endpoint-four has ratio at least $1 + 1/\sqrt{2}$. Endpoint-five has residual $< .02$ and numerator $> .4$, hence ratio > 2.8, even for own zero. Left near-three with $\gamma \geq \varphi/2$ has ratio

$$\frac{\sigma}{\sqrt{\delta}} > \frac{.8 + 5/11}{\sqrt{6/11}} > 31/20.$$

Its extra own-four case has $\gamma > .7$, $\sqrt{\delta} < .74$, ratio $> 635/407 > 31/20$. Next-to-end-three has $\delta \leq 2/11$ and $\gamma > .8$. Positive-own endpoint-four has ratio at least $\sqrt{11}/2 > 31/20$. Every positive-own bridge has ratio > 1.9: even its weakest numerator is $19/24$ over $\sqrt{1/6}$. Own-zero bridges have the actual affine cycle.

The only unresolved left far-three is the literal packet $v_D = \gamma\Omega_i + \Omega_1/2$. Weakness gives $\gamma < (31/20)\sqrt{6/11} < 1.147$; the own removals give $202/125 < \varphi \leq 2\gamma < 12/5$. Lemma 5 excludes it at eleven and twelve, including foreign contacts. That lemma is not applied to the right-fork packet.

We also need all isolated-within-D caps, without assuming $Q = 0$. Their remaining own satisfies $\gamma \geq \varphi/2$ by the ALL-foreign-cost affine removals already given. Put $D_i = 2i$ and $u = (1 - Q)/\gamma^2$. For $0 < u < 1$, the actual positive core

$$\xi = c + \beta e + \frac{1 - Q}{\gamma} n_i$$

has selected products zero except $\xi \cdot n_i = \gamma(u-1) < 0$, square $-\gamma^2 u(1-u)$, and positive time. The original maximum gives, using $x \cdot n_i > 0$,

$$x \cdot \Omega_i > \sqrt{D_i - u} - \sqrt{u(1-u)}.$$

Since $q_i - \sqrt{D_i - u} < .108u$, its loss is at most

$$\frac{.108 + \sqrt{1 + .108^2}}{2} < .557 < .61.$$

At $u = 1$ the same core is a forbidden feasible null. For $Q > 0, u > 1$, positively project onto the ACTUAL finite set c, n_i, e, with residual $\gamma^2(u-1) > 0$. All correction coefficients are positive; clearing the old-e contact leaves a pure-D displacement. Here

$$u \le \frac{3}{\varphi^2} = 6 - 3\varphi < 7/6, \qquad \sqrt{u-1} \le 2 - \varphi < .4, \qquad q_i - \sqrt{D_i - u} < .13,$$

so the full loss is $< .53$.

For $Q = 0$, native five uses the LEFT actual H_4 correction

$$w_5 = \gamma(3n_i/2 + n_{i-1} + n_{i-2}/2), \qquad \delta_5 = 1 - 3\gamma^2/2 > 0.$$

Its actual finite projection gives $x \cdot (\gamma\Omega_i - w_5) > A - \sqrt{\delta_5}$. The divided displacement is $3\Omega_{i+1}/4 + \Omega_{i-3}/4$; long-arm concavity bounds its pressure above by $x \cdot \Omega_i$. With $u = \gamma^{-2} < 25/16$, the loss is $< 25/144 + 1/4 < .61$. Native seven uses the positive LEFT A_2 negative core with correction $\gamma(4n_i/3 + 2n_{i-1}/3)$. Its divided displacement is $2\Omega_{i+1}/3 + \Omega_{i-2}/3$. Using $1.21 < u < 100/81$, its full loss is $< .14 + .36 = .5$. All finite natives at least eight use the actual singleton projection, with loss $< .14 + \tan(\pi/8) < .555$. Dotted own uses the preceding core and loss $< .557$. All corrections use original roots, with $i + 1 \le 13$; neither fork is treated as another long root.

Thus a new weak tail can ONLY be the pure packet

$$v_D = \gamma\Omega_i + \Omega_f/2, \qquad i = 11 \text{ or } 12, \quad f = 14 \text{ or } 15, \quad Q = 0.$$

Weakness gives $\gamma + 1/4 < (31/20)\sqrt{5/8} < 1.24$. For $t = 2\gamma$,

$$202/125 < t < 1.98 < 12/5.$$

Normalize the old D roots to raw square two over the ACTUAL field K. The actual simple-three fork contact permits the same normalization for the cap; its actual path cycle makes t integral. The raw inverse entries $i, i/2, 15/4$ give

$$4i\, t_\sigma(t_\sigma + 1) + 15 < 8, \qquad -1 < t_\sigma < 0$$

at EVERY nonphysical embedding. Every embedding of $k = \mathbb{Q}(t)$ extends to K. The nonzero integer $t(t+1)$ has physical value $< 204/25$ and other conjugates of magnitude at most $1/4$. Degree at least three would give a nonzero integral norm of magnitude $< 51/100$. In degree two the trace is one; trace two would require $t > 2$, contrary to the sharper weak strip. Then $-2 < t(1-t) < -.995456 < 0$ forces norm -1, and $t = \varphi$. There is no physical rational integer in the strip. The new weak tail is therefore native five. Arithmetic positivity by itself does not remove this packet.

For that packet put

$$\alpha = \sqrt{2}(\gamma + 1/4), \quad \alpha^2 = \varphi + 5/8 > 56/25, \quad \delta = (5 - 3\varphi)/8, \quad \frac{\sqrt{\delta}}{\alpha} < .091.$$

The last comparison follows, for example, from $\delta/\alpha^2 < 73/8972 < .091^2$. The full time square is

$$A^2 = i\alpha^2 - \kappa, \qquad \kappa = 1/2 \ (i = 11), \quad \kappa = 5/8 \ (i = 12).$$

The LEFT H_4 correction gives the actual finite point $p+(A/\delta)(c+w_5)$. It has positive future time, negative square and zero selected products. EVERY omitted original wall, including the OTHER fork and old e, pairs nonpositively termwise. For $Y = v_D - w_5$, $V = Y/\alpha$, $K = i - 3$, the SAME original maximum yields

$$x \cdot V > G = \sqrt{i - \kappa/\alpha^2} - .091, \qquad V_j = 1 \quad (1 \le j \le K),$$
$$V^2 = (A^2 + \delta)/\alpha^2 < i.$$

All fifteen coordinates are retained in this norm.

Choose the shared positive prefix reference w on its first $K = 8$ or 9 coordinates. If five is strong it uses knots $5, \dots, K$. Otherwise use the actual weak-five endpoint, strong six and terminal K. Both have $w^2 \ge E_K$, $w \cdot P_K \le f_K$, and terminal INDIVIDUAL coefficient respectively

$$\sqrt{K} - \sqrt{K-1} \quad \text{or} \quad \frac{1}{\sqrt{K} + \sqrt{6}},$$

both greater than $1/6$. Pay that coefficient BEFORE using the terminal floor: $W = w+(V-P_K)/6$. The FULL cross with w is zero since $V - P_K$ vanishes on its first K coordinates. Its residual norm is $< i - K = 3$. One square gives

$$e_D > E_K + \frac{G - f_K}{3} - \frac{1}{12}.$$

At eleven, $3.282^2 < 11 - 25/112$ gives $G > 3.191$; $f_8 < 2.398$, $E_8 > .76864$ yield $e_D > .94964$. At twelve, $3.423^2 < 12 - 125/448$ gives $G > 3.332$; $f_9 < 2.569$, $E_9 > .79805$ yield $e_D > .96905$. If either tail is weak its body bound is stronger than $E_{12} > .86989$. If neither is weak, the ordinary or weak-five reference through twelve gives this last bound directly. We have proved the UNCONDITIONAL conclusion

$$e_D > .86989. \tag{R1D}$$

We have not proved unconditional strength of eleven or twelve.

Let d now be the actual opposite of the old rank-one edge. If it contacts D_{15}, classification of the FULL white diagram $D_{15} + d$ permits only the D_{16} completion: simple three at the long endpoint. Thus

$$d = Ap - \Omega_1/2 - \gamma e, \qquad Q_D = 1/2, \qquad A^2 = \gamma^2 - 1/2.$$

For $\gamma < 1$, the positive actual projection

$$p + \frac{A}{1 - \gamma^2}(d + \gamma e)$$

is future and feasible against ALL original walls. All retained products vanish, and its omitted n_1 product is negative. It is a finite endpoint on old long edge one, so the SAME d closes both that edge and the orthogonal old rank-one direction. This contradicts the original-farthest orthogonal common-opposite obstruction proved above. For $\gamma = 1$, $d + e$ is the forbidden future feasible null. Hence $\gamma > 1$ and $e_1 > 3 - 2\sqrt{2} > .17$. Together with (R1D) this exceeds one. The ACTUAL rank-one opposite is isolated from the ENTIRE D_{15}.

E.9 One ordinary rank-one argument for B_{15} and D_{15}

For B_{15} use actual roots $n_j = (e_j - e_{j+1})/\sqrt{2}$, $j < 15$, $n_{15} = e_{15}$; for D_{15} retain the preceding roots. The long duals through thirteen agree. Their actual old rank-one opposites have been isolated by separate proofs and therefore have the form

$$d = \sqrt{h}\, p - \sqrt{1+h}\, e, \qquad h > 0.$$

Ordinary rank one is the range $h \geq 1/2$, INCLUDING its boundary, so its original edge gives

$$e_1 \geq q(h) = \frac{\sqrt{1+h}-1}{\sqrt{1+h}+1} > .1, \qquad q(1/2) = 5 - 2\sqrt{6} > .1.$$

This disjoint original energy is counted once.

The B_{15} short-layer inventory gives strong $6, \ldots, 13$, with $e_B > E_{13} > .88989$ in either initial-five case. If $h \geq .57$, then $\sqrt{1+h} > 5/4$ and $e_1 > 1/9 > .111$, a contradiction. Thus $h < .57$. For D_{15}, (R1D) instead gives $h < .7$: otherwise $e_1 > 3/23$, whereas

$$.86989 + 3/23 - 1 = \frac{747}{2300000} > 0.$$

In this ORDINARY branch only, either weak tail is now impossible: its pure-body bound $> .94964$ or $> .96905$ already exceeds the remaining $1 - .1$ budget. Thus D_{15} has strong $6, \ldots, 12$.

Before selecting the head, eliminate weak five in either component. Its actual pure-component point gives $x_1 > .5965$; strong six gives $S_6 > f_6 > 2.01815$. Interpolate through these pressures and EVERY later strong knot up to $J = 13$ in B, $J = 12$ in D. The first-six energy is

$$.5965^2 + \frac{(2.01815 - .5965)^2}{5} = .7600299945 > .7600.$$

Its middle slope is $> .2843$, above the next ordinary increment $< .197$, so all individual duals are positive. As $E_6 < .6971$, this SINGLE rebuilt reference gives $e_T > E_J + .0629$, $T = B, D$. Even for D this is $> .93279$; adding $e_1 > .1$ is impossible. Five is therefore strong.

Fix the positive ordinary reference w through all knots $5, \ldots, J$:

T	J	E_J lower bound	h upper bound
B	13	.88989	.57
D	12	.86989	.7

and

$$w_1 = \cdots = w_5 = b = f_5/5 < .361, \qquad \text{individual } P_5 \text{ coefficient} = b - (\sqrt{6} - \sqrt{5}) > .14.$$

No previous tail or weak-five gain is added to this w. For $.49 \leq X \leq .643$, replace its first five coordinates by $X, (f_5 - X)/4, \ldots, (f_5 - X)/4$. The other four stay $> .2904$, above the next coordinate $< .214$. Every individual dual remains nonnegative and later knots are unchanged. Whenever ACTUAL $x_1 > X$, the single reference square gives

$$e_T > E_J + \frac{5}{4}(X - b)^2 > E_J + \frac{5}{4}(X - .361)^2. \tag{R1G}$$

Let c be the ACTUAL opposite of old long edge one. It is distinct from d, which has no B/D component. Deleting one retains B_{14} or D_{14} and old e. The COMPLETE finite extension

in that connected white tree is no contact or simple three at its LONG endpoint two. For D_{14}, fork attachment gives nonfinite arms $11, 2, 1$, and an internal attachment adds a second fork. For B_{14}, another weighted bond or fork cannot give a finite classical extension. Two contacts to either tree form a nonpositive cycle; exceptional finite ranks are too small. Foreign contacts to e are still retained.

In the nonisolated branch, with $g = -c \cdot n_1$ and $Q = \beta^2 = (-c \cdot e)^2$, the unit inverse Gram gives

$$c = Ap - g\Omega_1 - \Omega_2/2 - \beta e, \quad A^2 = 2g^2 + 2g + Q, \quad C_1 = \frac{2g+1}{\sqrt{1-2Q}}, \quad Q < 1/2.$$

Own zero and $Q = 0$ has zero time. Own zero and $Q > 0$ has the ACTUAL positive null

$$c + \Omega_2/2 = \sqrt{Q}\, p - \beta e.$$

Its clearing expansions must be kept separate:

$$\frac{\Omega_2}{2} = \begin{cases} n_1 + 2(n_2 + \cdots + n_{14}) + \sqrt{2} n_{15}, & B, \\ n_1 + 2(n_2 + \cdots + n_{13}) + n_{14} + n_{15}, & D. \end{cases}$$

Thus BOTH D forks and the ACTUAL B short root have their correct coefficients. Selected products vanish; the old-e product and every unknown original-wall product are nonpositive. The null is future feasible, contrary to compactness. Own three has the affine triangle c, n_1, n_2. Every other positive own has $g \geq 1/\sqrt{2}$, so $C_1 \geq 1 + \sqrt{2}$ and $x_1 > \sqrt{\sqrt{2} - 1} > .643$. The legal graft (R1G) adds $> .099405$; even the D total is $.86989 + .099405 + .1 = 1.069295 > 1$. The head is therefore isolated WITHIN its entire classical component, but it may still contact old e.

Its remaining form is

$$c = Ap - g\Omega_1 - \beta e, \quad A^2 = 2g^2 + Q - 1, \quad C_1^2 = \frac{2g^2}{1-Q} = 1 + \frac{A^2}{1-Q}.$$

Because its actual finite endpoint retains e, the COMPLETE foreign alphabet splits into

$$Q = 0, \qquad Q = 1/4, \qquad 1/2 \leq Q < 1.$$

The last interval includes EVERY finite foreign label at least four; no typed-seven alphabet is imported. Actual acuteness with d gives

$$\lambda = -c \cdot d = A\sqrt{h} - \beta\sqrt{1+h} \geq 0, \qquad h(2g^2 - 1) \geq Q. \tag{R1N}$$

If $\lambda > 0$, the ACTUAL native gap gives $\lambda \geq 1/2$, including parallel and dotted mutuals. The squared inequality excludes $g < 1/\sqrt{2}$; equality requires $Q = A = 0$. Owns zero, three and four are therefore impossible. Own six has the affine $6, 3$ kernel $(\sqrt{3}, 2, 1)$ on c, n_1, n_2, and parallel own has the future feasible null $c + n_1$. Both apply with every foreign contact present. For $Q = 0$, native five has $A^2 = (\varphi - 1)/2 < .31$; even the weaker D cut $h < .7$ gives

$$0 < \lambda^2 = A^2 h < .217 < 1/4,$$

forbidden by the native gap. The remaining PURE owns are finite labels at least seven and the ENTIRE dotted interval.

With $u = g^{-2}$, their actual pressure table is

own	Z	H	$R = Z^2$
finite ≥ 7	$(1 - u/2)(e_1 + e_2) + (u-1)e_3$	$[\sqrt{2-u} - \sqrt{7u - 4 - 3u^2}]/\sqrt{2}$	$3 - 4u + 3u^2/2$
dotted	$(1 - u/2)e_1 + (u/2)e_2$	$[\sqrt{2-u} - \sqrt{u(1-u)}]/\sqrt{2}$	$1 - u + u^2/2.$

Both Z are nonnegative, sum to one, and use genuine first-three long-root coordinates. For the finite row use the positive combination

$$\xi = c + g^{-1}n_1 + 2(g^{-1} - g)n_2, \qquad \xi^2 = -(7 - 4g^2 - 3g^{-2}) < 0.$$

Its selected products are $0, 0, (3 - 4g^2)/(2g) < 0$. For the dotted row use $\xi = c + g^{-1}n_1$, of square $-(1-u) < 0$. Both have positive time A. All unselected original walls, including e, d and BOTH D forks, pair nonpositively termwise. Their future normalizations and the SAME original maximum give $x \cdot Z \geq H$.

We prove the scalar estimate on the FULL domain:

$$.8H - .16R > .2893. \tag{R1S}$$

For native seven the elementary cubic gives $.9009 < g < .901$. Positive rational squaring gives $A > .7894$, $\sqrt{7 - 4g^2 - 3/g^2} < .2395$, $g\sqrt{2} < 1.2743$, hence $H > .4315$. Also $1.2318 < u < 1.233$, where R decreases, so $R < .349$, and the expression is $> .28936$.

For every finite native at least eight, $1 < u \leq 4 - 2\sqrt{2} < 1.172$. Put $D = (u-1)(4-3u)$ and $a = 2\sqrt{2}/5$. For $F = .8H - .16R$,

$$F'(u) = a\left(-\frac{1}{2\sqrt{2-u}} - \frac{7-6u}{2\sqrt{D}}\right) + .64 - .48u.$$

On $(1, 7/6]$ the first contribution is $< -.28$, the second is nonpositive and the last is at most .16. On $[7/6, 1.172)$, $D > .08066$, so $\sqrt{D} > .284$; the now positive second contribution is $< .032$, while the others total $< -.28 + .08$. Thus F decreases. At the native-eight endpoint, $\sqrt{2-u} > .910$, $\sqrt{D} < .289$, $R < .373$, giving $H > .4388$ and $F > .29136$.

For dotted $0 < u < 1$, put $a = .8/\sqrt{2}$, so $.5656 < a < .5657$. For $u \leq 1/2$, concavity gives $\sqrt{2-u} \geq 1.4142 - .379u$, $R \leq 1 - .75u$. Consequently

$$F > .63987152 - .0944u - .5657\sqrt{u(1-u)} > .30587152.$$

Indeed $\sqrt{.0944^2 + .5657^2} < .5736$, and

$$bu + c\sqrt{u(1-u)} \leq \frac{b + \sqrt{b^2 + c^2}}{2}.$$

For $u \geq 1/2$, use the chord $\sqrt{2-u} \geq 1.4494 - .4494u$, $R \leq .75 - .25u$; then

$$F > .69978064 - .2142u - .5657\sqrt{u(1-u)} > .29023064,$$

using $\sqrt{.2142^2 + .5657^2} < .6049$. The two chord endpoint inequalities are positive rational squarings. This covers the unbounded dotted-own interval and proves (R1S).

Use ONE vector

$$W = w + \frac{2}{5}(Z - P_5/5).$$

Its debit $2/25 = .08$ is below the INDIVIDUAL P_5 coefficient $> .14$, paid BEFORE the floor. The residual has zero sum on the first five coordinates, hence FULL cross zero with w, and norm $R - 1/5$. The actual pressure, remaining individual floors and one square give

$$\begin{aligned} e_T &\geq 2x \cdot W - W^2 \\ &> E_J + .8H - .16R - .8b + .032 \\ &> E_J + .2893 - .8(.361) + .032. \end{aligned}$$

Even for D this is $> .90239$. Adding original $e_1 > .1$ is impossible. This PURE displacement is never used for a mixed head.

For EVERY $Q \geq 1/2$, (R1N) gives

$$C_1^2 > \frac{1+Q/h_*}{1-Q}, \qquad h_* = \begin{cases} .57, & B, \\ .7, & D. \end{cases}$$

The expression increases on $[1/2, 1)$. For B it is at least $214/57 > 3$, so $C_1 > 41/25$, $x_1^2 > 8/33 > .49^2$. For D it is at least $24/7 > (37/20)^2$, so $x_1^2 > 17/57 > .54^2$. Their legal graft gains are

$$\frac{5}{4}(.49 - .361)^2 = .02080125 \quad (B), \qquad \frac{5}{4}(.54 - .361)^2 = .04005125 \quad (D).$$

Their original totals exceed 1.01069125 and 1.00994125, respectively. This covers ALL finite foreign labels at least four, finite/dotted head owns and zero or positive mutuals.

For $Q = 1/4, \lambda > 0$, the native gap gives

$$A\sqrt{h} \geq 1/2 + (1/2)\sqrt{1+h} > 1.1.$$

For B, $1.45^2(.57) < 1.21$ gives $A > 1.45$, hence $C_1 > 1.95$, $x_1^2 > 19/59 > .56^2$. Its legal gain .04950125 is contradictory. For D, $1.3^2(.7) < 1.21$ gives $A > 1.3$, $C_1^2 > 244/75 > (9/5)^2$, $x_1^2 > 2/7 > .53^2$. Its legal gain .03570125 makes the total > 1.00559125.

Finally $Q = 1/4, \lambda = 0$ gives

$$g^2 = \frac{1 + 1/(4h)}{2}.$$

For B, $41/57 < g^2 \leq 3/4$; for D, $19/28 < g^2 \leq 3/4$. Thus $\cos(\pi/5) < .81 < g \leq \cos(\pi/6)$; more precisely $g > .84$ in B, $g > .82$ in D. The full native alphabet leaves only own six at the INCLUDED $h = 1/2$ boundary. Its actual affine null was already excluded. Parallel and dotted $g \geq 1$ are also impossible.

This covers every original wall, head own, foreign contact and mutual parameter for $h \geq 1/2$. Neither B_{15} nor D_{15} can occur with an ORDINARY nonshort rank-one component. The actual short-I cases $0 < h < 1/2$ still need their separate arguments; this ordinary bootstrap does not discard them.

E.10 The short-I cycle identity and the B_{15} long window

We now enter $0 < h < 1/2$. Keep the ACTUAL isolated opposite $d = \sqrt{h}\,p - \sqrt{1+h}\,e$, and put $V = 4h$. Its actual two-cycle and strict nonphysical positivity give

$$V \in \mathcal{O}_K, \qquad 0 < V < 2, \qquad -4 < V_\sigma < 0 \quad \text{at EVERY nonphysical place.}$$

In particular $V/4$ cannot be integral: every conjugate of that nonzero number has magnitude < 1.

For ANY actual classical-edge opposite $c = Ap - v - \beta e$, put $Q = \beta^2 < 1$ and $A_0^2 = A^2 - Q = \|v\|^2 - 1$. Actual acuteness gives $\lambda = -c \cdot d = A\sqrt{h} - \beta\sqrt{1+h} \geq 0$. The actual cycles

$$L_c = 4\lambda^2, \qquad W_I = 4\beta^2, \qquad T_c = 8\beta\sqrt{1+h}\lambda$$

are integral. The full radical cancellation is

$$L_c + T_c + W_I = V(A^2 - \beta^2) = V A_0^2 \in \mathcal{O}_K. \tag{IC}$$

This retains $\beta = 0$, $\lambda = 0$ and every positive foreign contact. It does not assume integrality of an individual Cartan entry or identify disconnected root scales.

For B_{15}, strong $6, \dots, 13$ are already available. If five is weak, its COMPLETE geometric profile is pure:

$$v = \gamma\Omega_5 + \Omega_1/2, \qquad t = 2\gamma \in \mathcal{O}_K, \quad 202/125 < t < 12/5, \quad A^2 = (5/2)t^2 + t - 1/2.$$

Every other conjugate of t obeys $5t_\sigma^2 + 2t_\sigma - 1 < 0$, hence $-.69 < t_\sigma < .29$. Degree at least four contradicts $2.4(.69)^3 < 1$. In degree three the norm is a unit; a positive small conjugate would make its magnitude $< 2.4(.29)(.69) < 1$. Writing the other conjugates $-a, -b$, the trace is one: it is $> 1.616 - 1.38 > 0$ and at most $t - 2/\sqrt{t} < 2$. Then $a + b = t - 1 < 1.38$, $t < 2.38$, and the integer second coefficient $1/t - t(t-1)$ is -1 or -2. The corresponding polynomials

$$X^3 - X^2 - X - 1, \qquad X^3 - X^2 - 2X - 1$$

have discriminants $-44, -31$, so are not totally real. In degree two the norm is ± 1 and the trace one or two. Trace two gives $t(2-t) \in (-.96, .621)$, impossible; trace one gives $t = \varphi$. The only rational value, two, is an actual parallel null. Thus weak five is golden.

Its connected raw old-B/cap Gram is defined over $\mathbb{Q}(\varphi)$, with old long and cap squares two and old short square one. It is physically Lorentzian. A proper actual-field extension would give a nonphysical embedding fixing that indefinite principal. Hence ACTUAL $K = \mathbb{Q}(\varphi)$. Here $A^2 = (4 + 7\varphi)/2$, and $\mathrm{N}(4 + 7\varphi) = -5$ is odd. Equation (IC) makes $V/2$ integral at every dyadic place, and therefore everywhere. The strict strips force $\mathrm{N}(V) = -4$. For $T = V/2$, $\mathrm{N}(T) = -1$, $0 < T < 1$, $-2 < T' < 0$. Thus $T' = -1/T$, $T > 1/2$, and the integral trace $T - 1/T \in (-1.5, 0)$ is -1. It follows that

$$V = 2\varphi - 2, \quad h = (\varphi - 1)/2 > .3, \quad q(h) > .065.$$

But the same weak-five reference gives $e_B > E_{13} + .7600 - .6971 > .95279$. Adding $e_I > .065$ is impossible. Thus five is strong, without a long-head hypothesis.

We next classify the ACTUAL long-fourteen opposite. Its retained white pieces are A_{13} on one through thirteen, the genuine short root n_{15}, and old e. The COMPLETE finite possibilities within B are: isolation; left-only endpoint-three/four or next-to-end-three; short-only any finite native contact; or a left endpoint-three bridge to short-three/four. Two left contacts form a cycle. A next-to-end branch joined to short has nonfinite arms $1, 2, 11$; a left-four bridge puts its double bond internally in rank fifteen; short-five or larger cannot complete that long finite path. Every old-e contact remains, including an old-e three arm on a short-three bridge.

Write $\gamma = -c \cdot n_{14}$, $Q = \beta^2$. Weakness at fourteen implies

$$\frac{\sigma}{\sqrt{\delta - Q}} \le T(\sqrt{28}) < \frac{25}{16}.$$

Indeed $\sqrt{28} < 5.292$, T increases, and the rational comparison at 5.292 has positive cleared numerator 76368925.

For isolation, own zero has $A^2 = Q - 1 < 0$. Own three has the affine star on $c, n_{14}, n_{13}, n_{15}$ with kernel $(1, 2, 1, \sqrt{2})$; own four has the $4, 4$ null, own six the $6, 3$ null, and parallel has $c + n_{14}$. Every kernel is future feasible against ALL original walls, regardless of foreign contacts. For the rest put $u = (1 - Q)/\gamma^2$. For $0 < u < 1$, the actual core $c + \beta e + \gamma u n_{14}$ gives

$$x \cdot \Omega_{14} > \sqrt{28 - u} - \sqrt{u(1-u)}.$$

As $(\sqrt{28}+\sqrt{27})^{-1} < .1$, its loss is $< (.1+1.005)/2 < .61$. The $u = 1$ core is null. For $Q > 0, u > 1$, $Q \geq 1/4$, $\gamma \geq \varphi/2 > .8$, so $u < 75/64$. The actual finite projection onto c, e, n_{14} gives pressure $> \sqrt{27} - \sqrt{u-1}$, with loss $< .1 + 5/12 < .61$. For pure finite owns at least seven the singleton projection has loss $< .1 + \tan(\pi/7) < .6$; one may use $\tan(\pi/7) < 1078/2159 < 1/2$. Native five uses the actual LEFT H_4 correction on n_{14}, n_{13}, n_{12}, with residual $(5-3\varphi)/8 < .136^2$. Since $\gamma > .8$, $\gamma^{-2} < 25/16$, its loss is $< 5/32 + .17 < .61$. Thus all isolated fourteen profiles are strong.

For left-three at $j = 1, 13, 2, 12$,

$$\delta = 1 - \frac{j(14-j)}{28}, \qquad \sigma = \gamma + j/28.$$

For left-four at $j = 1, 13$, $\delta = 1/14$, $\sigma = \gamma + j/(14\sqrt{2})$. Subtract Q in each denominator. At left-three $j \geq 2$, own zero has the actual affine B kernel

$$c + n_{j-1} + 2(n_j + \cdots + n_{14}) + \sqrt{2}n_{15}.$$

At $j = 1$ finite classification permits only $\beta = 0, 1/2, 1/\sqrt{2}$, hence $A^2 = Q - 1/2 \leq 0$. Own-zero left-four uses the positive affine C kernel $c + \sqrt{2}(n_1 + \cdots + n_{14}) + n_{15}$ at one, and the local squared-cosine star $1/2, 1/4, 1/4$ at thirteen. For a left-three, own three has its actual affine cycle and own four the $4, 4$ null. At far one, own six has the $6, 3$ null; parallel always has its null pair. The remaining far weakness has precisely $\gamma\Omega_{14} + \Omega_1/2$ and $202/125 < 2\gamma < 12/5$; Lemma 5 excludes it, including $Q > 0$. At near thirteen every remaining own is at least $1/\sqrt{2}$, with ratio $> (.707+13/28)/.732 > 25/16$. At next-to-end two/twelve the ratio is $> (.7+1/14)/\sqrt{1/7} > 25/16$. All positive-own left-four rows have ratio at least $(.5+1/(14\sqrt{2}))\sqrt{14} > 25/16$. Thus every left-only role is strong.

For short-only contact $\eta = -c \cdot n_{15} > 0$, introduce the actual integral cycles

$$B = 4\gamma^2, \quad W = 4\eta^2, \quad T = 4\sqrt{2}\gamma\eta, \qquad A_0^2 = 7B + 7T + 15W/4 - 1.$$

Equation (IC) implies $WV/4$ integral, because fifteen is odd. For short labels three, five, six and seven, W is respectively $1, \varphi^2, 3$, or the unit root of $X^3 - 5X^2 + 6X - 1$. Each is a unit at every dyadic place. Thus $V/4$ is integral, impossible. This argument includes EVERY own, including zero, and every old-e contact. For short four, only $V/2$ follows. Own zero has the actual $4, 4$ null. For own three, normalize the cap to raw square two; both products with old fourteen and fifteen are -1. Its rational old-B/cap principal is physically Lorentzian, with Schur residual -55. Hence actual $K = \mathbb{Q}$, making $0 < V/2 < 1$ an integer, impossible. Every other positive own has

$$\frac{\sigma}{\sqrt{\delta - Q}} \geq \frac{\gamma + 1/2}{\sqrt{1/2}} \geq 1 + 1/\sqrt{2} > 25/16.$$

For every short label at least eight, even own zero has ratio at least $\cot(\pi/8)/\sqrt{2} = 1+1/\sqrt{2} > 25/16$. All higher finite and unbounded dotted owns are included.

A short-four bridge has residual $1/28$ and $\sigma = \gamma + j/28 + 1/2$, $j = 1, 13$; even own zero has ratio > 2.8. For a short-three bridge,

$$v = \gamma\Omega_{14} + \Omega_j/2 + \Omega_{15}/2.$$

Its original cycles make $a = 2\gamma$ integral and $\sqrt{2}$ integral and K-valued. Directly

$$4A_0^2 = M = 28a^2 + (4j + 28\sqrt{2})a + 2j + 11 + 2j\sqrt{2} \in 1 + 2\mathcal{O}_K.$$

Equation (IC) gives $MV/4$ integral. Writing $M = 1 + 2Z$ and multiplying by $1 - 2Z$ makes $V/4$ integral. This is valid with arbitrary dyadic ramification; no quotient ring is presumed a field. It includes $a = 0$, both bridge orientations, and the retained old-e three arm.

We have proved strong EVERY $5, \dots, 14$ and

$$e_B > E_{14} > .9084, \quad b = f_5/5 < .361, \quad f_{14} < 3.311. \tag{BW}$$

The genuine reference has individual P_5 price $> .14$, terminal P_{14} price $\sqrt{14} - \sqrt{13} > .13$, and $w_{15} = 0$. Its first-coordinate graft is legal for $.49 \le X \le .67$: the other first-five coordinates are $> .2836 > .214$. Therefore

$$x_1 > X \quad \Longrightarrow \quad e_B > .9084 + \frac{5}{4}(X - .361)^2. \tag{BG}$$

E.11 The actual B_{15} head in the short-I branch

The finite B_{14} head list and the own-zero/triangle clearings in Section E.9 apply without any ordinary-h premise. A nonisolated head would have $x_1 > .643$. Now (BG) alone gives $e_B > .9084 + .099405 > 1$, without charging I. Thus the ACTUAL head is isolated within B_{15}:

$$c_H = A_H p - g_H \Omega_1 - \beta_H e, \quad A_H^2 = 2g_H^2 + Q_H - 1, \quad Q_H = \beta_H^2.$$

As before,

$$\lambda_H = A_H \sqrt{h} - \beta_H \sqrt{1+h} \ge 0, \qquad (2g_H^2 - 1)h \ge Q_H.$$

This removes owns zero, three and four. The actual $6, 3$ and parallel nulls remove six and one. Pure native five would give $0 < \lambda_H^2 < .31(.5) < 1/4$, impossible.

For pure finite own at least seven, use the actual finite head point and the SINGLE paid reference of Section E.9:

$$e_B > E_{14} + F(u) - .8b + .032, \qquad F(u) = .8H - .16R > .2893.$$

This is not a separately optimized head energy. Since $g_H < 1$, its positive mutual satisfies $\lambda_H < 1/\sqrt{2}$; the native alphabet forces $\lambda_H = 1/2$. Thus $h = [4(2g_H^2 - 1)]^{-1} > 1/4$. At native seven or eight, $h \ge \sqrt{2}/4$, so $q(h) > .0755$: use $\sqrt{2} > 1.4142$, $\sqrt{1+h} > 1.1634$ and $.1634/2.1634 > .0755$. The original total is then > 1.0164. For all finite natives at least nine, the largest-root cubic gives $g_H > .939$, hence $u < 1.135$. The already proved monotonicity of the finite scalar gives

$$F(u) > F(1.135) > .302416;$$

at that rational endpoint, $\sqrt{2-u} > .93$, $\sqrt{(u-1)(4-3u)} < .284$, $H > .4565$, $R < .3924$. Also $q(h) > .0557$, since $\sqrt{1.25} > 1.118$. The single original total is > 1.009716. Thus the only pure head left is DOTTED.

For $Q_H \ge 1/2$, acuteness gives $C_H^2 \ge 2 + 1/h$. If $h < .2$, then $C_H^2 > 7$, $x_1 > .67$, and (BG) gives body > 1.02775125. If $.2 \le h < .5$, then $C_H > 2$, $x_1 > .57$, and $q(h) > .045$, using $\sqrt{1.2} > 1.095$. The original total is > 1.00800125. For $Q_H = 1/4, \lambda_H > 0$, native gap gives $A_H\sqrt{h} > 1$. For $h < .2$ this again gives $C_H^2 > 7$. For $h \ge .2$, use $\sqrt{1+h} > 1.08$ to get $A_H\sqrt{h} > 1.04$, so

$$C_H^2 = 1 + (4/3)A_H^2 > 1 + (4/3)(1.04^2/.5) > 1.95^2.$$

Then $x_1 > .56$; its graft plus $q(h) > .045$ makes the total > 1.00290125. These intervals cover all positive foreign labels and every positive finite/parallel/dotted mutual.

The only mixed possibility left is $\beta_H = 1/2, \lambda_H = 0$. For $U = 4g_H^2$ its exact relation is

$$V(U-2) = 2, \qquad U > 3. \tag{BQ}$$

Initially both finite and dotted own are retained here.

We shall need a point valid for BOTH the pure dotted and quarter-foreign heads. Put $\delta_H = 1 - Q_H$, $u = \delta_H/g_H^2$. For the pure dotted head $0 < u < 1$. For the quarter head, (BQ) and $0 < h < 1/2$ give $g_H^2 > 3/4$, again $0 < u < 1$. The positive ACTUAL combination

$$\xi = c_H + \beta_H e + \frac{\delta_H}{g_H} n_1$$

has square $-\delta_H(1-u) < 0$, products zero with c_H, e, and product $-g_H + \delta_H/g_H < 0$ with n_1. Its time is $A_H > 0$. All omitted original walls, including d, have nonpositive product termwise. Its old-e displacement cancels EXACTLY. Its normalized pressure is the SAME dotted-row Z, H, R used in (R1S), now genuinely derived for the mixed head as well.

If this quarter head had finite own, then $u > 3/4$, $h > 1/4$. For the dotted-row scalar on $[3/4, 1)$, the derivative comparison reduces to $-4u^3 + 13u^2 - 10u + 2 > 0$. This polynomial is $1/8$ at $3/4$ and increases; the remaining derivative term $.16(1-u)$ is positive. Hence

$$F(u) > F(3/4) = \frac{2\sqrt{10} - \sqrt{6}}{10} - .085 > .3024.$$

Together with $q(h) > .0557$ and the same paid head reference, the total exceeds 1.0097. Thus BOTH remaining head families are dotted.

For any actual short-fifteen pressure $S_{15} > T$, use the SINGLE vector

$$W = w + .4(Z - P_5/5) + .13e_{15}.$$

The individual debits are .08 at P_5 and .13 at P_{14}, paid before using either floor. The head residual is zero-sum on the first five coordinates and hence orthogonal to w. The terminal e_{15} is orthogonal to both. The FULL square gives

$$e_B > E_{14} + F(u) - .8b + .032 + .26(T - f_{14}) - .0169. \tag{BT}$$

If the actual short cosh were at least fifteen, its original edge would give $S_{15} \geq \sqrt{105/8} > 3.62$, and the body alone would exceed $.9084 + .0325 + .06344 = 1.00434$. Thus $C_S < 15$. Also the legal $X = .632$ head graft would give $.9084 + (5/4)(.632 - .361)^2 = 1.00020125$; hence $x_1 < .632$, and the actual head comparison gives $C_H < 7/3$. These are necessary ACTUAL edge bounds, not cutoffs imposed on formal roots.

E.12 Every actual short-fifteen white profile

The finite endpoint of the actual short-fifteen edge retains A_{14} on one through fourteen and old e. Finite classification gives exactly SEVEN profiles, with both endpoint orientations kept:

contact in A_{14}	v_B	allowed β
none	γP_{15}	any finite native or zero
endpoint three $j = 1, 14$	$\gamma P_{15} + P_j/\sqrt{2}$	$0, 1/2, 1/\sqrt{2}$
next-to-end three $j = 2, 13$	$\gamma P_{15} + P_j/\sqrt{2}$	0
endpoint four $j = 1, 14$	$\gamma P_{15} + P_j$	0.

An extra old-e arm at a next-to-end branch gives nonfinite long E arms. At a white four it puts the double bond internally in rank sixteen. An endpoint-three old-e arm permits only three or four,

giving A_{16} or B_{16}. Thus the displayed foreign list is complete. The own $\gamma = -c \cdot n_{15}$ is initially zero, finite native, parallel or dotted.

Own zero is impossible in every row. Isolation has $A^2 = Q - 1 < 0$. Endpoint-three at one has $A^2 = Q - 1/2 \leq 0$. For every other white-three $j \geq 2$, use the actual affine B null

$$c + n_{j-1} + 2(n_j + \cdots + n_{14}) + \sqrt{2}n_{15}.$$

For white-four at one use $c + \sqrt{2}(n_1 + \cdots + n_{14}) + n_{15}$; at fourteen use $c + \sqrt{2}n_{14} + n_{15}$. Every selected product vanishes; its positive cap coefficient gives future time, and all unselected original products are nonpositive termwise. Parallel own has the universal null $c + n_{15}$.

For positive own the time squares, before adding Q, are

profile	A_0^2
isolated	$15\gamma^2 - 1$
white three, $a = 2\sqrt{2}\gamma$	$(15a^2 + 4ja + 4j - 8)/8$
white four, $t = 2\gamma$	$(15t^2 + 4jt + 4j - 4)/4$.

These are positive for every $\gamma \geq 1/2$. Normalize the old B long roots to square two and short root to square one. A white-three cap has square two, white product -1 and short product $-a/2$; a white-four cap has square one, white product -1 and short product $-t/2$. These scalings follow from the actual native bonds over K; no disconnected scale is guessed. The old-B/cap principal is physically Lorentzian and defined over $\mathbb{Q}(a)$ or $\mathbb{Q}(t)$. A proper actual-field extension would fix that indefinite principal at a nonphysical embedding. Hence actual K equals that generating field. For isolation normalize c/γ to get actual $K = \mathbb{Q}(B)$, $B = 4\gamma^2$, in the same way.

For an isolated finite own, $0 < B_\sigma < 4/15$ at every nonphysical place. For $B = 2 + 2\cos(2\pi/m)$, degree $d = \phi(m)/2$ and trace $2d + \mu(m)$ give $26d < 56 - 15\mu(m)$, hence $d \leq 2$. The complete finite list is $m = 3, 4, 5, 6, 8, 10, 12$. Rational cases three/four/six force actual $K = \mathbb{Q}$, excluded by the compact WHOLE-form Meyer obstruction. For five/eight/ten the other B values

$$(3 - \sqrt{5})/2, \quad 2 - \sqrt{2}, \quad (5 - \sqrt{5})/2$$

all exceed $4/15$. For twelve the other value $2 - \sqrt{3} > 4/15$, using $\sqrt{3} < 26/15$. Thus no isolated finite own survives, regardless of β.

For white three, the actual path makes a integral; the own two-cycle makes $a^2/2$ integral. Write $M_j = 15a^2 + 4ja + 4j - 8$. The cycle identity gives $M_j V/8$ integral. Let $t = 2\cos(\pi/m)$. The root-of-unity dyadic lemma proved in Section E.3 says t is a unit at every dyadic place unless m is a power of two. For a non-power-two label, $a = \sqrt{2}t$ has valuation $1/2$, normalized by $v(2) = 1$. The leading term of M_j has valuation one; the other terms have valuation at least two. Thus $v(M_j) = 1$, forcing $V/4$ integral, impossible. For $m = 4$, $a = 2$ gives a rational Lorentzian principal, so actual $K = \mathbb{Q}$. For $m = 2^k \geq 8$, put $B = t^2$. Its even degree d and norm two imply $\mathrm{N}(2B) = 2^{d+1}$ is nonsquare. Hence $\mathbb{Q}(a)/\mathbb{Q}(B)$ has the sign automorphism $a \mapsto -a$. At this nonphysical embedding,

$$M_j(-a) \geq 15a^2 - 56a + 48 > 0$$

because $a > 12/5$, the right polynomial vanishes at $12/5$ and increases thereafter. This contradicts positive definiteness.

For white four the actual native-four path makes $2t$ K-valued, and the own two-cycle makes t^2 integral, so $t \in \mathcal{O}_K$. For $M_j = 15t^2 + 4jt + 4j - 4$, the cycle identity gives $M_j V/4$ integral. A non-power-two own makes t, and hence M_j, a dyadic unit, again impossible. For the remaining power-two owns, the near $j = 14$ short edge already has $C \geq 15/2 + 14 > 15$. At far one the sign

automorphism $t \mapsto -t$ exists for every power-two $m \geq 4$, and its time numerator $15t^2 - 4t > 0$ contradicts admissibility. Thus EVERY finite own has been removed.

For completeness, the actual cosh with a white contact of unit magnitude η is

$$C_S = \frac{\sqrt{15}\gamma + \sqrt{2}\eta j/\sqrt{15}}{\sqrt{\delta - Q}}, \qquad \delta = 1 - \frac{2\eta^2 j(15-j)}{15}.$$

White-four has pure $C_S = 15\gamma + j$; its dotted range violates $C_S < 15$ at both ends. White-three at near next-to-end thirteen has pure $C_S = 15a/4 + j/2 > 17$. A foreign cost only increases these coshes.

For the remaining far next-to-end three at $j = 2$,

$$2\sqrt{2} < a < 56/15, \qquad -8/15 < a_\sigma < 0, \qquad K = \mathbb{Q}(a),\ a \in \mathcal{O}_K.$$

Degree at least four contradicts $(56/15)(8/15)^3 < 1$. In degree three the norm has magnitude < 2, hence is one. For other roots $-b, -c$, $bc = 1/a$ and $b, c < 8/15$ force $a > 225/64$. The trace is then three, so $b + c = a - 3 < 11/15$, whereas $(b+c)^2 \geq 4/a > 15/14$, impossible. Degree one is excluded by the whole-form rational obstruction. Degree two has norm -1, trace three and $a = (3 + \sqrt{13})/2$. Its time numerator $15a^2 + 8a = 53a + 15$ has odd norm -199. It is a unit at EVERY dyadic place; $M_j V/8$ would make $V/8$ integral.

Only end-three dotted roles remain. Far one has

$$2\sqrt{2} < a < 118/15, \qquad -2/3 < a_\sigma < 2/5.$$

For near fourteen put $S = a + 2$, giving time numerator $15S^2 - 4S - 4$ and

$$2 + 2\sqrt{2} < S < 122/15, \qquad -2/5 < S_\sigma < 2/3.$$

Both a, S are integral and have square divided by two integral. If their actual degree $d \geq 3$, their integer norms are divisible by four, while their magnitudes are

$$< (122/15)(2/3)^{d-1} \leq 488/135 < 4,$$

impossible. In degree two both trace and norm are EVEN: $\mathrm{N}(z)^2/4$ integral forces even norm, and $\mathrm{Tr}(z^2/2) = \mathrm{Tr}(z)^2/2 - \mathrm{N}(z)$ forces even trace.

The far strips allow trace $4, 6, 8$, with $-6 < \mathrm{N}(a) < 4$, nonzero. The near strips allow trace $6, 8$, with $-4 < \mathrm{N}(S) < 6$. The COMPLETE finite hand tables are

far trace	$N = -4$	$N = -2$	$N = 2$
4	$2 + 2\sqrt{2}$ (wrong conjugate)	$2 + \sqrt{6}$	$2 + \sqrt{2}$ (wrong conjugate)
6	$3 + \sqrt{13}$	$3 + \sqrt{11}$	$3 + \sqrt{7}$
8	$4 + 2\sqrt{5}$ (too large)	$4 + 3\sqrt{2}$ (too large)	$4 + \sqrt{14}$

and

near S trace	$N = -2$	$N = 2$	$N = 4$
6	$3 + \sqrt{11}$	$3 + \sqrt{7}$	$3 + \sqrt{5}$ (wrong conjugate)
8	$4 + 3\sqrt{2}$ (too large)	$4 + \sqrt{14}$	$4 + 2\sqrt{3}$.

For their nine retained rows, the integral time-numerator factor $M_j/2^k$ and its odd norm are

role and a	k	$M_j/2^k$	$\mathrm{N}(M_j/2^k)$
far $2+\sqrt{6}$	1	$77+32\sqrt{6}$	-215
far $3+\sqrt{13}$	2	$(169+47\sqrt{13})/2$	-39
far $3+\sqrt{11}$	1	$154+47\sqrt{11}$	-583
far $3+\sqrt{7}$	1	$124+47\sqrt{7}$	-87
far $4+\sqrt{14}$	1	$231+62\sqrt{14}$	-455
near $1+\sqrt{11}$	1	$142+43\sqrt{11}$	-175
near $1+\sqrt{7}$	1	$112+43\sqrt{7}$	-399
near $2+\sqrt{14}$	1	$215+58\sqrt{14}$	-871
near $2+2\sqrt{3}$	3	$50+29\sqrt{3}$	-23.

Every factor is a unit at ALL dyadic places, including ramified places in the actual field. The seven $k=1$ rows make $V/4$ integral. The $k=2$ row makes $V/2$ integral. In a quadratic field such a $W=V/2$ has $0<W<1$, $-2<W'<0$, norm -1 and trace -1 or zero. Trace zero is reducible or a boundary; trace -1 gives $\mathbb{Q}(\sqrt{5})$, not the actual $\mathbb{Q}(\sqrt{13})$.

Only the final near row remains. Here actual $K=\mathbb{Q}(\sqrt{3})$ and $\mathcal{O}_K=\mathbb{Z}[\sqrt{3}]$. The full strict V strips give $V\in\{\sqrt{3}-1,\sqrt{3}\}$: writing $V=a+b\sqrt{3}$ gives $0<b<\sqrt{3}<2$, so $b=1$, then $a=-1,0$. Hence $h>.18$, $q(h)>.04$. Its pure short cosh is $15(2+2\sqrt{3})/8+7/2>13$; foreign Q only increases it. Thus $S_{15}>\sqrt{90/7}>3.58$. The SAME joint vector (BT), with I counted once, gives

$$e_B+e_I>.9084+.0325+.05304+.04=1.03394>1.$$

Every nonisolated short profile is excluded. The actual short-fifteen opposite is ISOLATED within B_{15} and DOTTED.

E.13 The quarter-foreign head cannot meet an isolated short cap

Suppose $\beta_H=1/2,\lambda_H=0$. Let $U=4g_H^2>4$, so $V(U-2)=2$, and write the isolated dotted short cap

$$c_S=A_Sp-g_SP_{15}-\beta_Se,\qquad B=4g_S^2>4,\quad Q=\beta_S^2.$$

The actual short edge and the nonphysical principal give

$$B<60(1-Q),\qquad 0<B_\sigma<\frac{4(1-Q_\sigma)}{15}.$$

All finite native β_S, including zero, remain.

The native-three head/old-e link fixes the actual K-root squares:

$$r=\frac{c_H}{\sqrt{2}g_H},\quad f=\frac{e}{\sqrt{2}g_H},\quad r^2=f^2=\frac{2}{U},\quad r\cdot f=-1/U,\quad s=\frac{c_S}{g_S},\quad s^2=4/B.$$

The old short product of s is -1. Its old-e coefficient $q_0\in K$ is nonnegative, with $s\cdot f=-2q_0/U$, and

$$q_0^2=\frac{2UQ}{B}\text{ is a FULL }K\text{-square},\qquad r\cdot s=1+q_0/U-J=-D,\quad J>0,\ D\geq 0.$$

These are actual-root normalizations, not assigned relative lengths for disconnected old components.

The ENTIRE long old-B path cycle, the two-cycle, the head/old-e/short triangle and the long old-e/head/old-B/short cycle give, respectively,

$$G=UBD,\quad L_c=UBD^2/2,\quad T_c=q_0BD,\quad P_c=2q_0B\quad\text{integral.}$$

All formulas retain mutual zero. The two time squares are $(2U-3)/(2U)$ and $15+4(Q-1)/B$. Expanding $J = D+1+q_0/U$ gives

$$4(Bq_0 + G + T_c + L_c) = 28UB - 45B + 8UQ - 16Q - 8U + 12. \tag{BQ2}$$

Since $4Bq_0 = 2P_c$, $4Q$ is integral and forty-five is odd, this equality forces $B/2$ integral at every dyadic place, with arbitrary ramification. For $Q = 0$ its stronger divisibility forces $B/4$ integral. Degree at least three is impossible because

$$0 < \mathrm{N}(B/2) < 30(2/15)^{d-1} \leq 8/15 < 1.$$

Degree one is excluded by compact rational anisotropy. In degree two, $Q = 0$ likewise gives $0 < \mathrm{N}(B/4) < 1$. Only positive Q remains.

The actual head principal is Lorentzian over $\mathbb{Q}(U)$, so actual $K = \mathbb{Q}(U)$. The earlier actual head bound $C_H < 7/3$ gives

$$4 < U < 49/6, \qquad 0 < U' < 3/2.$$

From $V = 2/(U-2)$ integral, $\mathrm{N}(U-2)$ is a negative divisor of four, of magnitude greater than one. It is -2 or -4, and the trace of U is $5, \ldots, 9$. The COMPLETE surviving irreducible rows are

$\mathrm{N}(U-2)$	U	$\mathrm{N}(U)$
-2	$3+\sqrt{3}$	6
-2	$(7+\sqrt{17})/2$	8
-4	$(5+\sqrt{17})/2$	2
-4	$3+\sqrt{5}$	4
-4	$4+2\sqrt{2}$	8
-4	$(9+\sqrt{41})/2$	10.

For verification, writing $T = \mathrm{Tr}(U)$ gives $\mathrm{N}(U) = 2T-6$ or $2T-8$. The other $T = 5, \ldots, 9$ rows are rational, boundaries, or violate $U' < 3/2$. Thus this is a derived list of actual quadratic fields, not a prior field restriction.

Put $Z = B/2$, $M_0 = \mathrm{N}(Z) \in \mathbb{Z}_{>0}$. The full short bounds give $M_0 < 4\mathrm{N}(1-Q)$. A finite β_S in a quadratic field has label $3, 4, 5, 6, 8, 10, 12$. For labels four, six, eight, ten and twelve, the right side is respectively at most $1, 1/4, 1/2, 1/4, 1/4$, and the inequality is strict. Only

$$Q = 1/4,\ M_0 = 1 \text{ or } 2; \qquad Q = \varphi^2/4,\ M_0 = 1,\ K = \mathbb{Q}(\sqrt{5})$$

remain; native five has $4\mathrm{N}(1-Q) = 5/4$. In both branches $\mathrm{N}(Q) = 1/16$. The FULL square $q_0^2 = UQ/Z$ requires $\mathrm{N}(U)/M_0$ to be a rational square. In the entire six-row table this leaves only $\mathbb{Q}(\sqrt{2})$ or $\mathbb{Q}(\sqrt{17})$ with $M_0 = 2$, or $\mathbb{Q}(\sqrt{5})$, $U = 3+\sqrt{5}$, $M_0 = 1$.

For $Q = 1/4, M_0 = 2$, the full bounds give $20 < Z < 45/2$, $0 < Z' < 1/10$. Its integer trace is twenty-one or twenty-two, with discriminant 433 or 476. Neither has squareclass two or seventeen. For $\mathbb{Q}(\sqrt{5}), M_0 = 1, Q = 1/4$,

$$10 < Z < 45/2, \qquad 0 < Z' < 1/10.$$

The trace is $11, \ldots, 22$. For $T^2 - 4$ to be five times a square, only $T = 12, 13, 17, 18, 22$ need inspection; the quotients are $28, 33, 57, 64, 96$. Only eighteen survives, giving

$$Z = 9 + 4\sqrt{5} = \varphi^6, \quad U = 3 + \sqrt{5} = 2\varphi^2, \quad q_0^2 = \frac{1}{2\varphi^4}.$$

This is not a square in $\mathbb{Q}(\sqrt{5})$, because two is not a square there. Finally $Q = \varphi^2/4, M_0 = 1$ gives $8.29 < Z < 10.365$, $0 < Z' < 1/8$: use $Z < 30(3-\varphi)/4$, $Z' < (2+\varphi)/30$, and $1.618 < \varphi < 1.6181$. Its only integer traces are nine and ten; discriminants $77, 96$ are not five times squares. Every mixed quarter-head branch is excluded.

E.14 The remaining pure dotted pair: a complete norm argument

The actual remaining head is PURE dotted and the short cap is isolated-within-B dotted:

$$c_H = A_H p - \sqrt{2} g_H e_1, \qquad c_S = A_S p - g_S P_{15} - \beta e, \qquad g_H, g_S > 1.$$

Put $U = 4g_H^2$, $B = 4g_S^2$, $Q = \beta^2$, $\delta = 1 - Q$. The original Lorentzian connected principals give $K = \mathbb{Q}(U) = \mathbb{Q}(B)$. At all places the complete necessary strips are

$$4 < U < 98/9, \quad 4 < B < 60\delta, \quad 0 < V < 2,$$
$$0 < U_\sigma < 2, \quad 0 < B_\sigma < 4\delta_\sigma/15, \quad -4 < V_\sigma < 0, \quad 0 < \delta_\sigma \le 1.$$

All U, B, V are integral. Equation (IC) gives

$$UV/2,\ BV/4 \in \mathcal{O}_K.$$

For example subtracting the integral V term from $V(U/2 - 1)$ gives the first statement; for $V(15B/4 - 1)$, the odd coefficient fifteen is invertible at dyadic places and all other places already allow division by four. For the nonzero integer $Y = BV/4$,

$$0 < Y < 30, \qquad |Y_\sigma| < 4/15.$$

Degree at least four contradicts $30(4/15)^3 < 1$. The whole-form rational obstruction removes degree one. Thus actual degree is two or three. Also $K = \mathbb{Q}(V)$: any proper extension would have a nonphysical embedding fixing positive physical V, contrary to the strict $V_\sigma < 0$ strip.

The one paid head reference already proves

$$e_B > .9409.$$

The same head-plus-short vector gives

$$e_B > .9084 + .0325 + .26(T - 3.311) - .0169 \quad \text{if } S_{15} > T. \tag{BP}$$

In particular $S_{15} > 3.61$ forces $e_B > 1.00174$. There are two uniform dispatches. If $V > 6/5$, then $h > .3$, $q(h) > .065$, and $e_B + e_I > 1.0059$. If $B > 225/4$, the ACTUAL isolated short edge has

$$C_S^2 = \frac{15B}{4\delta} \ge 15B/4 > 3375/16 > (29/2)^2.$$

Its original comparison gives $S_{15} > \sqrt{405/31} > 3.61$, contradicting (BP) without charging I. These dispatches use one reference and its two distinct individual anchor prices.

In degree three every conjugate of B is positive, so

$$\mathrm{N}(B) \in \{1, 2, 3, 4\}, \qquad \mathrm{N}(V) \in \{1, \dots, 31\}.$$

Indeed $\mathrm{N}(B) < 60(4/15)^2 = 64/15$, and V has two negative conjugates with norm < 32. Integrality of $BV/4$ requires $64 \mid \mathrm{N}(B)\mathrm{N}(V)$. The entire finite list forces $\mathrm{N}(B) = 4, \mathrm{N}(V) = 16$. As both nonphysical B values are $< 4/15$, the physical $B > 4/(4/15)^2 = 225/4$. The high-short-edge dispatch excludes the entire cubic branch, with every finite β retained.

In degree two write $K = \mathbb{Q}(\sqrt{D})$, $D > 1$ squarefree. In the full quadratic order, the gap between V, V' is < 6. For $D \equiv 1 \pmod 4$, write $V = (a + b\sqrt{D})/2$, same-parity integers, $b > 0$; then $b\sqrt{D} < 6$. The complete possible D are $5, 13, 17, 21, 29, 33$. Otherwise $V = a + b\sqrt{D}$, $b > 0$,

and $b\sqrt{D} < 3$, so $D = 2, 3, 6, 7$. Only $D = 5$, respectively $D = 2$, admits $b = 2$; all other possible b equal one. Imposing BOTH strict strips gives exactly

D	V	$-\mathrm{N}(V)$
2	$\sqrt{2}-1,\ \sqrt{2},\ 2\sqrt{2}-1$	1, 2, 7
3	$\sqrt{3}-1,\ \sqrt{3}$	2, 3
5	$(\sqrt{5}-1)/2,\ (\sqrt{5}+1)/2,\ \sqrt{5}-1$	1, 1, 4
6	$\sqrt{6}-1$	5
7	$\sqrt{7}-1$	6
13	$(\sqrt{13}-3)/2,\ (\sqrt{13}-1)/2$	1, 3
17	$(\sqrt{17}-3)/2,\ (\sqrt{17}-1)/2$	2, 4
21	$(\sqrt{21}-3)/2,\ (\sqrt{21}-1)/2$	3, 5.

For full coverage: at $D = 5, b = 1$, $a = -1, 1$; at $b = 2$, $a = -2$. At $D = 13, 17, 21$, $a = -3, -1$. At $D = 29, 33$ the strips would require both odd $a \le -3$ and odd $a \ge -1$, impossible. At $D = 2, b = 1$, $a = -1, 0$, and at $b = 2$, $a = -1$. At $D = 3$, $a = -1, 0$, and at $D = 6, 7$, $a = -1$. Thus no coefficient, actual field or strict boundary has been silently omitted.

Now $0 < \mathrm{N}(B) < 16$. Writing $n = -\mathrm{N}(V)$, integrality of $BV/4$ requires $16 \mid \mathrm{N}(B)n$. Odd n is impossible. For $n = 2, 6$, $\mathrm{N}(B) = 8$; for $n = 4$, it is $4, 8, 12$. Only SIX V rows remain:

$$\sqrt{2}, \quad \sqrt{3}-1, \quad \sqrt{5}-1, \quad \sqrt{7}-1, \quad (\sqrt{17}-3)/2, \quad (\sqrt{17}-1)/2.$$

Four exceed $6/5$, and are excluded by the head-plus-I dispatch. The entire quadratic branch is reduced to

$$K = \mathbb{Q}(\sqrt{3}),\ V = \sqrt{3}-1,\ \mathrm{N}(B) = 8; \qquad K = \mathbb{Q}(\sqrt{17}),\ V = (\sqrt{17}-3)/2,\ \mathrm{N}(B) = 8.$$

In $\mathbb{Z}[\sqrt{3}]$, norm eight would require $a^2 - 3b^2 = 8$, impossible modulo three. In the full $\sqrt{17}$ order put $B = (T + s\sqrt{17})/2$, same-parity integers, $s > 0$. Its norm and strict strip give

$$30 < B < 60, \quad 31 \le T \le 60, \quad T^2 - 17s^2 = 32.$$

Modulo seventeen $T \equiv 7, 10$. In that COMPLETE integer interval the only candidates are $T = 41, 44, 58$, giving $s^2 = 97, 112, 196$. Only the last is a square. Thus $B = 29 + 7\sqrt{17} > 57 > 225/4$, and the high-short-edge dispatch is contradictory. This excludes BOTH quadratic remnants and the entire pure dotted pair.

Together with the ordinary branch, the full actual $B_{15}+A_1$ family is now excluded by the written arguments, whether the isolated rank-one wall has ordinary $h \ge 1/2$ or actual short $0 < h < 1/2$. No numerical-output premise, field census or realization is used in this closure.

E.15 The odd-D fork cycle and the actual D_{15} centre

Retain the original D_{15} roots, both signed forks and the actual short I opposite. We use (IC), with every foreign β and its radical-cancelling triangle retained.

First, in WHOLE $D_r + A_1$ with $r \ge 5$ ODD, an actual long-edge opposite cannot have exactly one extra D contact, at a fork with ANY finite native magnitude η. Indeed for $v_D = \gamma\Omega_i + \eta\Omega_f$ put

$$U = 4\gamma^2, \qquad B = 4\eta^2, \qquad Z = 4\gamma\eta.$$

Actual two-cycles and the entire old-three path cycle make these integral, also at $\gamma = 0$. The exact inverse products give

$$8A_0^2 = 4iU + 4iZ + rB - 8 =: N.$$

At every dyadic place, normalized by $v(2) = 1$, a finite native B has $0 \le v(B) \le 1$. To see this in its minimal field, write $B = (1+\zeta)(1+\zeta^{-1})$, $\mathrm{N}(B) = \Phi_m(-1)$. This norm is odd unless m is a power of two, when it equals two; the elementary factorization $n = \prod_{d|n,d>1} \Phi_d(1)$ proves the assertion. The valuation bound persists in EVERY actual extension. As r is odd, rB is the unique lowest term in N. Thus $v(N) = v(B) \le 1$. Integrality of VA_0^2 forces $v(V) \ge 2$ at every dyadic place, hence $V/4$ integral, impossible. This covers own zero, every finite/parallel/dotted own and every old-I contact. Native four is included: its own two-cycle is essential and must not be dropped.

At D_{15}, this removes the ONLY weak right-fork packets at eleven and twelve already classified in Section E.8. Together with the shared bulk, strong $6, \dots, 12$ are available without ordinary rank-one energy. It remains to prove thirteen directly.

Deleting thirteen leaves A_{12}, BOTH forks, and old e. Finite classification gives the COMPLETE table:

left contact	fork contacts
none	none; one arbitrary finite label; $(3,3), (3,4), (3,5)$
endpoint three at $1, 12$	none; one fork three/four; both forks three
next-to-end three at $2, 11$	none
endpoint four at $1, 12$	none.

Two left contacts form a cycle. Next-to-end with a new fork has nonfinite unequal long arms. With no left contact, two finite fork contacts must form A_3, B_3, H_3; with an endpoint contact, finite long $A/B/D$ completions give precisely the displayed three/four choices. Adding old e can restrict but cannot enlarge this list. For left coefficient η_L at j, fork coefficients a, b, and foreign $Q = \beta^2$, put

$$q_L = \frac{2\eta_L^2 j(13-j)}{13}, \quad \Delta = 1 - q_L - a^2 - b^2 - Q > 0, \quad \sigma = \gamma + \frac{\eta_L j}{13} + \frac{a+b}{2}.$$

The actual inverse formula is $A^2 = 26\sigma^2 - \Delta$, $C_{13} = \sqrt{26}\sigma/\sqrt{\Delta}$. The exact threshold satisfies

$$T(\sqrt{26}) < \frac{100}{61} - \frac{4490}{48909} < \frac{31}{20},$$

with final positive cleared numerator 146719. Foreign Q only increases every direct ratio.

For isolation, own zero has negative time. Own three has the affine four-leaf-three star on $c, n_{13}, n_{12}, n_{14}, n_{15}$; own four has the $4, 3, 3$ star; own six the $6, 3$ path; parallel its actual null pair. All apply with arbitrary foreign cost. Every remaining finite own is excluded arithmetically. For $B_0 = 4\gamma^2$, actual reflection normalization gives old raw squares two, cap square $2B_0$ and marked product $-B_0$. The original connected principal has Schur residual $B_0(2 - 13B_0)$. Thus every nonphysical B_0 obeys $0 < (B_0)_\sigma < 2/13$. Cyclotomic trace $2d + \mu(m)$ and physical $B_0 < 4$ force $d \le 2$. The complete labels are $3, 4, 5, 6, 8, 10, 12$. The nonrational other values at five/eight/ten/twelve,

$$(3-\sqrt{5})/2, \quad 2-\sqrt{2}, \quad (5-\sqrt{5})/2, \quad 2-\sqrt{3},$$

all exceed $1/4 > 2/13$. The rational cases were already null. For every dotted isolated own, the actual core $c + \beta e + \gamma u n_{13}$, $u = (1-Q)/\gamma^2 \in (0,1)$, gives

$$x \cdot \Omega_{13} > \sqrt{26-u} - \sqrt{u(1-u)}.$$

Its full loss is $< (.1 + \sqrt{1+.1^2})/2 < .553$. Every isolated row is therefore strong or impossible.

With a left contact and no fork, far-three own zero has $A^2 = Q - 1/2$, $Q < 7/13$, so its native costs $0, 1/4, 1/2$ give nonpositive time. Far-four own zero forces $Q = 0$. Near/next-three own zero

has affine D trunk coefficients two and cap/previous-node/both-fork coefficients one. Near-four own zero has the affine $4,3,3$ star. Positive own three has its all-three cycle, own four the fork star, own six the fork $6,3$ path, and parallel its null pair. The remaining near/next-three owns have $\gamma \ge \varphi/2 > .8$, with ratios at least

$$\frac{.8 + 6/13}{\sqrt{7/13}} > 1.7, \qquad \frac{.8}{\sqrt{2/13}} > 2.$$

Positive endpoint-four has ratio at least $\sqrt{13}/2 > 1.8$. The only left-only possible weak role is the literal far helper at thirteen, with

$$202/125 < 2\gamma < 2(31/20)\sqrt{7/13} < 12/5.$$

Lemma 5 excludes it including foreign contacts.

One fork and no left contact is excluded by the odd-D cycle just proved. For fork pair $(3,3)$, own zero is the actual four-cycle and own three the local triangle; all other positive owns have ratio $\ge 1 + 1/\sqrt{2} > 31/20$. For pair $(3,5)$, even own zero has ratio $\varphi^3/2 > 2.1$. For $(3,4)$, every positive own has ratio > 2.2. At own zero finite whiteness forces $Q = 0$. The original four-cycle puts $\sqrt{2}$ in K, and

$$A_0^2 = (37 + 26\sqrt{2})/8.$$

The numerator has odd norm seventeen, so is a unit at EVERY dyadic place, also in proper extensions. The cycle identity forces $V/4$ integral.

For a left endpoint-three and one fork-three, $\Delta \le 15/52$, $\sigma \ge \gamma + 15/52$. Own zero has its actual cycle, own three its triangle; all other owns have ratio > 1.8. For one fork-four, $\Delta \le 1/26$, and even own zero has ratio $(1/26 + 1/(2\sqrt{2}))\sqrt{26} > 1.9$. For both forks-three, $\Delta \le 1/26$, $\sigma \ge 7/13$, ratio > 2.7. Both left and fork orientations are included. Each affine kernel has positive cap coefficient and time, with all omitted ORIGINAL walls nonpositive termwise. No foreign wall was discarded. This exhausts the table and proves strong thirteen.

Hence $6, \dots, 13$ are strong. Both initial-five references give $e_D > E_{13} > .88989$. If five is weak, its curved reference gives the stronger $e_D > E_{13} + .7600 - .6971 > .95279$. Therefore weak five implies $h < .21$, since $q(.21) = 1/21$ already exceeds the remaining budget. Its literal far-helper inverse entries are $5,1,1$, the SAME entries used in Section E.10. The complete minimal-degree and actual-field golden argument there therefore applies over old D_{15}, without using a B rank conclusion. It forces $V = 2\varphi - 2$, $h > .3$, contrary to $h < .21$. Five is strong. Fix the genuine reference through $5, \dots, 13$:

$$w^2 = E_{13} > .88989, \quad b < .361, \quad f_{13} < 3.175, \quad w_{14} = w_{15} = 0.$$

Its INDIVIDUAL P_5 and P_{13} prices exceed .14. The first-five graft is legal on $.49 \le X \le .67$, as in the B calculation.

E.16 The actual D_{15} head with short I

The COMPLETE actual D_{14} white head list and both-fork positive clearing were proved in Section E.9. For a nonisolated head,

$$C_1^2 = \frac{(2g+1)^2}{1-2Q} = 1 + \frac{2A^2}{1-2Q}.$$

If $Q = 0$, native gap with d gives $A^2 h \ge 1/4$. If $Q > 0$, acuteness gives $A^2 h \ge Q(1+h) \ge 1/4$. For $h < .05$, $C_1^2 > 11 > 7$, $x_1 > .67$; the legal graft gives $e_D > 1.00924125$. For $h \ge .05$, the

complete own-zero and own-three nulls leave $g \geq 1/\sqrt{2}$, hence $x_1 > .643$. Now $q(h) > .012$, using $\sqrt{1.05} > 1.0245$. The original total is $.88989 + .099405 + .012 = 1.001295 > 1$. Thus the head is isolated WITHIN D_{15}.

Its form and full foreign split are

$$c_H = A_H p - g_H \Omega_1 - \beta_H e, \quad A_H^2 = 2g_H^2 + Q_H - 1, \quad Q_H = 0,\ 1/4,\ \text{or } [1/2, 1),$$

with $(2g_H^2 - 1)h \geq Q_H$. The original nulls and native gap remove owns zero, three, four, six, parallel, and pure five. For a pure finite head use the SAME proved actual finite point and the ONE paid head reference:

$$e_D > .88989 + F(u) - .2888 + .032.$$

For finite own, native gap forces $h = [4(2g_H^2 - 1)]^{-1} > 1/4$. The complete four native cases are

native	F lower bound	$q(h)$ lower bound	original total lower bound
7	.2893	.083	1.00539
8	.291984	.0755	1.000574
9	.302416	.069	1.004506
≥ 10	.31690336	.0557	1.00569336.

Here are the tight fresh scalar details. For seven, $g_H < .901$ and $4(2(.901)^2 - 1)(.4) < 1$, so $h > .4$. For eight, put $s = \sqrt{2}$,

$$H_8 = \frac{\sqrt{2s - 2} - \sqrt{34s - 48}}{s}, \quad R_8 = 23 - 16s, \quad h_8 = s/4.$$

The brackets $1.4142135 < s < 1.4142136$ give $\sqrt{2s - 2} > .91017$, $\sqrt{34s - 48} < .28856$, $H_8 > .4395$, $R_8 < .3726$. Thus $F_8 > .291984$, and $\sqrt{1 + h_8} > 1.1634$ gives $q(h_8) > .0755$. For nine, the cubic gives $.939 < g_H < .94$, $u < 1.135$, $h > .32$. The earlier finite scalar at 1.135 gives $F > .302416$; $\sqrt{1.32} > 1.1488$ gives $q(h) > .069$. For EVERY native at least ten, $u \leq 2 - (2/5)\sqrt{5} < 1.106$. At that rational bound, $\sqrt{.894} > .9455$, $\sqrt{.072292} < .269$, $H > .4783$, $R = .410854$, giving $F > .31690336$. The finite scalar's already proved monotonicity supplies the entire infinite native tail. Each table row applies to ONE reference and charges I once.

For $Q_H \geq 1/2$, $C_1^2 \geq 2 + 1/h$. The complete h split is

h range	x_1 bound	$q(h)$ bound	total lower bound
$(0, .2)$	.67	unused	1.00924125
$[.2, .3)$	.62	.045	1.01874125
$[.3, .5)$	.57	.065	1.00949125.

The edge squares are respectively > 7, $> 16/3 > (9/4)^2$, and > 4. For $Q_H = 1/4, \lambda_H > 0$, native gap gives $A_H\sqrt{h} \geq .5 + .5\sqrt{1 + h}$. The same three ranges give, respectively, $x_1 > .67, .63, .57$, and totals $> 1.00924125, 1.02534125, 1.00949125$. For the middle row use $A_H\sqrt{h} > 1.04$, $C_1^2 > 49/9$; for the last use $A_H\sqrt{h} > 1.07$, $C_1^2 > 4$. The I bounds follow from $\sqrt{1.2} > 1.095, \sqrt{1.3} > 1.14$. All grafts lie in the PROVED .49, .67 interval; these are alternative contradictions, never stacked.

Thus the only remaining heads are pure dotted or $\beta_H = 1/2, \lambda_H = 0, V(U_H - 2) = 2$. The second still includes finite own. For BOTH families $u = (1 - Q_H)/g_H^2 \in (0, 1)$, and the actual cleared point

$$c_H + \beta_H e + \frac{1 - Q_H}{g_H} n_1$$

is future negative, selected-wall nonpositive and feasible against EVERY original wall, including both forks and d. The detailed computation in Section E.11 uses only these first long roots and applies here verbatim. It gives the pure first-two-coordinate pressure $x \cdot Z \geq H$, $F = .8H - .16R > .2893$. It does not declare the finite quarter-head dotted.

E.17 Both signed forks and the complete isolated-cap argument

The genuine fork duals are $\Omega_\pm = (P_{14} \pm e_{15})/\sqrt{2}$; their squared norm is $15/2$. The ORIGINAL edge comparison gives, independently of the opposite cap's contact pattern,

$$T_\pm = x \cdot (P_{14} \pm e_{15}) \geq \sqrt{15}\sqrt{(C_\pm - 1)/(C_\pm + 1)}.$$

For either ACTUAL signed pressure $T_\pm > T$, use

$$W_\pm = w + .4(Z - P_5/5) + .14(P_{14} \pm e_{15} - P_{13}).$$

Its individual debits are .08 from P_5 and .14 from P_{13}, both affordable BEFORE using their floors. The head residual is zero-sum on the first five coordinates. The signed residual $e_{14} \pm e_{15}$ has norm two and is disjoint from the head residual. Both are orthogonal to w. The FULL one-vector square therefore gives

$$e_D > .88989 + .0325 + .28(T - 3.175) - .0392. \tag{DF}$$

If either $C_\pm \geq 14$, then $T_\pm \geq \sqrt{13} > 3.605$, making $e_D > 1.00359$. Thus BOTH actual fork coshes are < 14, without assuming a fork-cap whitelist.

Fix either genuine fork f, and suppose its actual opposite is isolated within D_{15}:

$$c = Ap - g\Omega_f - \beta e, \quad Q = \beta^2 < 1.$$

Own zero has negative time; parallel own has the actual future feasible null $c + n_f$. For every remaining positive own put $U = 4g^2$; it is integral and $A_0^2 = 15U/8 - 1 > 0$. Here is an explicit actual K-normalization, valid for finite as well as dotted own. Choose old raw roots $\alpha_i = \sqrt{2}n_i$. The actual reflection difference

$$r = s_c(\alpha_f) - \alpha_f = 2\sqrt{2}g\,c$$

is a K-vector on the actual cap line. Its Gram has $r^2 = 2U$, $r \cdot \alpha_f = -U$, other old products zero. The connected original principal is defined over $\mathbb{Q}(U)$, with physical Schur residual

$$2U - (15/4)U^2 = -2UA_0^2 < 0.$$

Any proper actual extension would have a nonphysical embedding fixing that indefinite principal. Hence ACTUAL $K = \mathbb{Q}(U)$, and every nonphysical place has $0 < U_\sigma < 8/15$. The actual finite fork endpoint gives

$$C_f^2 = \frac{15U}{8(1-Q)} < 196, \qquad 0 < U < 1568/15.$$

The cycle identity makes $15UV/8$ integral; as $15 = 16 - 1$, $Y = UV/8$ is integral too. Its full strips are

$$0 < Y < 392/15, \qquad |Y_\sigma| < 4/15.$$

Degree at least four contradicts $(392/15)(4/15)^3 < 1$. Degree one is excluded by the compact WHOLE-form rational-anisotropy obstruction.

In degree three, $0 < \mathrm{N}(U) < 100352/3375 < 32$ and $0 < \mathrm{N}(V) < 32$. Integrality of $UV/8$ requires $512 \mid \mathrm{N}(U)\mathrm{N}(V)$. Each positive integer below thirty-two has ordinary two-adic valuation

at most four, so their product cannot have valuation nine. This excludes EVERY cubic field and ramification pattern.

In degree two $0 < \mathrm{N}(U) < 12544/225 < 56$. The actual V generates K, and the COMPLETE quadratic strict-I list is precisely the sixteen values already derived in Section E.14; that derivation used only $0 < V < 2, -4 < V' < 0$, not a B geometric hypothesis. Write $n = -\mathrm{N}(V)$. Now $64 \mid \mathrm{N}(U)n$. Odd n is impossible. For $n = 2, 6$, $\mathrm{N}(U) = 32$; for $n = 4$, it is $16, 32, 48$. The same SIX surviving V values are

$$\sqrt{2}, \quad \sqrt{3}-1, \quad \sqrt{5}-1, \quad \sqrt{7}-1, \quad (\sqrt{17}-3)/2, \quad (\sqrt{17}-1)/2.$$

If $\mathrm{N}(U) \geq 32$, the strict conjugate strip gives physical $U > 60$. Then $C_f^2 > 112.5 > 10.6^2$, and $T_f > \sqrt{360/29} > 3.52$. Equation (DF) gives $e_D > .97979$. Every displayed V is $> 14/25$, hence $h > .14$, $\sqrt{1+h} > 16/15$, $q(h) > 1/31 > .032$. The original total is > 1.01179. If $\mathrm{N}(U) = 16$, necessarily $n = 4$, so $V = \sqrt{5}-1$ or $(\sqrt{17}-1)/2$, both $> 6/5$. Thus $q(h) > .065$. Also $U > 30$, $C_f > 15/2$, $T_f > \sqrt{195/17} > 3.38$. The SAME signed reference gives $e_D > .94059$ and total > 1.00559. No head or fork energy optimized separately has been added to this reference.

Thus neither ACTUAL fork opposite can be isolated within D_{15}, with its entire finite old-I contact and every own range retained. The remaining nonisolated cases are now classified at their own actual finite endpoints.

E.18 Every nonisolated fork cap

Fix fork f and denote the other fork by k. Deleting f retains the ACTUAL A_{14} path $n_1, \ldots, n_{13}, n_k$, plus old e. The COMPLETE finite contact and foreign table is

retained D contact	allowed β
simple three at n_1 or n_k	$0, 1/2, 1/\sqrt{2}$
simple three at n_2 or n_{13}	0
native four at n_1 or n_k	$0.$

Two path contacts form a cycle. Other branches would need finite exceptional rank at most eight, while this completion has rank fifteen. A next-to-end old-I arm creates nonfinite long E arms. An old-I arm at white four makes its double bond internal. At endpoint three only old-I labels three or four give finite A_{16}/B_{16}. This retains BOTH path ends and next-to-end sites.

Write $v_D = g\Omega_f + \eta\Omega_j$. Own zero is completely dispatched as follows. Far-three at n_1 has $A^2 = Q - 1/2 \leq 0$; next-three at n_2 has $A^2 = 0$. Near-next-three at n_{13} has the actual affine D_4 null

$$c + n_{12} + n_f + n_k + 2n_{13}.$$

Near-end-three at n_k has affine E_8 kernel

$$2c + 4n_k + 6n_{13} + 3n_f + 5n_{12} + 4n_{11} + 3n_{10} + 2n_9 + n_8.$$

Its arms have lengths $1, 2, 5$. Far-four at one and near-four at k have respectively

$$\sqrt{2}c + 2(n_1 + \cdots + n_{13}) + n_f + n_k, \qquad \sqrt{2}c + 2n_k + 2n_{13} + n_f + n_{12}.$$

Every selected product vanishes; its cap coefficient gives positive future time. Every unselected ORIGINAL wall, including the entire old-I arm, has nonpositive product termwise. Thus these are forbidden feasible nulls. Parallel own has $c + n_f$.

For white three put $t = 2g$. The actual closed three-path cycle makes t integral and K-valued. Actual reflection differences normalize cap and old D roots to square two, with own product $-t$ and white product -1. The physically Lorentzian old-D/cap principal then proves ACTUAL $K = \mathbb{Q}(t)$, with time numerators

$$\begin{array}{c|c} j = 1, 2, 13 & 8A_0^2 = 15t^2 + 4jt + 4j - 8 \\ j = k & 8A_0^2 = 15t^2 + 26t + 7. \end{array}$$

For white four put $a = 2\sqrt{2}g$. The original reflection difference at its white-four neighbour equals $2c$, fixing the cap's K-line. Use raw cap square one, white product -1 and own product $-a/2$. The actual cycles give $a \in \mathcal{O}_K$, $a^2/2 \in \mathcal{O}_K$, and actual $K = \mathbb{Q}(a)$. Now

$$\begin{array}{c|c} j = 1 & 16A_0^2 = 15a^2 + 8a \\ j = k & 16A_0^2 = 15a^2 + 52a + 44. \end{array}$$

All these A_0^2 are positive for positive native or dotted own, so the actual-field descent uses an original Lorentzian principal in its precise domain.

For white three with own three, the original old-three path closes an all-three affine cycle; its all-ones kernel remains future feasible with foreign contacts. For every other FINITE own, the dyadic lemma of Section E.3 gives $v(t), v(t+1) \leq 1/2$ at EVERY dyadic place. In the long-j numerator the term $15t^2$ has unique smallest valuation ≤ 1. At $j = k$, putting $s = t + 1$ gives $15s^2 - 4s - 4$, with the same unique-lowest-term argument. Integrality of VA_0^2 forces $v(V) \geq 2$ at all dyadic places, contrary to $V/4$ nonintegral.

For white four and a non-power-two finite own, $t = 2\cos(\pi/m)$ is a dyadic unit, so $v(a) = 1/2$. The leading term in either numerator has unique valuation one; the others have valuation at least two. The cycle identity makes $V/8$ integral, impossible. For $m = 4$, $a = 2$ gives a rational Lorentzian principal, so actual $K = \mathbb{Q}$, impossible. For higher powers of two, the SAME norm-two half-square argument used in Section E.12 gives an automorphism $a \mapsto -a$. The far conjugate numerator $15a^2 - 8a$ is positive. The near numerator $15a^2 - 52a + 44$ has roots $22/15, 2$ and is positive since $a^2 \geq 4 + 2\sqrt{2} > 6 > (12/5)^2$. Both contradict strict nonphysical positivity. Every finite own is now excluded.

Only $g > 1$ remains. The complete pure fork coshes, which only increase on adding foreign Q, are

white contact	C_f at $Q = 0$
n_1 three	$(15t + 2)/8$
n_k three	$(15t + 13)/8$
n_2 three	$(15t + 4)/4$
n_{13} three	$(15t + 26)/4$
n_1 four	$15a/4 + 1$
n_k four	$15a/4 + 13/2$.

They follow from $\Omega_f^2 = 15/2$, $\Omega_f \cdot \Omega_k = 13/2$, $\Omega_f \cdot \Omega_j = j$, and retained-path inverse costs $7/15, 13/15, 14/15$. Since $C_f < 14$, $t > 2$, $a > 2\sqrt{2}$, near-next-three and near-four are immediately impossible.

Far-four has

$$2\sqrt{2} < a < 52/15, \qquad -8/15 < a_\sigma < 0.$$

Degree at least three contradicts $(52/15)(8/15)^2 = 3328/3375 < 1$. In degree two the norm magnitude is < 2 but must be EVEN since $a^2/2$ is integral. In degree one an even integer in the physical interval does not exist. Far-next-three has $t = 2g$ in the FULL dotted strip $2 < t < 52/15$, with

$-8/15 < t_\sigma < 0$ at every nonphysical place. No half-square condition or $2\sqrt{2}$ lower bound is imposed on t. Degree at least three is again impossible; degree one is rational, and degree two has norm -1, trace two or three. Thus $t = 1 + \sqrt{2}$ or $(3 + \sqrt{13})/2$. Their time numerators $15t^2 + 8t$ have odd norms $-79, -199$, so are units at all dyadic places. The cycle identity makes $V/8$ integral. Both next-three and both white-four dotted roles are now exhausted.

The only roles left are endpoint-three dotted. For the far one put $z = t$, and for the near one put $z = t + 1$. The COMPLETE actual strips and time numerators are

role	physical z	every other z_σ	$M = 8A_0^2$
far	$2 < z < 22/3$	$-2/3 < z_\sigma < 2/5$	$15z^2 + 4z - 4$
near	$3 < z < 38/5$	$-2/5 < z_\sigma < 2/3$	$15z^2 - 4z - 4$.

Here actual $K = \mathbb{Q}(z)$, z is integral, and $MV/8$ is integral. If $\mathrm{N}(z)$ were odd, z and M would be units at every dyadic place, forcing $V/8$ integral. Thus its nonzero integer norm is EVEN.

We need a small ideal argument to exclude absolute norm two WITHOUT assuming a half-square or unramified dyadic primes. If $|\mathrm{N}(z)| = 2$, then $(z) = \mathfrak{p}$ for one prime of norm two, with exponent one. Let its ramification index be e. For $e \geq 2$, $v_\mathfrak{p}(z) = 1/e$; the leading $15z^2$ in M has unique smallest valuation $2/e \leq 1$. Thus $v_\mathfrak{p}(V) \geq 3 - 2/e \geq 2$. At all other dyadic primes z, hence M, is a unit, giving $v(V) \geq 3$. So $V/4$ is integral. For $e = 1$, the other dyadic primes have total ramification-residue degree $d - 1$. At each, M is a unit and $\mathrm{ord}_\mathfrak{q}(V) \geq 3e_\mathfrak{q}$. Hence

$$2^{3(d-1)} \mid |\mathrm{N}(V)|, \qquad 0 < |\mathrm{N}(V)| < 2\,4^{d-1} = 2^{2d-1}.$$

For $d \geq 2$, $3(d-1) \geq 2d - 1$, a contradiction. This proves the norm-two exclusion in EVERY ramification pattern using the Dedekind ideal norm and $\sum e_\mathfrak{p} f_\mathfrak{p} = d$.

For $d \geq 5$, the full strips give $|\mathrm{N}(z)| < (38/5)(2/3)^4 < 2$, contrary to nonzero even norm. For $d = 3, 4$ its magnitude is $< 152/45 < 4$; only two could remain, already excluded. Degree one is rational and impossible. Thus actual degree is two, with $|\mathrm{N}(z)| = 4$. Its trace is EVEN as well. If odd, z cannot be nonunit at every dyadic prime: conjugation permutes those primes, making both z, z' nonunit there and their integer trace even. Norm four also ensures a nonunit prime exists. A quadratic field would therefore need TWO dyadic primes, both with $e = f = 1$, one unit and one nonunit for z. At the unit prime M is a unit, so eight divides $|\mathrm{N}(V)|$, contrary to its strict quadratic bound < 8.

For far z, positive norm four is impossible since $zz' < (22/3)(2/5) < 4$. Thus norm is -4; $z' > -2/3$ gives $z > 6$. Its even trace is six, so

$$z = 3 + \sqrt{13}, \qquad M = 4(169 + 47\sqrt{13})/2.$$

The displayed factor in the FULL quadratic order has odd norm -39. Thus $MV/8$ integral makes $V/2$ integral. As proved in the short-B analysis, an integral $W = V/2$ with $0 < W < 1, -2 < W' < 0$ must generate $\mathbb{Q}(\sqrt{5})$, not the actual $\mathbb{Q}(\sqrt{13})$. No far endpoint-three dotted cap exists.

For near z, negative norm -4 is impossible since $|zz'| < (38/5)(2/5) < 4$. Norm is four, $z' > 0$, $z > 6$; its even trace is eight. Thus

$$z = 4 + 2\sqrt{3}, \quad t = 3 + 2\sqrt{3}, \quad M = 8(50 + 29\sqrt{3}), \quad K = \mathbb{Q}(\sqrt{3}).$$

The full V strips in $\mathbb{Z}[\sqrt{3}]$ give $V = \sqrt{3} - 1$ or $\sqrt{3}$, so $h > .18$, $q(h) > .04$. Its actual pure fork cosh is

$$(15t + 13)/8 = (29 + 15\sqrt{3})/4 > 13.$$

Any foreign Q increases it. The original signed edge gives $T_f > \sqrt{90/7} > 3.58$. The SINGLE reference (DF) gives $e_D > .99659$ and $e_D + e_I > 1.03659$. This excludes the LAST endpoint-three dotted role.

Every actual fork opposite is either isolated or in the complete nonisolated table. Both classes have now been contradicted for either signed fork, both remaining head families and every finite foreign old-I contact. This proves the short-I D_{15} branch. With its separately written ordinary argument and the unconditional actual rank-one isolation, the whole $D_{15} + A_1$ family is covered for every $h > 0$. These arguments introduce no field-census output premise.

F Actual typed-seven spectators beside B_{14} and D_{14}

Here the ENTIRE old forest is $B_{14} + Z$ or $D_{14} + Z$, where Z is an ACTUAL N_7 or O_7 profile. We do not apply these conclusions to a bare old seven bond. The actual defect field descent in Section 2.1 gives $K = K_7 = \mathbb{Q}(\theta)$, where

$$\theta = 2\cos(2\pi/7), \quad \theta^3 + \theta^2 - 2\theta - 1 = 0, \quad 4g_7^2 = 2 + \theta, \quad g_7 = \cos(\pi/7).$$

Indeed the ACTUAL old-plus-defect-wall $[7, 3]$ principal is Lorentzian over this cubic field; any nonphysical embedding fixing it would preserve its indefinite Gram. The normalized defect wall c_T is orthogonal to the WHOLE classical component, with time squares

$$h_N = (\theta - 1)/3, \qquad h_O = \frac{\theta - 1}{2 - \theta}.$$

Keep the one original decomposition $z = E(p - x - y)$, where x is classical and y is the entire old rank-two projection. Then

$$e_T + e_Z = \|x\|^2 + \|y\|^2 < 1, \qquad e_Z > .09.$$

The already proved actual-defect geometry and propagation are reused at their precise whole-component domain; no duplicate propagation theorem is introduced.

The three cubic roots lie in

$$(31/25, 5/4), \quad (-1/2, 0), \quad (-2, -9/5).$$

The polynomial signs at those six endpoints are $-559/15625, 1/64, 1/8, -1, -1, 1/125$. Thus $0 < h_N < 1/12$, $0 < h_O < 1/3$. The order discriminant is forty-nine, so $[\mathcal{O}_K : \mathbb{Z}[\theta]]$ is odd and K_7 is unramified at three.

For an opposite retaining the whole old spectator, let Q be its FULL inverse unit-Gram cost. The COMPLETE possible list is

$$O_7 : \ Q = 0; \qquad N_7 : \ Q = 0, 1/3, 2/3. \tag{TQ}$$

Even one old-$I_2(7)$ contact costs $1/(4\sin^2(\pi/7)) > 1$. For old A_2, two contacts contain the affine all-three triangle or a stronger nonpositive principal. A single contact permits only labels three/four/five, with costs $1/3, 2/3, \varphi^2/3$. Five is incompatible with the cubic actual field. These are whole-spectator costs, not isolated bond estimates.

F.1 A common actual head argument

The fresh rank-fourteen prefix gives strong $6, \dots, 12$ for B, and $6, \dots, 11$ for D. The B conclusion uses its actual B3/F4 and B2 sign-conjugate rows, not a larger-rank tail. For D, the actual D4/D3 sites are ten and eleven; their only extra weak fork packets have the actual left-H_4 ONE-reference body bounds $> .92531$ and $> .94697$. Adding the separate spectator $> .09$ excludes them. The exact local inventory and reference proof are in Proposition 12; their terminal indices remain ten and eleven here. No floor at the unclassified D centre twelve is assumed.

If five is weak, its actual pure point gives $x_1 > .5965$. Interpolate through that pressure, strong six and every later genuine strong knot. The first-six energy is $> .7600$, versus $E_6 < .6971$, with middle slope $> .2843$, above the next $< .197$. Even the weaker D total is $.84816 + .0629 + .09 = 1.00106 > 1$. Five is therefore strong BEFORE selecting the head. The positive reference w, zero after J, has

T	J	$w^2 = E_J$
B	12	$> .86989$,
D	11	$> .84816$

$b = f_5/5 < .361$, individual P_5 coefficient $> .14$.

For $X = .64, .643$, the first-five graft is legal, preserves every later knot and gives gain $> (5/4)(X - .361)^2$.

Deleting the actual long-head direction retains B_{13} or D_{13} and the ENTIRE spectator. The COMPLETE classical white list is no contact, or simple three at its long endpoint two. In the latter branch

$$c = Ap - g\omega_1 - \omega_2/2 - v_Z, \quad A^2 = 2g^2 + 2g + Q, \quad C_1 = \frac{2g+1}{\sqrt{1-2Q}}, \quad Q < 1/2.$$

Own zero at $Q = 0$ has zero time; at $Q = 1/3$ it has the actual future feasible null $c + \omega_2/2 = \sqrt{Q}\, p - v_Z$. Its original clearing is

$$\frac{\omega_2}{2} = \begin{cases} n_1 + 2(n_2 + \cdots + n_{13}) + \sqrt{2} n_{14}, & B, \\ n_1 + 2(n_2 + \cdots + n_{12}) + n_{13} + n_{14}, & D. \end{cases}$$

The foreign products are nonpositive because its spatial foreign part is the negative of a positive inverse-Gram combination. All omitted ORIGINAL wall products are nonpositive termwise. Own three has its actual affine triangle. Every other positive own gives $C_1 \geq 1 + \sqrt{2}$, $x_1 > .643$. Its legal graft adds $> .099405$, already exceeding one with $e_Z > .09$, even at E_{11}. The head is isolated WITHIN its entire classical component, while foreign contacts are retained.

Write the remaining head

$$c = Ap - g\omega_1 - v_Z, \quad A^2 = 2g^2 + Q - 1, \quad C_1^2 = \frac{2g^2}{1-Q}, \quad U = 4g^2 \in \mathcal{O}_K.$$

Own zero has negative time. Own three has negative time at $Q = 0, 1/3$; at $Q = 2/3$, the ACTUAL chain n_2, n_1, c, z_1, z_2, labels $3, 3, 4, 3$, has affine kernel $(1, 2, 3, 2\sqrt{2}, \sqrt{2})$. Own four at $Q = 0$ has zero time; at $Q = 1/3$, the $3, 4, 3, 3$ chain has kernel $(\sqrt{2}, 2\sqrt{2}, 3, 2, 1)$; at $Q = 2/3$, n_1, c, z_1 is the $4, 4$ null. Own five is absent in K_7; own six has the $6, 3$ kernel $(\sqrt{3}, 2, 1)$; parallel has $c + n_1$. Every affine combination has positive cap coefficient, positive future time and nonpositive product with ALL omitted original walls. Thus only finite owns at least seven or dotted remain.

The COMPLETE finite native invariant alphabet in K_7 is

$$4\cos^2(\pi/m) \in \{1, 2, 3, 2 + \theta, 1 + \theta + \theta^2\}, \qquad m = 3, 4, 6, 7, 14. \tag{TA}$$

Indeed $\phi(m)/2$ divides three, giving the complete totient list $3, 4, 6, 7, 9, 14, 18$. The nine/eighteen field is ramified at three: its polynomial $X^3 - 3X + 1$, translated by minus one, becomes Eisenstein $X^3 - 3X^2 + 3$. It cannot equal K_7, whose discriminant divides forty-nine. The last invariant follows from $2\cos(\pi/7) = \theta^2 + \theta - 1$. Native four is retained; an unsquared $\sqrt{2}$ need not belong to K when root-line squareclasses differ.

For $Q > 0$, a full nonphysical Schur complement requires $U_\sigma/2 + Q < 1$. Normalize the head to raw square $2U$, with own product $-U$ against an old square-two long root; the full Schur complement is $2U(1 - U/2 - Q)$. This compares no disconnected root scales. The remaining finite own seven has a conjugate $U_\sigma > 3/2$, contradicting $Q \geq 1/3$. Fourteen has one $U_\sigma > 61/25 > 2$. Every finite mixed head is excluded.

For a dotted mixed head with $Q = 2/3$, $C_1^2 > 6$, so $C_1 > 12/5$, $x_1 > .64$. For $Q = 1/3$ write $v_Z = \omega_b/2$. The actual N_7 wall gives

$$c \cdot c_T = q - A\sqrt{h_N}, \qquad q = 2g_7/3 \text{ in the same old } A_2 \text{ direction}, \quad q = g_7/3 \text{ in the other.}$$

In the first orientation acuteness yields $A^2 h_N \geq q^2 > 16/45$, so $A^2 > 64/15$. The CORRECT identity is

$$C_1^2 = \frac{3}{2}(A^2 + 2/3) > 37/5.$$

In the other orientation $g > 1$ gives

$$A^2 h_N - q^2 > \frac{4}{3} h_N - \frac{2+\theta}{36} = \frac{5\theta - 6}{12} > 0.$$

The product is strictly negative. Its actual native gap forces $A\sqrt{h_N} \geq q + 1/2 > .8$, hence $A^2 > 192/25$, $C_1^2 > 313/25$. Both orientations give $x_1 > .64$. The one legal graft and $e_Z > .09$ contradict even the D total:

$$.84816 + .09 + \frac{5}{4}(.64 - .361)^2 > 1.03546.$$

Every remaining head is PURE.

For a pure head its product with the actual defect wall is $-A\sqrt{h_T} < 0$. Their actual two-cycle gives

$$M = 4A^2 h_T = 2(U - 2)h_T \in \mathcal{O}_K, \qquad M \geq 1.$$

For N_7, $h_N < 1/12$ forces $U > 8$. For O_7, $M \geq 4$ likewise forces $U > 8$. For its finite mutual alphabet, use (TA). Put $k = \theta^2 + 2\theta - 1$. The cubic gives $1/h_O = k$, so $U = 2 + Mk/2$. The ODD order index means integrality of $Mk/2$ requires its three polynomial coefficients to be even. The COMPLETE table is

M	$[1]Mk$	$[\theta]Mk$	$[\theta^2]Mk$
1	-1	2	1
2	-2	4	2
3	-3	6	3
$2 + \theta$	-1	5	3
$1 + \theta + \theta^2$	1	6	2.

Only $M = 2$ remains. Then $U = (1 + \theta)^2$ and

$$.79 < 64/81 < u = 4/U < 625/784 < .8.$$

All pure heads, including every original finite own, have thus reduced to DOTTED owns with complete domains

$$0 < u < 1/2 \quad \text{or} \quad .79 < u < .8.$$

The actual positive point $c + g^{-1}n_1$ has square $-(1-u)$, positive time, product zero with c, negative product with n_1, and nonpositive product with EVERY other original wall, including c_T. It gives the same dotted-row Z, H, R pressure as before. On these TWO full intervals its scalar improves to

$$F = .8H - .16R > .3055.$$

For $u \le 1/2$, use $a = 2\sqrt{2}/5$, $.56568 < a < .5657$, the chord $\sqrt{2-u} > 1.4142 - .379u$ and $R \le 1 - .75u$. The circle inequality gives

$$F > .639984656 - \frac{.0944 + \sqrt{.0944^2 + .5657^2}}{2} > .305984656,$$

since $.0944^2 + .5657^2 < .5736^2$. For $.79 < u < .8$, direct positive squares give $\sqrt{2-u} > 1.0954$, $\sqrt{u(1-u)} < .4074$, $H > .4864$, $R < .52205$. Thus $F > .305592$. Both continuous domains, including unbounded dotted own, are covered.

Pay ONLY .08 from the individual P_5 price: $Y_H = Z - P_5/5$, $W = w + .4Y_H$. The residual is zero-sum on the first five coordinates, has FULL cross zero with w, and norm $R - 1/5$. The effective gain is

$$F - .8b + .032 > .0487. \tag{TG}$$

For B_{14}, adding spectator ONCE already gives $.86989 + .0487 + .09 = 1.00859 > 1$. This completes BOTH typed B_{14} cases. For D_{14}, the weaker E_{11} reference still needs its own centre-twelve proof.

F.2 The full D_{14} centre at its actual rank

Use actual roots $n_i = (e_i - e_{i+1})/\sqrt{2}$, $i \le 13$, $n_{14} = (e_{13} + e_{14})/\sqrt{2}$. The centre is twelve, forks thirteen and fourteen. The signed coordinate x_{14} is retained. At this point ONLY $5, \dots, 11$ are strong; E_{12} is a scalar value, not yet an available floor. The reference through eleven has individual terminal price $\sqrt{11} - \sqrt{10} > 1/7$, $f_{11} < 2.886$, and the head residual Y_H is supported on the first five coordinates.

Deleting twelve retains A_{11}, both genuine fork A_1 roots and the entire old spectator. The COMPLETE finite-white list is: without forks, isolation, simple three at $j = 1, 2, 10, 11$, or endpoint four at $j = 1, 11$; with left contact and forks, only endpoint-three at one/eleven with one fork-three/four or both forks-three; without left contact, one arbitrary finite fork label, or fork pairs $(3,3), (3,4), (3,5)$. Five is absent in K_7. Two left contacts or two contacts within old A_2 form a nonpositive cycle. Next-to-end plus a fork has nonfinite arms $1, 2, 9$; a four internal to a long path is nonfinite. Every left/fork bridge is foreign-sealed: an extra old-A_2 arm makes arms $11, 1, 2$ or adds another branch to a full D_{14} tree. All other foreign contacts retain (TQ).

For the actual opposite write

$$c = Ap - g\omega_{12} - \eta\omega_j - \delta\omega_{13} - \epsilon\omega_{14} - v_Z.$$

A missing contact has coefficient zero. The exact actual Schur and outgoing data are

$$q_L = \frac{2\eta^2 j(12-j)}{12}, \quad \Delta = 1 - q_L - \delta^2 - \epsilon^2 - Q > 0,$$
$$\sigma = g + \eta j/12 + (\delta + \epsilon)/2, \qquad C_{12} = \sqrt{24}\sigma/\sqrt{\Delta}.$$

The .61 sufficient ratio is $< 25/16$. For $a = \sqrt{24}$, $4.8 < a < 5$, its exact expression is

$$\frac{2 - 2(.61)/a + (.61/a)^2}{1.22 - .3721/a} < \frac{1.773}{1.142} < \frac{25}{16}.$$

Thus any ratio above $25/16$ proves a true strong-twelve floor.

For no fork and a left contact, own-zero far-three would require $1/2 < Q < 13/24$, next-to-end-three would require $0 < Q < 1/6$, and far-four would require $0 < Q < 1/12$. None occurs in the COMPLETE foreign list; the latter two have zero time at $Q = 0$. Near/next-three own zero has the affine double-fork principal: preceding old node, path to twelve, both forks and cap. Near-four has the actual kernel $(\sqrt{2}, 2, 2, 1, 1)$ on $c, n_{11}, n_{12}, n_{13}, n_{14}$. Own three with white three has the actual cycle; own four the star at twelve with leaves cap and both forks; own six the $6, 3$ path to a fork; parallel the null pair. Own five is absent. Positive endpoint-four ratios are at least $\sqrt{12}/2 + 1/\sqrt{24} > 25/16$; remaining next-to-end-three ratios exceed 2.4, and near-three ratios exceed 1.84. A possible weak far-three is precisely $g\omega_{12} + \omega_1/2$, with $.9 < g < 6/5$. It is excluded by Lemma 5 at twelve, which includes arbitrary foreign contacts.

For no fork and isolation, own zero has negative time. Own three has the actual four-leaf-three star; four/six/parallel have the preceding star/path/null, and five is absent. For $Q > 0$ after these removals, or for dotted $Q = 0$, put $u = (1-Q)/g^2 < 1$. The original positive clearing $c + v_Z + gun_{12}$ is future negative and gives

$$x \cdot (\omega_{12} - un_{12}) \geq \sqrt{24 - u} - \sqrt{u(1-u)}.$$

Here v_Z is a positive inverse-Gram combination of ACTUAL old spectator normals. At $u = 1$ this is a forbidden feasible null. For $0 < u < 1$, the loss below $\sqrt{24}$ is $< .104u + \sqrt{u(1-u)} < .555 < .61$; use $\sqrt{24} + \sqrt{23} > 9.69$ and $\sqrt{1 + .104^2} < 1.006$. Because $x \cdot n_{12} > 0$, this proves strength. For every pure finite own at least seven, $1 < u = g^{-2} < 4/3$, use instead

$$\xi = c + g^{-1}n_{12} + 2(g^{-1} - g)n_{11}.$$

It has square $-g^2(7u - 4 - 3u^2) < 0$, products zero with cap and twelve, negative with eleven and both forks, positive future time, and all omitted original-wall products nonpositive termwise. Its displacement is $\omega_{12} - un_{12} - 2(u-1)n_{11}$; the pressure is $\sqrt{24 - u} - \sqrt{7u - 4 - 3u^2}$. The first radical loses $< .139$, while

$$7u - 4 - 3u^2 = 1/12 - 3(u - 7/6)^2 < .289^2.$$

The total loss is $< .428 < .61$. This covers the ENTIRE finite native tail.

For left endpoint-three plus forks, $Q = 0$. With both forks-three, $\Delta = 1/24$, $\sigma \geq 1/2 + 1/24$; with one fork-four, $\Delta = 1/24$, $\sigma \geq 1/(2\sqrt{2}) + 1/24$. Both are directly strong even at own zero. With one fork-three, $\Delta = 7/24$; own zero has its actual cycle, own three its contacted-fork triangle, and every own at least four is directly strong. The far orientation is used in each lower bound, so near orientations are also covered.

With no left contact and positive own, a single fork-three/own-three has its affine triangle. Single fork at least six with own three is directly strong. Single fork-four/own-three with $Q > 0$ has $\Delta \leq 1/6$ and is strong; its $Q = 0$ case is retained below. Own four has the star at twelve with leaves cap, eleven and the UNCONTACTED fork. Own six has the $6, 3$ path through eleven; parallel has its null pair. For every remaining own at least seven or dotted, fork at least four is directly strong. Fork-three with $Q > 0$ has $\Delta \leq 5/12$ and is strong. At $Q = 0$, a possible weak fork-three has $.9 < g < 6/5$. Its ACTUAL triangle makes $t = 2g$ integral. Normalize the connected simple-three cap to raw square two; all nonphysical K_7 embeddings obey

$$48t_\sigma(t_\sigma + 1) + 14 < 8, \qquad -1 < t_\sigma < 0.$$

Its nonzero integer $t(t+1)$ has norm $< (204/25)(1/4)^2 = .51 < 1$, impossible. For two forks-three, own three has its triangle and owns at least four are directly strong. Pair three/four is directly strong for every positive own.

With no left contact and own zero, single fork-three has the actual affine E_8 kernel: centre twelve coefficient six, uncontacted fork three, contacted fork four, cap two, and old eleven through seven coefficients five through one. Single fork-four has kernel $(\sqrt{2}, 2, 2, 1, 1)$ on cap, contacted fork, twelve, eleven and other fork. Single fork-six has the $6, 3$ principal. Every null has positive cap time and correct signs against ALL omitted ORIGINAL walls. Single fork-five is absent. For any finite fork label at least seven, $\delta > .9$; finite whiteness $\delta^2 + Q < 1$ forces $Q = 0$. Two forks-three have the actual four-cycle; pair three/four has $Q = 0$ and remains below. Thus the ONLY nonstrong possibilities are: own-three/single-fork-four $Q = 0$; own-zero/pair-three-four; own-zero/single-finite-fork-at-least-seven. All other own, foreign and unbounded dotted cases have been covered.

F.3 Reciprocal endpoints and the remaining continuous fork family

For own-three/single-fork-four, the SAME actual cap is also the contacted fork's opposite. Deleting that fork retains A_{13}; adding the cap at its next-to-end node by three gives finite D_{14}. For own-zero/pair-three-four, release the fork-three direction instead; the retained A_{13} and cap give finite B_{14}. In both cases keep the ENTIRE orthogonal spectator. Positive inverse finite-Gram projection from p gives an actual future vertex on that original edge. Every unknown wall remains strict: its product with p is strict and all correction coefficients are nonnegative. The omitted old fork is strict because the cap coefficient and that fork contact are positive. This proves an ACTUAL endpoint, not a formal completion or a new farthest vertex.

Both reciprocal fork coshes are

$$C_f = 6 + 7\sqrt{2}/2 > 10.$$

In the first case $A^2 = (17 + 12\sqrt{2})/2$, $v \cdot \omega_f = 6 + 7/\sqrt{2}$, $(v \cdot \omega_f)^2 - 7A^2 = 1$. In the second, $A^2 = (17 + 12\sqrt{2})/4$, $v \cdot \omega_f = 7/2 + 3\sqrt{2}$, and the difference is $1/2$. These give the displayed cosh by the original edge formula. For either sign, the genuine spatial direction $V = P_{13} \pm e_{14}$, $V^2 = 14$, gives

$$x \cdot V > \sqrt{126/11} > 3.38.$$

Pay $\mu = 1/7$ at the individual P_{11} coefficient $> 1/7$. The residual $V - P_{11}$ vanishes on all first eleven coordinates, has square three and full cross zero with w. One original square WITHOUT any head credit gives

$$e_D + e_Z > .84816 + .09 + \frac{2(3.38 - 2.886)}{7} - \frac{3}{49} > 1.018.$$

Both exceptional reciprocal rows are excluded.

Finally let own be zero and the single finite fork contact have square $b_f = \delta^2 \in (.81, 1)$. Put $u = 1/b_f \in (1, 100/81)$; $Q = 0$ was proved, rather than assumed. The actual positive combination

$$\xi = c + \delta^{-1} n_f + 2(\delta^{-1} - \delta) n_{12}$$

has zero cap and contacted-fork products, negative centre and other-fork products, and square

$$-k, \qquad k = \frac{(1 - b_f)(4b_f - 3)}{b_f} > 0.$$

Its time is positive and every unknown original-wall product is nonpositive termwise. It is therefore a feasible future point of the ENTIRE polytope. For $\alpha = \delta/\sqrt{2}$, its divided spatial displacement and pressure are

$$V = (1^{11}, 3 - 2u, u - 1, \ \pm(1 - u)),$$
$$x \cdot V \geq G = \sqrt{14 - 2u} - \sqrt{-8 + 14u - 6u^2},$$
$$R = V^2 = 22 - 16u + 6u^2.$$

This retains the ACTUAL fork sign; it never assumes $x_{14} > 0$. The residual $V - P_{11}$ vanishes on the entire reference support and is orthogonal to Y_H.

On the ENTIRE interval,

$$14G - R > 30.58.$$

Here is a continuous rational seal. The chord $\sqrt{14 - 2u} \geq 3.756 - .292u$ has positive endpoint-square margins 11/15625, 574919/410062500. Put $z = u - 7/6$. Then

$$-8 + 14u - 6u^2 = 1/6 - 6z^2, \qquad R = 23/2 - 2z + 6z^2,$$

and $\sqrt{1/6 - 6z^2} \leq 1/\sqrt{6} - 3\sqrt{6}z^2$. Consequently

$$\begin{aligned} 14G - R &\geq 13618/375 - 14/\sqrt{6} - 2.088z + (42\sqrt{6} - 6)z^2 \\ &> 13618/375 - 5.716 - 2.088z + 96.8748z^2 \\ &> 13618/375 - 5.716 - .012 = 2294/75 > 30.58. \end{aligned}$$

Use $\sqrt{6} > 2.4494$, $5.716(2.4494) > 14$ and $2.088^2 < .048(96.8748)$ for the final completed square.

Now combine the TWO primitive pressures BEFORE taking the norm:

$$W = w + .4Y_H + (V - P_{11})/7.$$

Its INDIVIDUAL prices are .08 at P_5 and 1/7 at P_{11}. Both remaining coefficients are positive. Both residuals are orthogonal to w AND to each other. Using the head gain (TG) and the same spectator ONCE gives

$$\begin{aligned} e_D + e_Z &> E_{11} + .09 + .0487 + \frac{14G - R - 14f_{11} + 11}{49} \\ &> .84816 + .09 + .0487 + \frac{30.58 - 40.404 + 11}{49} = 1.01086 > 1. \end{aligned}$$

This is one original reference, not the addition of independent head and centre optima.

Every ACTUAL centre-twelve cap is now strong or impossible. The positive curve through twelve is available with $E_{12} > .86989$. Its first-five mean and individual P_5 price are unchanged. Apply the SAME actual head point and its primitive .08 payment once to this NEW reference:

$$e_D + e_Z > .86989 + .0487 + .09 = 1.00859 > 1.$$

This excludes BOTH D_{14} actual typed-seven branches. The order was fresh $6, \ldots, 11$, then five, then the actual head, THEN centre twelve and its reciprocal/continuous exceptions. Neither fork was assigned an unsupported prefix floor, and no actual short-I barrier or arithmeticity of a face was used.

G A whole D_{13} component with any rank-three remainder

This argument covers EVERY finite rank-three spectator forest beside a whole D_{13} at the original farthest vertex in dimension sixteen. It does not depend on subtracting rows of a previous frontier. Retain actual unit D_{13} roots, long spine one through eleven, forks twelve and thirteen, and signed coordinates $x_1 > \cdots > x_{12} > |x_{13}|$. Write $e_D + e_S < 1$ for the SAME original vector and its entire orthogonal rank-three spectator.

Every such spectator has

$$e_S > .21. \tag{S3}$$

Here is the COMPLETE partition and exception argument. With ordinary components only, partitions three, two-plus-one and one-plus-one-plus-one cost $> 1/4$, $> 1/7 + .1$ and $> .3$. For actual I, the remaining rank two is one ordinary irreducible T_2, or two ordinary A_1's. The existing rank-free joint costs are $I + T_2 > .21$ and $I + A_1 > .20204$; the latter plus one further A_1 is $> .30204$. The actual I is counted once. A J pair plus the last ordinary A_1 costs $> .34 + .1$. An actual typed rank-two defect costs $> .09$. For N_7, existing propagation gives the last $A_1 > .2$; for O_7, it gives $A_1 > 2/15$. The compatibility theorem permits no additional I/J beside a typed defect and at most the single stated short-rank exception. These are every rank-three partition and exception. The propagation/joint theorems are the SAME whole component statements already used in the shared seals; no new field refinement is needed for this cost.

For D_{13}, choose generic parameter $M = 7$. Its exact rank hypothesis $13 = M + 6$ holds. Deleting $i = 5, \dots, 8$ retains $A_{i-1} + D_{13-i}$, with both necessary side ranks at least four/five. The full finite connected replacement has rank thirteen, so exceptional high-rank alternatives do not arise; the existing small E_6, E_7, E_8 fork attachments are retained in the local list. The complete generic weak row is the pure far helper, excluded at $i \geq 6$ by Lemma 5. Thus six, seven and eight are strong before either short layer or strength at five.

We need only two earlier anchors, $K = 6, 7$. If five is strong use the positive ordinary reference with norm E_K. If five is weak, its actual pure point gives $x_1 > .5965$, $S_6 > 2.01815$. At $K = 6$ use coordinates $(.5965, .28433, \dots, .28433)$. Their norm is $.7600299945 > E_6$, their sum $2.01815 < f_6$, and terminal coefficient $.28433 > 1/6$. At seven append $\sqrt{7} - \sqrt{6}$; the endpoint is $< f_7$, energy $> E_7$ and terminal coefficient $1/(\sqrt{7} + \sqrt{6}) > 1/6$. All individual duals are nonnegative. The strong-five terminal coefficients at six/seven are also $> 1/6$. In EITHER initial branch we therefore have an actual supported reference w on the first K coordinates satisfying

$$w^2 \geq E_K, \quad w \cdot P_K \leq f_K, \quad \text{individual } P_K \text{ coefficient} > 1/6.$$

The exact scalar seals are

$$E_6 > .69685,\ E_7 > .73531,\ E_{10} > .82436, \qquad f_6 < 2.019,\ f_7 < 2.215.$$

The fresh short layers are $i = 9$, with white $A_8 + D_4$, and $i = 10$, with white $A_9 + D_3$. Their COMPLETE left-only finite extensions are endpoint-three/four and next-to-end-three. Ranks nine/ten preclude a new exceptional type. Two contacts to either retained component form a forbidden cycle. The shared local inverse formulas and original spectator clearing apply at these ACTUAL side ranks. We give the right and bridge cases explicitly:

right role	δ before Q	σ
D_4 long three	$1/2$	$\gamma + 1/2$
D_4 fork three	$1/2$	$\gamma + 1/4$
D_3 middle three	$1/2$	$\gamma + 1/2$
D_3 end three	$5/8$	$\gamma + 1/4$
D_3 end four	$1/4$	$\gamma + 1/(2\sqrt{2})$
D_3 end five	$(5 - 3\varphi)/8$	$\gamma + \varphi/4$.

Every foreign Q is the ENTIRE spectator cost, not an individual bond, and subtracts from the residual.

The long/middle-three own-zero rows have the actual two-fork/four-leaf affine kernel, own three their triangle; every remaining own at least four has ratio $\geq 1 + 1/\sqrt{2} > 100/61$. End-four own zero has the actual star of squared costs $1/4, 1/4, 1/2$; every positive own has the same direct lower

ratio. End-five is the sealed H_4, strong even at own zero, since its residual is $< 1/40$ and numerator $> 2/5$. A left endpoint-three/right-simple-three bridge completes D_{13}, or, for D_3 end-three, A_{13}. The former has residual $1/(2i)$ and positive-own numerator at least $\gamma + 1/4 + 1/(2i)$; the latter at ten has residual $7/40$ and positive-own numerator $> .75$. Both ratios exceed $100/61$. Their own-zero cycles are affine. The bridge cap is internal in a rank-thirteen classical tree, so has no finite spectator arm; the A_{13} bridge's old arms have lengths nine and three. An end-four/five cannot join the long left path: its non-three bond would be internal in a nonfinite rank-thirteen tree.

For the left-only small owns, internal-three and endpoint-four own zero retain their actual affine fork principals. Own three with a white-three contact has its cycle. Own four with no right contact has the affine principal obtained from the retained right D_5/D_4 INCLUDING n_i: its unit long-leaf inverse diagonal is two, so a four attachment has cost one. Own six has its local $6, 3$ path. Parallel always has its future feasible null pair. Far-left-three at own zero with foreign contacts has $A^2 = Q - 1/2 < 1/(2i) < 1/4$. The original small-time shadow argument proved in the dihedral section would make the SAME cap close an orthogonal spectator direction, contrary to the original-farthest common-opposite obstruction. For $Q = 0$ its time square is negative. Thus that own-zero row also is impossible. Every null here is a positive combination of ACTUAL walls with positive cap time and nonpositive product at EVERY omitted original wall. The nonzero left-only ratios and foreign clearings are the legal shared prefix formulas at $i = 9, 10$. The only potentially weak left row is the literal far helper and is removed by Lemma 5.

For isolation, the original singleton and LEFT H_4/A_2 projected points and negative cores remain available through $i = r - 3$, with $D_i = 2i \geq 18$. They cover arbitrary foreign cost and all finite/parallel/dotted own, without a fictitious long root. For clarity their scalar margins still suffice here: the $0 < u < 1$ core has full loss

$$< \frac{.12 + \sqrt{1 + .12^2}}{2} < .57 < .61,$$

using $(\sqrt{18} + \sqrt{17})^{-1} < .12$. For foreign $u > 1$, $u < 7/6$, the radical loss is $< .14 + .4 < .61$. Pure native five's LEFT H_4 correction has loss $< .2 + .25 < .61$; native seven's LEFT A_2 correction has loss $< .15 + .36 < .61$; all finite natives at least eight use the singleton loss $< .15 + .415 < .61$. The underlying divided displacements and concavity are exactly those printed in Section E.8; only these freshly stated $D_i \geq 18$ scalar bounds are used, not its $i = 11, 12$ estimates.

For the retained single-fork-three/end-three roles, own zero, three, four and six have respectively the actual E_8, cycle, F_4 and $6, 3$ principals; all required old arms exist at nine/ten. Parallel has its actual null pair. Thus $\gamma \geq \varphi/2$. If $Q > 0$, then $\delta \leq 3/8$, $\sigma > 21/20$, giving ratio $> 42/25 > 100/61$. For $Q = 0, \gamma \geq 6/5$, the ratio again exceeds $100/61$. ALL finite and dotted own values in between are retained. The ONLY possible weak short caps are therefore

$$v = \gamma\omega_i + \omega_f/2, \quad Q = 0, \quad i = 9, 10, \ f = 12, 13, \quad 202/125 < 2\gamma < 12/5.$$

Their actual simple-three contact normalizes the cap to raw square two over ACTUAL K. The original path cycle makes $t = 2\gamma$ integral. At EVERY nonphysical embedding,

$$4it_\sigma(t_\sigma + 1) + 13 < 8.$$

Hence $-1 < t_\sigma < 0$. All embeddings of the minimal field lift to K. The integer $t(t+1)$ has physical value $< 204/25$ and other absolute conjugates at most $1/4$. Degree at least three contradicts its nonzero integral norm bound $< .51$. In degree two the trace is one or two. Trace one gives norm -1 and $t = \varphi$; trace two gives norm in $(-24/25, 0)$, impossible. The rational value two is an actual

parallel null. Every weak fork helper is therefore golden, without a field census or a scale comparison with spectators.

For each golden packet use the ACTUAL LEFT finite H_4 correction

$$w_{\mathrm{old}} = \gamma(3n_i/2 + n_{i-1} + n_{i-2}/2), \quad \alpha^2 = \varphi + 5/8 > 56/25, \quad \delta = (5 - 3\varphi)/8, \quad \sqrt{\delta}/\alpha < .091.$$

The actual finite point is the normalization of $p + (A/\delta)(c + w_{\mathrm{old}})$. It has square $-1 - A^2/\delta$, positive future time and zero selected products. Every omitted original-wall product is nonpositive by actual acuteness and the positive coefficients. The correction alone is not a hyperbolic point. For $Y = v - w_{\mathrm{old}}$, $V = Y/\alpha$, $K = i - 3$, the fresh inverse calculations give

$$A^2 = i\alpha^2 - \kappa, \quad \kappa = 1/2 \ (i = 9), \quad \kappa = 5/8 \ (i = 10),$$
$$V_j = 1 \ (j \le K), \quad V^2 < i, \quad x_D \cdot V > G = \sqrt{i - \kappa/\alpha^2} - .091.$$

Use the already established $K = 6/7$ reference in EITHER initial-five branch. Pay its INDIVIDUAL terminal coefficient $1/6$ BEFORE using its floor. For $W = w + (V - P_K)/6$, the residual vanishes on the first K coordinates, so its FULL cross with w is zero, and its norm is < 3. One square gives

$$e_D > E_K + \frac{G - f_K}{3} - \frac{1}{12}.$$

The COMPLETE fresh table is

layer	i	K	$G >$	$f_K <$	$e_D >$
D_4	9	6	2.871	2.019	.8975
D_3	10	7	3.026	2.215	.92231.

The two radical seals are $2.962^2 < 9 - 25/112$, $3.117^2 < 10 - 125/448$. Adding ONLY the actual disjoint spectator (S3) contradicts the original norm in BOTH cases. Thus all $6, \dots, 10$ are now genuinely strong.

The final ordinary or legitimate weak-five reference through ten gives $e_D > E_{10} > .82436$. Consequently

$$e_D + e_S > .82436 + .21 = 1.03436 > 1.$$

This excludes WHOLE D_{13} with EVERY rank-three spectator forest at the actual farthest vertex in dimension sixteen. Every pressure belonged to one ORIGINAL vector; the spectator was counted once. No actual-I barrier, typed field lock, field census, arithmeticity of a face or unexecuted computation was used to enlarge this theorem's scope.

H Whole B_{16}: geometry before the finite pair domain

Now the ENTIRE incident forest is B_{16}; there is no spectator to delete or to charge. Use actual unit roots $n_i = (e_i - e_{i+1})/\sqrt{2}$, $i \le 15$, $n_{16} = e_{16}$, and the original vector $z = E(p - x)$, $x_1 > \cdots > x_{16} > 0$, $\|x\| < 1$. The legal fresh bulk and B3/B2 terminal checks give strong $6, \dots, 14$ independently of five and either endpoint. We first prove all physical reductions needed to enter the exact quadratic algorithm.

H.1 The actual head and a tight short-edge length bound

For $a = .643$ or $.648$, the reference through $(1, a), (6, f_6), \ldots, (14, f_{14})$ has positive decreasing coordinates

$$a, \quad (f_6 - a)/5 \text{ five times}, \quad \sqrt{j} - \sqrt{j-1}\ (7 \le j \le 14), \quad 0, 0.$$

Its first middle slope exceeds .274, hence the next square-root slope $< .2$. If actual $x_1 > a$, ONE primitive square gives

$$\|x\|^2 > a^2 + (f_6 - a)^2/5 + \sum_{i=6}^{13} (\sqrt{i+1} - \sqrt{i})^2.$$

The last sum is $> .2117$. A direct rational proof avoids a numerical receipt: for $A = 4i + 2$, put $L = (A^2 - 1)/(A(A^2 - 2))$. The small root $d = (\sqrt{i+1} - \sqrt{i})^2$ satisfies $d(A - d) = 1$, whereas

$$L(A - L) - 1 = -\frac{1}{A^2(A^2 - 2)^2} < 0.$$

Thus $d > L$. For $i = 6, \ldots, 13$, strict lower numerators of L at denominator 10^7 are

$$385186, 333704, 294372, 263340, 238230, 217494, 200080, 185248.$$

Their sum is 2117654. With $f_6 > 2.01815$, the two reference norms are > 1.0033565045, > 1.0070662045.

The ACTUAL head's retained white diagram is B_{15}. Finite classification permits isolation or simple three at its LONG endpoint two, with no symmetric short-end alternative. For the latter, $A_H^2 = 2g^2 + 2g$, $C_H = 2g + 1$. Own zero has zero time, own three has its affine triangle, and parallel has the actual null pair. Every remaining own has $g \ge 1/\sqrt{2}$, so $x_1 > \sqrt{\sqrt{2} - 1} > .643$, impossible by the first reference. The head is isolated:

$$c_H = A_H p - \sqrt{2} g_H e_1, \quad U = 4g_H^2 \in \mathcal{O}_K, \quad A_H^2 = U/2 - 1, \quad U > 2.$$

If $U \ge 12$, its ACTUAL edge has $C_H = \sqrt{U/2} \ge \sqrt{6}$, giving $x_1 > .648$, impossible by the second reference. The strict rational seal uses $\sqrt{6} > 2.449$, $.648^2(3.449) < 1.449$. Thus $2 < U < 12$.

Let C_S be the ACTUAL short-sixteen edge cosh. Its actual finite endpoint and original comparison give $S_{16} \ge 4\sqrt{(C_S - 1)/(C_S + 1)}$. If $C_S \ge 16$, this is > 3.7573. For strong five, interpolate through true knots five through eight and this terminal pressure. The last slope is $< .172 < \sqrt{8} - \sqrt{7}$. Using

$$E_8 > .69685 + .0385186 + .0333704 = .768739, \qquad f_8 < 2.3971,$$

its norm is

$$> .768739 + 1.3602^2/8 = 1.000007005.$$

The upper seal for f_8 follows from $\sqrt{8} < 2.828428$, $\sqrt{2} < 1.414214$, $.431328(1.414214) < .61$; also $f_8 > 2.39$ gives the claimed last-slope bound. For weak five use the actual pressures $x_1 > .59$, $S_6 > 2$ and $S_{16} > 3.7573$. The decreasing slopes $.59, .282, .17573$ give one reference of norm

$$.59^2 + 1.41^2/5 + 1.7573^2/10 = 1.054530329 > 1.$$

Both alternatives contradict the SAME original norm without adding a head gain. Therefore $C_S < 16$.

H.2 All seven actual short-wall profiles

The retained ACTUAL A_{15} finite-white list is isolation; helper three at $j = 1, 2, 14, 15$; or helper four at $j = 1, 15$. Two contacts give a cycle. Other simple branches or non-three bonds cannot give a finite rank-sixteen tree. There are no foreign components.

Normalize old long roots to raw square two and the genuine short root to square one: $\alpha_i = e_i - e_{i+1}$, $s_{16} = e_{16}$. All e_i are then K-valued. A helper-three cap normalizes to square two, helper product -1, and short product $-T/2$, where $T = 2\sqrt{2}\gamma$. The ACTUAL cycles give $T, T^2/2 \in \mathcal{O}_K$, not $T/2$ integral. Its exact formulas are

$$c = Ap - \gamma P_{16} - P_j/\sqrt{2}, \quad 2A^2 = F_j(T) = 4T^2 + jT + j - 2,$$
$$C_S = \frac{8T + j}{\sqrt{32 - j(16 - j)}}.$$

The original old-plus-cap Gram is Lorentzian over $\mathbb{Q}(T)$, so actual $K = \mathbb{Q}(T)$, and every other $F_j(T_\sigma) < 0$. A helper-four cap has raw square one and helper product -1, with

$$A^2 = 16\gamma^2 + 2j\gamma + j - 1, \quad C_S = 16\gamma + j, \quad K = \mathbb{Q}(\gamma).$$

These field equalities follow from nonphysical embeddings fixing an original indefinite principal, not from arithmeticity of a face.

Own zero is retained before dispatch. Helper-three at one/two has $A^2 = -1/2, 0$. At fifteen its actual null is $c + n_{14} + 2n_{15} + \sqrt{2}n_{16}$; at fourteen it is $c + n_{13} + 2n_{14} + 2n_{15} + \sqrt{2}n_{16}$. Helper-four at one has zero time; at fifteen use $c + \sqrt{2}n_{15} + n_{16}$. Isolation has $A^2 = -1$. Every null has positive cap time and selected products zero; ALL omitted original-wall signs are nonpositive termwise. Parallel own always has $c + n_{16}$.

Near helper-four positive own has $C_S \geq 23$. Far helper-four has $\gamma < 15/16$, so its complete positive native alphabet is three through eight. The native-nine cubic at $15/16$ is $-17/512$, so all larger finite labels, parallel and dotted are beyond this cutoff. Own three is rational and excluded by WHOLE-form compact anisotropy/Meyer. For owns four through eight use respectively the nonphysical conjugates

$$-\sqrt{2}/2, \quad (1 - \varphi)/2, \quad -\sqrt{3}/2, \quad \cos(3\pi/7), \quad -\cos(\pi/8).$$

The resulting time squares are $8 - \sqrt{2}$, $9 - 5\varphi$, $12 - \sqrt{3}$, $16\gamma_\sigma^2 + 2\gamma_\sigma$, and $16\gamma_\sigma^2 - 2\gamma_\sigma$, all positive. They contradict admissibility in the ACTUAL generating field.

Far next-end-three has $T < 15/4$, every other $-1/2 < T_\sigma < 0$. In degree $d \geq 2$, its nonzero integral norm has magnitude $< 15/8 < 2$, hence one. But $T^2/2$ integral would make $\mathrm{N}(T)^2/2^d$ an integer, impossible. Degree one is rational. Near next-end-three has $T < 9/4$, leaving only own three $T = \sqrt{2}$ or own four $T = 2$. Own five already has $T^2 = 2\varphi^2 > 81/16$. The rational case is impossible; the other conjugate time numerator is $20 - 14\sqrt{2} > 0$. Both next-end orientations and both helper-four roles are exhausted.

For the remaining far-end-three put $S = T$; for near-end-three put $S = T + 2$. Their time polynomials are $4S^2 + S - 1$ and $4S^2 - S - 1$, with respective nonphysical intervals

$$\frac{-1 - \sqrt{17}}{8} < S_\sigma < \frac{-1 + \sqrt{17}}{8}, \qquad \frac{1 - \sqrt{17}}{8} < S_\sigma < \frac{1 + \sqrt{17}}{8}.$$

Both have $0 < S < 42/5$, $|S_\sigma| < .641$ and $S^2/2$ integral. For degree $d \geq 3$, the nonzero integer norm is divisible by four but has magnitude $< (42/5)(.641)^2 = 3.4514004 < 4$. Degree one is rational;

actual degree is two. Trace and norm are EVEN by the same half-square argument already proved above.

The hand table uses the EXACT physical short-edge cutoff $C_S < 16$, rather than only the coarser $S < 42/5$ used to bound the degree:

$$\begin{array}{r|l} \text{far} & \sqrt{2} \le S < (16\sqrt{17}-1)/8 < 65/8 \\ \text{near} & 2+\sqrt{2} \le S < (16\sqrt{17}+1)/8 < 67/8. \end{array}$$

Indeed the actual edge formulas are $C_S = (8S+1)/\sqrt{17}$ and $C_S = (8S-1)/\sqrt{17}$, respectively; the lower bounds retain the native own gap. For example the far integer $4+3\sqrt{2}$ passes the coarse bound but violates this exact physical cutoff and is not a table row.

The far strips give traces $2, 4, 6, 8$, $-6 < \mathrm{N}(S) < 4$. The near strips give traces $4, 6, 8$, $-4 < \mathrm{N}(S) < 6$. Solving ALL even trace/norm pairs and imposing the two strict strips gives the COMPLETE eleven rows:

role	T	$\mathrm{N}(F_j(T))$
far	$2+\sqrt{6}$	-53
	$3+\sqrt{13}$	-25
	$3+\sqrt{11}$	-151
	$3+\sqrt{7}$	-19
	$4+\sqrt{14}$	-117
near	$\sqrt{2}$	-9
	$1+\sqrt{11}$	-43
	$1+\sqrt{7}$	-103
	$2+3\sqrt{2}$	-137
	$2+\sqrt{14}$	-229
	$2+2\sqrt{3}$	$-83.$

For explicit coverage, the far norm columns are $-4, -2, 2$. Trace two gives only inadmissible conjugates or nonreal roots; trace four retains $2+\sqrt{6}$; trace six retains all three $3+\sqrt{13}$, $3+\sqrt{11}$, $3+\sqrt{7}$; trace eight retains only $4+\sqrt{14}$. The near S norm columns are $-2, 2, 4$. Trace four retains only $2+\sqrt{2}$; trace six retains $3+\sqrt{11}, 3+\sqrt{7}$; trace eight retains $4+3\sqrt{2}, 4+\sqrt{14}, 4+2\sqrt{3}$. Subtract two to recover the printed near T. The omitted rows violate a strict conjugate/physical bound or are repeated rational roots. All displayed norms are negative and their physical times positive, so the other time signs are exactly the required negative ones. These are necessary actual models, never examples.

H.3 The original connected-basis squares remove every helper

Normalize the actual head to the K-root $h = c_H/(\sqrt{2}g_H)$, with $h \cdot \alpha_1 = -1$, $h^2 = 2/U$. Every nonphysical head Schur complement gives $0 < U_\sigma < 2$. For any of the eleven helper-three rows, the sixteen old raw roots and $r = \sqrt{2}c$ are an ORIGINAL K-basis. The coefficient of r in h, computed from its positive time coefficients, therefore gives

$$R_0 = \frac{U-2}{UF_j(T)} \in K^2.$$

This derives a common time-line square in one actual connected basis. It does not put separate unit roots or square roots in K.

For quadratic K, the full head strip gives

$$0 < \mathrm{N}(U) < 24, \qquad 0 < -\mathrm{N}(U-2) < 20.$$

For all five NEAR DOTTED rows and the far $\sqrt{6}, \sqrt{11}$ rows, the absolute time norm is a prime at least forty-three. Its valuation in $\mathrm{N}(R_0)$ is odd because neither of the two bounded integer norms can contain it. For far $\sqrt{7}$, the FULL ring $\mathbb{Z}[\sqrt{7}]$ gives the complete head list

$$U = 3+\sqrt{7},\ 4+\sqrt{7},\ 6+2\sqrt{7}, \quad \mathrm{N}(U) = 2, 9, 8, \quad \mathrm{N}(U-2) = -6, -3, -12.$$

Writing $U = a + b\sqrt{7}$ shows $b = 1, 2$; the strict head strips give precisely these a's. None contains prime nineteen. For far $\sqrt{14}$, its full strip similarly gives

$$U = 4+\sqrt{14},\ 5+\sqrt{14}, \quad \mathrm{N}(U) = 2, 11, \quad \mathrm{N}(U-2) = -10, -5.$$

The odd prime thirteen in time norm -117 therefore cannot cancel. These are nine eliminated rows.

The near $T = \sqrt{2}$ row needs its FULL square test rather than a norm-only rejection. Here actual $K = \mathbb{Q}(\sqrt{2})$, because its connected old-plus-cap Gram is physically Lorentzian over that field. Put $s = \sqrt{2}$, $\epsilon = 1 + s$. The ACTUAL cap has

$$A_F^2 = (21+15s)/2 = 3\epsilon^3/2, \qquad c_F = A_F p - P_{16}/2 - P_{15}/s.$$

The complete head strip in $\mathbb{Z}[s]$ is

$$U = 2+s,\ 3+s,\ 3+2s,\ 4+2s,\ 5+3s,\ 6+3s,\ 6+4s.$$

Indeed $U = a + bs$ has $1 \le b \le 4$; the allowed pairs are $b = 1 : a = 2, 3$; $b = 2 : a = 3, 4$; $b = 3 : a = 5, 6$; $b = 4 : a = 6, 7$, with the last $a = 7$ beyond twelve. For $R_0 = (U-2)/(3\epsilon^3 U)$, the norms are

$$1/9,\ 1/63,\ 7/9,\ 1/18,\ 1/7,\ 1/81,\ 4/9.$$

The four nonsquares are impossible. At $U = 2 + s$, $R_0 = 1/(3\epsilon^4)$; at $U = 6 + 4s$, $R_0 = 2/(3\epsilon^4)$. Neither $1/3$ nor $2/3$ is a square in $\mathbb{Q}(s)$: a rational square in that field requires the rational number itself or half of it to be a rational square. The last row $U = 6 + 3s$ REALLY gives a square, with its positive physical root $k = 1/(3\epsilon)$. For the ACTUAL mutual magnitude $M = -c_H \cdot c_F$,

$$A_H A_F / g_H = \epsilon^2, \qquad M/g_H = (4+3s)/2, \qquad 4M^2 = (174+123s)/2.$$

Its rational norm is $9/2$, so it is not integral. This contradicts the actual two-cycle. The positive time sign is fixed by actual outward roots, not an optional square-root sign. Thus the near own-three model also is excluded.

The far $T = 3 + \sqrt{13}$ model is last. Put $r = \sqrt{13}$, $\eta = (3+r)/2$. Its time polynomial is $F_1(T) = 5\eta^3$. Use the FULL ring $\mathbb{Z}[(1+r)/2]$. The complete head strip gives the SIX rows

U	$\mathrm{N}(U)$	$\mathrm{N}(U-2)$	$\mathrm{N}(R_0)$
$(5+r)/2$	3	-3	$1/25$
$(7+r)/2$	9	-1	$1/225$
$4+r$	3	-9	$3/25$
$5+r$	12	-4	$1/75$
$(11+3r)/2$	1	-17	$17/25$
$(13+3r)/2$	13	-9	$9/325$.

Writing $U = (a+br)/2$, same-parity integers, the strip gives $b = 1, 2, 3$ with respectively $a = 5, 7;\ 8, 10;\ 11, 13$. The last four norm entries are nonsquares. For the first, $R_0 = (101 - 28r)/15$. A full field square would have rational coefficient square $52/15$ or $49/15$, neither rational-square. For the second, multiplying by the square η^2 gives

$$\eta^2 R_0 = (7-r)/90.$$

Its rational coefficient square would be $13/180$ or $1/180$, again impossible. No norm-square is treated as sufficient. All seven actual white profiles are now exhausted except isolation.

H.4 The actual isolated pair and the entire cubic branch

The actual short cap is $c_S = A_S p - g_S P_{16}$. Own zero and parallel are impossible as above. Put $V = 4g_S^2$. Rescale the ACTUAL K-root to raw norm V, old-short product $-V/2$: physically $F_S = \sqrt{V}\, c_S$, spatial $(V/2)P_{16}$, time square $V(4V-1)$. Its original Lorentzian Gram over $\mathbb{Q}(V)$ gives ACTUAL $K = \mathbb{Q}(V)$, and EVERY nonphysical place has $0 < V_\sigma < 1/4$. The short edge bound gives $V < 64$. Rational K is excluded by WHOLE-form Meyer. Finite own would give $V < 4$ and $0 < \mathrm{N}(V) < 4(1/4)^{d-1} \le 1$, impossible. Thus $V > 4$, the short own is dotted. Degree at least four gives $\mathrm{N}(V) < 64(1/4)^3 = 1$. Hence actual degree is two or three.

Normalize $h = c_H/(\sqrt{2}g_H)$, $r_F = c_S/g_S$, so $h^2 = 2/U$, $r_F^2 = 4/V$, spatial parts e_1, P_{16}. The original old sixteen roots plus r_F form a K-basis. The positive coefficient of r_F in h gives

$$k^2 = \frac{(U-2)V}{4U(4V-1)} \in K^2, \qquad k > 0. \tag{B16K}$$

Let $H = Uh$, $F = (V/2)r_F$. The ACTUAL full long-path cycle and two-cycle give

$$\begin{gathered} G = -2H \cdot F = 4Uk(4V-1) - UV \ge 0, \qquad L = \frac{G^2}{2UV}, \qquad G, L \in \mathcal{O}_K, \\ (G+UV)^2 = 4UV(U-2)(4V-1), \\ L + G = ((15U-32)/2)V - 2U + 4. \end{gathered} \tag{B16C}$$

The old-path factors are -1, followed by $-V, -G/V$, whose complete product is G. Orthogonal $G = L = 0$ is retained. The last identity, subtracting seven integral copies of UV, implies $UV/2$ integral. This is rank sixteen's own half-product, not the odd-rank-seventeen census invariant. Thus every actual pair has the COMPLETE domain

$$\begin{gathered} K = \mathbb{Q}(U) = \mathbb{Q}(V), \quad d = 2 \text{ or } 3, \quad U, V, UV/2 \in \mathcal{O}_K, \\ 2 < U < 12, \quad 4 < V < 64, \qquad 0 < U_\sigma < 2, \quad 0 < V_\sigma < 1/4. \end{gathered}$$

Suppose $d = 3$. The positive norm $\mathrm{N}(V)$ is one, two or three. For odd norm, V is a unit at every dyadic prime, so $UV/2$ integral makes $U/2$ integral. For norm two, (V) is a SINGLE prime $\mathfrak{p}$ of norm two, exponent one. Use integer prime valuations and let $e = \mathrm{ord}_\mathfrak{p}(2)$. If $U/2$ were nonintegral, its only deficient prime could be $\mathfrak{p}$, with $\mathrm{ord}_\mathfrak{p}(U) = e - 1$. Then $\mathrm{ord}_\mathfrak{p}(U-2) = e-1$, $\mathrm{ord}_\mathfrak{p}(4V-1) = 0$. The nonzero field square (B16K) would have valuation

$$(e-1) + 1 - 2e - (e-1) = 1 - 2e,$$

ODD, impossible for a square. Thus $W = U/2$ is integral in every case, with physical $1 < W < 6$ and both other conjugates strictly between zero and one.

We prove the small cubic-integer fact needed here. Any totally real cubic integer with physical root $w > 1$ and other roots $a, b \in (0,1)$ has $w > 5$. Assume $w \le 5$. Put integer trace $S = w + a + b \le 6$, positive integer norms $A = wab$, $B = (w-1)(1-a)(1-b)$. Its monic polynomial is

$$P(X) = X^3 - SX^2 + (A + S - B - 1)X - A.$$

It is positive between the two distinct small roots. For $0 < X < 1$, it increases with S and decreases with A, B. If $A \ge 2$, then

$$P(X) \le X^3 - 6X^2 + 6X - 2 < -5(X - 3/5)^2 - 1/5 < 0.$$

If $B \geq 2$, similarly $P(X) < -5(X-2/5)^2 - 1/5 < 0$. Thus $A = B = 1$. For $S \leq 5$, $P(X) \leq X^3 - 5X^2 + 4X - 1 < -(2X-1)^2 \leq 0$. Only $S = 6$ remains, giving $P(X) = X^3 - 6X^2 + 5X - 1$. But $P(5) = -1 = (5-w)(5-a)(5-b)$ forces $w > 5$, a contradiction. Therefore $U > 10$.

The ACTUAL head edge now gives $x_1 > (\sqrt{5}-1)/2 > .618$. The strict conjugate short bounds give $V > 16\,\mathrm{N}(V) \geq 16$, so its ACTUAL short edge gives $S_{16} > 4\sqrt{7}/3 > 3.5276$. Together with unchanged $S_6 > 2.01815$, use ONE positive reference

$$w_0 = .33797P_1 + .129085P_6 + .150945P_{16}.$$

Its coordinates are .618, then five .28003's, then ten .150945's. The primitive square on the SAME original x gives

$$\|x\|^2 > .618^2 + 1.40015^2/5 + 1.50945^2/10 = 1.00185193475 > 1.$$

This excludes EVERY actual cubic field, with all dyadic ramification allowed and no polynomial field census. Actual K is consequently quadratic.

H.5 Exact inclusion in the distinct whole-B_{16} census

Every hypothetical whole B_{16} now satisfies EXACTLY the actual-root strips and cycles of Proposition 10(iv), with the original scales $U = 4g_H^2, V = 4g_S^2$. The square in (B16K) is a FULL K-square, with its PHYSICAL positive root. The exact G, L are (B16C). Acuteness gives $G \geq 0$. For $G = 0$, $L = L' = 0$; otherwise strict nonphysical two-root positivity gives $0 < L' < 4$. The own U and mutual L lie in the actual quadratic finite native alphabet or are > 4. Parallel four has its actual feasible null and is excluded; zero mutual is retained. These are necessary conditions of ACTUAL walls, not conditions assigned to a formal projection.

For completeness, this algorithm has a finite full-order domain justified by the physical strip. For the maximal basis $\omega = (1+\sqrt{D})/2$ or $\sqrt{D}$, $U - U' = b\sqrt{\Delta} < 12$, hence fundamental discriminant $\Delta < 144$. Trial $2 \leq D \leq 143$, exact squarefreeness and this discriminant test exhaust all actual fields. The FULL safe integer boxes are

$$U = a + b\omega: \quad 1 \leq a \leq 7,\ 1 \leq b \leq 5; \qquad V = a + b\omega: \quad 1 \leq a \leq 32,\ 1 \leq b \leq 28.$$

For U, the gap gives $b\sqrt{D} < 12$ in the half basis and $b\sqrt{D} < 6$ otherwise; $0 < U' < 2$ bounds a. For V, replace twelve/six by sixty-four/ thirty-two and use $0 < V' < 1/4$. These give the printed boxes, followed by EXACT strict two-place strips. At most ten head and twenty-eight short models occur per field. Visit EVERY ordered head/short pair. Exact rational-pair arithmetic, both signed-norm square-root branches, verified full square identities, integrality, conjugate and native predicates are the algorithm of Section 3. All loops are finite; there is no unbounded coefficient search or numerical root cutoff. The three disjoint discriminant intervals $[0,48), [48,96), [96,144)$ cover the domain. A separate complete combine recomputes field and pair records; it is a check with the SAME implementation, not an independent algorithm.

At this precisely proved domain we use the recorded finite-output proposition 10(iv) as an EXPLICIT computed premise. Its 43 fields,100 head models,85 short models and273 pairs leave one necessary pair,

$$U = 5 + 2\sqrt{6}, \qquad V = 10 + 4\sqrt{6}.$$

It is not an empty census or a realized polytope. For this pair the ACTUAL head/short edges have

$$C_H = (2+\sqrt{6})/2 > 11/5, \qquad C_S = 2(2+\sqrt{6}) > 44/5.$$

They give $x_1 > .61$, $S_{16} > 3.56$. With unchanged $S_6 > 2.01815$, ONE reference

$$w = .32837P_1 + .127445P_6 + .154185P_{16}$$

has positive coefficients and norm

$$.61^2 + 1.40815^2/5 + 1.54185^2/10 = 1.00640742675 > 1.$$

The original completed square is contradictory. Every actual short profile, all ground fields and the complete nonempty quadratic pair domain have now been covered for whole B_{16}. The finite output remains the declared numerical premise in Proposition 10(iv).

I Whole classical rank seventeen: actual fields and census inputs

The two rank-seventeen classical forests require DISTINCT geometric reductions. Their final quadratic parameter domain happens to agree, but this will be proved from each actual original-root configuration. We first record the common isolated-five lattice interface used in their head reductions.

I.1 The actual golden long-end star and all invariant lattices

We use the following explicitly EXTERNAL cutoff: for the literal golden lattice

$$L = \mathbb{Z}[\varphi]^{n+1}, \qquad f = -\varphi t^2 + x_1^2 + \cdots + x_n^2, \qquad n \geq 8,$$

the full integral reflection arrangement has no compact chamber [3]. Where infinite covolume of its reflection group is used, that stronger published literal cutoff is an external input as identified in the source ledger. The all-lattice step below is INTERNAL; similarity or commensurability alone would not justify it.

Put $u = \varphi - 1$ and use the unit-equivalent form $f = -ut^2 + \sum x_i^2$. A whole D_n long-end-five star has old raw roots $-e_i + e_{i+1}$, $-e_{n-1} - e_n$, and cap $r = e_0 + \varphi e_1$, all square two. Its full Lorentzian Gram forces actual $K = \mathbb{Q}(\varphi)$: any larger field has a nonphysical embedding fixing this indefinite principal. The old group gives all even spatial sign changes and permutations. Conjugating the cap reflection gives maps R_i whose only nonidentity block, on zero and i, is

$$\begin{pmatrix} \varphi & -\varphi \\ 1 & -\varphi \end{pmatrix}.$$

A whole B_n head-five star has the SAME configuration and contains those even signed permutations, so the argument also applies there. Arithmeticity supplies a full invariant $\mathcal{O}_K$-lattice Λ, for example by summing finitely many translates of a lattice stable under an arithmetic finite-index subgroup.

Let $\mathcal{A}$ be the algebra generated by these lattice-preserving maps. For distinct spatial i, j, k, even sign changes d_{ij} give $(1 - d_{ij})(1 - d_{ik}) = 4E_{ii}$. Permutations give all spatial $4E_{ij}$. The displayed R_i and the unit φ then give $4E_{i0}, 4E_{0i}, 4E_{00}$; explicitly

$$\begin{aligned} 4E_{ii}R_i &= 4E_{i0} - 4\varphi E_{ii}, \\ R_i 4E_{ii} &= -4\varphi E_{0i} - 4\varphi E_{ii}, \\ R_i 4E_{i0} &= -4\varphi E_{00} - 4\varphi E_{i0}. \end{aligned}$$

The golden ring is a PID, for example by its Minkowski bound $\sqrt{5}/2 < 2$. Scale the coordinate-content ideal of Λ to the unit ideal. The matrix units imply $4L \subset \Lambda \subset L$. The FOURfold sandwich is essential for odd n.

Modulo two, $\mathcal{O}_K/2 = \mathbb{F}_4$. Put

$$\ell(t,x) = \varphi t + \sum x_i, \qquad H = \ker \ell, \qquad v = (\varphi^2, 1, \ldots, 1).$$

The only invariant subspaces are $0, \langle v\rangle, H, L/2L$. Indeed any unequal spatial coordinates give $e_i + e_j$ by a transposition difference; their orbit spans the spatial augmentation hyperplane, and R_i supplies a time direction, giving H. If every spatial coordinate is equal, R_i-invariance forces the line $\langle v\rangle$. Here $v \in H$ exactly for odd n.

For the two residue flags

$$W_1 = (\Lambda + 2L)/2L, \qquad W_2 = ((\Lambda \cap 2L)/2)/2L, \qquad W_1 \subset W_2,$$

the content normalization makes $W_1 \neq 0$. If W_1 is full, $\Lambda = L$. If $W_1 = H$, differences under even sign changes of a lift of $e_i + e_k$, divided by two, put a coordinate vector in W_2, forcing it full. If $W_1 = \langle v\rangle$, the same differences force $H \subset W_2$; for even n, W_2 is full, while for odd n it is H or full. Thus, writing E for the inverse image of H, the COMPLETE lattice list is

$$L, \quad E, \quad \mathcal{O}_K v + 2L, \quad \mathcal{O}_K(v + 2z) + 2E \quad (n \text{ odd},\ z \in L).$$

This includes every lift, not merely every residue subspace.

Intrinsically,

$$E = \{x \in L : f(x) \in 2\mathcal{O}_K\}, \qquad E^* = L + \mathcal{O}_K(v/2), \qquad f(v) = n - 1 - 2\varphi.$$

The latter norm has dyadic valuation zero for even n, exactly one for odd n. Writing $x = y + av/2$ shows

$$\{x \in E^* : f(x) \in \mathcal{O}_K\} = L :$$

if a is not divisible by two, the final norm term has negative valuation and the other terms are integral. Hence every isometry of E, and by duality of $\mathcal{O}_K v + 2L = 2E^*$, preserves L. For a deep odd lattice put $\lambda = v + 2z$; its norm still has valuation one. Writing $x = a\lambda/2 + e$, $e \in E$, gives

$$\{x \in \Lambda/2 : f(x) \in \mathcal{O}_K\} = E.$$

Its isometries therefore preserve E and then L. Every invariant lattice in the COMPLETE list has $O(f,\Lambda) \subset O(f,L)$. Thus the actual reflection group lies inside the literal integral group. Its arrangement chamber refines the assumed compact chamber, contradicting the external literal no-compact-chamber cutoff (or the stronger infinite-covolume statement). This excludes the actual whole-B/D isolated long-end-five configuration, with every compatible invariant lattice retained.

I.2 Whole B_{17}: the fresh window and actual short wall

Use actual B_{17} roots and $x_1 > \cdots > x_{17} > 0$ from one original farthest vector. The generic parameter $M = 12$ proves the bulk through thirteen; literal far helpers at indices at least six are excluded arithmetically. At fourteen the right B_3 extra short-three extension is the sealed F_4: own zero has its affine principal, every positive own has ratio $\geq 1 + 1/\sqrt{2}$. At fifteen the right B_2 extra own-three/ short-three profile has the opposite-$\sqrt{2}$ Schur cost $15(3 - 2\sqrt{2})/4 + 1/2 > 1$. Its zero/four/six and parallel owns have actual nulls; larger owns and the full B_{17} bridge are directly

strong. The bridge residual is $1/(2i)$, and even own zero has ratio $1/\sqrt{2i}+\sqrt{i}/2>100/61$. All required left corrections use genuine long node sixteen. Thus six through fifteen are strong.

If five is weak with finite own, its ACTUAL finite helper point gives $x_1>.65$. The reference through one, six and every later strong knot gives

$$e_B>E_{15}+.65^2+(2.01815-.65)^2/5-E_6>.92564+.7968-.7>1.$$

For dotted weak five put $t=2g\in(2,12/5)$. Its actual path cycle makes t integral, with all other conjugates in $(-.69,.29)$. Degree at least four contradicts $2.4(.69)^3<1$; degree one has no integer; degree two has trace two and norm $t(2-t)\in(-24/25,0)$. In degree three a positive small conjugate gives norm magnitude <1. Two negative small conjugates force norm one; trace two violates $(t-2)^2\geq 4/t$. Trace one forces $t>9/4$, where the integer second coefficient $1/t+t-t^2$ lies strictly between $-3,-2$. Thus five is strong too.

The ACTUAL retained B_{16} head list is isolation or long-endpoint-three. The nonisolated case gives $x_1>.643$, whose legal first-five graft gains $>.099405$ over $E_{15}>.92564$, impossible. For an isolated head, small own/nulls remove zero, three, four, six and parallel; the golden long-star lemma removes five. For finite own at least seven or dotted, the already proved actual first-three-coordinate point gives $x\cdot Z\geq H$ and $F=.8H-.16R>.2893$. Paying .08 at the INDIVIDUAL P_5 coefficient gives gain $>.0325$.

Independently, a legal first-five graft at $x_1\geq .605$ would give $E_{15}+(5/4)(.605-.361)^2>1.00006$. Hence $x_1<.605$. For the actual head parameter $U=4g_H^2$,

$$K=\mathbb{Q}(U),\quad U\in\mathcal{O}_K,\quad 2<U<28/3,\quad 0<U_\sigma<2.$$

The raw head has square $2U$, spatial Ue_1, time square $U(U-2)$. This original full-rank Lorentzian principal proves ACTUAL field equality. FINITE HEADS ARE RETAINED until excluded below.

We first derive a preliminary short bound without using that head. If actual $C_S\geq 15$, then $S_{17}>3.856$. For strong five, the positive reference at $(5,1.804),(17,3.856)$ has norm 1.0017752. For weak five, use $(1,.59),(6,2),(17,3.856)$; its norm is $.74572+3.444736/11>1.05$. Therefore $C_S<15$. The actual retained A_{16} short-white list is exactly isolation, helper three at $j=1,2,15,16$, or helper four at $j=1,16$.

For helper three put $a=2\sqrt{2}\gamma$, raw cap square two and helper product -1. Actual cycles make $a,a^2/2$ integral. Its complete formulas are

$$A_j^2=(17a^2+4ja+4j-8)/4,\qquad C_S=\frac{17a+2j}{2\sqrt{34-j(17-j)}}.$$

Here A_j is the RAW cap time coefficient. For helper four use raw cap square one, time square $17\gamma^2+2j\gamma+j-1$, $C_S=17\gamma+j$. The original connected Lorentzian principal gives ACTUAL $K=\mathbb{Q}(a)$ or $\mathbb{Q}(\gamma)$.

Own zero at helper-three one/two has negative/ zero time; near fifteen/sixteen has the actual positive null

$$c+n_{14}+2n_{15}+2n_{16}+\sqrt{2}n_{17},\qquad c+n_{15}+2n_{16}+\sqrt{2}n_{17},$$

respectively. Helper-four own zero at one has zero time; near sixteen already violates the length bound. Parallel always has the actual future feasible null pair. Every null is feasible against ALL original walls termwise.

At helper-three two, $a<56/17$, every other $-8/17<a_\sigma<0$. Its norm is a unit, incompatible with $a^2/2$ integral. Near next-end fifteen leaves only own three $a=\sqrt{2}$, with inadmissible conjugate $(86-60\sqrt{2})/4>0$. Far helper-four has $\gamma<14/17$, leaving only owns three/four/five. The rational

case is impossible; the other time conjugates are $17/2 - \sqrt{2} > 0$, $(38 - 21\varphi)/4 > 0$. Thus all four non-end roles are excluded.

For far-end-three put $v = a$; for near-end-three put $v = a + 2$. Their nonphysical strips have absolute bound .617, with smaller one-sided bound .382; physical $v < 7.7$, and $v^2/2$ is integral. Degree at least three would give nonzero integer norm of magnitude $< 2.932 < 3$, but its half-square norm requires divisibility incompatible with norms one or two. Hence actual degree is two. Even trace/norm and the exact negative-time intervals give the COMPLETE tables

role	$(\mathrm{Tr}\, v, \mathrm{N}(v))$	a
far	$(4,-2),(6,2),(6,-2),(6,-4)$	$2+\sqrt{6},\ 3+\sqrt{7},\ 3+\sqrt{11},\ 3+\sqrt{13}$
near	$(4,2),(6,2),(6,-2),(8,4)$	$\sqrt{2},\ 1+\sqrt{7},\ 1+\sqrt{11},\ 2+2\sqrt{3}.$

For coverage: far traces $2, 4, 6$ and norms $-4, -2, 2$ exhaust the quadratic formula; near traces $4, 6, 8$ and norms $-2, 2, 4$ do likewise. All other rows have wrong physical/conjugate signs or are rational/nonreal. The exact time signs use

$$174 - 72\sqrt{6},\ 382 - 106\sqrt{13},\ 348 - 106\sqrt{11},\ 280 - 106\sqrt{7} < 0$$

for far roots, and

$$90 - 64\sqrt{2},\ 324 - 98\sqrt{11},\ 256 - 98\sqrt{7},\ 456 - 264\sqrt{3} < 0$$

for near v's. Thus the full exact intervals, not just safe decimal overstrips, are imposed.

An isolated short cap has invariant $V = 4\gamma_S^2$, raw square V, spatial $(V/2)P_{17}$, time square $V(17V - 4)/4$. Its original Lorentzian principal gives $K = \mathbb{Q}(V)$, $0 < V_\sigma < 4/17$. Finite own would have nonzero integral norm $< 16/17$; parallel is its null pair. Thus it is dotted. The preliminary $C_S < 15$ gives $V < 900/17$; degree at least four gives norm $< 57600/83521 < 1$. So the ACTUAL field, whether the short cap is isolated or one of the helper models, has degree two or three. The rational field was separately excluded by the WHOLE rank-eighteen Meyer obstruction.

I.3 Whole B_{17}: exact head/short compatibility

Before excluding any finite head, its actual degree bound and $0 < U_\sigma < 2$ give the COMPLETE remaining finite list

native	U
7	$U^3 - 5U^2 + 6U - 1 = 0$
8	$2 + \sqrt{2}$
10	$(5 + \sqrt{5})/2$
12	$2 + \sqrt{3}$
18	$U^3 - 6U^2 + 9U - 3 = 0.$

Indeed $\phi(m) \le 6$ gives $3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 18$; the earlier null/golden exclusions remove three/four/five/six. Nine and fourteen have a nonphysical $U > 2$. In particular finite native eight in $\mathbb{Q}(\sqrt{2})$ MUST still be retained.

The individual P_5, P_{15} prices exceed .14, .13. For any actual short pressure $S_{17} > T$, ONE vector

$$w_{15} + .4(Z - P_5/5) + .13(P_{17} - P_{15})$$

spends .08, .13 before using either floor. The head residual is zero-sum on the first five; the terminal residual is disjoint with square two. All FULL crosses with the reference and each other vanish. Thus

$$\|x\|^2 > E_{15} + .0325 + .26(T - f_{15}) - .0169(2).$$

If $C_S \geq 11$, the actual short edge gives $S_{17} \geq \sqrt{85/6} > 3.763$. With $E_{15} > .92564, f_{15} < 3.442$, the right side is > 1.0078. Therefore ACTUAL $C_S < 11$.

Of the eight short-helper models, the actual coshes $(17a+2)/(6\sqrt{2})$ and $(17a+32)/(6\sqrt{2})$ leave only far $a = 2+\sqrt{6}$ or near $a = \sqrt{2}$. For both remaining rows the raw cap time norm has prime twenty-three to odd valuation:

role	H_c^2	$-16\mathrm{N}(H_c^2)$
far	$(87+36\sqrt{6})/2$	828
near	$(45+32\sqrt{2})/2$	92.

The ACTUAL field equals that of a. The COMPLETE compatible dotted head strips $4 < U < 11, 0 < U' < 2$, with their $h_0 = -\mathrm{N}(U(U-2))$, are

K	U	h_0
$\mathbb{Q}(\sqrt{6})$	$3+\sqrt{6},\ 4+\sqrt{6},\ 5+2\sqrt{6},\ 6+2\sqrt{6}$	$15, 20, 15, 96$
$\mathbb{Q}(\sqrt{2})$	$3+\sqrt{2},\ 3+2\sqrt{2},\ 4+2\sqrt{2},\ 5+3\sqrt{2},\ 6+3\sqrt{2}$	$7, 7, 32, 63, 36$
$\mathbb{Q}(\sqrt{2})$ finite	$2+\sqrt{2}$	4.

The full rings are the displayed square-root orders. Writing $U = A+B\sqrt{d}$ gives $B < 11/(2\sqrt{d})$, and $A = \lfloor B\sqrt{d} \rfloor + 1, +2$; the physical strip gives exactly the rows. There is no finite head in $\mathbb{Q}(\sqrt{6})$; the finite $\sqrt{2}$ head is explicitly included. Both raw roots occupy the SAME actual K-time line, so the rational time-norm ratio $p/(16h_0)$ must be a square. Its twenty-three valuation is odd in EVERY row. No nonisolated short cap survives. The actual short cap is now isolated dotted:

$$K = \mathbb{Q}(V), \quad 4 < V < 484/17, \quad 0 < V_\sigma < 4/17.$$

Derive the B17 cycles in THEIR OWN actual scales: head raw square $2U$, time square $U(U-2)$; short raw square V, time square $V(17V-4)/4$. The actual long path makes $a = 2(r_H, r_S)$ integral and its actual two-cycle makes $L = a^2/(2UV)$ integral. The original common time line gives

$$\begin{gathered} R_0 = U(U-2)V(17V-4) \in K^2, \qquad a = UV - \sqrt{R_0}, \quad \sqrt{R_0} > 0 \text{ physically}, \\ L - a = 8UV - 17V - 2U + 4 =: d_0, \qquad a^2 = 2UV(a+d_0). \end{gathered} \tag{B17C}$$

No D17 fork normalization was used.

First exclude ALL finite heads. In degree three, $0 < \mathrm{N}(V) < (484/17)(4/17)^2 < 2$, so V is a unit. At every dyadic prime, $d_0 \equiv -V$ is a unit; the cycle equation forces a into the prime and then

$$2\,\mathrm{ord}(a) = \mathrm{ord}(2) + \mathrm{ord}(U).$$

For finite cubic heads seven/eighteen, the actual field has odd discriminant forty-nine/eighty-one, and $\mathrm{N}(U) = 1, 3$. Thus two is unramified and U a dyadic unit; the equality reads an even number equals one. Both finite cubic heads are excluded without assuming $U > 4$.

For finite quadratic heads eight/ten/twelve, the FULL short strips have exactly these seven models and necessary rational time-norm ratios:

head	V	$\mathrm{N}(H_S^2/H_H^2)$
8	$3+2\sqrt{2}$	103/64
8	$10+7\sqrt{2}$	383/16
12	$7+4\sqrt{3}$	647/48
12	$14+8\sqrt{3}$	61
10	$2+3\varphi$	171/80
10	$5+8\varphi$	919/80
10	$7+11\varphi$	239/16.

Coverage follows from the full-order coefficient ranges $B = 1, \ldots, 10$ in $\sqrt{2}$, $1, \ldots, 8$ in $\sqrt{3}$, $1, \ldots, 12$ in φ. The corresponding complete ceil arrays for A are

$$2, 3, 5, 6, 8, 9, 10, 12, 13, 15;$$
$$2, 4, 6, 7, 9, 11, 13, 14;$$
$$1, 2, 2, 3, 4, 4, 5, 5, 6, 7, 7, 8.$$

Exact $0 < V' < 4/17$ retains only $B = 2, 7;\ 4, 8;\ 3, 8, 11$, respectively. Rational brackets for the square roots and φ decide every ceil and endpoint. If $n = \mathrm{N}(V)$, $T = \mathrm{Tr}(V)$, and $h_0 = -\mathrm{N}(U(U-2))$, each ratio is

$$\frac{n(68T - 289n - 16)}{16h_0}.$$

None in the COMPLETE table is a rational square: the reduced denominator has odd three/five valuation where indicated, and the remaining integer numerators $103, 383, 61, 239$ are nonsquares. Thus every finite quadratic head is excluded. Only NOW conclude $4 < U < 28/3$.

In degree three the unit V and cycle valuation remain valid. If two is unramified they force $U/2$ integral. Its physical value is in $(2, 14/3)$, both other conjugates in $(0, 1)$, contradicting the complete cubic-integer lemma of Section H.4, which requires physical value > 5. If two ramifies, V's monic unit polynomial $X^3 - AX^2 + BX - 1$ has a repeated factor modulo two, forcing A, B ODD. Its two small positive roots in $(0, 4/17)$ and physical $V < 484/17$ give

$$A = 19, 21, 23, 25, 27, \qquad B^2 > 4(A - 8/17), \qquad B < \frac{272A + 4849}{1156}.$$

The least and largest possible odd B are, respectively,

$$(9, 7),\ (11, 9),\ (11, 9),\ (11, 9),\ (11, 9).$$

Every row is empty. This excludes the ENTIRE cubic branch in the actual B17 domain without a field census. The actual field is quadratic.

I.4 Whole B_{17}'s actual inclusion in the shared quadratic domain

Rename ONLY the algebraic variable $W = V$, $g = a$ in (B17C). We have proved from the ACTUAL B17 geometry

$$K = \mathbb{Q}(U) = \mathbb{Q}(W), \quad U, W \in \mathcal{O}_K,$$
$$4 < U < 11, \quad 0 < U' < 2, \qquad 2 < W < 676/17, \quad 0 < W' < 4/17,$$
$$g = UW - \sqrt{U(U-2)W(17W-4)}, \quad g, L = g^2/(2UW) \in \mathcal{O}_K.$$

The physical g is strictly negative: $d_0 > (8U - 17)2 - 2U + 4 = 14U - 30 > 0$, so $R_0 - (UW)^2 = 2UWd_0 > 0$. Thus $L > 0$. At the other embedding actual positive definiteness gives $0 < L' < 4$. The physical L is a true finite native invariant or > 4; parallel four has its actual null pair. Every predicate of Proposition 10(iii) is now satisfied. This domain inclusion uses B17's short raw square V, not D17's fork square $4W$. The common algebraic formula is a proved coincidence of necessary invariants.

The complete full-order algorithm and its explicit output premise are already stated in Section 3. Its discriminant bound $\Delta < 121$, two conjugate-strip candidates for each U coefficient, one for each W, and $b_U \le 4, b_W \le 17$ cover every field and ordered pair and make every loop finite. Both signed-norm square branches are tested; a rational-square norm alone is not sufficient. The old 36-field/120-pair output leaves TWO necessary pairs, in $\sqrt{13}$ and $\sqrt{17}$. The separate old combine authenticates

coverage/ counters and survivors; it does NOT independently recompute all118 rejected square tests. Those remain part of the explicit source-bound numerical-output premise.

For the two actual surviving heads, $u = 4/U$ lies in $(.92, .94)$ and $(.719, .72)$. Their actual dotted head points give, respectively,

$$H > .535,\ R < .51; \qquad H > .482,\ R < .54.$$

Positive rational squaring proves the radical bounds $\sqrt{2-u} > 1.0295, 1.1313$ and $\sqrt{u(1-u)} < .272, .4495$, respectively. Paying .36 at the head costs ONLY .072 from P_5, and gives gains $> .085$, $> .043$. Both actual W's exceed 104/5; the B17 ACTUAL short edge has $C_S^2 = 17W/4 > 442/5$, hence $S_{17} > \sqrt{357/26} > 3.7$.

Use ONE reference through fourteen:

$$w_{14} + .36(Z - P_5/5) + .13(P_{17} - P_{14}).$$

The individual prices $.072, .13$ are legal. The two residuals are disjoint, orthogonal to the reference and each other; terminal norm is three. Thus the SAME original vector would satisfy

$$\|x\|^2 > .9084 + .043 + .26(3.7 - 3.311) - .0169(3) = 1.00184 > 1.$$

Both complete census survivors are excluded. This closes whole B_{17}, conditional only on the explicitly identified external facts and the recorded complete quadratic output premise.

I.5 Whole D_{17}: fresh geometry before any field census

Use ACTUAL unit D_{17} roots, long spine one through fifteen, forks sixteen/seventeen, and

$$x_1 > \cdots > x_{16} > |x_{17}|, \qquad \|x\| < 1.$$

The signed coordinate is retained. The legal generic $M = 11$ window gives strong six through twelve. At thirteen/fourteen the COMPLETE right D4/D3 lists leave only the pure own-five/fork-three or own-five/end-three packets. Here completeness uses the SAME actual finite role list printed in Section G, with the FRESH indices thirteen/fourteen. The D4 long-three and D3 middle-three roles have residual 1/2, shift 1/2; fork/end-three has residual 1/2 or 5/8, shift 1/4. D3 end-four has residual 1/4, shift $1/(2\sqrt{2})$; sealed end-five has residual $(5-3\varphi)/8$, shift $\varphi/4$. The D_{17} bridge residual is $1/(2i)$; the A_{17} end-three bridge at fourteen has residual $1/8 + 1/28 = 9/56$. Their positive-own numerators make them strong, and their own-zero cycles are affine. For single-fork/end-three, own zero, three, four and six retain the actual affine E_8, cycle, F_4 and $6, 3$ principals. The own-three cycle has FIVE actual nodes at D4 and FOUR at D3 end-three; it is not replaced by a triangle. The far-helper arithmetic lemma removes the other nonisolated possibility. The original LEFT isolated-core/H4/A2/singleton proof applies with $2i \geq 26$, so its previously proved $2i \geq 18$ loss margins suffice. The remaining fork-helper raw Schur constant is ACTUAL seventeen; the complete norm/trace argument leaves only its golden own. There is no spectator or extra long root here. Their actual LEFT H_4 points have divided displacement V, flat one through $K = i - 3$, $V^2 < i$, and pressure

$$x \cdot V > G > \sqrt{i - \kappa/(\varphi + 5/8)} - .091, \quad \kappa = 1/2,\ 5/8.$$

Every point uses positive actual walls, positive future time and all original-wall signs. The relevant rows are

i	K	μ	$G >$	$f_K <$
13	10	.16	3.48	2.732
14	11	.15	3.61	2.886.

The radical seals are $13 - 25/112 > 3.571^2$, $14 - 125/448 > 3.701^2$. The INDIVIDUAL terminal prices exceed $.16, .15$, in both initial-five references. The residual $V - P_K$ vanishes on the whole reference support and has norm < 3.

If five is weak, its actual endpoint reference has energies $L_{10} > .8870, L_{11} > .9100$. The two paid squares are

$$.8870 + .32(3.48 - 2.732) - .0256(3) > 1.0495, \qquad .9100 + .30(3.61 - 2.886) - .0225(3) > 1.0597.$$

If five is strong, first classify the ACTUAL head: the retained D_{16} admits only isolation or simple three at its long endpoint two. The nonisolated row gives $x_1 > .643$, hence legal first-five gain $> .0994$. An isolated head has small owns/nulls excluded as before, including five by Section I.1. For finite own at least seven, its actual finite head point gives gain $> .0135$ at cost .1 from P_5. Explicitly for $1 < u < 1.232$,

$$\eta = H - .361 - (R - .2)/4 > .0135.$$

On $u \in [1, 1.1], [1.1, 1.2], [1.2, 1.232]$, the respective rational seals are

$$H > .6708 - .1871, \quad .63245 - .204125, \quad .6196 - .2, \qquad R \le .5, .415, .36.$$

The middle second-radical square has maximum $1/24$; all bounds are positive squarings. They give gains $> .0477, .013575, .0186$.

For a dotted head use its ACTUAL positive core, $Z = (1 - u/2)e_1 + (u/2)e_2$, $0 < u < 1$. Paying .36 at P_5, cost .072, gives gain

$$.72(H - b) - .1296(R - 1/5) > .04. \tag{D17H}$$

Here is the full continuous seal rather than a borrowed actual-I theorem. On $[0, 1/2]$, $[1/2, 3/4]$ and $[3/4, 1]$, use lower chords for $\sqrt{2 - u}$

$$1.4142 - .379u, \quad 1.4381 - .4268u, \quad 1.4721 - .4721u$$

and upper chords for R

$$1 - .75u, \quad .8125 - .375u, \quad .625 - .125u.$$

With $.5091 < 9\sqrt{2}/25 < .5092$, the three gain bounds reduce to

$$\begin{aligned} &.35636922 - .0958u - .5092\sqrt{u(1-u)}, \\ &.39283671 - .16873u - .5092\sqrt{u(1-u)}, \\ &.43444611 - .2242u - .5092\sqrt{u(1-u)}. \end{aligned}$$

The circle inequality and squared-root upper bounds $.5182, .5365, .5564$ give minima $> .04936922, .04022171, .04414611$. No I wall or energy enters.

Every head residual is zero-sum on the first five and orthogonal to the H4 residual beyond $K = 10/11$. The bare strong-five H4 bounds are $.98692, .99786$. Adding even the smallest LEGALLY funded head gain $> .0135$ excludes both. All six through fourteen are consequently strong BEFORE deciding five or the final head.

Weak five with finite own gives actual $x_1 > .65$; its rebuilt six-through-fourteen reference exceeds one. Its dotted case is precisely the complete minimal-degree norm/trace argument already proved for the SAME inverse entries in Section I.2. Thus five is strong. A nonisolated head now gives

$E_{14} + .0994 > 1$, so it is isolated. The legal graft at $x_1 \geq .632$ gives $E_{14} + (5/4)(.632 - .361)^2 > 1.00020125$. Therefore $x_1 < .632$. The original head raw normalization gives

$$r_H^2 = 2U, \quad \text{spatial } r_H = Ue_1, \quad H_0^2 = U(U-2), \quad K = \mathbb{Q}(U), \quad 2 < U < 11, \quad 0 < U_\sigma < 2.$$

Finite heads remain possible at this stage.

For completeness the conjugate interval itself bounds their COMPLETE native alphabet. For $U = 2 + 2\cos(2\pi/m)$, a coprime exponent $1 < a < m/4$ gives a forbidden conjugate > 2. For odd $m \geq 9$ use two. For even $m \geq 14$ not divisible by three use three. For $m = 6k$, $k = 4, 5$, use five/seven; for larger even k use $k-1$, or $k-3$ if $k \equiv 1 \pmod 3$; for larger odd k use $k-2$, or $k-4$ if $k \equiv 2 \pmod 3$. Each is coprime and below $3k/2$. The remaining labels are $3, 4, 5, 6, 7, 8, 10, 12, 18$. The prior null/golden exclusions leave exactly seven/eight/ten/twelve/eighteen, with the parameters already printed for B_{17}.

For either actual signed fork direction $V_f = P_{16} \pm e_{17}$, if $C_f \geq 13$, the ORIGINAL edge gives $x \cdot V_f \geq \sqrt{102/7} > 3.817$. Use the SAME head gain and a separately funded $.13(V_f - P_{14})$, whose residual norm is three and cross with the head/reference is zero. Then

$$\|x\|^2 > .9084 + .0135 + .26(3.817 - 3.311) - .0169(3) = 1.00276 > 1.$$

Thus BOTH actual fork coshes are < 13, without equating their parameters.

I.6 The complete actual D_{17} fork table

Fix either genuine fork f. Deleting it retains ACTUAL A_{16}. Finite classification gives isolation, helper three at $j = 1, 2, 15, 16$, or helper four at $j = 1, 16$. For three put $t = 2\gamma$, raw cap square two, white product -1, own product $-t$. For four put $a = 2\sqrt{2}\gamma$, raw square one, white product -1, own product $-a/2$. Actual cycles make the parameter integral, and for four also $a^2/2$ integral. The original Lorentzian principal gives ACTUAL field equality with that parameter. The COMPLETE raw time and actual edge table is

helper	H_c^2	C_f
3@1	$(17t^2 + 4t - 4)/4$	$(17t + 2)/(6\sqrt{2})$
3@2	$(17t^2 + 8t)/4$	$(17t + 4)/4$
3@15	$(17t^2 + 60t + 52)/4$	$(17t + 30)/4$
3@16	$(17t^2 + 30t + 9)/4$	$(17t + 15)/(6\sqrt{2})$
4@1	$(17a^2 + 8a)/16$	$(17a + 4)/4$
4@16	$(17a^2 + 60a + 52)/16$	$(17a + 30)/4$.

The near endpoint THREE formula differs from the B17 short formula and is not substituted for it. Every other conjugate has negative displayed time square. Every null below uses UNIT ACTUAL cap normals; its positive time and all original-wall signs follow termwise.

The 3@15 own-zero kernel is $n_{14} + 2n_{15} + n_{16} + n_{17} + c$; own three has its triangle. All other owns have $t \geq \sqrt{2}$, $C_f > 13$. The 4@16 own-zero kernel is $n_{14} + 2n_{15} + 2n_{16} + n_{17} + \sqrt{2}c$; every positive $a \geq \sqrt{2}$ gives $C_f > 13$. At 3@2, own zero has zero time; positive owns satisfy $1 \leq t < 48/17$, $-8/17 < t_\sigma < 0$. Degree at least three has nonzero integer norm magnitude < 1. Degree two leaves norm -1, trace one or two; the golden root violates the conjugate strip, leaving $1 + \sqrt{2}$. Its raw time norm is $-47/16$. For a finite head the only compatible head is eight, of time norm -4, and the ratio $47/64$ is nonsquare. For a DOTTED head, the complete actual $\sqrt{2}$ head strip gives time norms $-7, -7, -32, -63, -36$. Prime forty-seven again has odd ratio valuation. Thus this row is excluded under EVERY head. At 4@1 the same norm strips leave only $a = 1 + \sqrt{2}$, but $a^2/2 = 3/2 + \sqrt{2}$ is

not integral. Its own zero has zero time. Every four non-end role is exhausted, with finite, parallel and dotted own retained; parallel everywhere has $c+n_f$. Far-end own zero has negative time, while near-end own zero gives the forbidden rational actual field.

For the two remaining end-three roles put $v = t$ far, $v = t+1$ near. Their raw times are

$$H_c^2 = (17v^2 \pm 4v - 4)/4,$$

with upper sign far. The COMPLETE strips are

role	physical v	every other v_σ
far	$1 \le v < 6.372$	$-.617 < v_\sigma < .382$
near	$2 \le v < 6.607$	$-.382 < v_\sigma < .617.$

The exact nonphysical endpoints are $(\mp 2 \pm 6\sqrt{2})/17$, and the physical upper endpoints are $(78\sqrt{2} \mp 2)/17$. These exact signs are retained whenever a safe decimal overstrip admits an extra row.

Let g_0 be the ACTUAL raw head/helper product. Its actual triangle or complete old-chain cycle makes it integral. The common original time-line identities are

$$4g_0^2 - 4g_0U(t+2) + U^2(t+2)^2 = U(U-2)(17t^2+4t-4),$$
$$4g_0^2 - 4g_0U(t+1) + U^2(t+1)^2 = U(U-2)(17t^2+30t+9).$$

Reducing modulo $4\mathcal{O}_K$ gives the necessary parity statement

$$Uv^2 \in 2\mathcal{O}_K. \tag{D17P}$$

It does NOT justify $Uv/2$ integral at ramified dyadic primes. The physical and conjugate bounds belong to actual $K = \mathbb{Q}(v)$, not merely an unexamined smaller parameter field.

I.7 Whole D_{17}: every finite head is excluded

Assume one of the FIVE finite heads retained in Section I.5. Its actual field is the corresponding quadratic or cubic field, so the complete fork table applies with that field equality.

For heads seven/ten/eighteen, U is a dyadic unit and two is unramified, by discriminants forty-nine/five/eighty-one. Parity (D17P) makes $v/2$ integral. In the far role its nonzero norm has magnitude $< 3.186(.3085)^{d-1} < 1$ for $d = 2, 3$. For the near role the cubic norm is $< 3.304(.309)^2 < 1$. In the remaining quadratic golden near case, $v/2 = a + b\varphi$ has physical value < 3.304, other value in $(-.191, .309)$. The gap gives $b \le 1$, $b > 0$. For $b = 1$ the interval requires $.427 < a < .927$, no integer. Both end-helper roles are impossible.

For head eight use $K = \mathbb{Q}(s)$, $s = \sqrt{2}$, $U = 2 + s$. At its dyadic prime $\mathrm{ord}(U) = 1$, $\mathrm{ord}(2) = 2$. Parity requires $s \mid v$. The COMPLETE far strip has $t = 1 + s, 3 + 2s$, neither divisible by s. The complete near strip has $v = 2 + s, 3 + 2s$; only $2 + s$ survives, giving $t = 1 + s$. Its raw time square has norm $-23/4$, with nonsquare ratio $23/16$ to the actual head time norm -4. For head twelve, $U = 2 + \sqrt{3}$ is a dyadic unit although two ramifies. Parity requires v in its dyadic prime, where $\sqrt{3} = 1$ in the residue field. The COMPLETE far/near strips leave only $t = 2 + \sqrt{3}$, respectively $v = 2 + \sqrt{3}$; both have nonzero residue one. Thus BOTH actual forks are isolated under EVERY finite head. The assertion is presently conditional on a finite head, not carried into the dotted case.

For an isolated positive fork put $V = 4\gamma_f^2$. Its ACTUAL raw root has square $2V$, sole old product $-V$, time square

$$H_f^2 = V(17V - 8)/4, \quad K = \mathbb{Q}(V), \quad 0 < V_\sigma < 8/17.$$

Own zero has no spatial wall part. Own three has the actual affine E_8 arms one/two/five; own four has positive affine vector $2n_{15} + 2n_f + n_{14} + n_k + \sqrt{2}c$, where k is the other fork. Own six has its $6, 3$ path; parallel has the actual null pair. All are future feasible against EVERY original wall. For finite labels at least seven, degree at least three contradicts $\mathrm{N}(V) < 4(1/2)^2 \le 1$. In degree two the strict strip forces norm one, trace four, namely native twelve. Native five is the other finite candidate. The cubic/silver head fields cannot equal those of these candidates. At golden head ten the fork-five time norm is $-55/16$, giving nonsquare ratio $11/16$ to head norm -5. At head twelve the fork-twelve time norm is $-191/16$, ratio $191/48$ to head norm -3, also nonsquare. Both forks are DOTTED under every finite head.

The actual head/fork LONG cycle gives $g = (r_H, r_f)$ integral; the actual two-cycle gives $L = g^2/(UV)$ integral. Their spatial product is $UV/2$. The SAME original time line gives

$$(g - UV/2)^2 = U(U-2)V(17V-8)/4, \qquad L - g = 4UV - 17V/2 - 2U + 4.$$

Hence $W = V/2$ is integral, by subtracting eight copies of V from the integral $17V/2$. This uses BOTH cycles without a dyadic-unit assumption. The actual fork bound gives

$$2 < W < 676/17, \quad 0 < W_\sigma < 4/17, \quad H_f^2 = W(17W-4). \tag{D17W}$$

For the three quadratic finite head fields, the COMPLETE full-order strips and time-norm ratios are

head	W	$\mathrm{N}(H_f^2/H_0^2)$
8	$3+2\sqrt{2},\ 10+7\sqrt{2},\ 17+12\sqrt{2}$	$103/4,\ 383,\ 2007/4$
10	$2+3\varphi,\ 5+8\varphi,\ 7+11\varphi,\ 10+16\varphi$	$171/5,\ 919/5,\ 239,\ 5104/5$
12	$7+4\sqrt{3},\ 14+8\sqrt{3}$	$647/3,\ 976.$

For coverage, the positive coefficient bounds are fourteen/seventeen/eleven, and for each coefficient the small-conjugate width $4/17 < 1$ permits at most one integer constant term. The physical bound removes the otherwise small-conjugate $20 + 14\sqrt{2}$. For $n = \mathrm{N}(W), T = \mathrm{Tr}(W)$, $\mathrm{N}(17W-4) = 289n - 68T + 16$ gives every printed ratio. Each is a nonsquare by an odd denominator three/five valuation or the nonsquare integer numerator; $976 = 16(61)$. The original common K-time line therefore excludes every row.

For finite heads seven and eighteen the FULL rings are $\mathbb{Z}[U]$. Their polynomials have discriminants $49, 81$; the former translated by four is seven-Eisenstein, the latter is three-Eisenstein. The distinct-valuation uniformizer argument already proved for the cubic bases in Section D.4 excludes the only possible index prime. Both are irreducible modulo two, so two is inert. The positive norm of W is $< 10816/4913 < 3$. An even norm would be divisible by eight; hence $\mathrm{N}(W) = 1$, $W > 289/16$.

Let α, β, χ be physical and two small conjugates of the actual finite-head generator. For head seven the exact root and derivative bounds are

$$\begin{gathered} 3.246 < \alpha < 3.248, \quad 1.554 < \beta < 1.556, \quad .198 < \chi < .199, \\ 5.149 < f'(\alpha) < 5.167, \quad |f'(\beta)| > 2.289, \quad f'(\chi) > 4.128, \\ 1.752 < \beta + \chi < 1.754, \qquad \beta - \chi > 1.355. \end{gathered}$$

They follow by polynomial signs and products of the disjoint root gaps. Writing $W = A + BU + CU^2$ in the FULL ring, Lagrange interpolation gives

$$C = \sum \frac{W_i}{f'(U_i)}, \qquad B + C(\beta + \chi) = \frac{W_\beta - W_\chi}{\beta - \chi}. \tag{D17L}$$

The complete strip implies $3 < C < 8$ and $|B + C(\beta + \chi)| < .174$. At $C = 4, 5, 6, 7$, only $C = 4, B = -7$ survives. Then $7\chi - 4\chi^2 \in (1.229, 1.235)$, so the small strip requires $1.229 < A < 1.471$, no integer. For head eighteen the corresponding bounds are

$$3.879 < \alpha < 3.880, \quad 1.652 < \beta < 1.653, \quad .467 < \chi < .469,$$
$$7.590 < f'(\alpha) < 7.605, \quad |f'(\beta)| > 2.63, \quad f'(\chi) > 4, \quad 2.120 < \beta + \chi < 2.121, \quad \beta - \chi > 1.183.$$

Interpolation gives $2 < C < 6$, $|B + C(\beta + \chi)| < .199$. For $C = 3, 4, 5$, the products lie in $(6.360, 6.363), (8.480, 8.484), (10.600, 10.605)$; none is within .199 of an integer. No coefficient B exists. Both cubic finite heads are excluded.

All FIVE finite heads have now been eliminated at their ACTUAL fields. Only NOW conclude that the head is DOTTED, with $4 < U < 11$, $0 < U_\sigma < 2$. Fork isolation and the dotted fork assumptions proved conditionally in this subsection are NOT exported to that remaining head. We next prove them anew in the dotted case.

I.8 Whole D_{17}: degree reduction under a dotted head

The complete fork table remains valid. For an end helper, the integral v has norm magnitude $< 6.607(.617)^{d-1}$. Degree at least five gives norm < 1. At degree four it is a unit; parity (D17P) then makes $Y = U/2$ integral. With physical $2 < Y < 11/2$ and three other conjugates in $(0, 1)$,

$$0 < |\mathrm{N}(Y(1 - Y))| < 99/256 < 1,$$

impossible. For an isolated fork the actual LONG+TWO cycles above remain valid under a dotted head BEFORE assigning its own type. They again give integral $W = V/2$, with $0 < W < 676/17$ and $0 < W_\sigma < 4/17$ at every other embedding. The lower bound $2 < W$ is not used before the finite-own exclusion below. Degree at least four contradicts $(676/17)(4/17)^3 < 1$. Thus actual degree is two or three. Rational degree was excluded by the named WHOLE-form Meyer input.

We first remove cubic nonisolated forks. The end-helper norm is $\pm 1, \pm 2$. Opposite signs of the two small roots give magnitude $< 6.607(.617)(.382) < 2$. Two positive small roots in the far case, or two negative in the near case, give magnitude < 1. Norm magnitude two can therefore occur only as norm $+2$ in the following two patterns. For far roots $t, -a, -b$, $tab = 2$, $0 < a, b < .617$, the trace is five, $6 < t < 6.234$, and the integer second coefficient is -6 or -7. The polynomials $X^3 - 5X^2 - 6X - 2$, $X^3 - 5X^2 - 7X - 2$ have discriminants $-424, 229$. The first is not totally real. At $-.617$, the second has value $.180669887 > 0$, inconsistent with both negative roots above that lower endpoint. For near roots v, a, b with $vab = 2$, the trace is seven, $5.766 < v < 6$, and the second coefficient is seven. Its polynomial $X^3 - 7X^2 + 7X - 2$ has discriminant -59, impossible. Thus a remaining cubic end-helper is a unit, so $U/2$ integral.

Here is the full EVEN-cubic-head conclusion, also reused below. For integral $Y = U/2$, physical $2 < Y < 11/2$ and other roots in $(0, 1)$ give

$$0 < |\mathrm{N}(Y(1 - Y))| < 99/64 < 2.$$

Thus $\mathrm{N}(Y) = 1, \mathrm{N}(1 - Y) = -1$. Its polynomial is $X^3 - TX^2 + (T - 1)X - 1$, $T = 3, \ldots, 7$. The COMPLETE discriminants are $-23, -31, -23, 49, 257$. The first three are not totally real; at $T = 7$, evaluation at six is -1, forcing physical root > 6. Only $T = 6$ remains. Its reciprocal generates the cyclic field K_7, with full basis $1, u, u^2$ for $u^3 - 5u^2 + 6u - 1 = 0$. The square discriminant gives cyclicity, so choose its PHYSICAL largest root as generator. The full-order proof and root/derivative bounds are those just proved above and are algebraic facts of this field, not a finite-head hypothesis.

For a cubic end-helper $v = A+Bu+Cu^2$, the full far/near strips and (D17L) imply $-1 < C < 2$, hence $C = 0, 1$, and $|B + C(\beta + \chi)| < .738$. At $C = 0$, $B = 0$, and the small strip forces $A = 0$, contrary to physical $v \geq 1$. At $C = 1$, only $B = -2$ is possible. Since $2\chi - \chi^2 \in (.356, .359)$, the small strip gives $A = 0$ in either role. But $v_\beta = \beta(\beta - 2) < -.689$, outside BOTH strips. This exhausts every coefficient triple. Cubic end helpers are impossible, so BOTH actual cubic forks are isolated dotted. Finite own in degree three would violate $\mathrm{N}(V) < 4(8/17)^2 < 1$. Each integral W satisfies (D17W), with norm one or two; no equality of the two fork parameters has been assumed.

I.9 Whole D_{17}: the entire isolated cubic field domain

Choose ONE actual isolated cubic fork; the other may have a different parameter. Write its monic polynomial $P(X) = X^3 - AX^2 + BX - n$, $n = \mathrm{N}(W) = 1, 2$. Its two other positive roots lie in $(0, q)$, $q = 4/17$, physical $W < 676/17$, hence integer trace $A \leq 40$. Since $P(q) < 0$,

$$64 - 272A + 1156B - 4913n < 0.$$

For $n = 2$, physical $W > 289/8$, and the small-root arithmetic-geometric inequality gives $B \geq 2\sqrt{2W} + 2/W > 17$. The polynomial inequality gives $B < 20642/1156 < 18$, impossible. Thus W is a UNIT.

For the actual head/fork products in the normalization $r_H^2 = 2U, r_f^2 = 4W$, the LONG and TWO cycles give integral g, L, with

$$L - g = (8U - 17)W - 2U + 4 =: d_0, \qquad g^2 = 2UW(g + d_0).$$

At every dyadic prime, $d_0 \equiv -W$ is a unit. The equation forces g into the prime and then $2\,\mathrm{ord}(g) = \mathrm{ord}(2)+\mathrm{ord}(U)$. If two is unramified this makes $U/2$ integral. The even-cubic-head conclusion above gives actual field K_7. Apply the SAME full-order interpolation as for finite head seven, now merely as algebra of the field and the CURRENT unit-W strip. Its complete $C = 4, \ldots, 7$ tests leave only $C = 4, B = -7$, then no integer A. Thus the unramified case is impossible, without assuming the head itself is finite.

If two ramifies, the unit polynomial has a repeated factor modulo two. Its derivative $X^2 + B$ shows this requires BOTH A, B odd. The physical root and two small-root bounds give

$$A = 19, 21, \ldots, 39, \qquad B^2 > 4(A - 8/17), \qquad B < \frac{272A + 4849}{1156}.$$

The COMPLETE lower/upper odd-integer tests are

A	19	21	23	25	27	29	31	33	35	37	39
$B_{\min}$	9	11	11	11	11	11	13	13	13	13	13
$B_{\max}$	7	9	9	9	9	11	11	11	11	11	13.

Only

$$P_{29} = X^3 - 29X^2 + 11X - 1, \qquad P_{39} = X^3 - 39X^2 + 13X - 1$$

remain. Their discriminants are $4(3)(383)$, $4(5021)$, with squarefree odd parts. For an elementary full-order certificate, trial primes below $\sqrt{383}$ are $2, 3, 5, 7, 11, 13, 17, 19$, none dividing383. For5021, all primes below its square root are

$$2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67;$$

the respective remainders are

$$1, 2, 1, 2, 5, 3, 6, 5, 7, 4, 30, 26, 19, 33, 39, 39, 6, 19, 63.$$

The shifted polynomials are two-Eisenstein:

$$P_{29}(X+1) = X^3 - 26X^2 - 44X - 18, \qquad P_{39}(X+1) = X^3 - 36X^2 - 62X - 26.$$

Only two can divide the power-order index. At its uniformizer $W-1$, valuations of $a_0 + a_1\pi + a_2\pi^2$ are distinct modulo three; a numerator not all divisible by two has unique least valuation at most two. Dividing by two cannot be integral. Thus $\mathcal{O}_K = \mathbb{Z}[W]$ in BOTH cases.

The exact small-root brackets for P_{29} are $.14 < u < .15$, $.23 < v < .235$; its physical root satisfies $28.615 < w < 28.63$. They follow from values $-.025656, .000875, .008067, -.003547125$ at those four endpoints. The physical derivative is > 800, both small derivative magnitudes > 2.27. For P_{39}, values $-.040569, .000128, .019361, -.016952$ give $.11 < u < .12$, $.21 < v < .22$, $38.66 < w < 38.68$, physical derivative > 1480, small derivative magnitudes > 3.45.

Write the ACTUAL head $U = a + bW + cW^2$ in these proved FULL rings. Its current strips are physical $(4, 11)$, other values $(0, 2)$. Lagrange interpolation bounds c respectively by

$$-2/2.27 < c < 11/800 + 2/2.27, \qquad -2/3.45 < c < 11/1480 + 2/3.45.$$

Both intervals are strictly inside $(-1, 1)$, so $c = 0$. Then $b = (U(w) - U(u))/(w - u)$ satisfies $0 < b < 11/(w-u) < 1$, impossible for an integer. Every ramified cubic case has been excluded. The ACTUAL field is QUADRATIC.

I.10 Whole D_{17}: complete quadratic end-helper exclusion

The dotted-head fork table already excludes all four non-end white roles. For the remaining far/near end-three roles write v's polynomial $X^2 - TX + n$. The COMPLETE far ranges are $T = 1, \ldots, 6$, $n = -3, -2, -1, 1, 2$; near ranges $T = 2, \ldots, 7$, $n = -2, -1, 1, 2, 3, 4$. The quadratic formula and EXACT time signs leave the following fourteen and fifteen rows:

(T, n, d) far	(T, n, d) near
$(2, -1, 2)$	$(3, -1, 13)$
$(3, -1, 13)$	$(3, 1, 5)$
$(3, -2, 17)$	$(4, -1, 5)$
$(4, -1, 5)$	$(4, 1, 3)$
$(4, -2, 6)$	$(4, 2, 2)$
$(4, 1, 3)$	$(5, -1, 29)$
$(5, -1, 29)$	$(5, -2, 33)$
$(5, -2, 33)$	$(5, 1, 21)$
$(5, -3, 37)$	$(5, 2, 17)$
$(5, 1, 21)$	$(6, -1, 10)$
$(6, -1, 10)$	$(6, -2, 11)$
$(6, -2, 11)$	$(6, 1, 2)$
$(6, 1, 2)$	$(6, 2, 7)$
$(6, 2, 7)$	$(6, 3, 6)$
	$(7, 3, 37)$.

Here actual $K = \mathbb{Q}(\sqrt{d})$. At each bounded pair T, n, substitute the larger root and its conjugate into the FULL strips. Every missing pair is rational, nonreal or violates an endpoint. The safe far decimal

overstrip admits one extra $(3,1)$, $v=\varphi^2$. Its exact conjugate time numerator is $(123-55\sqrt{5})/2>0$, excluded by $123^2-5(55)^2=4$. Thus the exact, rather than decimal, domain is fully enforced.

For odd n, v is a unit at EVERY dyadic prime, so parity (D17P) makes $U/2$ integral even with ramification. The FULL quadratic head strip for $Y=U/2$ is $2<Y<11/2$, $0<Y'<1$. Its integer trace is three through six and norm $1\le k\le T-2$; at trace six the physical bound removes $k=1,2$. The COMPLETE eight actual heads and $h=-\mathrm{N}(U(U-2))$ are

d	U	h
5	$2\varphi^2,\ 4\varphi^2$	$16,64$
2	$4+2\sqrt{2}$	32
3	$4+2\sqrt{3}$	32
13	$5+\sqrt{13}$	48
17	$5+\sqrt{17}$	64
21	$5+\sqrt{21}$	48
6	$6+2\sqrt{6}$	96.

Odd-norm helper rows in fields29/10/37 are immediately incompatible. For all others, put $p=-16\mathrm{N}(H_c^2)>0$. The complete time-norm tests are

role	(T,n,d)	p	h
far	$(2,-1,2)$	287	32
	$(3,-1,13)$	711	48
	$(4,-1,5)$	1271	$16,64$
	$(4,1,3)$	423	32
	$(5,1,21)$	983	48
	$(6,1,2)$	1679	32
near	$(3,-1,13)$	207	48
	$(3,1,5)$	311	$16,64$
	$(4,-1,5)$	599	$16,64$
	$(4,1,3)$	839	32
	$(5,1,21)$	1503	48
	$(6,1,2)$	2303	32
	$(6,3,6)$	503	96.

All $p/(16h)$ are nonsquares. At $h=32,96$ the odd numerator gives odd two-adic denominator valuation. The four $h=48$ reduced cases are $237/256, 983/768, 69/256, 501/256$; the middle denominator has odd three valuation, the other numerators are nonsquares. For $h=16,64$, the numerators $1271, 311, 599$ are nonsquares. These are NECESSARY actual common-time-line square tests, not sufficiency of a norm-square.

For even-norm rows DO NOT assume $U/2$ integral. Their complete actual fields, full head strips and h's are

d	U	h
2	$3+\sqrt{2},\ 3+2\sqrt{2},\ 4+2\sqrt{2},\ 5+3\sqrt{2},\ 6+3\sqrt{2}$	$7,7,32,63,36$
6	$3+\sqrt{6},\ 4+\sqrt{6},\ 5+2\sqrt{6},\ 6+2\sqrt{6}$	$15,20,15,96$
7	$3+\sqrt{7},\ 4+\sqrt{7}$	$12,27$
11	$4+\sqrt{11},\ 5+\sqrt{11}$	$35,28$
17	$2+\eta,\ 3+\eta,\ 4+2\eta,\ 5+2\eta$	$8,16,64,19$
33	$3+\xi,\ 4+\xi$	$24,24.$

Here $\eta = (1+\sqrt{17})/2$, $\xi = (1+\sqrt{33})/2$. The FULL integral bases and $0 < U' < 2, 4 < U < 11$ give coefficient bounds respectively $3, 2, 2, 1, 2, 1$. At fixed positive coefficient, the width-two conjugate interval gives two consecutive constants; the physical strip removes the excess. For example at $\sqrt{7}$, coefficient two already has least physical $6 + 2\sqrt{7} > 11$. Thus no full-order head was omitted.

The COMPLETE remaining even-norm time rows are

role	(T, n, d)	p	all h
far	$(3, -2, 17)$	200	$8, 16, 64, 19$
	$(4, -2, 6)$	828	$15, 20, 15, 96$
	$(5, -2, 33)$	1592	24
	$(6, -2, 11)$	2492	$35, 28$
	$(6, 2, 7)$	252	$12, 27$
near	$(4, 2, 2)$	92	$7, 7, 32, 63, 36$
	$(5, -2, 33)$	72	24
	$(5, 2, 17)$	824	$8, 16, 64, 19$
	$(6, -2, 11)$	668	$35, 28$
	$(6, 2, 7)$	1692	$12, 27.$

Every entry follows by expanding

$$16\mathrm{N}(H_c^2) = (17T \pm 4)^2 n + (17T \pm 4)(-17n - 4)T + (-17n - 4)^2.$$

Every ratio $p/(16h)$ is nonsquare EXCEPT far $(T, n, d) = (3, -2, 17)$, $h = 8$. The far nonexceptional rows retain odd valuations at $23, 199, 89, 7$. In the exceptional field the other ratios are $25/32, 25/128, 25/38$, nonsquares. The near rows retain odd valuations at $23, 3, 103, 167, 47$. Thus a genuine square-norm exception is kept.

For that exception

$$t = (3+\sqrt{17})/2, \quad U = (5+\sqrt{17})/2,$$

the ACTUAL time squares are

$$H_c^2 = (225 + 55\sqrt{17})/8, \qquad H_0^2 = (11 + 3\sqrt{17})/2.$$

Their ratio $R = (165 + 35\sqrt{17})/64$ has norm $25/16$. If $R = z^2$ in K, then $\mathrm{N}(z) = \pm 5/4$, and

$$\mathrm{Tr}(z)^2 = 245/32 \quad \text{or} \quad 85/32.$$

Neither is a rational square, by its odd two-adic denominator valuation. But the trace is rational. The last actual common-time square fails. All quadratic end-helper rows are now excluded, in both signed orientations.

Both ACTUAL forks are therefore isolated under the dotted quadratic head. Their original LONG+TWO cycles, applied BEFORE assuming own dotted, give integral $W = V/2$ with $0 < W < 676/17$ and $0 < W' < 4/17$. The lower bound $2 < W$ is deferred until finite owns have been excluded. Finite own would have $0 < W < 2$, other $0 < W' < 4/17$, hence nonzero integer norm $0 < \mathrm{N}(W) < 8/17$, impossible. Parallel has its actual null pair. Both forks are isolated DOTTED, without equating their two W parameters.

I.11 Whole D_{17}'s distinct actual inclusion and final budget

Choose ONE of the two actual isolated dotted forks. Its raw square is $4W$, sole old product $-2W$, spatial $W(P_{16} \pm e_{17})$, time square $W(17W - 4)$. The actual head has square $2U$, spatial Ue_1, time square $U(U - 2)$. The original K-time line gives

$$R_0 = U(U-2)W(17W-4) \in K^2, \qquad g = UW - \sqrt{R_0},$$

with PHYSICAL positive square root. The original LONG path and TWO cycles give

$$g,\ L = \frac{g^2}{2UW} \in \mathcal{O}_K.$$

All physical/conjugate strips are now

$$K = \mathbb{Q}(U) = \mathbb{Q}(W), \quad 4 < U < 11, \quad 0 < U' < 2, \quad 2 < W < 676/17, \quad 0 < W' < 4/17.$$

The actual independent conjugate root lines give $0 < L' < 4$ when $g \neq 0$. Here the current physical strips force $d_0 = (8U - 17)W - 2U + 4 > 0$, hence $g < 0$, so no orthogonal exception is silently lost. The true physical mutual L is finite native or dotted, while parallel four has its forbidden actual null pair. Thus EVERY predicate of Proposition 10(iii) holds for this ORIGINAL D17 geometry.

This inclusion is separate from Section I.4: D17 raw fork norm is $4W$; B17 raw short norm was V. No equation of different root scales or arithmeticity of a face supplied either inclusion. The existing full-order algorithm covers every $\Delta < 121$, every strict-strip model and ordered pair with both field-square norm signs. Its 36-field/120-pair output, explicitly ASSUMED as the recorded computational proposition, has the same TWO necessary survivors. The field/counter/survivor combine does not independently recompute the118 rejected tests.

For either actual survivor, use its actual head point and ONE chosen signed fork. The head gain at cost .072 from P_5 is $> .043$, by the full rational seals already printed in Section I.4. The actual fork has $C_f^2 = 17W/4 > 442/5$, giving

$$x \cdot (P_{16} \pm e_{17}) > 3.7.$$

Pay .13 from the DISTINCT P_{14} price:

$$y = w_{14} + .36(Z - P_5/5) + .13(P_{16} \pm e_{17} - P_{14}).$$

The signed residual has norm three and vanishes on all reference coordinates; it is orthogonal to the head residual. Both individual payments are legal BEFORE using their original floors. ONE square on the SAME original x gives

$$\|x\|^2 > .9084 + .043 + .26(3.7 - 3.311) - .0169(3) = 1.00184 > 1.$$

Every actual field and actual fork type has now been covered. This closes whole D_{17} at the stated external and explicitly computed premises. The argument needed one chosen fork for the last contradiction, while preserving both signed walls and all original constraints.

J Chain component bounds at ranks ten through fourteen

We now finish the smaller whole-A consumers. The shared mechanism is unchanged: classify an ACTUAL central opposite, constrain a pure helper arithmetically, and absorb EVERY retained isolated foot in ONE reference. The index ranges, finite side exceptions, helper packing and coefficient capacities are proved at their own N; no numerical theorem is reranked.

J.1 A spectator-safe A_{14} bound greater than $4/5$

Use the zero-sum space in $\mathbb{R}^{15}$,

$$n_i = (e_i - e_{i+1})/\sqrt{2}, \quad \Omega_i = \sqrt{2}(P_i - iP_{15}/15), \quad D_i = 2i(15 - i)/15, \quad h_i = x_A \cdot \Omega_i.$$

All spectators are kept; the SAME original farthest vector has $\|x\| < 1$. For $i = 5, \dots, 10$ take loss $\ell = .597$. Here $D_5 = D_{10} = 20/3$, $D_6 = D_9 = 36/5$, $D_7 = D_8 = 112/15$. The exact ordinary sufficient ratio

$$R_\ell(q) = 1/\ell - (q - \ell)/(q(2q - \ell))$$

increases on this interval. Since $q < 11/4$,

$$R_\ell(q) < 1000/597 - 8612/53933 < 761/500;$$

the last positive cleared numerator is 106860761. The older coarse $q < 3$ bound is not used.

Deleting a central node retains two paths of rank at least four. The FULL side-order-$m \geq 5$ finite-white list is no contact, permitted endpoint/D/E simple three, endpoint four, or two end-threes joining the paths. The connected replacement rank is fourteen, so no high-rank exceptional type is missing. The complete own-zero repairs retain internal affine arms, stars and two-contact cycles; far/next-three use the ENTIRE foreign cost and reciprocal original-wall law; endpoint-four has no positive-time foreign completion. All isolated own-three/four/six/parallel nulls apply for EVERY foreign cost BEFORE splitting it, as proved in the common chain section.

The ordinary nonisolated ratios exceed 761/500, as do the complete foreign far/near bounds $227/(25\sqrt{35})$ and $\sqrt{121/35}$. The only extra near-four orientations are

$$v_A = \Omega_i/\sqrt{2} + \Omega_{i-1}/2, \qquad i = 5, 6, 7$$

and reflections, with zero foreign cost. Their conjugate time numerators at $N = 15$ are $114 - 80\sqrt{2}, 128 - 90\sqrt{2}, 136 - 96\sqrt{2} > 0$. The actual simple-three link puts $\sqrt{2}$ in K; its sign-reversing embedding makes the original Schur complement negative. Near-four of side order at least eight is already strong, so both helper orientations are covered.

For an isolated-within-A cap write $c = Ap - \gamma\Omega_i - v_S$, $Q = \|v_S\|^2 < 1$, $u = (1 - Q)/\gamma^2$. For $Q > 0, u > 1$, the actual positive clearing has loss $< .202 + .382 = .584 < \ell$. At $u = 1$ it is a feasible null. For pure native five the actual H_4 loss is $< 25/76 + 5/24 < \ell$. For EVERY pure finite native at least seven the following original negative core gives a fresh stronger bound, including native eight:

$$\xi = c + \gamma^{-1} n_i + \frac{2(1 - \gamma^2)}{\gamma} n_j, \qquad j \text{ either old neighbour.}$$

Put $d = 1 - \gamma^2 < .19$. The cap and n_i products vanish; $\xi \cdot n_j = (2d - 1/2)/\gamma < 0$; its square is $-d(1 - 4d)/\gamma^2 < 0$. Positive cap time and actual acuteness check EVERY omitted original wall. Its normalized comparison gives loss

$$< \frac{10000}{39771} + \frac{25}{81} = \frac{22275}{39771} < .561 < \ell.$$

Here $\sqrt{d(1 - 4d)} \leq 1/4$, $\gamma > .9$, $q > 2.58$, and $\sqrt{D_i - 100/81} > 2.33$. Every remaining isolated foot is retained in the FULL interval $0 < u < 1$, pure or foreign; none is assigned an unsupported strong floor.

Every remaining weak nonisolated cap is a PURE endpoint helper with

$$202/125 < t = 2\gamma < 3139/1458 < 54/25.$$

The fresh cheapest negative/positive conjugate Schur costs are

$$Q_-(t) = 2it^2 + 4t + 2 - 2(it + 1)^2/15, \quad Q_+(t) = 2it^2 + 2 - 2(it - 1)^2/15.$$

At $i = 5$, $Q_-(-4/5) = 4$ EXACTLY, $Q_+(12/25) = 4 + 16/375$. They increase at the relevant signs over the half-window, so strict positivity gives every other conjugate in $(-4/5, 12/25)$. The COMPLETE all-degree norm/trace argument of Section B.2 uses exactly these physical/conjugate bounds and proves $t = \varphi$, actual $K = \mathbb{Q}(\varphi)$. No restriction is imposed in a branch without a helper.

The COMPLETE six-type $30A^2 = M$ table is

type	M	$\mathrm{N}(M)$
near 5	$34 + 70\varphi$	-1364
near 6	$38 + 72\varphi$	-1004
near 7	$40 + 72\varphi$	-704
far 5	$34 + 60\varphi$	-404
far 6	$38 + 66\varphi$	-404
far 7	$40 + 70\varphi$	$-500.$

Different norm-squareclasses cannot coexist on the SAME original connected-block K-time line. For the repeated far-five/six class, the ratio is

$$\frac{97 + 9\varphi}{101} = \frac{203 + 9\sqrt{5}}{202}.$$

Its norm is one. A full K-square would have rational coefficient square $405/404$ or $1/404$, both impossible at prime101. For reflected copies, the actual cyclic parameter has constant numerators $8, 1, -10$ modulo fifteen, so is nonintegral in $\mathbb{Z}[\varphi]$. Explicitly the near/far formulas are

$$\frac{(15 - 2i)(i + (i \pm 2)\varphi) - 17}{15}.$$

Every different and repeated type is covered; at a fixed index the ACTUAL opposite is unique. Thus at most ONE central index has a weak helper. The number and adjacency of isolated feet are UNRESTRICTED.

For each retained $0 < u < 1$ foot the positive original combination $c + v_S + \gamma u n_i$ gives

$$x_A \cdot (\Omega_i - u n_i) \geq \sqrt{D_i - u} - \sqrt{u(1 - u)}.$$

The raw loss satisfies $L_i \leq a_i u + \sqrt{u(1 - u)}$, where

$$a_{5,10} < .202, \quad a_{6,9} < .194, \quad a_{7,8} < .190.$$

Each follows from the ACTUAL $a_i = (\sqrt{D_i} + \sqrt{D_i - 1})^{-1}$. The full adjusted-loss inequality

$$L_i - \frac{\beta_i}{2}(2u - u^2) < .597$$

holds for capacities

$$\beta_{5,10} \geq .06, \qquad \beta_{6,9} \geq .026, \qquad \beta_{7,8} \geq .0246.$$

For outer sites, $u \leq .4$ has loss $< .5708$; above .4 the circle bound gives $.6111 - .03(.64) = .5919$. For sites six/nine, $u \leq .5$ gives $< .597$; above it use $(.194 + 1.01865)/2 - 3(.026)/8 = .596575$. For seven/eight, the corresponding bounds are .595 and $(.190 + 1.018)/2 - 3(.0246)/8 = .594775$. Thus EVERY pure/foreign foot parameter is covered.

Use symmetric rational heights

$$H_5 = H_{10} = A = 1.98498, \quad H_6 = H_9 = B = 2.08628, \quad H_7 = H_8 = C = 2.13551,$$

linear on the outer tails. Their exact floor squares are below the true ℓ-strong heights. The central dual coefficients are

$$.147848, .026035, .024615, .024615, .026035, .147848.$$

A single internal helper is omitted by its neighbour chord, making its coefficient zero and increasing neighbouring ones. The largest energy loss is $(2B - A - C)^2/4 = .000677821225$.

For $w = \sum \beta_i \Omega_i$, and EVERY retained foot with positive coefficient, put $Z = w - \sum \beta_i u_i n_i$. Primitive actual pressures give $x_A \cdot Z \geq E_H - \sum \beta_i (L_i - \ell)$. ALL adjacent-foot crosses have the correct sign $n_i \cdot n_j \leq 0$, so

$$Z^2 \leq E_H - \sum \beta_i^2 (2u_i - u_i^2).$$

ONE square and the adjusted losses give $e_A > E_H$. Strong constraints supply strictness when no feet remain. The full energy is $A^2/5 + (B-A)^2 + (C-B)^2 = .80071440298$; after the largest internal omission it is

$$> .800036581755 > 4/5.$$

It remains to cover an OUTER helper at five/ten. Reverse labels if necessary, without assuming a chamber symmetry. The actual H_4 positive correction is

$$b_c = \gamma n_5 + (2/3) n_e + (1/3) n_{\text{adj}}, \quad \delta = (c + b_c)^2 = (5 - 3\varphi)/12 < .111^2.$$

The positive FINITE projected point is used; the spacelike $c + b_c$ alone is not a point. For helper end $e = 1$ or fourteen, the signed endpoint $y = x_1$ or $-x_{15}$ obeys

$$y > \sqrt{2}(A_c - .111 - \gamma h_5).$$

The smaller actual time has $A_c^2 = (34 + 60\varphi)/30 > 2.09^2$; weakness gives $h_5 < 1.985$, $\gamma < .81$. Thus BOTH orientations have $y > 1.4(2.09 - .111 - .81(1.985)) = .51961 > .49$.

Omit knot five by the chord zero-to-six. Its reference energy is

$$E_{\text{end}} = \frac{1}{2}[B^2/6 + A^2/5 + (B - A)^2 + 2(C - B)^2] > .764.$$

At the actual helper end there is a flat signed block of length six or five, mean $b_0 = B/(6\sqrt{2}) \in (.24, .25)$ or $A/(5\sqrt{2}) \in (.25, .3)$. Graft the ACTUAL endpoint to .49, keeping the block sum fixed. The other entries $(Lb_0 - .49)/(L - 1)$ lie in $(.19, .2525)$, above the next old coordinate $< .073$. All duals remain positive. The changed central boundary coefficient is $> .117/1.415 > .06$, meeting its own capacity; all others retain theirs. The missing-five coefficient remains zero. The gain in this SAME reference is

$$\frac{L}{L - 1}(.49 - b_0)^2 > (6/5)(.19)^2 = .04332.$$

Absorb ALL other isolated feet with their individual parameters as above. Then $e_A > .764 + .04332 = .80732 > 4/5$. No second helper exists by packing. The proof covers every actual field and arbitrary finite spectators:

$$e_{A_{14}} > 4/5. \tag{A14U}$$

In particular its ordinary-two-A_1 and actual-I-plus-A_1 rank-sixteen consumers close by adding respectively $> .2$ or the rank-free JOINT cost $> .20204$, with I counted once. In rank seventeen, every rank-three remainder has cost $> .21$ by the rank-free partition proof of (S3), so the ENTIRE older A_{14} row closes at once.

J.2 A fresh spectator-safe A_{13} bound

Now $N = 14$, central sites five through nine, with

$$D_5 = D_9 = 45/7, \quad D_6 = D_8 = 48/7, \quad D_7 = 7, \qquad \ell = 1199/2000 = .5995.$$

Deleting a central site again retains two paths of rank at least four. The complete side-order-$m \geq 5$ white list is isolation, permitted simple-three/D/E, endpoint-four, or one endpoint-three on EACH retained path. Two contacts within one path form a cycle; the connected rank-thirteen completion is classical, with no missing exceptional type. The fresh sufficient ratio satisfies

$$R_\ell(q) < 2000/1199 - 4801/32403 < 761/500, \qquad 5/2 < q < 3.$$

This follows from the same positive derivative and a direct positive rational cross product.

Own zero is dispatched BEFORE positive-own ratios. Isolation has $A^2 = Q - 1 < 0$. Internal-three has the actual affine arms $1, 2, 5$ or $1, 3, 3$; internal-four has its weighted star, two threes their cycle. The only residual endpoint/next-three/four times are

A^2	white bound
$Q - 15/28$	$Q < 3/5$
$Q - 1/7$	$Q < 2/5$
$Q - 1/14$	$Q < 1/5.$

The first positive time is $< 9/140 < 1/4$, below any foreign contribution. The second has at most one foreign contribution, larger than A^2. Both violate the actual orthogonal reciprocal opposite law; the last admits no positive foreign cost. At $Q = 0$ all three times are negative. Every affine/null vector has positive cap time and correct signs against EVERY original wall.

The complete ordinary ratios for positive own exceed 761/500, including

$$\frac{.808 + .4}{\sqrt{.6}}, \quad \frac{.808 + .2}{\sqrt{.4}}, \quad \frac{.808}{\sqrt{.25}}, \quad q/2 + 1/q.$$

The complete foreign far/near bounds are still $227/(25\sqrt{35})$, $\sqrt{121/35}$. Thus each weak nonisolated cap is PURE: an endpoint helper or own-four/simple-three near helper of side order five/six/seven. Near-four of side order at least eight is already strong. We retain the smaller near-four roles rather than use an invalid N15 conjugate exclusion.

For a pure endpoint helper, the exact square ratio identity gives $t = 2\gamma < 54/25 < 9/4$. If its own is not five, $t > 9/5$. Actual simple-three normalization makes t integral. At every lifted nonphysical embedding, the cheapest signed old-A Schur costs are

$$2it^2 + 4t + 2 - 2(it + 1)^2/14, \qquad 2it^2 + 2 - 2(it - 1)^2/14.$$

At the relevant negative/positive endpoints they increase with i and N. Already at $N = 13, i = 5$, costs at $-.83, .5$ exceed four by $69/1625, 2/13$. Hence every other conjugate belongs to $(-.83, .5)$. Degree at least six contradicts $(9/4)(.83)^5 < 1$. In degrees three/four/five the norm magnitude is < 2; a positive small conjugate would make it < 1, so all are negative, of product magnitude $1/t$. AM–GM bounds the integer trace respectively in

$$(.14, 1), \quad (-.69, 0), \quad (-1.52, -1),$$

none containing an integer. The exact underlying comparisons are $t(t - 1)^2 < (9/4)(5/4)^2 < 4$, $t^4 < (9/4)^4 < 27$, $t(t + 1)^4 < (9/4)(13/4)^4 < 256$. In degree two trace two gives a nonzero norm

strictly between $-1, 1$; trace one leaves only norm -2, whose polynomial $X^2 - X - 2$ is reducible. Degree one leaves parallel two, an actual null. Therefore every weak endpoint helper is golden, with ACTUAL $K = \mathbb{Q}(\varphi)$ by its original Lorentzian principal.

The COMPLETE $28A^2 = M$ golden table is

type	M	$\mathrm{N}(M)$
near 5	$30 + 63\varphi$	-1179
near 6	$33 + 64\varphi$	-895
middle 7	$34 + 63\varphi$	-671
far 5	$30 + 55\varphi$	-475
far 6	$33 + 60\varphi$	$-531.$

All five norm-squareclasses differ, excluding unlike types on the common actual K-time line. Reflected same-type cyclic parameters are

$$\frac{(14 - 2i)(i + (i \pm 2)\varphi) - 16}{14}.$$

At $i = 5, 6$ their constant coefficients are $2/7, -2/7$, nonintegral. At the middle there is only one actual index. Thus at most ONE weak golden helper exists.

If such a helper occurs, EVERY other central site is .65-strong. Indeed $R_{.65}(3) < 1.4$, below all ordinary and near-four roles; the isolated raw loss bound below is $< .6135$. Any .65-weak endpoint helper is also .5995-weak, so packing excludes a second one. For the actual finite H_4 point use

$$U = v_A - \gamma n_i - (2/3)n_e - (1/3)n_{\mathrm{adj}}, \qquad \delta = (5 - 3\varphi)/12 < .111^2, \qquad x_A \cdot U > A - \sqrt{\delta}.$$

Positive finite projection, not the spacelike corrected normal alone, checks all original walls.

For helper index five, set $a = 1.885, b = 1.968$. Use heights $H_6 = H_7 = H_8 = b, H_9 = a$, linear on its two outer tails. The energy is $E_5 = (b^2/6 + (b - a)^2 + a^2/5)/2 = .681519$. Project U onto its flat blocks. For far/near helper the projections are

$$(5\gamma/6)\Omega_6 + \Omega_9/10, \qquad (5\gamma/6 + 1/12)\Omega_6.$$

Their reference products are < 1.5169. The smaller time is $A > 2.061$; the deficit is $> .4331$. The FULL orthogonal residual norms are $(2/3)\gamma^2 + 1/15$ and $(2/3)\gamma^2 + 1/12 + \gamma/3$, both $< .8$. Pay $\mu = .2$ from the INDIVIDUAL strong coefficients: the largest debit is $.2(5(.81)/6 + 1/12) < .151667 < b/12 = .164$, and the far other debit .02 is below $(a/5 - (b - a))/2 = .147$. ONE square gives $e_A > E_5 + .4(.4331) - .04(.8) = .822759$.

For helper at six/seven, use only $H_5 = H_9 = a$, with flat middle block of four coordinates. Energy $E_0 = a^2/5 = .710645$, both individual coefficients .1885. Writing $l = i - 5$, the flat-block projection is

$$\gamma[(4 - l)\Omega_5 + l\Omega_9]/4 + \Omega_{\text{helper end}}/10.$$

Here the added term is $\Omega_5/10$ for helper endpoint one, and $\Omega_9/10$ for helper endpoint thirteen; these are the flat-block boundaries, not the literal helper-node indices. Its reference product is $a(\gamma + .1) < 1.71535$. The smaller time is $A > 2.155$, so the deficit is $> .328$. The FULL residual norm is $\gamma^2[2l(4 - l)/4 - 1] + 1/15 < .724$. At $\mu = .1$, EACH actual debit is $< .091 < .1885$. ONE square gives

$$e_A > .710645 + .2(.328) - .01(.724) = .769005.$$

Every golden case therefore already exceeds the target energy below.

A weak near-four also costs too much, without a false arithmetic conjugate ban. Its actual simple-three mate $j = 4, 5, 6$ has the reciprocal finite endpoint data

$$\delta_j = \frac{1}{14 - j}, \quad \sigma_j = \frac{1}{2} + \frac{13 - j}{(14 - j)\sqrt{2}}, \quad C_j = \sigma_j \sqrt{(14 - j) D_j} > 31/4.$$

Positive finite inverse projection proves an ACTUAL original endpoint, with every omitted wall strict/nonpositive as required. Thus its normalized dual height is $> \sqrt{27/35} > .878$. At site nine every nongolden nonweak role has raw loss $< .618$, hence normalized height $> .75$. If nine itself is weak near-four, use its ACTUAL mate ten, whose height is $> .878$. The other orientation, with mate eight, has side order nine and was already strong. The two chosen unit duals have $j \le 6, k \ge 9$, correlation

$$\rho \le \sqrt{5/12} < 2/3.$$

Cauchy on their sum gives $e_A > (.878 + .75)^2/[2(1 + 2/3)] = .7951152$. Reflection only reverses index names.

It remains to absorb ALL isolated feet. Own-three/four/six/parallel have their ALL-foreign actual nulls before the Q split. Pure five's actual H_4 loss is $< 595/1056 < .57$. Every pure finite native at least seven uses the positive adjacent core already proved in Section J.1; at this rank its loss is

$$< \frac{4/3}{q + \sqrt{D - 4/3}} + 1/\sqrt{12} < .279 + .289 = .568 < \ell.$$

For $Q > 0, u > 1$, only own five remains, and positive finite clearing gives loss $< .206 + .383 = .589 < \ell$. The $u = 1$ vector is a forbidden feasible null. Retain EVERY $0 < u < 1$, pure or foreign, with its actual pressure

$$x_A \cdot (\Omega_i - u n_i) \ge \sqrt{D_i - u} - \sqrt{u(1 - u)}.$$

Its raw loss is $< a_i u + \sqrt{u(1 - u)}$, with $a_{5,9} < .206$, $a_{6,7,8} < .199$. The raw circle maximum is $< .6135$, justifying its earlier use.

Choose symmetric heights, linear on the ends,

$$H_5 = 1.9359, \quad H_6 = 2.0191, \quad H_7 = 2.0462.$$

The exact floor squares after adding ℓ are $2.5354^2 < 45/7$, $2.6186^2 < 48/7$, $2.6457^2 < 7$. Its central coefficients are

$$\beta_5 = \beta_9 = .15199, \quad \beta_6 = \beta_8 = .02805, \quad \beta_7 = .0271.$$

The energy is

$$E = 1.9359^2/5 + .0832^2 + .0271^2 = .757198412, \qquad E - 53/70 = \frac{97221}{1750000000} > 0.$$

Assume $e_A \le E$. The golden and near-four contradictions leave only genuine strong constraints or isolated ACTUAL feet at every paid index. Their FULL adjusted losses satisfy

$$a_i u + \sqrt{u(1 - u)} - (\beta_i/2)(2u - u^2) < .5995.$$

For outer sites the proof works already at $\beta = .08$: below .4 raw loss $< .5824$, above it $.6135 - .04(.64) = .5879$. For inner sites below .5, raw loss $< .5995$; above it use

$$(.199 + 1.01961)/2 - .0271(3/8) = .5991425 < .5995.$$

All root comparisons are positive rational squares, covering the ENTIRE intervals.

Form ONE vector $Z = w - \sum \beta_i u_i n_i$ over ALL retained weak feet. The actual pressures and every adjacent-foot cross give exactly the simultaneous-foot inequalities proved above. The single square implies $e_A > E$, contradicting the assumed $e_A \le E$. Strictness comes from strong floors if no foot is present, otherwise from each own-coefficient adjusted loss. Thus

$$e_{A_{13}} > .757198412 > 53/70 \tag{A13U}$$

with every actual field and arbitrary finite spectators. Only now add the disjoint original ordinary costs: $A_3 > 1/4$, or $T_2 > 1/7, A_1 > .1$. They close BOTH ordinary rank-three alternatives. The actual-I and actual-O_7 alternatives retain their separate hypotheses and proof.

J.3 The actual seven-field A_{13} consumer

For the actual $O_7 + A_1$ alternative, the ACTUAL field is K_7. We use a direct full-order integer strip, rather than any computed field census. Put $\vartheta = 2\cos(\pi/7)$, $f(T) = T^3 - T^2 - 2T + 1$. Its full ring is $\mathbb{Z}[\vartheta]$. This follows already from the proved full ring $\mathbb{Z}[u]$, $u^3 - 5u^2 + 6u - 1 = 0$:

$$\vartheta = u^2 - 3u + 1, \qquad u = \vartheta^2.$$

Thus no new external integral-basis assumption is required here.

There is NO algebraic integer $w \in K_7$ with physical $0 < w < 256/25$ and both other values in $(0, 28/45)$. To prove this, exact polynomial signs give roots

$$1.8019 < \vartheta < 1.802, \quad .445 < \alpha < .446, \quad -1.248 < \beta < -1.246.$$

Write $w = a + b\vartheta + c\vartheta^2$ in the FULL ring. For $d = b - (\vartheta - 1)c$,

$$w_\alpha - w_\beta = (\alpha - \beta)d, \qquad |d| < .369,$$

using $\alpha - \beta > 1.691$. Put $\Delta = w - (w_\alpha + w_\beta)/2$. The exact trace identities give

$$\Delta = \frac{(3\vartheta - 1)}{2}d + (3\vartheta^2 - 2\vartheta - 2)c.$$

The first coefficient is < 2.203, its product with d has magnitude $< .812907$; the second coefficient is > 4.136. Since $-28/45 < \Delta < 10.24$, this forces $c = 0, 1, 2$. For $c = 0$, $b = 0$, leaving a positive integer $a < 28/45$. For $c = 2$, the d interval gives $1.2348 < b < 1.973$, no integer. For $c = 1$, it gives $b = 1$. Now $.643025 < \alpha + \alpha^2 < .644916$; positivity of $w_\alpha = a + \alpha + \alpha^2$ forces $a \ge 0$, but then $w_\alpha > .643025 > 28/45$. This exhausts every full-order coefficient.

Reclassify the SAME N14 central sites at the NEW loss $\ell = .574$, not the previous .5995 list. The exact ratio is

$$R_\ell(q) < 500/287 - 1213/8139 < 8/5.$$

The finite-white list and own-zero reciprocal repairs are unchanged, with their full foreign costs. Native five is absent in K_7. Every own-four/simple-three near role, at any position and with any foreign contacts, has the actual $\sqrt{2}$ cycle and is absent in the cubic field. The remaining COMPLETE nonisolated comparisons against the larger threshold 8/5 are: near-three $> (1.3)/\sqrt{.6}$; D-position-three $> (1.1)/\sqrt{.4}$; E-position > 1.8; far-four $> .9\sqrt{5}$; near-four white contact has $\sigma > 53/50, \delta \le 1/5$; two-white-three has $q/2 + 1/q > 8/5$. The full FOREIGN far-three bound is

$$\frac{1.8m + 1}{\sqrt{m(m+2)}} \ge 10/\sqrt{35} > 8/5 \quad (m \ge 5).$$

It increases with m, and positive foreign cost only strengthens every other ratio. All own-three/four/six/parallel affine cases retain the original nulls in their proper roles. Thus any weak nonisolated cap is pure endpoint helper with $\gamma > .9$, $9/5 < t = 2\gamma < 57/25$. The signed Schur costs proved above still give every other conjugate in $(-.83, .5)$. In actual K_7, t has degree one or three. Degree three gives unit norm, both small conjugates negative, and integer trace

$$.14 < \mathrm{Tr}(t) \le t - 2/\sqrt{t} < 1,$$

using $t(t-1)^2 < 58368/15625 < 4$. Degree one leaves only parallel two. No weak nonisolated cap remains.

For an isolated cap keep its ENTIRE spectator projection. Its actual own cycle $w = 4\gamma^2$ is integral in K_7. Original K-rescaling gives new diagonal $2/w$, marked old product -1. At both other embeddings, the old-A Schur complement requires

$$0 < w_\sigma < 4/D_i \le 28/45.$$

Weakness would give $\gamma/\sqrt{1-Q} \le R_\ell(q) < 8/5$, hence physical $0 < w < 256/25$. This is precisely the impossible FULL-order strip just proved. It covers EVERY finite/dotted/pure/foreign isolated cap, not only generating w's. No weak feet remain.

All sites five through nine are now .574-strong. The symmetric positive reference with

$$H_5 = H_9 = 1.9614, \quad H_6 = H_8 = 2.0446, \quad H_7 = 2.0717$$

has genuine floor squares $2.5354^2 < 45/7$, $2.6186^2 < 48/7$, $2.6457^2 < 7$. Its duals are $.15454, .02805, .0271, .02805, .15454$, and ONE square gives

$$e_A > 1.9614^2/5 + .0832^2 + .0271^2 = .777074642 > 233/300.$$

The ACTUAL O_7 spectator costs $> .09$ and propagates the last ordinary $A_1 > 2/15$. Adding these disjoint original projections ONCE gives $233/300 + .09 + 2/15 = 1$, strictly exceeded. A bare old seven bond is never substituted for the actual O_7 profile.

J.4 The $A_{13} + I + T_2$ consumer with every remaining foot

Here I is the ACTUAL isolated short wall, and T_2 an ORDINARY irreducible dihedral. The strengthened ordinary theorem proved in Section A, together with the separate ordinary A_2 bound, gives $e_T > 3/19$. Use the actual rank-free I parameters from (3):

$$e_I \ge u^2, \quad 0 < u < 1/3, \quad t + 3u/2 > .767.$$

Both signed A_{13} endpoint coordinates satisfy

$$x_1, -x_{14} \ge \alpha t, \qquad \alpha = \sqrt{13/14} > .9636.$$

All these pressures concern the SAME original farthest vector. The I energy is counted once.

For endpoint means $b < .2738$, or $b < .2666$, the positive linear lower bounds are

$$\alpha t - b > .4652 - 1.446u, \qquad \alpha t - b > .4724 - 1.446u,$$

respectively. They follow from $.9636(.767 - 1.5u) = .7390812 - 1.4454u$. The following exact completion applies on the FULL continuous domain:

$$(1 + cB^2)[u^2 + c(A - Bu)^2] - cA^2 = [(1 + cB^2)u - cAB]^2.$$

For $B = 1.446$, the positive rational margins

$$(5/2)(.4652)^2 - .0868[1 + (5/2)B^2] = .000498828,$$

$$(5/4)(.4724)^2 - .077[1 + (5/4)B^2] = .000701535$$

therefore give the uniform bounds $u^2 + (5/2)(.4652 - Bu)^2 > .0868$ and $u^2 + (5/4)(.4724 - Bu)^2 > .077$.

The COMPLETE N14 helper/near-four classifications of Section J.2 do not depend on its contrary energy assumption. A golden helper at five gives $e_A > .822759$, which closes with the joint $I + T_2 > .21$. For a helper at six/seven, retain its SAME one-reference H_4 square of energy $L > .769005$, with price $< .091$ at each outer coefficient .1885. Choose the five-coordinate block OPPOSITE its helper endpoint. The actual H_4 displacement and its flat block projection are constant there; their residual vanishes POINTWISE on that block. Thus an endpoint residual G has

$$G^2 = 4/5, \quad w \cdot G = Y \cdot G = 0, \quad b = 1.885/(5\sqrt{2}) < .2666.$$

Its individual price is $\kappa/(5\sqrt{2})$, with remaining capacity $> .0975$. One JOINT pressure square gives

$$e_A > L + 2\kappa(\alpha t - b) - (4/5)\kappa^2.$$

For $u \geq .3$, $L + 3/19 + .09 > 1$. For $u \leq .2$, choose $\kappa = .2$; its price is $< .0283$. The total is $> L + 3/19 + u^2 + .4(.4724 - Bu) - .032$, decreasing to an endpoint additive value .08128, already contradictory. For $.2 \leq u \leq .3$, put $d = .4724 - Bu > 0$, $\kappa = 5d/4$; its price is $< .033 < .0975$. The total exceeds $L + 3/19 + .077 > 1$ by the completed-square certificate. Both helper ends and reflected indices are covered with ACTUAL orthogonal residuals.

Every weak near-four gives $e_A > .7951152$; adding joint $I + T_2 > .21$ also exceeds one. We are left with ONLY true strong sites or the actual isolated feet of Section J.2.

Use that section's full rational reference, energy $E = .757198412$, outer coefficients .15199, inner .02805, .0271, .02805. Its two endpoint residuals are

$$G_L = e_1 - P_5/5, \qquad G_R = -e_{14} + (e_{10} + \cdots + e_{14})/5.$$

They are disjoint, have norm 4/5, and are orthogonal to w. For $W = w + \kappa(G_L + G_R)$, EACH outer central coefficient becomes

$$\beta'_5 = \beta'_9 = .15199 - \kappa/(5\sqrt{2}).$$

The others are unchanged. For EVERY remaining foot parameter $v_i \in (0, 1)$, form ONE vector $T = W - \sum \beta'_i v_i n_i$. The primitive endpoint/strong/foot pressures, paid BEFORE taking its square, give

$$x_A \cdot T \geq E + 2\kappa(\alpha t - b) - \sum \beta'_i (L_i - \ell),$$

$$T^2 \leq E + (8/5)\kappa^2 - \sum (\beta'_i)^2 (2v_i - v_i^2).$$

Here $b = 1.9359/(5\sqrt{2}) < .2738$. Every endpoint/foot cross is incorporated through the REMAINING coefficient β'_i, and every adjacent-foot cross has the actual nonpositive old-root sign. The proved continuous adjusted-loss inequalities hold at outer coefficients at least .08 and the unchanged inner coefficients. Consequently

$$e_A > E + 4\kappa(\alpha t - b) - (8/5)\kappa^2.$$

For $u \geq .3$, $E + 3/19 + .09 > 1$. For $u \leq .2$, choose $\kappa = .2$; both outer coefficients remain $> .1236 > .08$. The total is $> E + 3/19 + u^2 + .8(.4652 - Bu) - .064$, decreasing to endpoint

additive value .1168. For $.2 \le u \le .3$, use $\kappa = 5(.4652 - Bu)/4 \le .22$; both outer coefficients remain $> .1199 > .08$. The exact completion gives total $> E + 3/19 + .0868 > 1$. All actual I parameters, ALL foreign/adjacent feet and both endpoints have been covered. This closes the last A_{13} rank-sixteen alternative with one original vector and one I energy.

J.5 A universal A_{12} bound from one paid endpoint

Set $N = 13$, central sites five through eight,

$$D_5 = D_8 = 80/13, \quad D_6 = D_7 = 84/13, \qquad \ell = .6005.$$

Keep arbitrary finite spectators and EVERY actual field. Assume $e_A \le 5/7$. The fresh exact sufficient ratio is

$$R_\ell(q) < 2000/1201 - 77980/458949 < 3/2, \qquad q < 51/20,$$

with positive margin $5105207/(2402 \cdot 458949)$. Deleting a central site leaves two paths of rank at least four. The COMPLETE white list is isolation, permitted endpoint/D/E simple three, endpoint four, or one endpoint-three on EACH path. Connected rank twelve precludes exceptional completions; its two long arms preclude a fork completion. Every spectator contribution remains in the actual retained finite Gram.

The complete own-zero residual repairs are

A^2	Q bound
$Q - 7/13$	$Q < 3/5$
$Q - 2/13$	$Q < 2/5$
$Q - 1/13$	$Q < 1/5.$

The first positive time is $< 4/65 < 1/4$, below any foreign contribution. The second permits at most one foreign component, with contribution $Q > A^2$. Both violate the actual reciprocal orthogonal opposite law. The last has no possible foreign contact. At $Q = 0$ their times are negative. Internal-three/four/two-contact cases have the actual affine arms/star/cycle. ALL isolated owns three/four/six and parallel retain their positive future nulls before splitting the foreign cost.

The ordinary nonisolated ratios exceed $38/25 > 3/2$, including the same first three side-order-five comparisons printed above, and two-path $q/2 + 1/q > 97/60$. Foreign far/near bounds are still $227/(25\sqrt{35})$, $\sqrt{121/35}$. Thus every weak nonisolated cap is PURE: an endpoint helper, or OWN-four/SIMPLE-three near helper, with BOTH orientations retained. The roles are not interchanged.

For an endpoint helper the exact ratio-square identity gives $t = 2\gamma < 17/8$. Non-five own has $t > 9/5$. At $N = 13, i = 5$, the signed conjugate Schur costs at $-.83, .5$ exceed four by $69/1625, 2/13$, and they increase across the half-window. Hence every other conjugate lies in $(-.83, .5)$. The COMPLETE all-degree trace/norm proof of Section J.2 uses only these bounds, $9/5 < t < 9/4$, actual integrality and all lifted embeddings. It excludes every non-five helper. A weak endpoint helper is therefore golden, actual $K = \mathbb{Q}(\varphi)$. A near-four simple-three normalization instead forces actual $K = \mathbb{Q}(\sqrt{2})$, so it cannot coexist with a golden helper.

The COMPLETE $26A^2$ golden time rows are

type	$26A^2$	$\mathrm{N}(26A^2)$
near 5	$26 + 56\varphi$	-1004
near 6	$28 + 56\varphi$	-784
far 5	$26 + 50\varphi$	-524
far 6	$28 + 54\varphi$	$-620.$

The four norm-squareclasses differ. Reflected cyclic parameters are $21\varphi/13, 9\varphi/13$ at five, and have constant $-9/13$, coefficient $8/13$ or $4/13$ at six. All are nonintegral in the full golden ring. Thus at most ONE golden helper occurs. Every other central site is .65-strong, by the complete geometric inventory at that loss; another weak helper would also be .6005-weak.

Use its ACTUAL H_4 correction and positive finite projected point as before, with $\delta = (5 - 3\varphi)/12 < .111^2$. For helper at five, take $a = 1.830, b = 1.891$, heights $H_6 = H_7 = b, H_8 = a$, linear tails. Its energy

$$E = (b^2/6 + a^2/5 + .061^2)/2 = .6347405833\ldots$$

has individual coefficients $b/12 > .15758, .0305, .1525$. The far/near flat-block projections are $(5\gamma/6)\Omega_6 + \Omega_8/10$ and $(5\gamma/6 + 1/12)\Omega_6$. The reference product is < 1.459425, smaller actual time > 2.027, deficit $> .45$. FULL residual norm is $< .8$. Pay $\mu = .2$: largest own price $< .151667$, far other price .02, both individually affordable. ONE square gives $e_A > E + .148 > 5/7$.

For helper at six, take only $H_5 = H_8 = a$. Energy $a^2/5 = .66978$, both outer prices .183. The flat middle block has three coordinates. The projections are $(2\gamma/3)\Omega_5 + (\gamma/3 + .1)\Omega_8$ or the reverse placement of the .1. The reference product is < 1.6653; smaller actual time > 2.106, deficit $> .329$. FULL residual norm is $\gamma^2/3 + 1/15 < .286$. At $\mu = .1$, each price is $< .064 < .183$. ONE square gives $.66978 + .2(.329) - .01(.286) = .73272 > 5/7$. Reflections merely reverse labels. No golden helper occurs under the contrary energy assumption.

For near-four side order m, its exact ratio

$$R_m = \frac{1/\sqrt{2} + (m-1)/(2m)}{\sqrt{(m+1)/(2m)}}$$

increases in m. At seven, $R_7^2 = 67/56 + 3\sqrt{2}/4 > 9/4$, by $42\sqrt{2} > 59$. Weak near-four therefore has only own/helper positions $5/4, 6/5$ or their reflections. Deleting the ACTUAL helper $j = 4, 5$ gives a finite B completion and reciprocal endpoint:

$$\delta_j = \frac{1}{13 - j}, \quad \sigma_j = \frac{1}{2} + \frac{12 - j}{(13 - j)\sqrt{2}}, \quad C_j = \sigma_j \sqrt{(13 - j)D_j} > 31/4.$$

Positive finite projection checks ALL original walls and the reciprocal law identifies the SAME actual opposite. Its normalized height is $> .878$. On the opposite side, index eight has height $> .75$ from loss $< .618$, or use its ACTUAL near-four mate nine with height $> .878$. The two normalized duals have correlation $\le 5/8$, so

$$e_A > \frac{(.878 + .75)^2}{2(1 + 5/8)} > 4/5.$$

No weak near-four remains.

For isolated caps keep the entire Q. Pure five has the actual LEFT H_4 correction with $u = \gamma^{-2} < 25/16$, $\sqrt{\delta_5}/\gamma < 5/24$. The divided displacement is the old concave combination at barycentre i, so it is bounded above by h_i. At this ACTUAL $D_i \ge 80/13$,

$$q + \sqrt{D_i - 25/16} > 12/5 + 2 = 22/5,$$

and its loss is

$$< 125/352 + 5/24 = 595/1056 < .57.$$

This proves the exact local bound used in the preceding N14 theorem; no larger-rank denominator is assumed.

EVERY pure finite native at least seven uses the positive adjacent core $c+\gamma^{-1}n_i+2(\gamma^{-1}-\gamma)n_j$. It is zero on cap and n_i, negative on n_j, future timelike and nonpositive at ALL omitted original walls. Its pressure loses

$$< \frac{4/3}{q+\sqrt{D_i-4/3}} + 1/\sqrt{12} < .286 + .289 = .575 < \ell.$$

Here $q > 2.48$, $\sqrt{D_i - 4/3} > 2.195$. For $Q > 0, u > 1$, only own five is possible, since every other remaining finite $\gamma > .9$ has $u \le .75/.81 < 1$. Positive finite clearing gives loss $< .212 + .383 = .595$. At $u = 1$ its actual vector is a forbidden future feasible null. Retain ALL $0 < u < 1$, pure/foreign, with pressure

$$x_A \cdot (\Omega_i - un_i) \ge \sqrt{D_i - u} - \sqrt{u(1-u)}.$$

Its loss is $< a_i u + \sqrt{u(1-u)}$, where $a_{5,8} < .212$, $a_{6,7} < .205$; the raw circle maximum $< .6175$ supplies the mate argument above.

Fix the symmetric reference

$$H_5 = H_8 = 1.8795, \quad H_6 = H_7 = 1.9405, \qquad \beta_{5,8} = .15745, \quad \beta_{6,7} = .0305, \quad E = .71022505.$$

All heights are strictly below the true strong floors. The FULL continuous adjusted-loss estimate

$$a_i u + \sqrt{u(1-u)} - (\beta_i/2)(2u - u^2) < .6005$$

holds even for ANY outer $\beta_i \ge .08$. For outer sites below .4, raw loss $\le .5848$; above it, $.6175 - .04(.64) = .5919$. For inner sites the expression is at most $.1745u + .01525u^2 + \sqrt{u(1-u)}$. On $(0, .5]$ it is $\le .5910625$. On $[.5, .7]$, $u^2 \le 1.2u - .35$ and the circle bound with coefficient .1928 give $< .6003125$. On $[.7, 1)$, $\sqrt{u(1-u)} < (3-u)/5$, by positive square difference $(2u-1)(13u-9)/25$, giving $< .6$. Thus EVERY endpoint of the continuous split and every foot parameter are covered.

We record the full JOINT endpoint/foot rule. Let $\bar{Z}$ sum to one, supported on the first at most FOUR coordinates, and let its ACTUAL pure pressure be $x_A \cdot Z \ge h$, where $Z = \bar{Z} - P_{13}/13$. Its flat-end projection is $\Pi Z = \Omega_5/(5\sqrt{2})$. For $Y = Z - \Pi Z$,

$$w \cdot Y = 0, \quad S = Y^2 = \|\bar{Z}\|^2 - 1/5, \quad b = w \cdot \Pi Z < .266.$$

For $0 \le \mu \le .4$, spend $\mu/(5\sqrt{2})$ from INDIVIDUAL β_5. Its remainder is $> .1008 > .08$; the opposite outer and both inner coefficients are unchanged. For EVERY weak foot use this REMAINING $\beta_i' = (w + \mu Y) \cdot n_i$, and form

$$T = w + \mu Y - \sum \beta_i' u_i n_i.$$

Primitive pressures and ALL old-root crosses give

$$x_A \cdot T \ge E + \mu(h - b) - \sum \beta_i'(L_i - \ell),$$
$$T^2 \le E + \mu^2 S - \sum (\beta_i')^2 (2u_i - u_i^2).$$

ONE square and the just-proved adjusted losses yield

$$e_A > E + 2\mu(h - b) - \mu^2 S. \tag{A12P}$$

Endpoint/foot crosses have not been discarded: they are paid through the REMAINING individual coefficients. Reflection gives the same rule at the other end.

If actual $x_1 \geq .34$, take $\bar{Z} = e_1$, $S = 4/5, \mu = .1$. Then $e_A > .71702505 > 5/7$, impossible. Likewise $-x_{13} < .34$. The two ACTUAL endpoint edges therefore have half-tangent square

$$< (13/12)(.34)^2 = 3757/30000 < 3/23,$$

hence cosh $< 13/10$. The strict rational margin is 3589/690000.

Deleting endpoint one retains A_{11} and ALL spectators. Its COMPLETE finite list is isolation, simple three at global $j = 2, 3, 11, 12$, or four at $j = 2, 12$. The six pure coshes for positive own γ are respectively

$$\frac{24\gamma + 11}{13},\ \frac{24\gamma + 10}{2\sqrt{13}},\ \frac{24\gamma + 2}{2\sqrt{13}},\ \frac{24\gamma + 1}{13},\ \frac{24\gamma + 11\sqrt{2}}{\sqrt{26}},\ \frac{24\gamma + \sqrt{2}}{\sqrt{26}}.$$

Foreign Q only increases them. At $\gamma \geq 1/2$, all but far-three exceed 1.3; far-three/own-three has its actual affine cycle, and every remaining own gives $(12\sqrt{2} + 1)/13 > 1.3$. For own zero with white three, the spatial directional square is $(13-j)^2/312 \leq 121/312$. The actual short length bound gives time square $< 69(121)/(169 \cdot 312) < 1/4$. Any positive foreign contribution violates the reciprocal rule. At $Q = 0$, the remaining times are negative or the actual affine $1, 2, 5$ principal. White-four at two has its affine star of squared weights $1/2, 1/4, 1/4$; far twelve has $Q < 1/12$, hence $Q = 0$ and negative time. Every own-zero and nonisolated role is excluded.

Each ACTUAL endpoint is therefore isolated within A_{12}, with foreign contacts retained:

$$c = Ap - \gamma\Omega_1 - v_S, \quad u = (1 - Q)/\gamma^2, \quad D_1 = 24/13, \quad C^2 = D_1/u < 169/100.$$

Thus $u > 2400/2197 > 1$, excluding the dotted/parallel/$u \leq 1$ domain by an ACTUAL edge bound. Own three has $Q < 1 - 600/2197 < 3/4$, at most two contacted spectator components. Their largest contribution is at least $Q/2$, and $Q/2 > A^2 = Q - 7/13$, forbidden by reciprocity; at $Q = 0$ time is negative. Own four has $Q < 1 - 1200/2197 < 1/2$, so at most one foreign contribution $Q > A^2 = Q - 1/13$. Own six has the actual $6, 3$ null. Only finite owns five or at least seven remain.

For pure five use the ACTUAL positive finite H_4 projection on c, n_1, n_2, n_3, with $\delta = (5-3\varphi)/8 > 0$. Its sum-one displacement is $\bar{Z} = P_4/4$, $S = 1/20$, and pressure

$$h = \frac{A^2 + \delta - \sqrt{\delta(A^2 + \delta)}}{A\gamma\sqrt{2}}, \quad A^2 = (24/13)\gamma^2 - 1, \quad A^2 + \delta = 9\gamma^2/26.$$

The exact brackets $.6545 < \gamma^2 < .65475$, $.2083 < A^2 < .2088$, $A < .457$, $.017875 < \delta < .01825$, $A^2 + \delta > .2265$ give $\sqrt{\delta(A^2 + \delta)} < .0644$ and $A\gamma\sqrt{2} < .525$. Hence $h > .308$. Payment (A12P) at $\mu = .4$ gains $> .0256$.

For EVERY pure finite own at least seven use the ACTUAL adjacent negative core at one/two. Its sum-one displacement and pressure are

$$\bar{Z} = (1 - u/2)(e_1 + e_2) + (u - 1)e_3, \quad h = \frac{\sqrt{D_1 - u} - \sqrt{7u - 4 - 3u^2}}{\sqrt{2}}, \quad S \leq .3.$$

Since $u < 100/81$, $D_1 - u > 644/1053 > .782^2$; the other radical is $< .289$. Thus $h > .348$, and $\mu = .4$ gains $> .8(.348 - .266) - .16(.3) = .0176$. This covers native eight and the whole finite tail without a label cutoff.

For EVERY positive-foreign endpoint, $u > 1$ forces native five. Clear ALL contacted spectator normals positively with n_1. The actual finite projection gives the simpler pure pressure

$$x_A \cdot (\Omega_1 - n_1) > \sqrt{D_1 - 1} - \sqrt{u - 1}.$$

Its stronger exact pressure has additional positive factor $\sqrt{D_1-1}/\sqrt{D_1-u}>1$, so the displayed lower bound is valid. Here $\bar{Z}=(e_1+e_2)/2$, $S=.3$, $\sqrt{11/13}>.919$, $\sqrt{u-1}<.383$, so $h>.378$. At $\mu=.3$, gain $>.0402$.

All these points have positive ACTUAL wall coefficients, correct selected products, positive future time and nonpositive products at EVERY omitted original wall. Every $\bar{Z}$ has the support required by the JOINT payment rule. Even the weakest endpoint gain gives the CONTRARY-branch bound $.71022505+.0176=.72782505>5/7$. Thus the assumed $e_A\le 5/7$ is impossible:

$$e_{A_{12}}>5/7 \tag{A12U}$$

with every actual field and arbitrary spectators. The number .72782505 is not asserted as an unconditional component rate.

This one stronger theorem closes the rank-sixteen ordinary-rank-four and actual-$I+T_3$ rows, whose disjoint or JOINT remainder costs $>2/7$. It also closes BOTH typed-seven rank-two rows: the actual defect costs $>.09$ and propagates the remaining ordinary $T_2>.2$, giving $5/7+.09+.2>1$. The weaker earlier typed-only component estimate therefore needs no separate chapter. In rank seventeen it closes every older A_{12} row by the actual rank-five spectator partitions and rank-free joint costs.

J.6 A rank-eleven chain: the complete local alternatives

We next prove a uniform estimate at the smaller path size, and then a joint estimate when an ACTUAL short isolated I is present. Both arguments use the original vector x; the second will rebuild its reference rather than add a new gain to an optimized first one. Write $N=12$, $A=A_{11}$, and use

$$n_i=(e_i-e_{i+1})/\sqrt{2},\qquad D_i=\frac{2i(12-i)}{12},\qquad C_{ij}=\frac{2i(12-j)}{12}\quad(i\le j).$$

Only sites $5,6,7$ will be paid. Put $\ell=.61$, $q_i=\sqrt{D_i}$. Then

$$D_5=D_7=35/6,\quad D_6=6,\qquad a_i=\frac{1}{q_i+\sqrt{D_i-1}}<.217. \tag{A11a}$$

Indeed $q_i>2.415$, $\sqrt{D_i-1}>2.198$, and $(2.415+2.198)(.217)=1.001021>1$. A site is strong if $h_i>q_i-\ell$.

Deleting one of these sites retains two paths of ranks at least four. The complete one-node finite extensions are: no white contact; one admissible endpoint, D- or E-position three; one terminal four; or two endpoint threes joining the paths. Two contacts to one path make a forbidden cycle. A connected completion has rank eleven, so exceptional finite diagrams cannot add a missing joined-path role. We retain the ENTIRE contacted spectator forest, writing its squared projection as Q; every positive whole-component contribution is at least $1/4$.

The own-zero cases are excluded before any positive-own ratio is used. An internal three contains the actual affine arms $1,2,5$ or $1,3,3$; an internal four contains the affine $3,3,4$ star; two white threes contain the affine cycle. The remaining far-three role, with side order $m\ge 5$, has time square $<1/(2m)$, below any necessary positive foreign contribution. The next-to-far three has $Q<2/m$, hence at most one contacted whole spectator, and time square $Q-1/6<Q$. Either would make this same actual wall close an edge in that orthogonal spectator, contrary to the actual reciprocal-opposite law. With $Q=0$ its time is negative. Far-four has $Q<1/m\le 1/5$, so $Q=0$ and negative time. These are all residual own-zero roles. The affine vectors have positive time through the actual cap and nonpositive product with every omitted original wall. They are forbidden feasible future nulls, even in the presence of foreign contacts.

For $q \ge 12/5$, the sufficient strong-edge ratio is

$$R_\ell(q) = \frac{100}{61} - \frac{q - .61}{q(2q - .61)}.$$

It increases with q, and

$$R_\ell(q) \le R_\ell(3) < 3/2, \qquad \frac{3}{2} - R_\ell(3) = \frac{1669}{197274} > 0.$$

The complete near-three, D/E, terminal-four and two-white bridge formulas have, respectively, the lower comparisons

$$\frac{.808 + .4}{\sqrt{.6}}, \quad \frac{.808 + .2}{\sqrt{.4}}, \quad \frac{.808}{\sqrt{.25}}, \quad q/2 + 1/q > 97/60.$$

All exceed $38/25 > 3/2$. The smaller positive owns have their actual affine cycle, star or 6–3 obstruction. For the two exceptional profiles WITH $Q > 0$, clearing the entire spectator gives ratio

$$> \frac{227}{25\sqrt{35}} > 38/25 \quad \text{or ratio square } > 121/35.$$

Consequently a weak NONISOLATED cap is PURE, and is either a simple-three END-helper

$$c = A_c p - \gamma \Omega_i - \Omega_e/2, \qquad e = 1 \text{ or } 11, \tag{A11helper}$$

or own-four/simple-three near helper

$$c = A_c p - \Omega_i/\sqrt{2} - \Omega_j/2, \qquad |i - j| = 1. \tag{A11near4}$$

Both orientations are retained. The near-four ratio increases with side order; from order eight it already exceeds $3/2$. Thus its complete list, up to reflection, is $(i, j) = (5, 4), (6, 5), (5, 6)$. It would be incorrect to exclude these three roles by the larger-path conjugate test.

For an isolated-within-A cap retain

$$c = A_c p - \gamma \Omega_i - v_S, \quad Q = \|v_S\|^2, \quad u = (1 - Q)/\gamma^2.$$

Before splitting Q, own-three, four, six and parallel have their actual affine E_7 arms $1, 3, 3$, weighted star, 6–3 path or positive null pair. These nulls are feasible against ALL foreign walls by acuteness. Hence every remaining positive own has $\gamma \ge \varphi/2$, $\varphi = (1 + \sqrt{5})/2$. For $Q > 0, u > 1$, positive clearing gives loss

$$< a_i + \sqrt{u - 1} < .217 + (2 - \varphi) < .599 < \ell.$$

The actual $u = 1$ vector is a forbidden future feasible null.

The PURE finite cases with $u > 1$ are handled by actual points at this rank. For own five the same positive H_4 projection used above has loss

$$< \frac{25/16}{q_i + \sqrt{D_i - 25/16}} + 5/24 < \frac{625}{1792} + \frac{5}{24} = \frac{2995}{5376} < .558.$$

Here the denominator is $> 2.415 + 2.066 > 4.48$; $2.066^2 < 205/48 = D_{\min} - 25/16$. For EVERY finite native at least seven put $d = 1 - \gamma^2$ and choose an old adjacent root n_j. The positive actual combination

$$\xi = c + \gamma^{-1} n_i + (2d/\gamma) n_j$$

is zero on c, n_i, negative on n_j, and has square $-d(1-4d)/\gamma^2 < 0$. It is future timelike and all omitted ORIGINAL wall products are nonpositive. Its comparison gives loss

$$< \frac{100/81}{q_i + \sqrt{D_i - 100/81}} + \frac{25}{81} < \frac{213975}{369279} < .58 < \ell.$$

We used $\gamma > .9$, $\sqrt{D_i - 100/81} > 2.144$ and $\sqrt{d(1-4d)} \le 1/4$. This includes native eight and the entire unbounded finite-native tail.

Retain EVERY remaining $0 < u < 1$, pure or foreign. Positive inverse-Gram clearing expresses v_S as a positive combination of all contacted old spectator roots. The actual vector

$$\eta = c + v_S + \gamma u n_i = \gamma\{\sqrt{D_i - u}\, p - (\Omega_i - u n_i)\}$$

has positive time, square $-\gamma^2 u(1-u)$, and zero products on the cap and cleared spectators. Its product on n_i is $-\gamma(1-u) < 0$; all other original walls have nonpositive product termwise. It therefore supplies the SAME-x pressure

$$x_A \cdot (\Omega_i - u n_i) \ge \sqrt{D_i - u} - \sqrt{u(1-u)} = q_i - L_i, \qquad L_i < .217u + \sqrt{u(1-u)}. \qquad \text{(A11foot)}$$

No restriction on the number or adjacency of these isolated feet is made.

J.7 The all-degree end-helper reduction at $N = 12$

For (A11helper), the exact side-order identity is

$$\frac{(\gamma + 1/(2m))^2}{(m+1)/(2m)} - (4\gamma - 2) = \frac{(\gamma - 1 - 1/(2m))^2}{(m+1)/(2m)} \ge 0.$$

Weakness gives $\varphi \le t = 2\gamma < 17/8$; if the own is not five then $t > 9/5$. The actual simple-three normalization makes t integral in K. For a lifted nonphysical embedding of $\mathbb{Q}(t)$, strict old-A Schur positivity gives the cheapest signed costs

$$Q_-(s) = 2is^2 + 4s + 2 - 2(is+1)^2/12, \qquad Q_+(s) = 2is^2 + 2 - 2(is-1)^2/12.$$

They increase with i on the relevant half-window. At $i = 5$,

$$Q_-(-17/20) - 4 = 31/480 > 0, \qquad Q_+(1/2) - 4 = 1/8 > 0.$$

Their values at zero are below four, and the outer quadratic branches increase away from their allowed interval. Thus EVERY other conjugate obeys

$$-17/20 < t_\sigma < 1/2. \qquad \text{(A11strip)}$$

We give the full degree argument, including degree five. In degree at least six a nonzero integral norm has magnitude $< (17/8)(17/20)^5 < 1$, impossible. In degrees three through five its magnitude is less than two. One positive small conjugate would instead make it $< (17/8)(1/2)(17/20)^{a-2} < 1$. All small conjugates are therefore negative and their absolute product is $1/t$. AM–GM bounds their trace above by $t - (a-1)t^{-1/(a-1)}$. The resulting strict trace intervals are

a	trace interval	upper comparison
3	$(0, 1)$	$t(t-1)^2 < (17/8)(9/8)^2 < 4$
4	$(-1, 0)$	$t^4 < (17/8)^4 < 27$
5	$(-2, -1)$	$t(t+1)^4 < (17/8)(25/8)^4 < 256.$

The lower bounds follow from $t > 9/5$ and each small conjugate $> -.85$. None contains an integer. In degree two the norm is 1 or -1. A negative small conjugate gives trace $t - 1/t \in (1, 2)$. A positive one forces $t > 2$ and trace $t + 1/t \in (2, 3)$. Degree one leaves only $t = 2$, the forbidden actual parallel pair. Thus every weak end-helper is native five.

Its actual connected old-path-plus-cap Gram, normalized by its simple three, is Lorentzian over $\mathbb{Q}(\varphi)$. A proper totally real extension would have a nonphysical embedding fixing this negative direction. Admissibility forbids it: the ACTUAL field is $\mathbb{Q}(\varphi)$. A near-four simple-three cap similarly forces ACTUAL $K = \mathbb{Q}(\sqrt{2})$. These restrictions apply only in the stated exceptional branches; no field list is assumed for the other caps.

J.8 Uniform A_{11} energy, paying a helper and all feet

We prove $e_{A_{11}} > 13/20$ with arbitrary finite spectators and every actual field. Assume contrariwise $e_A \le .65$. First remove the retained near-four cases by their OWN actual mate, not a field ban. For (A11near4) the three complete time squares, up to reflection, are

$$\frac{39 + 28\sqrt{2}}{12}, \qquad \frac{83 + 60\sqrt{2}}{24}, \qquad \frac{41 + 30\sqrt{2}}{12}. \tag{A11silver}$$

Each exceeds six. With $v = \Omega_i/\sqrt{2} + \Omega_j/2$ and $T_j = (v \cdot \Omega_j)/(A_c\sqrt{D_j})$, direct inverse-Gram multiplication gives

$$T_j^2 - 1 = \begin{cases} 1/((12 - j)A_c^2), & j = i - 1, \\ 1/(jA_c^2), & j = i + 1. \end{cases}$$

In all three roles and their reflections the integer factor is at least six. Consequently $1 < T_j^2 < 37/36 < (51/50)^2$. The full old-root replacement at j is positive definite, its final Schur complement being $(v \cdot \Omega_j/\sqrt{D_j})^2 - A_c^2 > 0$. Its actual off-diagonal entries are nonpositive. Positive finite inverse projection of the strict interior point gives a common point strict against every other ORIGINAL wall. Hence this cap is the ACTUAL opposite at j, rather than a formal cutter behind another facet. Its half-tangent pressure gives

$$x_A \cdot \frac{\Omega_j}{\sqrt{D_j}} \ge T_j - \sqrt{T_j^2 - 1} > \frac{51 - \sqrt{101}}{50} > .81.$$

Thus $e_A > .6561 > .65$, a contradiction. Every weak nonisolated exception is now golden.

The complete golden time table is

type	$24A_c^2$	$\mathrm{N}(24A_c^2)$
5, near	$22 + 49\varphi$	-839
6, either end	$23 + 48\varphi$	$-671 = -11 \cdot 61$
5, far	$22 + 45\varphi$	$-551 = -19 \cdot 29.$

(A11gold)

Both actual norm-two helper caps lie on the same K-time line; their time-square ratio must be a K-square. The three norm squareclasses are distinct, so unlike rows cannot coexist. Two center-six caps cannot be the distinct actual opposites of the same edge. For the reflected near-five and far-five pairs the actual closed helper paths give raw cyclic products $(2 - 7\varphi)/6$ and $(2 - 3\varphi)/6$. Neither is integral in the full ring $\mathbb{Z}[\varphi]$. Thus at most ONE central index has a weak golden helper. This does not limit the isolated feet of (A11foot).

Set $a = 1.805, b = 1.839$. With heights linear on the unmentioned outer tails, there are three reference cases:

helper	positive $\beta_i = w \cdot n_i$	$E = w^2$
none	$\beta_5 = \beta_7 = .1635,\ \beta_6 = .034$	.652761
6	$\beta_5 = \beta_7 = .1805$	.651605
5	$\beta_6 = .17025,\ \beta_7 = .1635$	.60820725.

(A11refs)

In the first case $H_5 = H_7 = a, H_6 = b$. For the second replace H_6 by a, so its helper coefficient is zero. For the third flatten heights on $0, \dots, 6$, with $H_6 = b, H_7 = a$; reflect a helper at seven. Its energy is exactly $(b^2/6 + (b-a)^2 + a^2/5)/2$. All paid heights are strictly below the strong floors, since $2.415^2 < 35/6$ and $2.449^2 < 6$.

Only the end-helper case needs a helper payment. For the actual helper at five put $g = \varphi/2$, let $e = 1$ or 11 and let $j = 2$ or 10 be its adjacent old root. The ACTUAL finite H_4 projection uses

$$w_{\text{old}} = gn_5 + (2/3)n_e + (1/3)n_j, \qquad \delta = 2/3 - g^2 > 0, \qquad U = v - w_{\text{old}}.$$

The positive vector

$$\xi = p + (A_c/\delta)(c + w_{\text{old}})$$

has selected products zero, square $-1 - A_c^2/\delta$, and positive time. Every unselected original wall has nonpositive product, by positivity and acuteness. Comparison with the ORIGINAL p, x gives

$$x_A \cdot U > A_c - \sqrt{\delta}, \qquad U \cdot n_k \geq 0 \quad \text{for ALL old } k. \tag{A11U}$$

The second conclusion follows directly from the selected zero and unselected nonpositive products of the actual projected point. The correction U alone is not claimed to be a hyperbolic point.

Let Π average the first six and last five coordinates, leaving coordinate seven fixed. Then w of the last row of (A11refs) lies in its range. The correction roots have zero block averages. For $Y = U - \Pi U$, direct block variance gives

	ΠU	Y^2
near	$(5g/6 + 1/12)\Omega_6$	$(2/3)g^2 + g/3 + 1/12$
far	$(5g/6)\Omega_6 + \Omega_7/10$	$(2/3)g^2 + 1/15.$

In particular $w \cdot Y = 0$, $Y^2 < .8$. For example the far variances of Ω_5, Ω_{11} are $5/3, 8/5$, their cross variance is zero, and the old correction contributes $-g^2 - 1/3$; the near cross variance contributes $g/3$. These are FULL residual norms. The smaller actual time square is $(35g^2 + 5g)/6 - 13/24$. Using $.809 < g < .81$ gives

$$A_c > 1.987, \quad \sqrt{\delta} < .111, \qquad w \cdot \Pi U \leq .675b + .1a = 1.421825$$

in the far case; in the near case the product is $< (91/120)b = 1.394575$. Both actual helper deficits therefore satisfy

$$\Delta = A_c - \sqrt{\delta} - w \cdot \Pi U > .454175 > .45.$$

Pay $\mu = .1$. Writing $\Pi U = \sum a_i \Omega_i$, define

$$W = w + \mu Y, \qquad b_i = \beta_i - \mu a_i.$$

The near debit at six is $< .075834$. The far debits are $< .0675$ at six and .01 at seven. Each remaining positive b_i exceeds .034, and no helper index is paid. Crucially (A11U) gives

$$W \cdot n_i = b_i + \mu U \cdot n_i \geq b_i. \tag{A11price}$$

This sign retains every helper/foot cross. It is not replaced by an unsupported orthogonality assertion. In the no-helper and center-helper cases set $W = w, b_i = \beta_i, \mu = 0$.

Let S contain ALL retained isolated feet at indices with positive b_i, and set

$$Z = W - \sum_{i \in S} b_i u_i n_i.$$

Before assigning any floor the helper debit has been removed from each individual coefficient. Combining its pressure with all strong and actual foot pressures gives

$$x_A \cdot Z \geq E + \mu\Delta - \sum_{i \in S} b_i (L_i - \ell).$$

Since distinct old normals have nonpositive products, (A11price) gives, INCLUDING adjacent-foot crosses,

$$Z^2 \leq E + \mu^2 Y^2 - \sum_{i \in S} b_i^2 (2u_i - u_i^2).$$

For $0 < u \leq 1/2$, raw loss is $< .217/2 + 1/2 = .6085$. For $1/2 \leq u < 1$, the circle bound and $\sqrt{1 + .217^2} < 1.024$ give

$$L_i - \frac{b_i}{2}(2u - u^2) < \frac{.217 + 1.024}{2} - \frac{3(.034)}{8} = .60775 < \ell.$$

One completed square therefore yields

$$e_A \geq 2x_A \cdot Z - Z^2 > E + 2\mu\Delta - \mu^2 Y^2.$$

If no foot occurs, strict paid strong/helper pressures give the same strictness. The three cases in (A11refs) give respectively $.652761, .651605$, and $.60820725 + .2(.45) - .01(.8) = .69020725$. Each exceeds .65, contradicting the contrary assumption. We have proved

$$e_{A_{11}} > 13/20. \tag{A11Urate}$$

The ordinary-rank-five remainder costs $> 5/14$, so $13/20 + 5/14 = 141/140 > 1$. The typed-seven plus ordinary A_3 alternative also closes directly: its distinct whole blocks cost $> .09$ and $> .26$. The actual-$I + T_4$ case needs the joint argument that follows.

J.9 Actual I: six time-denominator obstructions

Suppose an ACTUAL isolated short I spectator has opposite

$$c_I = \sqrt{h}\, p - \sqrt{1 + h}\, e_I, \qquad 0 < h < 1/2.$$

The common $N = 12$ local inventory above was proved before assuming this wall exists. Its only nonisolated weak exceptions are exactly the three golden times in (A11gold) and the three silver times in (A11silver). We now exclude ALL six by actual cycles.

For completeness write $A_c^2 = M/L$. Their fields, denominators and numerator norms are

type	K	M/L	N(M)
5, near	$\mathbb{Q}(\varphi)$	$(22 + 49\varphi)/24$	-839
6	$\mathbb{Q}(\varphi)$	$(23 + 48\varphi)/24$	-671
5, far	$\mathbb{Q}(\varphi)$	$(22 + 45\varphi)/24$	-551
$(5,4)$	$\mathbb{Q}(\sqrt{2})$	$(39 + 28\sqrt{2})/12$	-47
$(6,5)$	$\mathbb{Q}(\sqrt{2})$	$(83 + 60\sqrt{2})/24$	-311
$(5,6)$	$\mathbb{Q}(\sqrt{2})$	$(41 + 30\sqrt{2})/12$	-119.

The golden numerators follow from

$$M_N = i(12-i) - 13 + (i+2)(12-i)\varphi, \quad M_F = i(12-i) - 13 + i(14-i)\varphi;$$

the silver formula is $A_c^2 = D_i/2 + C_{ij}/\sqrt{2} + D_j/4 - 1$. The actual fields equal the displayed quadratic fields by the connected Lorentzian-principal argument already proved. The rings of integers are respectively $\mathbb{Z}[\varphi]$, $\mathbb{Z}[\sqrt{2}]$. In particular this is not a field assumption in the remaining no-exception branch.

The own two-cycle of c_I makes $v = 4h$ integral. Strict positivity of its two-root Gram at the other embedding gives $-4 < v' < 0$; physically $0 < v < 2$. Thus

$$0 < |\mathrm{N}(v)| < 8.$$

Every listed cap is PURE A_{11}, whereas c_I has spatial part only in I. Their product is $-A_c\sqrt{h}$. The scale-invariant actual two-cycle therefore makes $A_c^2 v = Mv/L$ integral. Each displayed $\mathrm{N}(M)$ is coprime to 6, while L is 12 or 24. No principal-ideal hypothesis is needed: multiply by the conjugate of M and use an integer Bezout relation for $\mathrm{N}(M)$ and L. It follows that $v \in L\mathcal{O}_K$. Its nonzero norm is then divisible by $L^2 \geq 144$, contradicting the bound eight. All six types and their reflected orientations are excluded. Consequently every paid central site is strong or has its actual directional foot (A11foot), with no restriction on the actual field in this remaining branch.

J.10 One actual endpoint and a joint $A_{11} + I$ square

The rank-free actual-I edge parameters are

$$u^2 = \frac{\sqrt{1+h} - 1}{\sqrt{1+h} + 1}, \qquad t = \sqrt{1+4h} - 2\sqrt{h}, \qquad 0 < u < 1/3.$$

The original I edge gives $E_I \geq u^2$. Let $d = A_d p - v_A - \beta e_I - v_{\mathrm{other}}$ be the ACTUAL opposite on the first A_{11} edge. It differs from c_I, which stays strictly negative on the entire outgoing A_{11} ray. Acuteness gives

$$-d \cdot c_I = A_d\sqrt{h} - \beta\sqrt{1+h} \geq 0.$$

If $\beta = 0$, this positive actual-wall mutual magnitude is at least $1/2$ by the native gap. If $\beta > 0$, its finite contact with the retained old I root has $\beta \geq 1/2$, so $A_d\sqrt{h} > 1/2$. Thus $A_d\sqrt{h} \geq 1/2$ in every white, own and foreign role. For the actual inward unit first-edge direction $U_1 = \Omega_1/\sqrt{11/6}$, the ENTIRE spatial vector of d has norm $\sqrt{1 + A_d^2}$. Therefore

$$\tanh L = \frac{A_d}{v_{\mathrm{total}} \cdot U_1} \geq \frac{A_d}{\sqrt{1 + A_d^2}} \geq \frac{1}{\sqrt{1+4h}}.$$

Taking the increasing half-tangent yields

$$x_A \cdot U_1 \geq t, \qquad x_1 \geq \alpha t, \quad \alpha = \sqrt{11/12}. \tag{A11Iend}$$

Only physical unit-normal geometry is used; no disconnected raw-root scales are identified.

The exact relation between u, t is

$$(1-u^2)(1-t^2) = 8ut.$$

It implies $t + (3/2)u > 3/4$. Indeed substituting $t_0 = 3/4 - 3u/2 > 0$ into left-minus-right and multiplying by sixteen gives

$$7 - 60u + 149u^2 - 36u^3 + 36u^4 > 7 - 60u + 137u^2 > 0.$$

The last quadratic has discriminant -236, and the relation decreases strictly in positive t. Also $t < 1$ and $t > \sqrt{3} - \sqrt{2} > .3$. These estimates hold throughout the full interval $0 < h < 1/2$.

Keep the EXACT central heights $H_5 = H_7 = 1.805, H_6 = 1.839$ and $E = .652761$. Put

$$b = \frac{361}{1000\sqrt{2}}, \quad c = \frac{.034}{\sqrt{2}}, \quad \eta = \alpha t, \quad r = (5b - \eta)/4,$$

and choose the NEW zero-sum reference

$$w = (\eta, r, r, r, r, c, -c, -b, -b, -b, -b, -b).$$

Its first-five sum is still $5b$, so its central dual heights are the three fixed H_i, while its first height $\sqrt{2}\eta$ is paid by the ACTUAL endpoint (A11Iend). The safe rational bounds are

$$.95 < \alpha < .958, \quad .255 < b < .256, \quad \eta > b, \quad \eta < .958, \quad r > .07925, \quad c < .025.$$

All follow by positive squares and $t > .3$. Hence $\eta > r > c > 0$, so all dual coefficients are nonnegative. The ONLY positive ones are

$$\beta_1 = (\eta - r)/\sqrt{2}, \quad \beta_5 = (r - c)/\sqrt{2} > 217/6000 > .034, \quad \beta_6 = .034, \quad \beta_7 = .1635.$$

In particular β_5 is the coefficient of this NEW reference, including its full endpoint/foot cross.

Let S be ANY set of retained feet among $5, 6, 7$ and form $Z = w - \sum_{i \in S} \beta_i s_i n_i$. Every new central coefficient is at least .034. The continuous adjusted-loss bound proved above therefore applies afresh:

$$L_i - \frac{\beta_i}{2}(2s_i - s_i^2) < .61.$$

Combining all foot pressures, strict strong central heights and the actual endpoint gives

$$x_A \cdot Z \geq w^2 - \sum_{i \in S} \beta_i (L_i - .61),$$
$$Z^2 \leq w^2 - \sum_{i \in S} \beta_i^2 (2s_i - s_i^2).$$

Every adjacent-foot cross is nonpositive. The single completed square proves $e_A > w^2$, including the empty-foot case by strict central floors. At the REFERENCE level, the change from the constant first-five block is $(\eta - b, -(\eta - b)/4, \ldots, -(\eta - b)/4)$; it has zero sum and zero cross with that old constant block. Thus exactly

$$w^2 = E + (5/4)(\eta - b)^2.$$

This identity is not added to an independently optimized old-foot norm.

Write $D = \eta - b > 0$. The exact I barrier and $\alpha < 1$ give

$$D + (3/2)u > (3/4)\alpha - b > 913/2000.$$

Weighted Cauchy gives

$$(D + 3u/2)^2 \leq \{(5/4)D^2 + u^2\}(4/5 + 9/4) = \frac{61}{20}\{(5/4)D^2 + u^2\}.$$

Since the ACTUAL I energy is counted ONCE, on the original vector,

$$e_{A_{11}} + E_I > \frac{652761}{1000000} + \frac{833569}{12200000} = \frac{43986266}{61000000} > \frac{18}{25}. \tag{A11Irate}$$

The positive final margin is 66266/61000000. Every ordinary $T_4 = A_4, B_4, D_4, F_4, H_4$ costs $> 2/7$, so $18/25 + 2/7 = 176/175 > 1$. This closes the entire actual-$I + T_4$ alternative with arbitrary other spectators and every actual field.

J.11 The fresh $N = 11$ window and its small-side repairs

The remaining rank-ten chain cases require a different finite-white list. At A_{10}, use zero-sum $\mathbb{R}^{11}$,

$$D_i = \frac{2i(11-i)}{11}, \qquad C_{ij} = \frac{2i(11-j)}{11} \quad (i \le j),$$

and ONLY sites $4, 5, 6, 7$. Here $D_i = 56/11$ or $60/11$, so $9/4 < q_i = \sqrt{D_i} < 12/5$. Write an actual cap as $c = A_c p - v_A - v_S$, with full spectator cost $Q = \|v_S\|^2 < 1$. Each positive whole-component contribution is at least $1/4$. The actual reciprocal-opposite law gives the stronger individual restriction

$$Q_C < A_c^2 \tag{A10foreign}$$

for each contacted whole spectator C. If $Q_C > A_c^2$, the original finite replacement on its contacted node has positive Schur residual $Q_C - A_c^2$. Positive finite projection checks EVERY original wall and would make c close two orthogonal components at the original farthest vertex. Equality instead gives the positive-cleared future null. Neither is possible. The cleared projection is never treated as an additional native facet normal.

Take loss $\ell = 13/20 = .65$. The sufficient ratio

$$T(q) = \frac{20}{13} - \frac{q - 13/20}{q(2q - 13/20)}$$

increases on the stated q-interval. Indeed its derivative has positive numerator $2q^2 - (13/5)q + 169/400$. The exact upper comparison is

$$T(q) < \frac{17645}{12948} < \frac{1363}{1000},$$

with cleared positive gap 3124 in the last cross product. Thus ratio at least 1.363 is strong.

Deleting a selected site leaves path ORDERS in $\{4, 5, 6, 7\}$. The COMPLETE finite one-node extensions are: isolation; one admissible endpoint/D/E white-three; terminal white-four; terminal white-five on an A_3, giving H_4; one endpoint-three in each path, giving A_{10}; or, at site four or its reflection seven, a middle-A_3/end-three join, giving D_{10}. This last role is a genuine new small-side case. Two contacts to one retained path make a cycle. Rank ten precludes exceptional joined completions; a weighted B_{10} join would need a rank-one end side, which is absent. An A_3 has no missing one-node F_4 extension with these unweighted old bonds.

At own zero, isolation has $A_c^2 = Q - 1 < 0$. Internal three has the affine E_8 arms $1, 2, 5$, or E_7 arms $1, 3, 3$; internal four has the weighted $4, 3, 3$ star; either two-three join has its actual all-three cycle. The positive E_7 kernel has central coefficient four, short arm two and the two other arms $3, 2, 1$. The E_8 kernel has center six and arms 3, $4, 2$, $5, 4, 3, 2, 1$. The weighted star has center $\sqrt{2}$, four-leaf one and three-leaves $1/\sqrt{2}$; cycle coefficients are all one. Every kernel is future through the positive cap time and nonpositive against all unselected ORIGINAL walls.

The residual single-contact rows are

A_c^2	strict finite-white bound
$Q-6/11$	$Q<5/8$
$Q-2/11$	$Q<1/2$
$Q-1/11$	$Q<1/4.$

In the first, positive time is $<7/88<1/4$, contradicting (A10foreign). The second permits at most one whole foreign contribution, larger than its time. The third has $Q=0$ and negative time. The white-five/own-zero case will instead be directly strong below.

Isolated owns three, four and six have the actual E_7, weighted-star or 6–3 null, BEFORE splitting Q. A parallel actual pair has its positive future null in every profile. For SINGLE white-three, own-three closes the actual all-three cycle; own-six uses an old neighbor different from the helper. Own-four has its star unless that helper is adjacent to the own site. The adjacent near-four role is retained in the next calculation. All other positive owns have $\gamma\ge\varphi/2>4/5$.

For a white-three at position t in a side of order l, discarding foreign cost only decreases the ratio, giving

$$\frac{\gamma+t/(2l)}{\sqrt{1-t(l-t)/(2l)}}.$$

Near-three exceeds $(4/5+3/8)/\sqrt{5/8}>1.363$. The D-position exceeds $(4/5+1/4)/\sqrt{1/2}>1.363$; its formula increases with $l\ge4$. Allowed E-positions have denominator at most $1/2$, hence ratio >1.6. Near own-four/simple-three has minimum at $l=4$, with

$$R_4^2=41/40+3\sqrt{2}/5>1.8734>1.363^2.$$

Its numerator increases and denominator decreases with l, so ALL these roles are strong without using a quadratic-field exclusion.

White-four has ratio $\ge\gamma\sqrt{l}$. For own at least four this is $\ge\sqrt{2}$. Near white-four/own-three already has ratio >2 at $l=4$. Far white-four/own-three has $\sqrt{l}/2+1/\sqrt{2l}$, increasing for $l\ge4$. At $l=5$ its square is $5/4+1/\sqrt{2}+1/10>1.363^2$. Only the far role at $i=4$, or its reflection seven, remains. It has $Q=0$ and

$$v_A=\Omega_4/2+\Omega_1/\sqrt{2},\qquad A_c^2=(13+7\sqrt{2})/11.$$

The actual own-three normalization and closed cycle put $\sqrt{2}$ in the field. At any embedding extending its sign reversal, the connected principal still has positive time square $(13-7\sqrt{2})/11>0$. It therefore retains a negative direction, contradicting strict nonphysical positivity. This argument needs only the contained quadratic field, not equality with it. It is the required small-side repair; the unavailable longer E_7 arm is not used to discard this role.

For white-five on A_3, put $\beta=\varphi/2>4/5$. Its white residual is $\delta_H=(7-3\sqrt{5})/16<1/50$. Even own zero has ratio $\beta/(4\sqrt{\delta_H})$, whose square exceeds two. All such profiles are strong. The A_{10} two-end-three join has ratio $\ge\gamma q_i+1/q_i>113/72$ for positive own. The D_{10} join has white cost $1/2+3/7=13/14$, so $Q=0$. Its two offsets are $19/28,9/28$; the positive-own ratio is at least $23\sqrt{14}/28>3$. All their own-zero cycles were handled above.

Finally a far-three helper WITH foreign cost has residual at most $3/8$ and numerator $>4/5+1/14=61/70$. Its ratio square exceeds $29768/14700>1.363^2$. Every possible weak NONISOLATED cap is therefore a PURE end-helper

$$v_A=\gamma\Omega_i+\Omega_e/2,\qquad e=1\text{ or }10.\tag{A10helper}$$

This conclusion is independent of both the actual-I and typed-seven hypotheses.

J.12 A small conjugate polynomial replaces the degree census

For (A10helper), the exact side-order square identity gives $4\gamma - 2 \le R^2$. Weakness and $T(q) < 1.363 < 1.37$ imply

$$\varphi \le t = 2\gamma < 38769/20000 < 39/20.$$

Actual simple-three normalization makes t integral. The old roots and cap have a common totally positive scale times the norm-two matrix with helper product -1 and own product $-t$. This full-rank connected principal is physically Lorentzian. Hence a nonphysical embedding fixing $\mathbb{Q}(t)$ would preserve a negative direction: ACTUAL $K = \mathbb{Q}(t)$.

At every other embedding the exact Schur inequality is

$$D_i t_\sigma^2 + 2C_{ie} t_\sigma + 20/11 < 4.$$

At $i = 4$, its cheapest signed costs are $(56s^2 + 28s + 20)/11$ for negative s, and $(56s^2 + 16s + 20)/11$ for positive s. At five the increases are $4(s^2 - s)/11$, $4(s^2 + s)/11$, respectively. They are positive in the stated half-windows. Boundary excesses $456/6875$ at $-24/25$ and $263/13750$ at $53/100$ imply

$$-24/25 < t_\sigma < 53/100.$$

Let $f(T) = T(T^2 - 1)$. The physical $f(t)$ is nonzero integral, with $0 < f(t) < 43719/8000$. On the ENTIRE small-conjugate interval,

$$|f(s)| \le \frac{2}{3\sqrt{3}} < \frac{77}{200};$$

the last comparison squares to $160000 < 160083$. Degree at least three would give a nonzero integral norm of magnitude

$$< \frac{43719}{8000}\left(\frac{77}{200}\right)^2 = \frac{259209951}{320000000} < 1,$$

and every further factor is smaller than one. The degree is at most two. Degree one has no integer in $[\varphi, 39/20)$. In degree two the nonzero norm has magnitude $< 1.95(.96) < 2$, hence is 1 or -1. A positive small conjugate gives integral trace $t + 1/t \in (2, 3)$, impossible. A negative one gives trace $t - 1/t \in [1, 2)$, hence exactly one. Thus

$$t = \varphi, \qquad K = \mathbb{Q}(\varphi). \tag{A10gold}$$

This is a complete all-degree argument. No field census or assumed cubic bound is used.

J.13 Every isolated A_{10} cap is strong

This part also needs neither spectator hypothesis. Keep $u = (1 - Q)/\gamma^2 > 0$. The small-own/null cases were excluded for all Q, so $\gamma \ge \varphi/2$. For $0 < u < 1$, the positive ACTUAL vector

$$\xi = c + v_S + \frac{1 - Q}{\gamma} n_i$$

has square $-(1 - Q)(1 - u) < 0$, positive time, zero selected products and negative product on n_i. The entire spectator correction is positive, so every original omitted wall is nonpositive. The original farthest comparison gives

$$h_i > \sqrt{D_i - u} - \sqrt{u(1 - u)}.$$

Here

$$a_i = \frac{1}{q_i + \sqrt{D_i - 1}} < \frac{100}{427} < \frac{47}{200},$$

since $q_i > 9/4$ and $\sqrt{D_i - 1} > 101/50$. The full continuous loss is

$$< \frac{.235 + \sqrt{1 + .235^2}}{2} < \frac{.235 + 1.028}{2} = .6315 < .65.$$

Equality $u = 1$ is the feasible future null. For $Q > 0, u > 1$, positive finite clearing has loss $< a_i + \sqrt{u - 1}$. Since $Q \geq 1/4$ and $\gamma \geq \varphi/2$, $u \leq 3/\varphi^2$, hence $\sqrt{u - 1} \leq 2 - \varphi < .4$. The loss is $< .635 < .65$.

For PURE native five the actual finite H_4 point has loss

$$< \frac{25/16}{q_i + \sqrt{D_i - 25/16}} + 5/24 < 625/1648 + 5/24 < .588.$$

The denominator is $> 2.25 + 1.87 = 4.12$. It is the positive finite projected point, not its spacelike correction, that checks all original walls. For EVERY finite native at least eight the finite two-root projected point gives, with $B = \sqrt{D_i - 1}$, $v = \sqrt{u - 1}$,

$$h_i > \frac{\sqrt{B^2 - v^2}}{1 + v/B} > B - v.$$

Its bracket is positive, and $v \leq \tan(\pi/8) = \sqrt{2} - 1 < .415$. Thus the full loss is $< .235 + .415 = .65$, covering the entire infinite native tail.

For native seven, the elementary cubic $8\gamma^3 - 4\gamma^2 - 4\gamma + 1$ and $\pi < 22/7$ give $.9 < \gamma < 10/11$. Consequently $121/100 < u < 100/81 < 4/3$. For an adjacent old root n_j, the positive actual core

$$\xi = c + \gamma u n_i + 2\gamma(u - 1)n_j$$

is zero on c, n_i, negative on n_j, and has square $-\gamma^2(7u - 4 - 3u^2) < 0$. It is future and nonpositive against all other ORIGINAL walls. Its pressure gives

$$h_i > \sqrt{D_i - u} - \sqrt{7u - 4 - 3u^2}.$$

The first loss is $< .294$, using $(100/81)/4.21 < .294$, $q_i > 2.25$, and $\sqrt{D_i - 100/81} > 1.96$. The second radicand decreases for $u > 7/6$; at $121/100$ it is $.0777 < .28^2$. Thus the total loss is $< .574 < .65$. All isolated finite, foreign and dotted profiles are strong; no foot is left unpaid.

J.14 The actual-I and typed-seven consumers of the same window

In the actual-I case, only the golden helpers (A10gold) still need to be excluded. The fresh COMPLETE time table, with reflected indices covering six and seven, is

i	e	$11A_c^2$	$\mathrm{N}(11A_c^2)$
4	1	$8 + 21\varphi$	-209
4	10	$8 + 18\varphi$	-116
5	1	$9 + 21\varphi$	-171
5	10	$9 + 20\varphi$	$-139.$

These are direct substitutions in

$$A_c^2 = (\varphi^2/4)D_i + (\varphi/2)C_{ie} + D_e/4 - 1, \qquad D_e = 20/11.$$

The first numerator norm has exactly one factor eleven; none of the others has one. For the ACTUAL isolated I, its own two-cycle gives $V = 4h \in \mathcal{O}_K$, $0 < V < 2$, and $-4 < V' < 0$. We are now in the actual quadratic field $K = \mathbb{Q}(\varphi)$, so $0 < |\mathrm{N}(V)| < 8$. The helper and I cap have disjoint spatial supports. Their actual two-cycle makes $A_c^2 V$ integral, hence $\mathrm{N}(A_c^2)\mathrm{N}(V)$ integral. Every displayed row forces eleven to divide $\mathrm{N}(V)$: its denominators are eleven or eleven squared, with numerators coprime to eleven after cancelling the one factor in the first row. This contradicts the nonzero norm bound eight. No relative primitive scale, PID assertion or extra I-energy charge is involved.

For EITHER actual typed-seven defect, the same generic local geometry and all-degree helper proof remain valid. The previously proved ACTUAL $[7,3]$ principal instead gives $K = K_7$, a cubic field with no quadratic subfield. Thus (A10gold) is impossible in this branch as well. This reuses the complete generic window; it does not transport the $N = 12$ inventory or infer arithmeticity of an arbitrary face. It also avoids a duplicated cubic-only trace argument for the same already excluded helper domain.

In both cases every site $4, \dots, 7$ is genuinely .65-strong. Take symmetric heights, linear on the tails,

$$H_4 = H_7 = 8/5, \qquad H_5 = H_6 = 42/25.$$

They lie strictly below the proved floors, because $(9/4)^2 < 56/11$ and $(233/100)^2 < 60/11$, with positive margins $5/176, 2821/110000$. The only positive dual coefficients are $\beta_4 = \beta_7 = 4/25$, $\beta_5 = \beta_6 = 1/25$. Every one is paid by an actual strong height. One square on the original vector yields

$$e_{A_{10}} > (8/5)^2/4 + (42/25 - 8/5)^2 = \frac{404}{625}. \tag{A10rate}$$

This rate has the stated actual-I OR actual typed-seven hypothesis; it is not asserted for a bare arbitrary-field A_{10}.

For the actual-I + (A_5 or D_5) alternative, add only the ordinary whole rank-five component: $404/625 + 5/14 = 8781/8750 > 1$. The positive I energy is not needed. For actual typed-seven plus ordinary T_4, count the defect once and use

$$404/625 + 9/100 + 2/7 = \frac{17887}{17500} > 1.$$

These are sums of disjoint whole-component energies of the SAME original x. They exclude both entire rank-sixteen A_{10} alternatives, without a field census or a newly executed finite calculation.

K Chain consumers with typed or ordinary spectators

K.1 The two actual typed-seven A_{14} forests

Here the WHOLE incident forest is A_{14} plus one actual rank-two defect, with no further spectators. Put $c = \cos(\pi/7)$, $\theta = 2c$. The actual $[7,3]$ principal proves $K = K_7 = \mathbb{Q}(\theta)$. Its full integer ring and root brackets are those proved in Section J.3. The defect wall is isolated from the ENTIRE old A_{14}, and has the form

	$d = \sqrt{h_D}\, p - v_D$	h_D
N_7	$v_D = c\,\Omega_e^{A_2}$	$(4/3)c^2 - 1$
O_7	$v_D = \Omega_e^{I_2(7)}/2$	$1/(4(1-c^2)) - 1.$

Both defects have disjoint energy $> .09$. We first strengthen the central floors, then use this ACTUAL wall to pay both ends.

At $N = 15$, use sites $5, \dots, 10$ and fresh loss $\ell = .55$. Their dual squares are $20/3, 36/5, 112/15$, in reflected order, with $5/2 < q < 11/4$. The increasing sufficient ratio obeys

$$R_\ell(q) < R_\ell(11/4) = 164/99 < 5/3.$$

The full finite-white list remains isolation, admissible endpoint/D/E three, endpoint four, or two endpoint threes joining paths. The retained paths have ranks at least four. The target-independent own-zero residuals are $Q - 8/15, Q - 2/15, Q - 1/15$, with bounds $Q < 3/5, 2/5, 1/5$. The first positive time is below any whole foreign contribution; the second permits at most one contribution greater than its time; the last permits none. The actual reciprocal law excludes those positive cases; at $Q = 0$ times are negative. Internal-three/four and two-three roles retain their all-wall positive affine kernels.

The actual field K_7 has no quadratic subfield. Native five is absent by its two-cycle; own-four/simple-three is absent by its UNSQUARED connected $\sqrt{2}$ cycle. All appropriate smaller owns and parallel retain their all-foreign actual nulls. For the other ordinary roles, the FRESH lower ratios are

role	lower ratio
near-three	$13/\sqrt{60} > 5/3$
D-position	$11/\sqrt{40} > 5/3$
E-position	1.8
far-four	$.9\sqrt{5}$
near white-four	$53\sqrt{5}/50$
two end-threes	$q/2 + 1/q \geq 13/\sqrt{60} > 5/3.$

These use $\gamma > .9$, side order $m \geq 5$, white-three residual $1 - j(m-j)/(2m)$ and offset $j/(2m)$; white-four has residual $1/m$, offset $j/(\sqrt{2}m)$. The last comparison uses the ACTUAL minimum $q^2 = 20/3$, not merely $q > 5/2$. Positive foreign cost strengthens all these ratios. For a foreign far-three helper complete clearing gives

$$\frac{1.8m + 1}{\sqrt{m(m+2)}} \geq 10/\sqrt{35} > 5/3,$$

increasing for $m \geq 5$. Thus a weak nonisolated cap is PURE end-helper with $\gamma > .9$.

Put $t = 2\gamma$. Its exact square-ratio identity gives

$$9/5 < t < 23249/9801 < 19/8.$$

It is integral in the ACTUAL cubic field. The signed Schur costs are

$$2it^2 + 4t + 2 - 2(it+1)^2/15, \qquad 2it^2 + 2 - 2(it-1)^2/15.$$

At $i = 5$, their values at $-4/5$, $12/25$ are four and $4 + 16/375$. They increase across the half-window, so both other conjugates lie in $(-.8, .48)$. In degree three the nonzero norm has magnitude $< (19/8)(.8)^2 < 2$. A positive small conjugate would make it $< (19/8)(.48)(.8) = .912 < 1$. Both are negative with absolute product $1/t$; the integral trace is one, being $> t - 1.6 > .2$ and $\leq t - 2/\sqrt{t} < 2$. Its second coefficient is

$$B = t - t^2 + 1/t, \qquad -3459/1216 < B < -199/225.$$

The function decreases, so $B = -2$ or -1. The candidate cubics $X^3 - X^2 - 2X - 1$, $X^3 - X^2 - X - 1$ have discriminants $-31, -44$, contradicting total reality. Degree one leaves only $t = 2$, its actual parallel null. No weak helper remains.

We now prove directly that no $w \in \mathcal{O}_{K_7}$ satisfies

$$0 < w < 11, \qquad 0 < w_\alpha, w_\beta < 3/5. \tag{A14strip}$$

Use the proved FULL basis $w = a + b\theta + c_0\theta^2$, with roots

$$1.8019 < \theta < 1.802, \quad .445 < \alpha < .446, \quad -1.248 < \beta < -1.246.$$

Set $d_0 = b - (\theta - 1)c_0$ and $\Delta = w - (w_\alpha + w_\beta)/2$. The exact identity is

$$\Delta = A_0 d_0 + B_0 c_0, \quad A_0 = (3\theta - 1)/2 < 2.203, \quad B_0 = 3\theta^2 - 2\theta - 2 > 4.136.$$

The small-value difference gives $|d_0| < .6/(\alpha - \beta) < .355$, hence $|A_0 d_0| < .782065$, while $-.6 < \Delta < 11$. Values $c_0 \le -1$ force $\Delta < -3.353935$; $c_0 \ge 3$ forces $\Delta > 11.625935$. Only $c_0 = 0, 1, 2$ remain. At zero, $b = 0$, then a would be an integer in $(0, .6)$. At two, $1.2488 < b < 1.959$, impossible. At one, $.4469 < b < 1.157$, so $b = 1$. But $.643025 < \alpha + \alpha^2 < .644916$; positivity forces $a \ge 0$, contradicting $w_\alpha < .6$. This proves the entire strip.

For ANY isolated-within-A cap retain its entire foreign cost and put $w = 4\gamma^2$. The actual own two-cycle makes w nonzero integral. Normalize its original root to own raw product -1, square $2/w$. The old inverse entry $D_i/2$ gives

$$0 < w_\sigma < 4/D_i \le 3/5$$

at BOTH other embeddings, even with foreign contacts. Weakness would imply

$$0 < w < 4(164/99)^2 = 107584/9801 < 11,$$

contradicting (A14strip). Every site $5, \ldots, 10$ is therefore genuinely .55-strong, with no helper or weak foot left to omit.

K.2 The actual typed wall forces both A_{14} ends

The increasing cubic $8c^3 - 4c^2 - 4c + 1$ has values $-.000569738168, .000257608$ at $.9009, .901$. Thus

$$.9009 < c < .901, \quad .08 < h_N < .083, \quad .32 < h_O < .3284 = 821/2500.$$

Substitution into the displayed height formulas gives these rational seals. The shared exact gap is

$$\psi = \frac{15(3c^2 - 2)}{28(4c^2 - 3)}, \qquad c^2 < \psi < (1 + c)/2 < 1. \tag{A14gap}$$

For the lower bound, $\psi = (1 + 1/(4h_O))/(28/15) > 21690/22988 > .9 > c^2$. For the upper, clearing positive denominators and using the cubic reduces to $39c^2 - 14c - 19 > 0$, increasing for $c > .9$ and equal to $.04061159 > 0$ at $.9009$. In an N_7 branch, h_O in this identity is only its scalar function of c.

The COMPLETE finite actual own-square alphabet in K_7 is

$$\gamma^2 \in \{0, 1/4, 1/2, 3/4, c^2, (1 + c)/2\}.$$

Cyclotomic degree first permits $2, 3, 4, 6, 7, 9, 14, 18$. The $9/18$ cubic field differs: $2\cos(\pi/9)$ has polynomial $X^3 - 3X - 1$, whose shift by one is Eisenstein at three, whereas K_7 has discriminant 49. If γ itself is K_7-valued, its finite list further reduces to $0, 1/2, c$; $\cos(\pi/14)$ has degree six.

For EITHER actual endpoint cap $c_A = A_c p - v_A - v'_D$, let $\delta > 0$ be its Schur residual against ALL retained old roots. The actual edge gives

$$C_{\text{end}}^2 = 1 + A_c^2/\delta, \quad 0 < \delta \le 1, \qquad y \ge \sqrt{14/15}\sqrt{\frac{C_{\text{end}} - 1}{C_{\text{end}} + 1}}. \tag{A14end}$$

The two signed coordinates are $x_{A,1}$, $-x_{A,15}$. This cap differs from the actual defect wall d, which cannot close its edge. The two endpoint caps need not differ from each other.

For O_7, positive retained $I_2(7)$ contact would cost at least $\csc^2(\pi/7)/4 > 1$, impossible. Thus $v'_D = 0$, and the positive ACTUAL mutual magnitude $\lambda = -c_A \cdot d = A_c\sqrt{h_O}$ is at least $1/2$. We exclude equality in every endpoint role. An isolated cap has $A_c^2 = (28/15)\gamma^2 - 1$; equality forces $\gamma^2 = \psi$. The finite alphabet, parallel one and dotted $\gamma > 1$ all avoid (A14gap).

For a nonisolated endpoint, retained A_{13} has the COMPLETE finite list: helper-three at $j = 2, 3, 13, 14$, or helper-four at $j = 2, 14$. The exact fresh time polynomials are

$(j, \text{helper label})$	$15A_c^2$
$(2, 3)$	$28\gamma^2 + 26\gamma - 2$
$(3, 3)$	$28\gamma^2 + 24\gamma + 3$
$(13, 3)$	$28\gamma^2 + 4\gamma - 2$
$(14, 3)$	$28\gamma^2 + 2\gamma - 8$
$(2, 4)$	$28\gamma^2 + 26\sqrt{2}\gamma + 11$
$(14, 4)$	$28\gamma^2 + 2\sqrt{2}\gamma - 1.$

(A14endtimes)

Simple-three normalization makes γ itself K_7-valued. At own zero only $j = 3$ is future, with time $1/5$, giving $\lambda^2 < 1/4$. At own $1/2$ the four times are $6/5, 22/15, 7/15, 0$. The first two have mutual square $> 1/4$, the third $< 1/4$, and the last is not future. For every $\gamma \ge c > .9$ all four polynomials increase and their minimum exceeds $16.48/15 > 1$. Thus their mutual squares exceed $1/4$. This covers every remaining finite and every dotted own; parallel has its actual null.

For helper-four, connected normalization gives $2\sqrt{2}\gamma \in K_7$. At finite own labels $3, 6, 7, 14$, the norms of $2w$, $w = 4\gamma^2$, are $8, 216, 8, 56$, all nonsquares. Only own four remains, besides zero, parallel or dotted. At zero, near time $11/15$ has mutual square $< 1/4$, and far time is negative. At own four, times $17/5, 1$ have mutual squares $> 1/4$. At $\gamma \ge 1$, both times exceed one. Hence equality $\lambda = 1/2$ is impossible in EVERY role. The next positive native magnitude is at least $1/\sqrt{2}$, so

$$C_{\text{end}}^2 \ge 1 + 1/(2h_O) > 2071/821 > (397/250)^2.$$

Equation (A14end) gives

$$y^2 > \frac{2058}{9705} > (23/50)^2, \qquad y > .46.$$

The two cleared rational margins are $40511, 11055$. Only connected products within A_{14} were used to normalize helpers; no equal scale of its disconnected defect is assumed.

For N_7, retain the ENTIRE foreign old A_2 block and write

$$\lambda = A_c\sqrt{h_N} - \rho \ge 0, \qquad \rho = v'_D \cdot v_D.$$

At $Q = 0$, the native gap gives $A_c^2 > 250/83$, hence $C_{\text{end}} > 2$. Positive foreign contact has the COMPLETE finite list

label	Q	ρ
3	$1/3$	$c/3$ or $2c/3$
4	$2/3$	$\sqrt{2}c/3$ or $2\sqrt{2}c/3$.

Two contacts make a cycle; six is affine; larger labels are nonfinite, and five is forbidden in K_7. If an A_{13} contact is also present, its full inverse cost is at least 13/28. Thus $\delta \le 17/84$. Acuteness gives $A_c\sqrt{h_N} \ge \rho > .3$, so $A_c^2 > 90/83$ and $A_c^2/\delta > 7560/1411 > 3$. Without an A_{13} contact, $\delta = 1 - Q$. At $Q = 2/3$, $\rho > .3\sqrt{2}$ gives $A_c^2/\delta > 540/83 > 3$. At $Q = 1/3, \rho = 2c/3$, the same bound follows from $\rho > .6$. These arguments retain EVERY own parameter.

Only $Q = 1/3, \rho = c/3$, with no A_{13} contact, remains. Positive λ gives $A_c\sqrt{h_N} \ge c/3 + 1/2 > .8$ and $A_c^2/\delta > 960/83 > 3$. At $\lambda = 0$, the exact equation $A_c^2 = (28/15)\gamma^2 - 2/3$ instead gives $\gamma^2 = \psi$. Its ACTUAL own two-cycle puts $4\gamma^2$ in K_7; the finite alphabet, parallel and dotted ranges again avoid the gap. Thus every N_7 endpoint has $C_{\text{end}} > 2$, giving $y > \sqrt{14/45} > .55 > .46$. The two strict .46 signed demands therefore hold for BOTH actual defect types.

K.3 Both typed endpoints in one legal reference

All central sites are .55-strong. Choose symmetric heights, linear on the tails,

$$H_5 = H_{10} = a = 2.03198, \quad H_6 = H_9 = b = 2.13328, \quad H_7 = H_8 = c_0 = 2.18251.$$

The floor tests are $2.58198^2 < 20/3$, $2.68328^2 < 36/5$, $2.73251^2 < 112/15$. The six central coefficients are $.152548, .026035, .024615, .024615, .026035, .152548$. The base norm is

$$E = a^2/5 + (b-a)^2 + (c_0 - b)^2 = .83847382698.$$

The first and last five coordinates are flat with signed mean $b_0 = a/(5\sqrt{2})$, $.28 < b_0 < .2875$. In EACH block replace the signed endpoint by .46, and redistribute its remaining sum constantly over the other four coordinates. Their value $r = (5b_0 - .46)/4$ lies in $(.235, .245)$, below the endpoint and above the next old coordinate $< .073$. The NEW reference stays concave, with positive endpoint and changed boundary coefficients. Every central dual height is unchanged and genuinely paid. There are no omitted weak feet or helpers.

The two residuals have disjoint block supports, zero block sums and zero cross with the original constant blocks. Each reference gain is exactly $(5/4)(.46 - b_0)^2 > .0371953125$. All actual paid pressures are strict. One square therefore gives

$$e_{A_{14}} > .83847382698 + 2(.0371953125) = .91286445198.$$

Only NOW add the disjoint defect energy $> .09$ of the SAME original vector, counted ONCE. The full norm exceeds 1.00286445198, a contradiction. Both actual typed-seven A_{14} forests are excluded. No K_7 condition is imposed on the separate ordinary rank-two branch.

K.4 The ordinary $A_{14} + T_2$ branch has one actual three-wall core

Now the WHOLE forest is $A_{14} + S$, where $S = I_2(m)$, $m \ge 3$, is ORDINARY: neither actual typed defect is present. The universal bound (A14U) gives

$$0 < e_S < 1/5.$$

Let $k = \cos(\pi/m) \ge 1/2$, $d_0 = (1 - k^2)^{-1}$. The old unit Gram inverse is $d_0 \left(\begin{smallmatrix} 1 & k \\ k & 1 \end{smallmatrix}\right)$. The positive unit incident duals u_1, u_2 have correlation k. For their ACTUAL edges put $\tau_i = \tanh(L_i/2)$, $r_i = \tanh L_i$. Original farthest comparison gives $x \cdot u_i \ge \tau_i > 0$; hence $\tau_i^2 < 1/5$ and each actual edge cosine is $< 3/2$.

The two edges have ONE actual opposite. Indeed Cauchy on u_1+u_2 gives $\tau_1+\tau_2 \le \sqrt{2e_S(1+k)}$, so $\tau_1\tau_2 < (1+k)/10$. For $0 < a, b < 1$,

$$\frac{2a}{1+a^2}\frac{2b}{1+b^2} \le \frac{4ab}{(1+ab)^2}.$$

The right side increases in $ab < 1$; therefore

$$r_1 r_2 < \frac{40(1+k)}{(11+k)^2} < k.$$

The last cleared polynomial is $k^3 + 22k^2 + 81k - 40$, increasing for $k \ge 1/2$ and equal to $49/8 > 0$ at $1/2$. The actual two-wall law for distinct opposites requires $r_1 r_2 \ge k$. Thus one wall d closes BOTH directions, including the old A_2 case.

Write

$$d = A_d p - v_S - v_A, \quad h = A_d^2 > 0, \quad b = -d \cdot n_1, \quad g = -d \cdot n_2, \quad Q = \|v_A\|^2,$$

so

$$h = d_0(b^2 + g^2 + 2kbg) + Q - 1.$$

Every positive b, g is finite native, being in a retained positive two-root Gram at the other actual endpoint. The whole A_{14} is retained there, so $Q < 1$; a positive Q is at least $7/15$. Both $b = g = 0$ would give negative time. If both are positive and $Q > 0$, the actual edge formula gives

$$C_1^2 = \frac{d_0(b+kg)^2}{1-g^2-Q} \ge \frac{(4/3)(3/4)^2}{3/4 - 7/15} = \frac{45}{17} > \frac{9}{4},$$

contrary to $C_1 < 3/2$.

If precisely one contact is positive, write $b = \beta, g = 0$. At the other endpoint the connected finite rank-sixteen diagram is classical. Its only possible foreign attachment is a SINGLE old A_{14} END-three, with $\beta = 1/2$ or $1/\sqrt{2}$. An internal attachment would give arms of lengths two and two others totaling thirteen, which are neither D_{16} arms nor exceptional finite arms. An end-four would become an internal weighted bond; a five bond cannot occur. Thus $Q = 7/15$ and $h = d_0\beta^2 - 8/15$. The smaller normalized derivative square is $k^2 d_0 \beta^2$. Since $r_i^2 < 5/9$, we get

$$\beta^2(4d_0 + 5) \le 24/5.$$

At $\beta = 1/\sqrt{2}$, this forces $d_0 \le 23/20 < 4/3$, impossible. At $\beta = 1/2$, $d_0 \le 71/20 < 4$, so $m = 3, 4, 5$. The first two times are $-1/5, -1/30$. At five, $d_0 = 2 + 2/\sqrt{5} < 3$, giving $0 < h < 13/60 < 1/4$. The actual small-height foreign-isolation lemma then forbids this remaining contact: the same wall would close an A_{14} direction in addition to its S directions. Consequently

$$Q = 0. \tag{A14coreiso}$$

This is isolation of an ACTUAL wall from the entire old component, not face heredity.

We classify every native S-contact. With one contact the same derivative law now gives $\beta^2(4d_0 + 5) \le 9$. Thus $d_0 \le 31/4$. For $m \ge 9$, $\sin(\pi/m) < \pi/m$ and $\pi^2 < 10$ give $d_0 > 81/10 > 31/4$. Hence $m = 3, \ldots, 8$ is complete. If $\beta \ge 1/\sqrt{2}$, then $d_0 \le 13/4$, so $m \le 5$. At four, positive time and the inequality leave only new five. At five, new three has negative time, new four remains, and new five or larger violates $\beta^2(4d_0 + 5) \le 9$. At six only new three remains, with zero time. At seven/eight only new three remains; old seven/new three is ACTUAL O_7 and excluded by ordinary

meaning. At old three, positivity and the inequality give $3/4 < \beta^2 \le 27/31$. Only new seven or eight remains: for all native nine or larger, $\cos(\pi/9) > 76/81$ and $(76/81)^2 > 27/31$. Old three/new seven is ACTUAL N_7 and is excluded. Thus precisely four one-contact rows survive.

With two positive contacts, every old $m \ge 5$ has

$$C_i \ge \frac{\cot(\pi/(2m))}{\sqrt{3}} \ge \frac{\cot(\pi/10)}{\sqrt{3}} > \sqrt{3} > 3/2.$$

At old four, either contact of at least four gives $C_i \ge \sqrt{3}$; both must be three. At old three, both contacts at least four give the same contradiction. One is therefore three; if the other $\beta \ge \cos(\pi/5) > 4/5$, the opposite edge has $C^2 = (1+\beta)/(3(1-\beta)) > 3$. The remaining possibilities are three/three, with zero time, or three/four in either direction orientation. This completes the classification.

Put $\varphi = (1+\sqrt{5})/2$, $s = \sqrt{2}$. The SIX necessary old-role models are

S	d contacts in S	h	K
$I_2(4)$	one 5	$(\varphi-1)/2$	$\mathbb{Q}(\varphi)$
$I_2(5)$	one 4	$1/\sqrt{5}$	$\mathbb{Q}(\varphi)$
A_2	one 8	$(s-1)/3$	$\mathbb{Q}(s)$
$I_2(8)$	one 3	$s/2$	$\mathbb{Q}(s)$
A_2	3, 4 in either orientation	$s/3$	$\mathbb{Q}(s)$
$I_2(4)$	3, 3	$s/2$	$\mathbb{Q}(s)$.

(A14six)

They are necessary actual Gram types, never realized chambers.

We justify the ACTUAL field column. The native two-cycles put the quadratic field inside K in the tree types $(2,4,5)$, $(2,3,8)$. For $(3,3,4)$, the essential actual THREE-cycle is $8kbg = \sqrt{2}$. Choose the degree-two root as base in a tree, and any root with both products nonzero in the triangle. Let its original K-root square be the totally positive q_0. Rescale the two other original roots by ratios of K-valued products so their products with the base are $-q_0$. Their squares become $q_0/k^2, q_0/b^2$; the remaining product is zero or $-q_0 g/(kb)$, whose coefficient is in the displayed field F because $g/(kb) = kbg/(k^2b^2)$. Thus the three ACTUAL root lines have Gram $q_0 M$, M over F. It is Lorentzian by $Q = 0, h > 0$. A proper totally real extension K/F would have a nonphysical embedding fixing this negative direction while preserving the positive scalar. Strict nonphysical positivity forbids it. Hence $K = F$, without identifying raw scales across the old components.

K.5 Every ordinary-core A_{14} endpoint exceeds .44

Name the five distinct heights in (A14six)

$$h_a = (\varphi-1)/2, \quad h_b = 1/\sqrt{5}, \quad h_c = (s-1)/3, \quad h_d = s/2, \quad h_e = s/3.$$

Every one is positive and at most $s/2$. For an actual endpoint cap $c = A_c p - v_A - v_F$, let $\gamma = -c \cdot n_1 \ge 0$, $Q = \|v_F\|^2$, and $\delta > 0$ be the full retained-white Schur residual. Its actual edge obeys (A14end). It suffices to prove $C_{\text{end}} > 31/20$, because then its signed coordinate satisfies

$$y^2 > \frac{154}{765} > .44^2.$$

This actual wall differs from the common spectator wall d, which is orthogonal to all A_{14} directions. Their mutual magnitude is

$$\lambda = -c \cdot d = A_c\sqrt{h} - \rho \ge 0, \qquad \rho = v_F \cdot v_S \ge 0. \tag{A14mut}$$

A positive λ is at least $1/2$. No individual unit product is declared K-valued: its invariant square is. The finite positive mutual labels are

K	finite positive labels
$\mathbb{Q}(\varphi)$	$3,4,5,6,10$
$\mathbb{Q}(s)$	$3,4,6,8.$

Indeed cyclotomic degree at most two first permits $3,4,5,6,8,10,12$; the $5/10, 8, 12$ fields are respectively $\mathbb{Q}(\sqrt{5}), \mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3})$. The full orders are $\mathbb{Z}[\varphi]$, $\mathbb{Z}[s]$.

First suppose c is PURE isolated: no retained A_{13} or S contact. Then $\delta = 1$, $A_c^2 = (28/15)\gamma^2 - 1$, and $\lambda = A_c\sqrt{h} > 0$. Its own two-cycle requires

$$4\gamma^2 = \frac{15}{7}(1 + \lambda^2/h) \in \mathcal{O}_K.$$

Here are ALL values $z = 1 + \lambda^2/h$ for the finite mutual alphabet:

	3	4	5	6	10
h_a	$(2+\varphi)/2$	$1+\varphi$	$(3+2\varphi)/2$	$(2+3\varphi)/2$	$(3+3\varphi)/2$
h_b	$(3+2\varphi)/4$	$(1+2\varphi)/2$	$(5+3\varphi)/4$	$(1+6\varphi)/4$	$(4+5\varphi)/4$

and

	3	4	6	8
h_c	$(7+3s)/4$	$(5+3s)/2$	$(13+9s)/4$	$(16+9s)/4$
h_d	$(4+s)/4$	$(2+s)/2$	$(4+3s)/4$	$(3+s)/2$
h_e	$(8+3s)/8$	$(4+3s)/4$	$(8+9s)/8$	$(7+3s)/4.$

Each dyadic numerator has a coefficient not divisible by seven. Multiplication by $15/7$ cannot make both full-order coefficients integral. Thus EVERY finite mutual is impossible. Retain parallel/dotted mutuals by the weaker conclusion $\lambda \geq 1$. Then $A_c^2 \geq 1/h \geq s$, so

$$C_{\text{end}}^2 \geq 1 + s > (31/20)^2.$$

For PURE nonisolated caps, the complete endpoint roles and times are (A14endtimes). Their residuals, in row order, are

$$15/28, \quad 1/7, \quad 1/7, \quad 15/28, \quad 1/14, \quad 1/14.$$

At own zero only row $j = 3$, helper-three, and near helper-four have positive time. The first has $\lambda^2 = h/5 < 1/4$, impossible. For the second $\lambda^2 = 11h/15$: it lies in $(0, 1/4)$ at h_a, h_c, in $(1/4, 1/2)$ at h_b, h_e, and in $(1/2, \cos^2(\pi/5))$ at h_d. These are all native gaps. The coarse exact brackets $1.6 < \varphi < 5/3$, $2 < \sqrt{5} < 5/2$, $1.4 < s < 1.5$ seal every strict interval. Every other own-zero time is nonpositive.

For ANY positive own $\gamma \geq 1/2$, the first three helper-three rows already give $C_{\text{end}} \geq 9/5$, $C_{\text{end}}^2 \geq 169/15$, $C_{\text{end}}^2 \geq 64/15$. Both helper-four rows give $C_{\text{end}}^2 > 113/15$. Only far helper-three at $j = 14$ remains. Its actual simple-three normalization gives $2\gamma \in K$. Thus finite own is $0, 1/2, \varphi/2$ in $\mathbb{Q}(\varphi)$, or $0, 1/2, s/2$ in $\mathbb{Q}(s)$. Zero is negative time and $1/2$ has zero time. At $\varphi/2$, time $(8\varphi - 1)/15$ gives mutual squares

$$(9 - \varphi)/30 \in (0, 1/4), \qquad (8\varphi - 1)/(15\sqrt{5}) \in (1/4, 1/2).$$

At $s/2$, time $(6 + s)/15$ gives, for h_c, h_d, h_e,

$$\frac{5s - 4}{45} \in (0, 1/4), \quad \frac{3s + 1}{15} \in (1/4, 1/2), \quad \frac{6s + 2}{45} \in (0, 1/4).$$

All are forbidden native gaps. Parallel own has its positive actual null. For EVERY dotted $\gamma > 1$ all time polynomials increase and the minimum far time exceeds $22/15$. With residual at most $15/28$ this gives $C_{\text{end}} > 29/15$. The entire unbounded own range is covered.

We now keep the ENTIRE foreign S block. At most one S contact is possible in the retained finite Gram; two would make a nonpositive cycle through its old bond. For old $I_2(4)$, $I_2(5)$, only native three is finite, of costs $1/2, d_5/4$, where $d_5 = 2 + 2/\sqrt{5}$. For old $I_2(8)$, any contact costs > 1. For old A_2, the finite labels are three, four, five; its actual $\mathbb{Q}(s)$ excludes five, leaving costs $1/3, 2/3$. White parallel/dotted products cannot occur.

If there is also an A_{13} contact, the full connected finite rank-sixteen diagram is classical. An old five/eight bond is impossible there. For old $S = A_2, I_2(4)$, the ONLY join is an A_{13} END-three and a native-three S contact; an internal or weighted attachment gives excessive arms, an internal double bond, or a cycle. The exact residuals are

$$S = I_2(4) : \ \delta = 1/28, \qquad S = A_2 : \ \delta = 17/84.$$

At old four, $\rho > 1/2, h < 3/4$, so acuteness alone gives $A_c^2 > 1/3$ and $C_{\text{end}}^2 > 31/3$. At old A_2, the complete inverse products give

$$\frac{\rho_{\min}^2}{h} = \frac{1}{3} + \frac{s}{4} > \frac{41}{60}.$$

Thus $C_{\text{end}}^2 > 1 + (41/60)(84/17) > 4$. These estimates retain all own parameters and require no finite mutual assumption.

It remains to consider foreign caps isolated WITHIN A_{14}, with

$$A_c^2 = (28/15)\gamma^2 + Q - 1, \qquad \delta = 1 - Q.$$

The full inverse products yield

S, h	foreign label	$\rho_{\min}^2/h$	$1 - Q$
$I_2(4), h_a$	3	$\varphi^3/4 > 1$	$1/2$
$I_2(4), h_d$	3	$(4 + 3s)/8 > 1$	$1/2$
$I_2(5), h_b$	3	$d_5(d_5 - 1)/(4(d_5 - 2)) > 3/2$	$1 - d_5/4 < 3/10$
A_2, h_c or h_e	4	$2(1/3 + s/4) > 4/3$	$1/3$.

The d_5 inequality reduces to $(d_5 - 3)(d_5 - 4) > 0$, with $2 < d_5 < 3$. Acuteness gives respectively $C_{\text{end}}^2 > 3, 3, 6, 5$, sufficient even when the mutual is zero.

Only old A_2, foreign-three, remains. At h_c, the two inverse products are $a_8/3, 2a_8/3$, $a_8 = \cos(\pi/8)$; at h_e, they are $(s+2)/6, (2s+1)/6$. These include BOTH old-node orientations. If $\lambda = 0$, the exact time equation forces the own two-cycle $4\gamma^2$ to be one of

$$\frac{15(4 + s)}{28}, \qquad \frac{15(2 + s)}{7}, \qquad \frac{15(8 + 3s)}{56}.$$

The first occurs for the smaller product in both models; the others are their respective larger products. None is integral in $\mathbb{Z}[s]$, by its nonzero coefficient residue at seven. Thus every zero-mutual exception is impossible for ANY finite or dotted own. If $\lambda > 0$, then $\lambda \geq 1/2$, while $h < 1/2, \rho > .3$. Hence $A_c^2 > (.5 + .3)^2/.5 = 32/25$ and

$$C_{\text{end}}^2 > 1 + \frac{32/25}{2/3} = 73/25 > (31/20)^2.$$

We have covered every white, own, foreign and zero-mutual role. Both actual endpoints therefore satisfy

$$x_{A,1} > .44, \qquad -x_{A,15} > .44. \tag{A14ordends}$$

Reversing index names proves the second demand without requiring a chamber symmetry or distinct opposite walls at the two ends.

K.6 Ordinary A_{14}: pay both ends and all feet afresh

We return to the COMPLETE .597 local inventory and directional pressures in Section J.1. There is at most ONE weak golden helper. All other weak central sites are the retained pure/foreign $0 < u_i < 1$ feet; no adjacency or packing restriction is imposed on them. Their full continuous adjusted-loss bounds hold whenever the ACTUAL remaining prices satisfy

$$\beta_{5,10} \geq .06, \quad \beta_{6,9} \geq .026, \quad \beta_{7,8} \geq .0246.$$

These are individual coefficient thresholds, not a total debit budget.

The reference must be rebuilt BEFORE any new energy is used. Let $v = \sum \beta_j \Omega_j$ be a concave reference whose endpoint prices are paid by actual endpoint demands, and whose central heights retain their strong floors. For ALL paid isolated feet put

$$Z = v - \sum \beta_i u_i n_i.$$

Its actual pressure and full norm obey

$$x_A \cdot Z \geq v^2 - \sum \beta_i (L_i - .597),$$
$$Z^2 \leq v^2 - \sum \beta_i^2 (2u_i - u_i^2).$$

Here $v \cdot n_i = \beta_i$ is the NEW coefficient, including every endpoint/foot cross. Every adjacent-foot cross is nonpositive. The previously proved full-interval loss bounds give $e_A > v^2$ by ONE square. Strict endpoint pressures give strictness also when no foot remains. We use orthogonality of the block changes only to compute the NEW reference norm, never to ignore their crosses with feet.

Use the base heights

$$H_5 = H_{10} = a = 1.98498, \quad H_6 = H_9 = b = 2.08628, \quad H_7 = H_8 = c_0 = 2.13551.$$

The full base norm is .80071440298. If one internal helper lies at six through nine, omit ONLY its knot by the neighbor chord, losing at most .000677821225. The neighboring coefficients increase. Thus the base norm is at least .800036581755, with its first and last five coordinate blocks still flat at signed mean

$$b_0 = \frac{a}{5\sqrt{2}}, \qquad .28 < b_0 < .3.$$

In each block replace the signed endpoint by its actual paid value .44, keeping the block sum fixed. The other four become $r = (5b_0 - .44)/4$, with $.24 < r < .265 < .44$. They exceed the next old signed coordinate, which is $< .073$. The modified reference is concave, its new endpoint coefficients are positive, and EACH changed outer central price is

$$> .167/1.415 > .118 > .06.$$

Every other paid central coefficient stays unchanged or increases; the helper price remains zero. All central heights are preserved. The two disjoint block gains give

$$v^2 > .800036581755 + 2(5/4)(.44 - .3)^2 = .849036581755. \tag{A14ord0}$$

Apply the simultaneous-foot square to THIS new reference.

For ONE outer helper, relabel it as site five. Its ACTUAL H_4 point already proved in Section J.1 gives the signed helper-end demand $> .51961$, in BOTH helper orientations. Pay that endpoint at .51, and the OTHER endpoint at .44 by (A14ordends). Omit only knot five by the chord zero-to-six. Its base norm is $> .764$. The left six-block mean lies in $(.24, .25)$; the right five-block mean lies in $(.25, .3)$. Distribute each remaining block sum constantly after replacing its endpoint.

If the helper end is near, the five other left coordinates exceed $(1.44 - .51)/5 = .186$, while the other block's four exceed $(1.25-.44)/4 = .2025$. If it is far, the right four exceed $(1.25-.51)/4 = .185$, and the left five exceed $(1.44 - .44)/5 = .2$. Each is positive, below its endpoint target and above the next old coordinate $< .073$. Thus both new boundary prices exceed $.112/1.415 > .079$. Site six needs only .026; site ten needs .06. All other prices retain their own thresholds; the missing-five price remains zero. No strong-five floor has been used.

For near helper the exact block-gain formula gives gains $> .08112, .0245$. For far helper they are $> .055125, .04332$. Consequently

$$v^2_{\text{near}} > .86962, \qquad v^2_{\text{far}} > .862445.$$

The SAME simultaneous-foot square applies, retaining every foot and all boundary crosses with these end changes.

The minimum of the three complete branches is (A14ord0), which exceeds 16/19: it exceeds .849, and $19(.849) > 16$. Only NOW add the disjoint ORDINARY T_2 energy $> 3/19$ proved in Proposition 19. The original norm exceeds one. Thus the entire ordinary $A_{14} + T_2$ branch is excluded. Together with the actual typed-seven proof and the already stated two-A_1 and joint-$I + A_1$ consumers, this covers every rank-sixteen A_{14} alternative. The proof never increases the ordinary dihedral floor or transfers a K_7 assumption to this quadratic-field branch.

K.7 The rank-seventeen A_{15} typed alternatives

Here the WHOLE forest is A_{15} plus one actual rank-two typed defect. Thus $K = K_7$, its disjoint defect energy is $> .09$, and the ENTIRE foreign cost is

$$Q \in \{0, 1/3, 2/3\}.$$

Positive values occur only for old A_2: finite foreign labels are three, four, five, and K_7 excludes five. For old $I_2(7)$, even a three costs $\csc^2(\pi/7)/4 > 1$. There are no further spectators.

Use $\vartheta = 2\cos(2\pi/7)$, satisfying $p(X) = X^3 + X^2 - 2X - 1$. The earlier generator $\theta = 2\cos(\pi/7)$ satisfies

$$\vartheta = \theta^2 - 2, \qquad \theta = \vartheta^2 + \vartheta - 1.$$

Hence the already proved FULL integer ring is also $\mathbb{Z}[\vartheta]$. Its roots and derivatives satisfy

$$1.246 < \vartheta_1 < 1.247, \quad -.446 < \vartheta_2 < -.445, \quad -1.802 < \vartheta_3 < -1.801,$$
$$5 < p'(\vartheta_1) < 6, \quad -3 < p'(\vartheta_2) < -2, \quad 4 < p'(\vartheta_3) < 5, \quad \vartheta_2 - \vartheta_3 > 1.355.$$

The three paired endpoint signs are $(-.005049064, .000105223)$, $(.002199464, -.000096125)$, $(-.000257608, .003875599)$; derivative bounds then follow on the brackets. For ANY integer $y = a + b\vartheta + c_0\vartheta^2$, Lagrange interpolation gives

$$c_0 = \sum_j \frac{y_j}{p'(\vartheta_j)}, \qquad \frac{y_2 - y_3}{\vartheta_2 - \vartheta_3} = b - (1 + \vartheta_1)c_0.$$

We prove three COMPLETE integer-strip exclusions.

There is no integer with

$$4 < y_1 < 16, \qquad 0 < y_2, y_3 < 32/55. \tag{A15stripA}$$

Interpolation gives $0 < c_0 < 4$ and $|b - (1 + \vartheta_1)c_0| < .43$. At $c_0 = 1, 2, 3$, respectively, $b = 2$, no integer, or $b = 7$. For the first case

$$-.358399 < \vartheta_3^2 + 2\vartheta_3 < -.356796$$

forces $.356796 < a < 32/55 + .358399 < 1$. For the last case

$$-2.876197 < 3\vartheta_3^2 + 7\vartheta_3 < -2.872388$$

forces $a = 3$, but the physical value then exceeds $3 + 7(1.246) + 3(1.246)^2 = 16.379548$.

There is no integer with

$$9/5 < y_1 < 68/25, \qquad -.8 < y_2, y_3 < .48. \tag{A15stripB}$$

Interpolation gives $-.14 < c_0 < 1.064$; hence $c_0 = 0, 1$. The difference bound is $< .945$. At zero $b = 0$, giving incompatible physical and small rational-integer strips. At one $b = 2, 3$; their values at ϑ_2, before adding a, lie in $(-.693084, -.691975)$, $(-1.139084, -1.136975)$. The small-value strip permits $a = 0, 1$ in the first case, $a = 1$ in the second. Their physical values exceed 4.044516, 6.290516, respectively.

Finally no integer has

$$2 < y_1 < 21/10, \qquad -7/5 < y_2, y_3 < -1/5. \tag{A15stripC}$$

Now $-1 < c_0 < 2$: the interpolation lower and upper bounds are $-1/60$, 1.12. The difference bound is $< .886$. At zero the rational case fails; at one $b = 2, 3$. The same ϑ_2 intervals force $a = 0$, with physical values $> 4.044516, 5.290516$. All three full-ring strips are proved.

At $N = 16$ use sites $5, \ldots, 11$ and loss $1/2$. Here $D_i = i(16 - i)/8 \geq 55/8$, $2.6 < q_i < 3$. The full white list is the two paths of ranks at least four, with the usual endpoint/D/E three, endpoint four, or two-end-three join. Rank fifteen excludes exceptional completions. The increasing sufficient ratio is

$$T(q) = 2 - \frac{q - 1/2}{q(2q - 1/2)} < 61/33.$$

Own-zero global end-three has time $Q - 17/32$, $Q < 3/5$, negative for the actual foreign-cost set. Next-end-three has time $Q - 1/8$, $Q = 0, 1/3$; its positive time is less than its foreign cost. End-four has $Q = 0$ and negative time. The actual reciprocal law, affine internal three/four and two-three kernels, and parallel nulls dispose of all those roles.

For a simple-three helper $t = 2\gamma$ is integral in K_7; the only positive finite own cosines themselves in this field are labels three and seven. Three closes the actual affine cycle. For native seven at

a NONEND helper node j, $D_j \geq 7/2$, and its positive conjugate $t_\sigma = 2\cos(3\pi/7) > 2/5$ violates Schur positivity already through

$$D_i t_\sigma^2 + D_j > (55/8)(4/25) + 7/2 = 23/5 > 4.$$

The omitted cross and foreign costs are nonnegative. For dotted owns the remaining lower ratios are $6/\sqrt{10} > 61/33$ at D-positions, $2\gamma + 3/8 > 19/8$ at E-positions, and

$$\frac{3m-1}{\sqrt{2m(m+1)}} \geq 17/\sqrt{84} > 61/33$$

for near-three of order $m \geq 6$. The last function increases; its cleared square margin at six is 2157. Endpoint-four/own-three would put $\sqrt{2}$ in K_7; other owns have ratio at least $6/\sqrt{10}$. The two-path bridge has $\gamma q + 1/q > .9q + 1/q > 61/33$. Foreign far-three has $Q \geq 1/3$ and ratio square

$$\geq \frac{3(2m\gamma+1)^2}{2m(m+3)} > 15/4 > (61/33)^2.$$

It increases with $m \geq 5$ at $\gamma > .9$. Only pure far helper or near dotted side-order-five helper remains.

For the pure far role the square identity gives $9/5 < t < 5899/2178 < 68/25$. The fresh $N = 16, i = 5$ signed Schur costs exceed four at $-.8, .48$ by $3/40, 59/1000$. Their monotonicity puts both other values in $(-.8, .48)$. Strip (A15stripB) excludes every such cap. For near dotted order five, $2 < t < 1363/660 < 21/10$. Its necessary conjugate inequality is $55t_\sigma^2 + 88t_\sigma + 16 < 0$. That quadratic equals $3/5 > 0$ at $-7/5, -1/5$, so both values lie between these endpoints. Strip (A15stripC) excludes the last nonisolated weak cap.

For isolation within A, the actual own two-cycle $B = 4\gamma^2$ is integral, and full conjugate Schur positivity gives

$$0 < B_\sigma < 4/D_i \leq 32/55.$$

Use the COMPLETE finite-square alphabet already proved in Section K.2. Small owns three/four/six have all-foreign actual nulls. The other finite labels seven/fourteen each have a conjugate > 1, using the stated brackets for θ^2, $2 + \theta$. Thus no finite isolated own survives. For a dotted own, $0 < u = (1 - Q)/\gamma^2 < 1$. Its actual all-wall foot has loss

$$< u/5 + \sqrt{u(1-u)},$$

since $(\sqrt{D} + \sqrt{D-1})^{-1} < 1/5$. At $u \leq 1/4$ this is $< 1/20 + \sqrt{3}/4 < .484 < 1/2$. Weakness therefore implies $u > 1/4$ and $4 < B = 4(1-Q)/u < 16$. Strip (A15stripA) excludes every such dotted foot, including both positive foreign costs.

Every site $5, \ldots, 11$ is now actually half-loss strong. Choose symmetric heights, linear on the tails,

$$H_5 = 2.122, \quad H_6 = 2.238, \quad H_7 = 2.306, \quad H_8 = 2.328.$$

The strict floor tests are $2.622^2 < 55/8$, $2.738^2 < 15/2$, $2.806^2 < 63/8$, $2.828^2 < 8$. The successive slopes $.4244, .116, .068, .022$ and their reflection decrease at all paid knots. One positive reference square gives

$$e_{A_{15}} > 2.122^2/5 + .116^2 + .068^2 + .022^2 = .9191408.$$

Only NOW add the disjoint actual defect energy $> .09$, obtaining > 1.0091408. Both actual typed-seven A_{15} alternatives are excluded, with no remaining weak foot, field census or independently added optimum.

K.8 The ordinary $A_{15} + T_2$ rank-seventeen consumer

Here the WHOLE forest is $A_{15} + T_2$, with $T_2 = I_2(m)$ ordinary. If $e_T > 1/6$, the already proved $e_{A_{15}} > .8348$ gives total $> .8348 + 1/6 > 1$. Suppose $e_T \le 1/6$. Put $k = \cos(\pi/m)$, $d = (1-k^2)^{-1}$. The same two-direction argument as above gives

$$\tanh L_1 \tanh L_2 \le \frac{48(1+k)}{(13+k)^2} < k.$$

The last cleared polynomial is $k^3 + 26k^2 + 121k - 48$, increasing on $k \ge 1/2$ and equal to $153/8$ there. Thus the two ACTUAL opposites are one wall.

Write its two T_2 contacts $b, g \ge 0$, its full A_{15} cost Q, and its time square $h > 0$. Any positive foreign cost is at least $15/32$. Two positive T_2 contacts are impossible. At old three, contacts three/three have the actual affine null. Otherwise one is at least native four; the edge coshes are at least $(1+2\sqrt{2})/3 > 5/4$, $(1+\sqrt{2})/\sqrt{3} > 29/21$. Their half-tangents exceed $1/3, 2/5$. The positive dual certificate at correlation $1/2$ gives energy $> 124/675 > 1/6$. At old four both coshes are at least $(1+\sqrt{2})/\sqrt{3}$, giving $e_T > 32/175 > 1/6$, using $k < 3/4$. For old $m \ge 5$, an edge cosine is at least $\cot(\pi/(2m))/\sqrt{3} > 7/5$; its individual energy already exceeds $1/6$. Positive foreign cost only increases these actual coshes.

There is exactly one T_2 contact $\beta > 0$. Put $q = d\beta^2$. The actual own/mate derivative law gives

$$h = q + Q - 1, \quad h \le (24/49)k^2 q, \qquad \beta^2(25d + 24) \le 49.$$

Necessarily $q > 1$: if $q \le 1, Q > 0$, its foreign reciprocal shadow makes the same actual wall close an orthogonal component, or gives a feasible null; $Q = 0$ instead has nonpositive time. For $m \le 5$, these inequalities give $Q < (24/49)k^2 \le 18/49$. For $m \ge 6$, $\beta^2 \ge 1/4, d \ge 4$ give $Q \le (172 - 25d)/196 \le 18/49$. Both contradict $Q \ge 15/32$ unless $Q = 0$. Also $d \le 172/25$, so $m \le 8$; for $m \ge 9$, $d > (63/22)^2 > 172/25$. The complete native table is

m	β^2 upper bound	$q > 1$ own labels
3	147/172	7, 8
4	49/74	5
5	$< 441/841$	4
6	49/124	none
7	$< 49/149 < 1/2$	3
8	$< 1/2$	3.

At old three, every native nine or larger has $\beta > 15/16$, from $\pi/9 < 22/63$, $\cos t > 1 - t^2/2$. The other rows use the consecutive native squares and $q > 1$. Parallel or dotted $\beta \ge 1$ violates the same table bounds. The two seven rows are exactly actual typed defects and absent by ordinary meaning. Only old/new pairs

$$(3,8), (8,3), (4,5), (5,4)$$

remain. Their ACTUAL fields are respectively $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\varphi)$, by the original connected Lorentzian-principal descent proved above for the three-wall core. No relative scale between old components has been asserted.

In both silver rows the mate edge has

$$C_s^2 = (3 + 2\sqrt{2})/3, \qquad e_T > 411/2500 = .1644.$$

For an exact seal, $\sqrt{2} > 707/500$ gives $C_s^2 > 1457/750 > (13937/10000)^2$; the half-tangent comparison with .1644 has positive cleared margin 4393. The golden rows have $C_g = \cot(\pi/5) > 11/8$, hence $e_T > 3/19$.

For silver $K = \mathbb{Q}(\sqrt{2})$, recalibrate the common $N = 16$ loss to $\ell = .6007$. Its proof is unchanged on the full rectangle $.6 \le \ell \le .601$: $R_\ell(q) \le R_{.6}(3) = 41/27 < 761/500$; all finite and foreign $u > 1$ seals printed in the common inventory are below .6, and near-four has the SAME three positive conjugate numerators. Every remaining weak pure helper would be golden and force the other actual field, so none exists. Retain ALL isolated feet, with $a_i < .1982$. Take symmetric heights, linear on the tails,

$$H_5 = 2.0213, \quad H_6 = 2.1373, \quad H_7 = 2.2053, \quad H_8 = 2.2273.$$

Every positive central price is at least .022, and the floor squares are unchanged from the complete N16 reference. Below $u = 1/2$, raw loss is $< .5991$. Above it, adjusted loss is

$$< (.1982 + 1.0195)/2 - 3(.022)/8 = .6006 < .6007.$$

The FULL simultaneous-foot square, including adjacent feet, therefore gives

$$e_A > 2.0213^2/5 + .116^2 + .068^2 + .022^2 = .835694738.$$

The total with the silver spectator exceeds 1.000094738.

For golden $K = \mathbb{Q}(\varphi)$, the exact old spectator is $I_2(4)$ or $I_2(5)$. Its ENTIRE possible foreign cost is

$$Q = 0, 1/2 \quad \text{or} \quad Q = 0, (5 + \sqrt{5})/10,$$

respectively. A retained finite cap has only one native-three contact; a four already costs at least one, and two contacts make a cycle. Use loss $\ell = .59$, sites $5, \dots, 11$. The fresh sufficient ratio is

$$R_{.59}(q) \le 148081/95757 < 31/20.$$

The complete finite-white list is unchanged, but we check its ordinary roles against THIS stronger target. Own-zero internal-three/four and two-three retain their affine kernels. The residual global end-three, next-three and end-four own costs are $15/32, 7/8, 15/16$, less than one. At $Q = 0$ their times are negative; at positive Q they are smaller than that foreign cost, violating the actual reciprocal law. Parallel is its actual positive null.

A simple-three helper makes 2γ integral in the actual golden field, so finite own is three or five. Three has its cycle; hence $\gamma \ge \varphi/2$. Endpoint-four/own-three would put $\sqrt{2}$ in K, so its remaining owns have $\gamma \ge 1/\sqrt{2}$. The complete nonfar lower comparisons are

D-position	$(\varphi/2 + 1/5)/\sqrt{2/5} > 31/20$
E-position	$2\gamma + 3/8 > 31/20$
near-three	$(\gamma + (m-1)/(2m))/\sqrt{(m+1)/(2m)} > 31/20$
endpoint-four	$6/\sqrt{10} > 31/20$
two-path bridge	$\gamma q + 1/q > 2.$

The near-three expression increases for $m \ge 5$; at five use $\varphi/2 > .809$, $\sqrt{3/5} < 31/40$, giving $1209/775 > 31/20$. Own-four/simple-three is also impossible by its unsquared connected cycle. For foreign far-three, $Q \ge 1/2$ leaves residual at most $1/(2m)$, and ratio $> (\gamma + 1/(2m))\sqrt{2m} > 31/20$. A weak nonisolated cap is therefore a PURE far helper.

Its square identity gives $\varphi \le t = 2\gamma < 1761/800 < 9/4$; the unchanged N16 signed Schur strip is $-.8 < t' < .48$. For $t = a + b\varphi$, these force $b = 0, 1$. At zero only parallel two fits physically. At one, the small strip gives $a = 0, 1$; one gives $\varphi^2 > 9/4$. Thus $t = \varphi$. The COMPLETE N16 golden table and cyclic packing in Section B.3 apply independently of weakness: at most ONE weak helper exists.

For isolated-within-A cap, $B = 4\gamma^2 \in \mathbb{Z}[\varphi]$, $0 < B' < 4/D_i \le 32/55$. Its complete finite native alphabet is labels $3, 4, 5, 6, 10$. The conjugate of label ten exceeds $32/55$; small three/four/six have their actual affine principals. Only finite own five remains. At $Q = 0$, its actual H_4 loss is $< 25/76 + 5/24 < .59$. At the two positive foreign costs the actual foot parameter is respectively

Q	u	loss upper bound
$1/2$	$4 - 2\varphi < .764$	.577
$(5 + \sqrt{5})/10$	$4 - 8\sqrt{5}/5 < .423$	.584.

For the first, $\sqrt{u(1-u)} = \sqrt{10\varphi - 16} < .425$; for the second use its full circle loss with $a_i < .1982$. The rational seals $809/500 < \varphi < 40451/25000$, $\sqrt{5} > 2.236$ suffice. Every finite isolated cap is strong.

A dotted foot with $u \le 1/4$ has loss $< 1/20 + \sqrt{3}/4 < .484$. A weak dotted foot therefore has $4 < B < 16$, $0 < B' < 32/55$. Writing $B = a + b\varphi$ gives $b = 2, \dots, 7$; the interval $b/\varphi < a < b/\varphi + 32/55$ leaves, in order, no integer, two, three, no integer, four, no integer. The COMPLETE list is

$$B = \varphi^4, \quad 3 + 4\varphi, \quad 2\varphi^4.$$

The last two have $u < 4/9$. For the first with $Q > 0$, $u < 1/3$. Their loss is $< (4/5 + \sqrt{20})/9 < 44/75 < .59$. Only PURE $B = \varphi^4$ can be weak, with $u_0 = 20 - 12\varphi \in (.58, .59)$.

In the actual connected old-A normalization, the raw dotted time square is

$$H_i^2 = \frac{B\{i(16-i)B - 32\}}{16}.$$

For $i = 5, 6, 7, 8$ its norms are

$$-8271/256, \quad -8816/256, \quad -9119/256, \quad -9216/256.$$

These four squareclasses differ. Here $8816 = 16 \cdot 19 \cdot 29$, while the others are nonzero modulo nineteen; $8684^2 < 8271 \cdot 9119 < 8685^2$; and $9216 = 96^2$, whereas $8271, 9119$ are not integer squares. Reflected copies have the actual K-valued two-cycle parameters

$$-7/4 - 45\varphi/8, \quad -1 - 9\varphi/2, \quad 1/4 - 21\varphi/8.$$

None is integral. Their squares must be integral by the actual two-cycle criterion, which for a K-element forces integrality of the element itself. Thus at most ONE weak dotted index exists.

A weak helper and weak dot cannot coexist. Compare only their ACTUAL old-A-linked root lines on the same K-time line. The following odd prime has odd valuation in each helper time norm, and does not divide ANY dotted numerator:

helper	$-\mathrm{N}(M)$	p	8271 mod p	8816 mod p	9119 mod p	9216 mod p
near 5	1559	1559	476	1021	1324	1421
near 6	1111	101	90	29	29	25
near 7	719	719	362	188	491	588
near 8	431	431	82	196	68	165
far 5	311	311	185	108	100	197
far 6	239	239	145	212	37	134
far 7	271	271	141	144	176	2.

All listed primes have the elementary trial-division checks; $1111 = 11 \cdot 101$. The norm ratio is not a rational square, so the time ratio cannot be a K-square. No freely normalized spectator time is used in this argument.

Take symmetric heights, linear on the tails,

$$H_5 = 2.032, \quad H_6 = 2.148, \quad H_7 = 2.216, \quad H_8 = 2.238.$$

Their exact floor sums are $2.622, 2.738, 2.806, 2.828$, and

$$E = .8443688.$$

There are precisely three complete weak-set cases. With no helper and no weak dot, one square gives $e_A > E$. With one weak dot, use its actual foot in the SAME reference. The loss is $< .61$. At outer five/eleven, $\beta = .1452$ and $2u_0 - u_0^2 > .8$, so its correction gain is positive. At an internal knot $\beta \le .024$, the energy debit is $< 2(.024)(.02) = .00096$. Thus $e_A > .8434088 > .8434$.

With one internal helper there is no weak dot and no second helper. Omitting just its knot loses at most $.048^2/4 = .000576$, giving $e_A > .8437928 > .8434$.

With an outer helper, relabel it as five. Its actual finite H_4 point gives, at its signed helper end,

$$y > \sqrt{2}(A_c - .111 - \gamma h_5) > .49.$$

Indeed the smaller actual time is $(38+65\varphi)/32 > 2.115^2$, $h_5 < 2.033, \gamma < .81$, and $1.4(2.115 - .111 - .81(2.033)) = .500178 > .49$. This recomputes the endpoint demand at the new loss; it does not borrow the older stronger height cutoff. Omit five by the chord zero-to-six. Its base reference norm is $> .805$, by the exact chord formula already printed for the SAME heights. At the actual helper end its flat block has length five or six, signed mean $.25 < b_0 < .3$. Graft that signed coordinate to .49, preserving the block sum. The other coordinates lie in $(.19, .262)$, above the next old coordinate $< .101$. The reference stays concave and paid; its gain exceeds .04332. No weak dot or other helper remains. Hence $e_A > .84832 > .8434$.

Every golden branch gives $e_A > .8434$. Only NOW add its disjoint ordinary spectator energy $> 3/19$, obtaining total $> 1 + 123/95000 > 1$. Together with the silver and high-energy branches this excludes the entire ordinary $A_{15} + T_2$ rank-seventeen forest. In particular no blanket $T_2 > 1/6$ claim or disconnected root-scale comparison has entered the proof.

K.9 A whole A_{16} with any rank-one spectator

Suppose the WHOLE forest is $A_{16} + A_1$. The A_1 is initially ordinary or actual short I; we do not identify the two hypotheses. The common component proof gives

$$e_A > .8703, \qquad e_1 < .1297 < .13.$$

Write the ACTUAL rank-one opposite as $c = A_c p - \gamma e - v_A$, with $h = A_c^2 = \gamma^2 + Q - 1 > 0$, $Q = \|v_A\|^2 < 1$. A finite one-node completion permits at most one A_{16} contact; a second makes a cycle, and rank seventeen is too large for exceptional finite types. Thus $Q > 0$ implies $v_A = b\Omega_j$, $b \ge 1/2$, and $Q \ge 8/17$.

In fact $Q < h$. If $\gamma < 1$, put $\delta = 1 - \gamma^2 = Q - h > 0$, $\eta_c = c + \gamma e$. The actual vector

$$y = \delta p + A_c \eta_c$$

has square $-\delta Q < 0$, positive time, zero products on c, e and every old A-root except n_j. It is a positive combination of p and ACTUAL roots, so every omitted original wall product is nonpositive.

Its normalization is therefore an actual finite endpoint on the outgoing A edge j, with the SAME opposite as the A_1 edge. The original orthogonal-component theorem forbids this. At $\gamma = 1$, $c + e$ is a feasible future null. Consequently $\gamma > 1$, giving $Q < h$.

Let $\tau = \tanh(L/2)$ on the actual rank-one edge. Its exact equation gives

$$\tanh^2 L = \frac{h}{1 + h - Q} > Q \geq 8/17, \qquad \tanh^2 L = \frac{4\tau^2}{(1+\tau^2)^2} < \frac{5200}{12769} < 8/17,$$

using $\tau^2 \leq e_1 < .13$. The last cross-product margin is 13752. Contradiction. Thus $Q = 0$, and the actual opposite is

$$c_1 = \sqrt{h}\, p - \sqrt{1+h}\, e, \qquad 0 < h < 5200/7569 < 7/10. \tag{A16r1}$$

The height bound inverts the same half-tangent equation. This isolated dotted wall has not been called a short I.

Put

$$s^2 = \frac{\sqrt{1+h} - 1}{\sqrt{1+h} + 1}, \quad t = \sqrt{1+4h} - 2\sqrt{h}, \quad \eta = \sqrt{16/17}.$$

Then $e_1 \geq s^2$. For ANY other actual edge opposite $d = A_d p - v - \beta e$, distinct from c_1, native gap at $\beta = 0$ or acuteness with $\beta \geq 1/2$ otherwise gives $A_d\sqrt{h} \geq 1/2$. Bounding its ENTIRE spatial derivative by $\sqrt{1 + A_d^2}$ gives the same physical half-tangent pressure as in the earlier actual-I argument:

$$x_1, -x_{17} \geq \eta t.$$

It uses only $h > 0$, not $h < 1/2$.

The exact scalar barrier $t + 3s/2 > .767$ for $0 < h < 1/2$ is the already proved rank-free barrier (3), with its companion proof in the rank-free joint-payment appendix subsection [4]. For $1/2 \leq h \leq .6$, monotonicity gives $t \geq t(.6) > .294$, $s \geq s(.5) > .317$, and hence sum $> .7695$. The rational seals are $\sqrt{3.4} < 1.844, \sqrt{.6} < .775$, $.294(3.394) < 1$, and $s(.5)^2 = 5 - 2\sqrt{6} > .101 > .317^2$. For $.6 \leq h < .7$, use $t > t(.7) > .275$, $s \geq s(.6) > .34$, giving sum $> .785$. Here $\sqrt{3.8} < 1.95, \sqrt{.7} < .837$, $.275(3.624) < 1$, and $\sqrt{1.6} > 1.264$ gives $s(.6)^2 > 33/283 > .34^2$. Thus on the ENTIRE widened interval

$$t + 3s/2 > .767.$$

Its actual own two-cycle gives

$$v = 4h \in \mathcal{O}_K, \quad 0 < v < 14/5, \qquad -4 < v_\sigma < 0$$

at EVERY nonphysical embedding. No field degree is assumed.

The complete $N = 17, \ell = .6$ inventory has strong sites, all retained isolated feet, or one pure golden helper. For the latter, ACTUAL $K = \mathbb{Q}(\varphi)$; its complete $34A^2_{\text{helper}} = M$ table is already printed in Section B.3. Its seven norm magnitudes $1764, 1216, 716, 324, 196, 36, 76$ are all coprime to seventeen. The actual helper/c_1 two-cycle makes $Mv/34$ integral. Conjugating M and using an integer Bezout relation gives $v \in 17\mathcal{O}_K$. But its nonzero quadratic norm has magnitude $< 56/5 < 17^2$, impossible. This uses the widened physical bound explicitly, rather than the short-I norm bound eight. All weak golden helpers disappear. After this branch the actual field is again UNRESTRICTED.

We prove that BOTH actual A endpoint edges have cosine $C < 2$. Use the same central heights $H_5 = 2.056, H_6 = 2.186, H_7 = 2.269, H_8 = H_9 = 2.310$, reflected with linear tails. Their base energy is .8708972; every positive central coefficient is at least .0205. Each five-coordinate end block has

signed mean $.2907 < b < .291$; its next coordinate is $< .092$. Suppose one actual signed endpoint is at least .55. Pay that end at .55. Pay the disjoint other end at

$$y = \min(.55, \max(b, \eta t)).$$

Both block sums are preserved by distributing their other four coordinates constantly. All remaining coordinates exceed $.225875 > .092$. Thus the reference is concave and EACH changed central boundary price still exceeds .0205. All other prices retain their values; all central heights remain fixed. The FULL simultaneous-foot square therefore applies to this NEW reference, including every foot/endpoint and adjacent-foot cross.

The first block gain exceeds $(5/4)(.55 - .291)^2 = .08385125$. The disjoint A_1 energy plus the other block gain exceeds .07. If $\eta t \geq .55$, the gain alone is larger. Otherwise put $z_0 = (\eta t - .291)_+$. Then $z_0 + (3/2)\eta s > .4529$. Weighted Cauchy gives

$$e_1 + \text{other gain} \geq s^2 + (5/4)z_0^2 > \frac{.4529^2}{36/17 + 4/5} > .07.$$

The final rational comparison is $.4529^2(85) = 17.43506485 > 17.36$. The SAME reference and A_1 energy would total > 1.02474845. Thus each signed actual endpoint is $< .55$, by two separate reflected contradictions, not an added pair of optimized bounds. If its actual $C \geq 2$, its signed coordinate square is $\geq 16/51 > .55^2$. Hence BOTH actual endpoint edges have

$$C < 2. \tag{A16r1C}$$

Suppose an endpoint cap is isolated WITHIN A:

$$c = A_c p - \gamma \Omega_1 - \beta e, \qquad 0 \leq \beta < 1.$$

Own zero has negative time. For $U = 4\gamma^2$, rescaling the original K-root against an old norm-two root gives raw square $2U$, own product $-U$. Its actual own two-cycle is integral; nonphysical old-A Schur positivity gives

$$U \in \mathcal{O}_K, \qquad 0 < U_\sigma < 17/8.$$

The actual edge equation and cutoff give $C^2 = 8U/(17(1 - \beta^2))$, hence $0 < U < 17/2$. The three DISTINCT actual walls c, e, c_1 have $\lambda = A_c\sqrt{h} - \beta\sqrt{1+h} \geq 0$. Their two/three cycles make

$$L = 4\lambda^2, \quad W = 8\beta\sqrt{1+h}\lambda, \quad V = 4\beta^2$$

integral, including possible zero cycles. Expanding ALL β-terms gives

$$L + W + V = \frac{(8U - 17)v}{17} =: \rho \in \mathcal{O}_K.$$

Therefore

$$z = \frac{Uv}{17} = 15\rho - (7U - 15)v \in \mathcal{O}_K.$$

Physically $0 < z < 7/5$; EVERY other conjugate lies in $(-1/2, 0)$. If $[K : \mathbb{Q}] \geq 2$, its nonzero integer norm has magnitude $< (7/5)(1/2)^{[K:\mathbb{Q}]-1} < 1$. At degree one, integral U, v have $U \leq 8, v \leq 2$, so $0 < z \leq 16/17 < 1$. Both cases are impossible. Neither endpoint cap can be isolated within A. This includes every finite old-A_1 contact and the rational field directly; no Meyer or field-degree bound is used.

For completeness we exhaust the remaining ACTUAL endpoint roles. Deleting n_1 retains A_{15} on nodes two through sixteen and old A_1. The complete nonisolated list is simple-three at $j = 2, 3, 15, 16$, or four at $j = 2, 16$. Only simple-three at $j = 2, 16$ permits positive old-A_1 contact, with $\beta = 1/2, 1/\sqrt{2}$. At $j = 3, 15$, appending that component would give nonfinite arms $1, 2, 13$; for helper-four it would make the four internal in a long path. Thus the other rows have $\beta = 0$.

For simple-three the exact retained residual and ACTUAL edge cosine are

$$\delta = 1 - \beta^2 - \frac{(j-1)(17-j)}{32}, \qquad C = \frac{32\gamma + 17 - j}{4\sqrt{34}\sqrt{\delta}}.$$

For helper-four replace $17-j$ in the numerator by $\sqrt{2}(17-j)$, and 32 in the subtracted denominator term by 16. The four non-end roles have

(j, helper)	C
$(3, 3)$	$(32\gamma + 14)/(2\sqrt{17})$
$(15, 3)$	$(32\gamma + 2)/(2\sqrt{17})$
$(2, 4)$	$(32\gamma + 15\sqrt{2})/\sqrt{34}$
$(16, 4)$	$(32\gamma + \sqrt{2})/\sqrt{34}$.

At positive own $\gamma \geq 1/2$ all exceed two. At zero, far-three and far-four have negative time, and near-four still has $C > 2$. The near next-three zero-own role has the actual affine E_8 ray

$$2n_1 + 4n_2 + 6n_3 + 5n_4 + 4n_5 + 3n_6 + 2n_7 + n_8 + 3c.$$

Every selected product and norm is zero; its time $3A_c > 0$ is future, and all omitted original walls are nonpositive. Compactness forbids it.

Near simple-three $j = 2$ has

$$C = \frac{32\gamma + 15}{17}, \quad \frac{32\gamma + 15}{3\sqrt{17}}, \quad \frac{32\gamma + 15}{\sqrt{17}}$$

at $\beta = 0, 1/2, 1/\sqrt{2}$. At zero foreign cost, own zero has negative time; under $C < 2$ only positive own three remains, with its actual affine triangle. At $\beta = 1/2$, every positive own gives $C \geq 31/(3\sqrt{17}) > 2$. Its own-zero role has the actual affine ray

$$3n_1 + 6n_2 + 5n_3 + 4n_4 + 3n_5 + 2n_6 + n_7 + 4c + 2e.$$

It has the same all-original-wall future null properties. At $\beta = 1/\sqrt{2}$, even own zero has $C \geq 15/\sqrt{17} > 2$.

Only far simple-three $j = 16$ remains. Put $t_0 = 2\gamma$. Its exact data are

$$A_c^2 = \frac{8t_0^2 + t_0 - 9}{17} + \beta^2, \qquad C = \frac{16t_0 + 1}{17}, \ \frac{16t_0 + 1}{3\sqrt{17}}, \ \frac{16t_0 + 1}{\sqrt{17}}$$

at the three foreign costs. Own zero has negative time in all three. The last row contradicts $C < 2$. In the middle row $t_0 < 3/2 < \varphi$; the only positive native values are one and $\sqrt{2}$. One gives its actual all-three cycle. At $\sqrt{2}$, simple-three normalization puts $\sqrt{2}$ in the ACTUAL field. The lifted sign reversal has positive time numerator $7 - \sqrt{2}$, hence a negative old-A Schur complement in a nonphysical positive form. This excludes that row. Thus $\beta = 0$, and

$$1 < t_0 < 33/16, \qquad 17A_c^2 = (t_0 - 1)(8t_0 + 9).$$

BOTH actual endpoint walls now have the form

$$c_L = A_c p - (u\Omega_1 + \Omega_{16})/2, \qquad c_R = B_c p - (v\Omega_{16} + \Omega_1)/2, \quad 1 < u, v < 33/16.$$

They are distinct: equality would force $u = v = 1$, contrary to positive time. Their spatial product is $[uv + 1 + 16(u + v)]/34$. Their time product is $< (8/17)uv$, because $8t_0^2 + t_0 - 9 < 8t_0^2$. Also $u + v > (32/33)uv$. Therefore

$$c_L \cdot c_R > uv/66 > 0,$$

contradicting acuteness of distinct ACTUAL walls. Every original wall, signed endpoint, foreign contact and field degree has been retained. The entire $A_{16}+A_1$ rank-seventeen alternative, ordinary or actual I, is excluded.

L Classical rank-fifteen and rank-sixteen spectator cases

L.1 Rank-fifteen classical typed consumers share the local head

The WHOLE forest is B_{15} or D_{15} plus actual N_7 or O_7. The actual field, defect wall and complete foreign costs are exactly (TQ), with disjoint defect energy $> .09$. We reuse the PROVED local head argument of Section F.1, checking all its rank-specific premises. No rank-fourteen whole-polytope conclusion is transported.

For B_{15}, the generic $M = 10$ proof applies through eleven: retained left A and right B ranks are at least four, and joined rank fifteen precludes exceptional completions. Its weak role is PURE far helper

$$v = g\omega_i + \omega_1/2, \quad Q = 0, \quad A_c^2 = 2ig^2 + 2g - 1/2, \quad 101/125 < g < 6/5.$$

Lemma 5 excludes these classified indices from six onward. At twelve the complete right-B_3 list has its actual short-three F_4 seal, with all other rows affine or strong. At thirteen the only extra weak B_2 role is PURE own-three/short-three. Its actual long-square-two/short-square-one normalization puts $\sqrt{2}$ in K; sign reversal gives old Schur cost

$$13(3 - 2\sqrt{2})/4 + 1/2 > 1.$$

The bound already holds at twelve by $\sqrt{2} < 17/12$. The full bridge has residual $1/(2i)$ and ratio $1/\sqrt{2i} + \sqrt{i}/2 > 100/61$; it admits no foreign arm because the new node is internal to the large B path. The F_4 piece likewise has no finite enlargement at its cap. All foreign/isolated seals retain the entire spectator. Thus six through thirteen are ACTUALLY strong.

For D_{15}, the generic $M = 9$ proof gives six through ten. The ACTUAL right-D_4, D_3 layers are eleven, twelve. Their rank-local complete lists in Proposition 12 retain all affine zero/three/four/six/parallel rows, direct end-four and sealed H_4 rows. Full D_{15} and long-arm A_{15} bridges have no finite foreign arm. The D_3 one-fork bridge has residual $1/8 + 1/(2i) \le 1/6$, positive-own numerator $> .75$, hence is strong; own zero has its actual all-three cycle. The only extra weak packet has $Q = 0$, own-five/right-three: the local arithmetic helper Schur calculation at these layers has $2\gamma < 12/5$ and leaves only the golden own. It is absent in ACTUAL K_7. Thus six through twelve are ACTUALLY strong, without a centre-thirteen or either fork floor.

In either component, weak five has its PURE actual foot $x_1 > .5965$, while strong six gives $S_6 > 2.01815$. Positive interpolation through one, six, eleven has energy

$$.7600299945 + \frac{(\sqrt{11} - \sqrt{6})^2}{5} > .9104.$$

Its slopes are positive and decreasing; the last term exceeds .150372482 from $\sqrt{11} > 3.3166, \sqrt{6} < 2.4495$. Adding the actual defect $> .09$ contradicts the original norm. Five is strong BEFORE the head. The ordinary positive prefix references through thirteen for B, twelve for D, therefore have energies $E_{13} > .88989, E_{12} > .86989$, first-five mean $b < .361$, and individual P_5 price $> .14$.

Deleting the actual head retains B_{14} or D_{14}. Their COMPLETE finite one-node extensions are isolation or simple-three at long leaf two: a second contact makes a cycle; another fork/weighted attachment gives nonfinite arms or an internal multiple bond. Consequently the exact local head formulas are those already proved:

	A_c^2	C_1
helper-three	$2g^2 + 2g + Q$	$(2g+1)/\sqrt{1-2Q}$
isolated within classical	$2g^2 + Q - 1$	$\sqrt{2g^2/(1-Q)}$.

The own-zero positive-Q null uses the FULL rank-fifteen old-root clearing

$$\frac{\omega_2}{2} = \begin{cases} n_1 + 2(n_2 + \cdots + n_{14}) + \sqrt{2}n_{15}, & B, \\ n_1 + 2(n_2 + \cdots + n_{13}) + n_{14} + n_{15}, & D. \end{cases}$$

All coefficients are on ACTUAL roots. Every omitted wall, including both signed D forks, has nonpositive product. All other small-own affine chains in that head proof use only c, n_1, n_2 and the two old spectator roots, present unchanged here.

The finite mixed-head Schur test uses only K_7 and the SAME rational full costs $0, 1/3, 2/3$. The actual N_7-wall mutual test uses only $g_7/3, 2g_7/3$, its two exact old-A_2 inverse products. The pure-head actual two-cycle, full mutual native alphabet and order parity are independent of classical rank. Thus its complete reduction to pure dotted $0 < u < 1/2$ or $.79 < u < .8$ applies to the PRESENT actual walls. Its actual pressure $x \cdot Z \geq H$ has support in the first three coordinates, sum one and full-interval scalar

$$.8H - .16Z^2 > .3055.$$

The chord/circle proof of this inequality is fully written in the cited SAME-part head argument; it is not a new external or computed premise.

Its primitive correction $.4(Z - P_5/5)$ pays ONLY .08 from the current available P_5 coefficient. Its full reference cross is zero. Every earlier graft excluding a mixed or nonisolated head preserves the first-five sum and all later knots. Thus that entire local proof supplies the SAME effective single-square gain

$$> .3055 - .8(.361) + .032 = .0487.$$

This is a completed-square gain, not the squared norm of an independently optimized head vector. It requires neither the outgoing short B wall nor the outgoing D centre or fork walls.

The two original-vector totals are

B_{15} + actual typed7	$.88989 + .0487 + .09 = 1.02859 > 1$
D_{15} + actual typed7	$.86989 + .0487 + .09 = 1.00859 > 1.$

Each actual defect is counted ONCE. No actual-I barrier, arbitrary-face theorem or additional finite output has been assumed.

L.2 The actual-I B_{16} row: a bound before classifying the short cap

Here the WHOLE forest is $B_{16} + I$, where I is the ACTUAL isolated short rank-one opposite, not an ordinary A_1. Keep unit long roots $n_i = (e_i - e_{i+1})/\sqrt{2}$, $i < 16$, actual short $n_{16} = e_{16}$, and orthogonal old I-root e. The SAME original vector has $x_1 > \cdots > x_{16} > 0$ and $e_B + e_I < 1$.

The fresh rank-sixteen bulk gives strong six through twelve. At thirteen the right-B_3 short-three F_4 seal applies. At fourteen the only extra B_2 role is own-three/short-three, whose actual sign-conjugate cost $14(3-2\sqrt{2})/4+1/2>1$ is impossible. The complete bridge is internally attached to full B_{16}, with no foreign finite arm; its zero-own ratio is already strong. Thus $S_i>f_i=\sqrt{i}-.61/\sqrt{2}$ for six through fourteen, independently of FIVE or the outgoing short wall.

The actual I parameters satisfy

$$e_I\ge u^2,\quad x_1\ge t,\quad t=\sqrt{1+4h}-2\sqrt{h},\quad 0<h<1/2,\qquad t+(8/5)u>197/250.$$

This is the previously proved rank-free actual-I barrier, with its original same-vector point domain. Take the reference linear on its first six coordinates and with actual knots six through fourteen. It has

$$E_0>.89026,\quad b=f_6/6<.3364,\quad f_6>2.01815,\quad f_{14}<3.311,\quad \text{individual } P_{14}\text{ price}>.13.$$

Indeed it is the ordinary reference of energy $E_{14}>.9084$ less

$$\frac{(6\sqrt{5}-5\sqrt{6}-.61/\sqrt{2})^2}{30}<.01814.$$

The positive radical bracket $.43133<.61/\sqrt{2}<.43134$ bounds the numerator's positive root by .73769. Lift its first coordinate to $\max(t,b)$, preserving the six-block sum. The other five remain $\ge(f_6-1)/5>.2036$, above the next slope $<.197$, so every residual coefficient stays nonnegative. The I energy plus this gain is

$$e_I+\frac{6}{5}(t-b)_+^2>\frac{.4516^2}{64/25+5/6}>.0601.$$

Thus the combined base already exceeds .95036, counting I once.

For the ACTUAL short edge of cosine C, original farthest comparison gives $S_{16}\ge 4\sqrt{(C-1)/(C+1)}$. If $C>21/2$, then $S_{16}>3.635$. Pay .13 at P_{14}, adding $.13(P_{16}-P_{14})$. It is orthogonal to the complete first-fourteen reference and its first-six adjustment, with square $2(.13)^2$. ONE square yields

$$\|x\|^2>.95036+.26(3.635-3.311)-.0338=1.0008>1.$$

Consequently

$$C\le 21/2 \tag{B16Icut}$$

for EVERY actual short opposite, before its own or foreign type is assigned.

Deleting the short root retains A_{15} and old I. The COMPLETE nonisolated white list is helper-three at $j=1,2,14,15$, or helper-four at $j=1,15$. For simple-three at two/fourteen, the retained D_{16} extension admits no finite old-I arm, so $\beta=0$. At one/fifteen it is A_{16}, allowing $\beta=0,1/2,1/\sqrt{2}$. For helper-four, an added old-I arm makes that four internal in a large path, so $\beta=0$.

For simple-three write

$$c=A_cp-\gamma P_{16}-P_j/\sqrt{2}-\beta e,\qquad a=2\sqrt{2}\gamma.$$

Positive own has $a\ge\sqrt{2}$, and

$$A_c^2=\frac{16a^2+4ja+4j-8}{8}+\beta^2,\quad b_j=\frac{16a+2j}{8\sqrt{2}},\quad \delta_j=1-\frac{j(16-j)}{32}-\beta^2,\quad C=b_j/\sqrt{\delta_j}.$$

Actual simple-three normalization keeps long roots and cap square two, short root square one, OVER K. The closed cycle and own two-cycle give

$$a \in \mathcal{O}_K, \qquad a^2/2 \in \mathcal{O}_K.$$

The connected old-B/cap Schur complement is $-(16a^2 + 4ja + 4j - 8)/4 < 0$. The Lorentzian-principal descent therefore gives ACTUAL $K = \mathbb{Q}(a)$, and

$$16a_\sigma^2 + 4ja_\sigma + 4j - 8 < 0.$$

Only AFTER this field identification is the rational branch excluded by the named whole-form Meyer/compact-anisotropy input. No field list or square-root extension was assumed.

Own zero at one/two has nonpositive time. At fourteen/fifteen its actual future null kernels are

$$n_{13} + 2n_{14} + 2n_{15} + \sqrt{2}n_{16} + c, \qquad n_{14} + 2n_{15} + \sqrt{2}n_{16} + c.$$

Their selected products vanish, time is positive through the cap, and every omitted original wall is nonpositive. Parallel own has its positive null pair. We may now treat positive own.

At $j = 1$, cutoff (B16Icut) gives $a < 53/10$, and the conjugate quadratic gives

$$-13/20 < a_\sigma < 25/64 < 2/5.$$

The nonzero integer $a^2/2$ is physically < 15, other values in $(0, 2/9)$. Degree at least three would give norm $< 15(2/9)^2 < 1$; degree is two. Its even nonzero norm $N(a)$ has magnitude < 4, hence is 2 or -2. For norm two, the small conjugate bound forces $a > 128/25$, and integral trace lies in $(5, 6)$. For norm minus two the integral trace is between two and six, and $\mathrm{Tr}(a^2/2) = (T^2 + 4)/2$ makes it even. Thus $T = 4$, $a = 2 + \sqrt{6}$. Positive β gives $\delta_j \le 9/32$, hence $C > 11 > 21/2$. At $\beta = 0$, $A_c^2 = (41 + 17\sqrt{6})/2$. The actual I invariant $J = 4(1 + h)$ has $4 < J < 6, 0 < J' < 4$. The full order $\mathbb{Z}[\sqrt{6}]$ leaves only $J = 3 + \sqrt{6}$: its positive radical coefficient is one, the physical strip leaves constants two/three, and conjugate positivity excludes two. The actual two-cycle

$$A_c^2(J - 4) = (61 + 24\sqrt{6})/2$$

is nonintegral. This excludes $j = 1$.

At $j = 2$, $\beta = 0, C = 4a + 1$, so $a \le 19/8$, while $-1/2 < a_\sigma < 0$. Degree at least three has norm < 1; degree two has nonzero even norm of magnitude $< 19/16 < 2$, impossible. At fourteen, positive own gives $C = 4a + 7 \ge 4\sqrt{2} + 7 > 21/2$. Its own-zero null was already printed.

At $j = 15$, $a < 18/5$ and $-12/5 < a_\sigma < -4/3$. For $z = a + 2$, $z^2/2$ is integral, physically < 18, other values $< 2/9$, hence degree is two. The trace of a is $0, 1, 2$; even norm and half-square trace parity leave zero or two. At trace two, $10/3 < a < 18/5$ gives $-6 < N(a) < -4$, whose sole integer is odd. At trace zero, a^2 is positive even integral, between $(4/3)^2$ and $(12/5)^2$, hence two or four. Four would be rational. Thus $a = \sqrt{2}, \gamma = 1/2$, $A_c^2 = (21 + 15\sqrt{2})/2 + \beta^2$. Positive β gives positive conjugate time since $43/2 - 15\sqrt{2} > 0$, contradicting the FULL old-B/I/cap Schur condition. At zero, the complete actual-I strip leaves $J = 3 + \sqrt{2}, 3 + 2\sqrt{2}, 4 + \sqrt{2}$. The actual two-cycle values are

$$(9 + 6\sqrt{2})/2, \quad (39 + 27\sqrt{2})/2, \quad (30 + 21\sqrt{2})/2,$$

all nonintegral in the full order.

For helper-four $\beta = 0$ and

$$A_c^2 = 16\gamma^2 + 2j\gamma + j - 1, \qquad C = 16\gamma + j.$$

At fifteen even own zero violates the cutoff. At one own zero has zero time; positive own has $\gamma \le 19/32$, leaving only native three. Its ACTUAL helper-four normalization keeps cap square one and old-short product $-1/2$; the whole connected old-B/cap Gram is rational Lorentzian. Thus ACTUAL $K = \mathbb{Q}$, excluded by the same whole-form Meyer input. All six nonisolated roles are exhausted.

The actual short cap is consequently

$$c_S = A_S p - \gamma P_{16} - \beta e, \quad B = 4\gamma^2, \quad A_S^2 = 4B + \beta^2 - 1.$$

Its actual finite endpoint has $\beta = 0$ or finite native < 1. Own zero has negative time; parallel has its positive null. The own two-cycle makes B integral. Its OVER-K normalization has raw square B, old-short product $-B/2$, and connected old-B/cap Schur complement $B - 4B^2 < 0$. Thus ACTUAL $K = \mathbb{Q}(B)$ and $0 < B_\sigma < 1/4$. A finite own has $B < 4$, giving nonzero integer norm < 1 in degree at least two; rational degree is already excluded after field descent. The own is dotted. The actual edge $C = 2\sqrt{B}/\sqrt{1-\beta^2}$ now gives

$$B \in \mathcal{O}_K, \quad 4 < B \le \frac{441}{16}(1-\beta^2), \quad 0 < B_\sigma < 1/4, \quad [K:\mathbb{Q}] = 2 \text{ or } 3. \tag{B16Istrip}$$

Degree at least four would have norm $< (441/16)(1/4)^3 < 1$. The cubic alternative is retained until its complete argument below.

L.3 The entire cubic $B_{16} + I$ short-cap branch

Assume degree three in (B16Istrip). Write its roots a, u, v, with a physical. Its nonzero positive norm satisfies $0 < auv < (441/16)(1/4)^2 < 2$, hence $auv = 1$ and $a > 16$. The minimal polynomial is

$$P(X) = X^3 - SX^2 + TX - 1, \qquad 17 \le S \le 28.$$

Because $u, v < 1/4 < a$, $P(1/4) < 0$; multiplying by sixty-four gives $4T - S \le 15$. At the root u,

$$0 = P(u) < -(S - 1/4)u^2 + Tu - 1.$$

The quadratic is positive there and has negative leading coefficient, so its discriminant is positive, giving $T^2 \ge 4S$. For $17 \le S \le 20$, the first bound gives $T \le 8$, contradicting $T^2 \ge 68$. For $21 \le S \le 24$, $T \le 9$ contradicts $T^2 \ge 84$. For $25 \le S \le 28$, $T \le 10$ forces $S = 25, T = 10$. Thus the COMPLETE cubic domain is

$$F(X) = X^3 - 25X^2 + 10X - 1.$$

Its signs at $0, 1/5, 1/4, 24, 25$ are $-1, 1/125, -3/64, -337, 249$. Its physical root lies in $(24, 25)$. Positive old-I contact would have $\beta \ge 1/2$, but then $a \le 1323/64 < 24$. Hence $\beta = 0$. The rational-root test proves irreducibility, and the polynomial discriminant is the squarefree $473 = 11 \cdot 43$. Therefore $\mathcal{O}_K = \mathbb{Z}[B]$, $\mathrm{N}(B) = 1$, $\mathrm{N}(4B - 1) = 3$. These are necessary algebraic data, not an example of an actual cap or chamber.

The ACTUAL I invariant $J = 4(1 + h)$ lies in this SAME field, with

$$4 < J_a < 6, \qquad 0 < J_u, J_v < 4.$$

Exact signs at $23/125, 369/2000$ and $441/2000, 111/500$ give

$$.184 < u < .1845, \quad .2205 < v < .222, \quad a > 24.5935, \quad v - u > .036.$$

The corresponding exact evaluations are

argument	numerator	denominator
23/125	−333	1953125
369/2000	2193409	8000000000
441/2000	1716121	8000000000
111/500	−144869	125000000.

Put $\eta = 1/B = B^2 - 25B + 10$. The integral bases $1, B, B^2$ and $1, B, \eta$ differ by determinant one. Thus write uniquely $J = l + mB + n\eta$. Since $auv = 1$,

$$\frac{J_a - J_u}{a - u} = m - nv, \qquad \frac{J_a - J_v}{a - v} = m - nu.$$

Both values lie in $(0, .247)$, because their numerators are in $(0, 6)$ and their denominators exceed 24.368. Hence $|n|(v - u) < .247$, so $|n| \le 6$. The COMPLETE integer check is as follows. For $n = 1, \dots, 4$, the first inequality gives $m \ge 1$, the second $m < .985$. For $n = 5, 6$, it gives $m \ge 2$, while the second gives $m < 1.354$. At zero, $m \in (0, .247)$, impossible. For $n = -k$, $1 \le k \le 4$, the second inequality gives $m \ge 0$; $k \ge 2$ violates the first because $kv > .441 > .247$. At $k = 1$ only $m = 0$ remains. For $k = 5, 6$, both inequalities force $m = -1$; at five the second is negative, at six the first exceeds $.323 > .247$. Thus only $m = 0, n = -1$ remains. Now $J = l - \eta$, with $l = 5, 6$ from the physical strip. Five violates $J_u > 0$, since $u < 1/5$. Consequently

$$J = 6 - 1/B.$$

The strip is solved completely; it was not falsely declared empty.

It forces $h = (2 - 1/a)/4 > 47/96$. The increasing relation between actual I energy floor r and height is $h = 4r/(1 - r)^2$. At $r = .099$, $396000/811801 < 47/96$, with cleared positive margin 138647. Thus $e_I > .099$. The isolated short cap has $\beta = 0$, so its actual edge has $C = 2\sqrt{a} > 9$, giving $S_{16} > 7/2$. Use the UNLIFTED reference of energy $> .89026$, with individual P_{14} price $> .13$, and insert $.13(P_{16} - P_{14})$. This residual has square $2(.13)^2$ and full reference cross zero. The SAME original-vector square gives

$$e_B > .89026 + .26(3.5 - 3.311) - .0338 = .90560.$$

Add I ONCE: total > 1.00460. The entire cubic branch is impossible. Only the quadratic short-cap branch remains.

L.4 The complete quadratic $B_{16} + I$ parameter domain

Assume the quadratic branch of (B16Istrip). The ACTUAL I invariant $J = 4(1 + h)$ is integral, $4 < J < 6$, $0 < J' < 4$, and cannot be rational. Write its polynomial $X^2 - sX + m$. The full trace/norm conditions are

$$5 \le s \le 9, \qquad 6s - 36 < m < 4s - 16, \quad m \ge 1.$$

The COMPLETE nineteen pairs are $s = 5 : m = 1, 2, 3$; $s = 6 : m = 1, \dots, 7$; $s = 7 : m = 7, \dots, 11$; $s = 8 : m = 13, 14, 15$; $s = 9 : m = 19$. The square discriminants at $(6, 5), (7, 10), (8, 15)$ are not

quadratic. The remaining sixteen actual-I models are as follows; $M = -\mathrm{N}(J(J-4)) > 0$:

K	J	M
$\mathbb{Q}(\sqrt{2})$	$3+\sqrt{2},\ 4+\sqrt{2},\ 3+2\sqrt{2}$	$7, 28, 7$
$\mathbb{Q}(\sqrt{3})$	$3+\sqrt{3},\ 4+\sqrt{3}$	$12, 39$
$\mathbb{Q}(\sqrt{5})$	$(7+\sqrt{5})/2,\ (9+\sqrt{5})/2,\ 3+\sqrt{5}$	$11, 19, 16$
$\mathbb{Q}(\sqrt{6})$	$3+\sqrt{6}$	15
$\mathbb{Q}(\sqrt{7})$	$3+\sqrt{7}$	12
$\mathbb{Q}(\sqrt{13})$	$(5+\sqrt{13})/2,\ (7+\sqrt{13})/2$	$3, 27$
$\mathbb{Q}(\sqrt{17})$	$(5+\sqrt{17})/2,\ (7+\sqrt{17})/2$	$4, 32$
$\mathbb{Q}(\sqrt{21})$	$(5+\sqrt{21})/2,\ (7+\sqrt{21})/2$	$3, 35.$

This field list follows from an ACTUAL wall strip; it is not a global field census.

For the short-cap integer B, put $T = \mathrm{Tr}(B)$, $N = \mathrm{N}(B)$. Its physical/conjugate strip gives $N = 1, \dots, 6$, $T \le 27$, $T \ge 4N+1$. For field discriminant d, write $T^2 - 4N = dk^2$. The full-order parity and small conjugate are exactly

$$T \equiv k \pmod 2 \quad (d \text{ odd}), \qquad T \equiv 0 \pmod 2 \quad (4 \mid d),$$
$$(2T-1)^2 < 4dk^2 < 4T^2.$$

Comparing these bounded integer squares gives the COMPLETE table

d	k full range	(T, k)
8	$1, \dots, 9$	$(6,2), (20,7)$
12	$1, \dots, 7$	$(14,4)$
5	$1, \dots, 12$	$(7,3), (18,8), (25,11)$
24	$1, \dots, 5$	$(10,2), (20,4)$
28	$1, \dots, 4$	$(16,3)$
13	$1, \dots, 7$	$(11,3), (22,6)$
17	$1, \dots, 6$	$(21,5)$
21	$1, \dots, 5$	$(5,1), (23,5).$

For $4 \mid d$, the even trace is at most twenty-six. All loop bounds are explicit integer inequalities; there is no Pell cutoff or unbounded coefficient search.

Here are the resulting fourteen models. Write $H = -\mathrm{N}(B(4B-1)) = N(4T - 16N - 1)$. The

last column keeps ALL possible positive old-I costs $Q=\beta^2$:

$\sqrt{D}$	B	(T,N)	H	$Q>0$
$\sqrt{2}$	$3+2\sqrt{2}$	$(6,1)$	7	$1/4$
	$10+7\sqrt{2}$	$(20,2)$	94	$1/4$
$\sqrt{3}$	$7+4\sqrt{3}$	$(14,1)$	39	$1/4$
$\sqrt{5}$	$(7+3\sqrt{5})/2$	$(7,1)$	11	$1/4,(3+\sqrt{5})/8$
	$9+4\sqrt{5}$	$(18,1)$	55	$1/4$
	$(25+11\sqrt{5})/2$	$(25,5)$	95	none
$\sqrt{6}$	$5+2\sqrt{6}$	$(10,1)$	23	$1/4,1/2$
	$10+4\sqrt{6}$	$(20,4)$	60	none
$\sqrt{7}$	$8+3\sqrt{7}$	$(16,1)$	47	$1/4$
$\sqrt{13}$	$(11+3\sqrt{13})/2$	$(11,1)$	27	$1/4,1/2$
	$11+3\sqrt{13}$	$(22,4)$	92	none
$\sqrt{17}$	$(21+5\sqrt{17})/2$	$(21,4)$	76	none
$\sqrt{21}$	$(5+\sqrt{21})/2$	$(5,1)$	3	none
	$(23+5\sqrt{21})/2$	$(23,1)$	75	none.

For completeness, positive Q initially has $Q<377/441$, so every native nine or larger is excluded. Native seven has cubic degree. Only labels $3,4,5,6,8$ remain; their own two-cycles must lie in K. Substituting their squares into the physical bound and FULL nonphysical Schur condition $4B'+Q'<1$ gives exactly the last column; in particular all six/eight contacts fail. The old I root has not been assigned a freely chosen raw length.

We eliminate EVERY positive Q. Let that old root's raw square be λ, while the old-B short root has square one. The short cap can be rescaled OVER K to square B, own product $-B/2$. If $Q>0$, its actual old-I product has square $B\lambda Q$, a K-square. The actual I opposite has raw square λJ and time square $\lambda J(J-4)/4$. Both actual cap times lie on the SAME K-vertex line. Taking their ratio and the known square $B\lambda Q$ proves the necessary FULL field-square condition

$$\frac{Q(4B+Q-1)}{J(J-4)}\in K^2.$$

No assumption that λ is a square is made.

For $Q=1/4$, the rational norm requires $(48T-256N-9)/M$ to be a rational square. In field order $\sqrt{2},\sqrt{3},\sqrt{5},\sqrt{6},\sqrt{7},\sqrt{13}$, its complete numerator list is

$$(23,439),\quad 407,\quad (71,599),\quad 215,\quad 503,\quad 263.$$

Against the complete M-table above, every ratio fails by odd prime valuation: seven for $\sqrt{2}$, three for $\sqrt{3},\sqrt{6},\sqrt{7},\sqrt{13}$ (after cancelling the common five in $215/15$); $71,599$ match none of the square-classes $11,19,1$. For $Q=1/2$, the numerator is $8T-64N-1$. At $\sqrt{13}$ it is twenty-three, incompatible with $M=3,27$. At $\sqrt{6}$ the norm passes, but the FULL expression is $B/4$, $B=5+2\sqrt{6}$. A square root of B would require rational square part three or two; neither is a rational square. The remaining golden $Q=(3+\sqrt{5})/8$ has norm $1/16$, while $\mathrm{N}(4B+Q-1)=-139/16$. It would require $139/M$ square for $M=11,19,16$, impossible. Thus the short cap is PURE:

$$\beta=0. \tag{B16Ibeta}$$

Its ACTUAL negative core $c_S+\gamma^{-1}n_{16}$ is future, zero on the cap, negative on the old short wall and nonpositive against every other original wall. With $u=4/B\in(0,1)$, it gives

$$V=P_{16}-ue_{16},\quad x\cdot V\ge G=\sqrt{16-u}-\sqrt{u(1-u)},\qquad R_{\mathrm{tail}}=(V-P_{14})^2=1+(1-u)^2.$$

Use the UNLIFTED reference w of energy $> .89026$. A signed first-coordinate demand $x_1 > X > b$ can be grafted in the first six block, preserving its sum. For EVERY $X \le 1$ the remaining five coordinates exceed $.2036 > .197$; all coefficients stay nonnegative. The gain is $(6/5)(X-b)^2$. Pay .13 at P_{14} before inserting the actual short-foot pressure. Its residual is orthogonal to the full reference and first-six change. ONE square gives

$$\|x\|^2 > .89026 + r_0 + \frac{6}{5}(X - .3364)^2 + g_S, \qquad g_S = .26(G - 3.311) - .0169R_{\text{tail}}, \qquad \text{(B16Ifinal)}$$

where $e_I > r_0$; omit the coordinate term if none is required.

The following COMPLETE elementary interval table seals the short term for the fourteen B-models, in their field order:

$\sqrt{D}$	u intervals	$G >$	$R_{\text{tail}} <$	$g_S >$
$\sqrt{2}$	$(.6856, .6864); (.1998, .2012)$	$3.448; 3.572$	$1.099; 1.641$	.017
$\sqrt{3}$	$(.28704, .2872)$	3.511	1.509	.0264
$\sqrt{5}$	$(.5834, .584); (.2224, .224); (.16116, .1616)$	$3.432; 3.554; 3.610$	$1.174; 1.605; 1.705$	.0116
$\sqrt{6}$	$(.404, .4048); (.202, .2024)$	$3.458; 3.571$	$1.356; 1.637$	.0153
$\sqrt{7}$	$(.2504, .2516)$	3.53	1.563	.0305
$\sqrt{13}$	$(.3664, .367); (.1832, .1835)$	$3.470; 3.588$	$1.402; 1.668$	.0176
$\sqrt{17}$	$(.192, .19225)$	3.581	1.653	.0422
$\sqrt{21}$	$(.8348, .835); (.174, .175)$	$3.521; 3.598$	$1.028; 1.683$	.0372.

All entries follow by positive squares from

$$\sqrt{2} \in (1.4142, 1.4143), \quad \sqrt{3} \in (1.73205, 1.73206),$$
$$\sqrt{5} \in (2.2360, 2.2361), \quad \sqrt{6} \in (2.4494, 2.4495),$$
$$\sqrt{7} \in (2.6457, 2.6458), \quad \sqrt{13} \in (3.6055, 3.6056),$$
$$\sqrt{17} \in (4.1231, 4.1232), \quad \sqrt{21} \in (4.5825, 4.5826).$$

Bound $u(1-u)$ at the interval end nearest $1/2$. For example $\sqrt{17}$ uses $\sqrt{16-u} > 3.9758$, $\sqrt{u(1-u)} < .3941$, giving $g_S > .0422643$. The smaller $\sqrt{2}$ model gives $g_S > .0170469$. These are entire interval seals, not sampled optimizer outputs.

L.5 The quadratic $B_{16} + I$ head and final same-vector table

Deleting the actual long head retains B_{15} and old I. Its ONLY nonisolated finite extension is simple-three at long endpoint two. A positive old-I contact there is only three. Thus

$$c_H = A_H p - g_H \omega_1 - \omega_2/2 - \beta_H e, \quad A_H^2 = 2g_H^2 + 2g_H + Q_H, \quad Q_H = 0, 1/4.$$

Own zero with $Q_H = 0$ has zero time; with $Q_H = 1/4$ its COMPLETE positive old-B clearing is the nonzero future null $(p-e)/2$. It is feasible against all original walls. Own three has its affine triangle. Every other positive own gives $C_H \ge 1 + \sqrt{2}$, $x_1 > .643$. The legal first-six graft alone yields

$$e_B > .89026 + (6/5)(.643 - .3364)^2 > 1.$$

The actual head is isolated WITHIN B.

Write

$$U = 4g_H^2 \in \mathcal{O}_K, \quad A_H^2 = U/2 + Q_H - 1, \quad C_H^2 = \frac{U}{2(1-Q_H)}.$$

Acuteness with the DISTINCT actual I opposite gives the useful complete foreign constraint

$$h(U/2-1) \geq Q_H.$$

Own zero/three/four is impossible by time and this constraint. Own six with $Q_H > 0$ contradicts $h < 1/2$; at zero it has its actual affine 6–3 path. Parallel has the positive null pair. The complete finite quadratic-own list therefore leaves only five/ten in $\mathbb{Q}(\sqrt{5})$, eight in $\mathbb{Q}(\sqrt{2})$, twelve in $\mathbb{Q}(\sqrt{3})$. Their necessary costs are $Q_H = 0$ for five, and $0, 1/4$ for the others. In the low-J $\sqrt{3}$ branch, twelve also has $Q_H = 0$, since $(\sqrt{3}-1)\sqrt{3}/8 < 1/4$.

For a dotted head the actual old-long normalization has raw square $2U$, own product $-U$. Its connected Schur complement is $-U(U-2) < 0$. Thus ACTUAL $K = \mathbb{Q}(U)$, $U > 4$, $0 < U' < 2$, and U is not rational. The short and head raw time squares, anchored in the SAME old B component, are

$$H_S^2 = B(4B-1), \qquad H_H^2 = U(U-2+2Q_H).$$

Subtracting their full K-valued old-forest projections leaves the SAME one-dimensional K-vertex line. Hence H_S^2/H_H^2 must be a FULL K-square. We do not compare the freely normalized I time after eliminating the short cap's old-I contact. For $z = a + b\sqrt{D}$ of norm q^2, a square requires one of $(a+q)/2, (a-q)/2$ to be a rational square; both signs will be checked. The actual short and head walls are distinct because only the latter has a positive old-head product.

For $\mathbb{Q}(\sqrt{2})$, finite-eight head time norms are $-4, -7/2$ at $Q_H = 0, 1/4$. Neither matches short norms $-7, -94$. The COMPLETE dotted U values below $5 + 3\sqrt{2}$ are

$$3+\sqrt{2}, \quad 3+2\sqrt{2}, \quad 4+2\sqrt{2}.$$

Their pure time norms are $-7, -7, -32$. All fail against short norm -94 by odd forty-seven valuation. Against $B = 3+2\sqrt{2}$, the first ratio is $(43+30\sqrt{2})/7$, norm one, whose rational square parts $25/7, 18/7$ both fail. The last fails the norm test. At $U = B$ the FULL time-square test PASSES, so a different actual condition is required. For $s = \sqrt{2}$, raw norms are $2B, B$, future time product $27 + 19s$, spatial product $B^2/2$. Their raw mutual product $g = -(37+26s)/2$ has $g/B = -(7+4s)/2$; the ACTUAL two-cycle is

$$\frac{4g^2}{2B^2} = \frac{81+56s}{2},$$

nonintegral. For low $J = 3+\sqrt{2}$, the foreign constraint excludes positive Q_H at the first two low U, with upper bounds $1/8, (3-\sqrt{2})/8$. At the next U, positive cost gives $C_H > 17/8$, $x_1 > .6$. The pure low values were all excluded, and the next pure U also gives that floor. For $J = 4+\sqrt{2}$, only the first low U could prevent $x_1 > .49$; its pure case failed, and positive cost gives $C_H > 5/3, x_1 > 1/2$. Every larger U has the same stronger floor. At the high $J = 3+2\sqrt{2}$, no head floor will be required.

For $\mathbb{Q}(\sqrt{3})$, low $J = 3+\sqrt{3}$ forces finite-twelve cost zero. Its time norm -3 mismatches short norm -39. The only dotted $U < 4+2\sqrt{3}$ is $3+\sqrt{3}$. Its pure norm -12 also fails. Positive cost gives $C_H > 71/40$, hence $x_1 > .527$. All larger U give a stronger floor. The high-J branch needs no head floor.

In $\mathbb{Q}(\sqrt{5})$, finite-five time norm is -1; finite-ten norms are $-5, -5/4$. None matches short norms $-11, -55, -95$. The COMPLETE dotted $U < (9+3\sqrt{5})/2$ are

$$3+\sqrt{5}, \quad 4+\sqrt{5}, \quad (7+3\sqrt{5})/2.$$

Their pure time norms are $-16, -11, -9$. Only $U = 4+\sqrt{5}$, smallest short $B = (7+3\sqrt{5})/2$, passes a norm test. Its time ratio is even a field square, but the actual raw products are

$$\text{time} = (69+31\sqrt{5})/2, \quad \text{spatial} = (43+19\sqrt{5})/4, \quad g = -(95+43\sqrt{5})/4.$$

With raw norms $2U, B$, $\mathrm{N}(U) = 11, \mathrm{N}(B) = 1, \mathrm{N}(g) = -55/4$. The ACTUAL two-cycle has norm $275/4$, not integral. At the lowest $U = 3 + \sqrt{5}$, the foreign constraint permits only $Q_H = 0, 1/4, 1/2, \varphi^2/4$. Their time norms are $-16, -11, -4, -19$. Only $Q_H = 1/4$, smallest short B, passes the norm test. Its field ratio $(103+45\sqrt{5})/22$ has norm one and rational square parts $125/44, 81/44$, both nonsquares. Thus the lowest U is impossible with EVERY foreign cost. At each of the next two low U, positive cost gives $x_1 > .57$; their pure cases failed above. Every larger U has that floor even at zero cost. Thus $x_1 > .57$ throughout the golden field.

The other fields have no finite head among five/eight/ten/twelve. Their complete low dotted cases are

$\sqrt{D}$	U requiring a test	$\mathrm{N}(H_H^2)$, $Q_H = 0$	$-\mathrm{N}(H_S^2)$	consequence
$\sqrt{6}$	none; $U \geq 3 + \sqrt{6}$	$--$	$23, 60$	$x_1 > .49$
$\sqrt{7}$	none needed	$--$	47	no head floor
$\sqrt{13}$	$(5 + \sqrt{13})/2$	-9	$27, 92$	high $J:\ x_1 > .48$
$\sqrt{17}$	$(5 + \sqrt{17})/2,\ (7 + \sqrt{17})/2$	$-8, -16$	76	$x_1 > .51$
$\sqrt{21}$	$(5 + \sqrt{21})/2$	-5	$3, 75$	$x_1 > .49.$

The displayed norm ratios are nonsquares $1/3, 9/92, 2/19, 4/19, 5/3, 1/15$. Positive Q_H at a listed low U already gives its required coordinate floor. The next U in the last three fields are respectively $(7+\sqrt{13})/2, 5+\sqrt{17}, (7+\sqrt{21})/2$; these suffice at zero cost and strengthen with cost. All these finite low-U lists are complete by writing $U = (T + k\sqrt{d})/2$ with full-order parity, $U > 4, 0 < U' < 2$: the stated physical threshold bounds $k > 0$, and the conjugate width gives the listed constants. For $\sqrt{6}$, $C_H > 80/49$ gives $x_1^2 > 31/129 > .49^2$. At $\sqrt{21}$, positive cost gives $C_H^2 > 8/3 > (80/49)^2$. The low-J $\sqrt{13}$ case instead uses the direct actual-I pressure $t > .5905 > .59$, from $h < .076$. No unbounded native or head-coefficient domain remains outside these monotone upper ranges.

Insert these ACTUAL floors into the single reference (B16Ifinal). Each r_0 follows by the increasing identity $h = 4r/(1-r)^2$ and the positive rational radical brackets already printed. The COMPLETE scalar dispatch is

$\sqrt{D},\ J$ branch	r_0	X	$g_S >$	total $>$
$\sqrt{2},\ 3 + \sqrt{2}$	.024	.6	.017	1.014641952
$\sqrt{2},\ 4 + \sqrt{2}$	.075	.49	.017	1.010571552
$\sqrt{2},\ 3 + 2\sqrt{2}$	.093	$--$	.017	1.00026
$\sqrt{3},\ 3 + \sqrt{3}$	.041	.527	.0264	1.001254032
$\sqrt{3},\ 4 + \sqrt{3}$	.089	$--$	.0264	1.00566
$\sqrt{5},$ all	.035	.57	.0116	1.002342752
$\sqrt{6}$	.076	.49	.0153	1.009871552
$\sqrt{7}$	.085	$--$	.0305	1.00576
$\sqrt{13},$ low	.018	.59	.0176	1.003035552
$\sqrt{13},$ high	.070	.48	.0176	1.002605152
$\sqrt{17},$ all	.032	.51	.0422	1.000624352
$\sqrt{21},$ all	.045	.49	.0372	1.000771552.

A dash means that the coordinate gain is OMITTED, not charged again. Every row strictly exceeds one. All old-I contacts, actual head roles, short models, field-square exceptions and positive time signs were retained. The quadratic branch is impossible. Together with the complete cubic and six-role reductions, this excludes the entire B_{16}+ACTUAL-I rank-seventeen forest on ONE original vector.

L.6 The ordinary $D_{16} + A_1$ branch and its shared coefficient

The WHOLE forest is $D_{16} + A_1$, with the actual rank-one edge NONSHORT: $L_e \geq \operatorname{arsinh}(1/\sqrt{2})$. This is the ordinary hypothesis, not a property of a bare A_1 diagram. Use genuine unit D-roots, centre fourteen, forks fifteen/sixteen, signed last coordinate, and orthogonal old unit root e. Write the original vector as (x, y), with $y > 0$ and $\|x\|^2 + y^2 < 1$. The actual nonshort edge gives

$$y^2 \geq 5 - 2\sqrt{6} > .1, \qquad y > s_0 = \sqrt{.1}, \quad .316 < s_0 < .317.$$

We will keep this ONE external coordinate inside any mixed head/centre square. For an opposite retaining e, its entire foreign contact is β, $Q = \beta^2 < 1$. Positive Q is native and at least $1/4$, with no value strictly between $1/4$ and $1/2$.

The generic $M = 10$ bulk applies through eleven at the ACTUAL D_{16} rank. Its only weak role is pure far helper, removed from six onward by the arithmetic literal-helper lemma. If five is weak, its actual pure point and strong six/eleven give the positive interpolation energy $> .9104$ already proved above. Adding the nonshort energy $> .1$ is impossible, so five is strong.

The FRESH D_4, D_3 layers are twelve, thirteen. Their full finite lists retain all one-sided foreign contacts. Full D_{16} or long-arm A_{16} bridges admit no extra old-A_1 arm; the latter would create inadmissible branch arms, the former another branch. Own zero/three/four/six/parallel have their actual affine/null or direct strong alternatives; end-four and sealed H_4 are strong. Positive $Q \geq 1/4$ strengthens the weak fork-three/end-three ratios enough to make own-five strong. At $Q = 0$, the full all-field helper norm/trace seal leaves only golden own.

For those two golden packets, the ACTUAL LEFT H_4 correction is $g(3n_i/2 + n_{i-1} + n_{i-2}/2)$, $g = \varphi/2$, with $\alpha^2 = \varphi + 5/8 > 56/25$, $\delta_5 = (5 - 3\varphi)/8$. Its finite PROJECTED POINT, positive and all-original-wall feasible, gives pure displacement V, flat through $i - 3$, $V^2 < i$, pressure $G > \sqrt{i - \kappa/\alpha^2} - .091$. The two complete payments are

	i	K	μ	$G >$	total $>$
D_4	12	9	1/6	3.34	$.79805 + .1 + (3.34 - 2.569)/3 - 3/36 > 1.0717$
D_3	13	10	.16	3.47	$.82436 + .1 + .32(3.47 - 2.732) - .0256(3) > 1.0837.$

Here $\kappa = 1/2, 5/8$, respectively. The available terminal slopes are $d_8 > 1/6, d_9 > .16$; each is paid BEFORE the actual pressure. The root seals are $3.431^2 < 12 - 25/112$, $3.561^2 < 13 - 125/448$. Every such packet is impossible. Thus ALL five through thirteen are actually strong, with

$$E_{11} > .84816, \quad E_{13} > .88989, \quad b = f_5/5 < .361, \quad P_5 \text{ price} > .14.$$

At the reference ending eleven, P_{11} price $> 1/7$, $f_{11} < 2.886$. The scalar $E_{14} > .9084$ is not yet an available centre floor.

Deleting the actual head retains D_{15} and old e. Its only nonisolated finite role is simple-three at long leaf two. The exact formula is

$$A_H^2 = 2g^2 + 2g + Q, \qquad C_H = (2g + 1)/\sqrt{1 - 2Q}.$$

Own zero at positive Q has the ACTUAL future null

$$c + \omega_2/2 = \sqrt{Q}\, p - \beta e, \qquad \omega_2/2 = n_1 + 2(n_2 + \cdots + n_{14}) + n_{15} + n_{16}.$$

All omitted wall signs are nonpositive. Own-three has its affine triangle. Other positive owns give $x_1 > .643$; the legal first-five graft gain $> .099405$, together with $E_{13} + .1$, exceeds one. The head is isolated WITHIN D, so

$$A_H^2 = 2g^2 + \beta_H^2 - 1, \qquad C_H^2 = 2g^2/(1 - \beta_H^2).$$

Own zero has negative time. For own three at positive time, its foreign shadow has $\beta_H/A_H > 1$, making it also close the orthogonal rank-one edge, impossible. Own four is zero time if pure, forbidden shadow equality if mixed. Own six/parallel has its actual affine/null principal with every foreign contact present. Only own five, finite at least seven, or dotted remains.

We record a compatible legal head gain

$$\eta_H = .0059. \tag{D16OH}$$

Every head residual is zero-sum on the first five coordinates. For a PURE finite own at least seven or dotted, the actual first-three-root point and FULL scalar (R1S) give gain $> .2893 - .8(.361) + .032 > .0325$ after paying .08 at P_5. No short or fork normal is used. For pure native five, the ACTUAL finite H_4 point has

$$x \cdot (P_4/4) \geq H_H = \frac{1}{2\sqrt{\varphi}} > .393.$$

Indeed its corrected square is $\delta_H = (5 - 3\varphi)/8$, time square $a = (\varphi - 1)/2$; the finite-point formula $H(A, \delta) = A/(1 + \sqrt{\delta/(A^2 + \delta)})$ divided by $\sqrt{2}g$ simplifies to the displayed $1/(2\sqrt{\varphi})$. Its strict rational seal follows from $\varphi > 1.618$. Pay .02 at P_5 for $.1(P_4/4 - P_5/5)$, of square .01/20. The gain is $> .2(.393 - .361) - .01/20 = .0059$.

For mixed finite own at least seven or dotted, $\beta_H \geq 1/2, g > .9$ gives $C_H^2 > (8/3)(.9)^2 > 1.46^2$, hence $x_1 > .43$. Its legal fixed-sum first-five graft gains $> (5/4)(.43 - .361)^2 = .00595125$. It has no external coordinate payment.

For mixed native five keep that coordinate. The ACTUAL H_4 point gives

$$x \cdot (P_4/4) + d_H y \geq H_H, \quad d_H = \frac{\beta_H}{\sqrt{2}g} < .875, \quad H_H = \frac{H(A_H, \delta_H)}{\sqrt{2}g}.$$

Here $A_H^2 = a + \beta_H^2$, $a = (\varphi - 1)/2$. The function $\sqrt{a + \beta^2} - \beta s_0$ increases for $\beta \geq 1/2$. Using $1.618 < \varphi < 1.619$ gives

$$A_H(1/2) > .7476, \quad \sqrt{\delta_H} < .136, \quad \sqrt{2}g < 1.145, \qquad H_H - d_H s_0 > \frac{.7476 - .136 - .5(.317)}{1.145} > .393.$$

Pay .02 at P_5 AND $.1d_H$ from the positive external coefficient s_0, retaining final coefficient s_0. The ONE ambient completed square gives the SAME gain .0059. A later centre pressure with coefficient 1/7 and external factor $d_C < .875$ spends $< .125$ more. The combined external payment is $< .0875 + .125 = .2125 < s_0$. Thus this head gain is LEGALLY compatible with that later mixed centre, without spending the rank-one energy twice.

L.7 The complete ordinary D_{16} centre

Deleting centre fourteen retains A_{13}, both ACTUAL fork roots and old e. Its COMPLETE finite-white list is: without fork, isolation or admissible endpoint/D-position three or endpoint four on the left; left endpoint-three with one fork-three/four; the full all-three endpoint/two-fork bridge; or no left with one arbitrary finite fork contact, or fork pairs $(3,3), (3,4), (3,5)$. An internal attachment plus fork gives inadmissible arms; a four internal to a long path is nonfinite. Rank excludes exceptional completions. Write the actual residual and ratio as

$$\Delta = 1 - q_L - \delta^2 - \epsilon^2 - Q > 0, \quad \sigma = g + \eta_L j/14 + (\delta + \epsilon)/2, \quad C_{14} = \sqrt{28}\sigma/\sqrt{\Delta}.$$

The sufficient .61 ratio target is $< 25/16$.

With no fork and a left contact, own-zero far-three requires $1/2 < Q < 15/28$, containing no native foreign square. Far D-position and far-four require $Q < 1/7, 1/14$, hence zero and zero time. Other own-zero roles have the actual affine fork/terminal-four principals. Own-three/white-three has the actual cycle; own-four the star at fourteen with cap and both forks; own-six its 6–3 principal; parallel its null. Positive endpoint-four, D-position and near-three are directly strong. Only PURE far-three can be weak, since the complete foreign-far clearing is strong at this index. Lemma 5 excludes that pure role at fourteen.

For isolation within D, $A_c^2 = 28g^2 + Q - 1$. Own zero is negative time. Small owns and parallel have the actual four-leaf/star/chain/null witnesses. For $Q > 0$ or dotted own, keep $u = (1-Q)/g^2$. At $0 < u < 1$, the actual positive clearing $c + \beta e + gun_{14}$ gives PURE-D pressure

$$h_{14} > \sqrt{28 - u} - \sqrt{u(1-u)}.$$

Its full loss is $< .1u + \sqrt{u(1-u)} < .5525 < .61$. At one the actual clearing is the forbidden future null. The finite mixed interval $u > 1$ MUST also be retained: it occurs only for own-five and $Q = 1/4$. The ACTUAL finite projection onto c, n_{14}, e, with positive corrections on all three roots, gives instead

$$h_{14} > \sqrt{28 - u} - \sqrt{u - 1}.$$

Here $u \le 3/\varphi^2 = 6 - 3\varphi < 7/6$, $\sqrt{u-1} \le 2 - \varphi < .382$. The first loss is $< (7/6)/10 < .117$, so total loss is $< .499 < .61$. The projected POINT, not its spacelike correction, provides the actual-wall comparison. Pure native five uses the left H_4 with loss $< .15 + .168$; pure native seven the left A_2 with loss $< .124 + .319$; every finite native at least eight the finite pair with loss $< .118 + .415$. All omitted original walls remain nonpositive by the same positive actual-root constructions. Thus every isolated own/foreign parameter is strong or impossible.

For left/fork bridges, full two-fork D_{16} is foreign-sealed and strong at own zero. Left/fork-four is sealed, residual $1/28$, and directly strong. Left/fork-three has pure residual $2/7$, own-zero cycle, own-three triangle and own at least four directly strong. Its POSSIBLE old-A_1-three arm is retained: $Q = 1/4$ leaves residual $1/28$ and strengthens the bound. No larger finite foreign label fits that long branch. This row is not incorrectly called sealed.

With no left and positive own, own-three/ fork-three has its affine triangle; own-three/fork at least six is strong. Own-three/fork-four or five is strong with $Q > 0$; its pure case is recognized below at the actual reciprocal fork. Own-four has the star on cap, old thirteen and the UNCONTACTED fork; own-six uses the preceding left node, and parallel its null. For own at least five and fork at least four, the direct ratio is $\ge \varphi/\sqrt{2} + 1/2 > 25/16$. Fork-three/own-five with $Q \ge 1/2$ is directly strong. For $Q = 0, 1/4$ take its ACTUAL left-H_4 correction, with $\alpha^2 = \varphi + 5/8$. At zero, its finite POINT is pure D, flat through eleven, pressure $G > 3.38$, squared norm < 14. At $Q = 1/4$, the positive FUTURE core

$$c + g(3n_{14}/2 + n_{13} + n_{12}/2) + \beta e$$

has square and cap product $\delta_5 - 1/4 < 0$, zero selected old and e-products, and all omitted wall products nonpositive. It clears e to the SAME pure D-displacement. Its pressure satisfies

$$G > \sqrt{14 - 1/(2\alpha^2)} - .322 > 3.38.$$

Use $\sqrt{14 - 25/112} > 3.711$ and $(1/4 - \delta_5)/\alpha^2 = 3(\varphi - 1)/(8\varphi + 5) < .322^2$. Paying $1/7$ at P_{11} gives

$$.84816 + .1 + \frac{2(3.38 - 2.886)}{7} - \frac{3}{49} > 1.028.$$

No head gain is needed for these rows. For every other own finite at least seven or dotted, positive Q is strong; $\gamma \geq 6/5$ is strong. The remaining PURE fork-three role has $202/125 < 2\gamma < 12/5$; the full all-field fork-helper proof in Lemma 5 leaves only the already excluded golden own-five.

With no left and own zero, single fork-three has the actual E_8 kernel: centre fourteen six, uncontacted fork three, contacted fork four, cap two, and old thirteen through nine coefficients five through one. Single fork-four has kernel $(\sqrt{2}, 2, 2, 1, 1)$ on cap, contacted fork, fourteen, thirteen, other fork. Single fork-six has the 6–3 principal. Every kernel has positive cap time and correct signs at ALL omitted walls. Fork-five allows exactly $Q = 0, 1/4$ and remains below. Every finite fork label at least seven has $\delta^2 > .81$; retained-white positivity forces $Q = 0$. Two forks/own-three has its actual triangle, own at least four is strong. Own-zero pair-three/three has the affine square; pairs-three/four and three/five have $Q = 0$. These are all remaining finite, native, parallel, dotted and foreign roles.

L.8 Ordinary D_{16}: the mixed finite point is paid jointly

For own-zero/single-fork-five, put $\delta = \varphi/2, \alpha = \delta/\sqrt{2}$. Select the ACTUAL H_4 on cap, contacted fork, fourteen, thirteen, with positive correction $\delta(3n_f/2 + n_{14} + n_{13}/2)$. Its finite projected POINT has divided D-displacement

$$V_D = (1^{12}, 1/2, 1/2, 1/2, \ \pm 1/2), \qquad V_D^2 = 13.$$

The sign is the genuine fork sign. At $Q = 0$, its demand is $G > \sqrt{8\varphi} - \sqrt{13 - 8\varphi} > 3.36$, from $\varphi > 1.618$. Pay $1/7$ at P_{11}; the residual norm is $13 - 11 = 2$. With $E_{11} + .1$ alone the total exceeds 1.0427.

At $Q = 1/4$, KEEP the actual external coordinate in the SAME point:

$$x \cdot V_D + d_C y \geq G, \qquad d_C = \sqrt{2}/\varphi < .875, \qquad G = \frac{\sqrt{4 + 6\varphi}}{1 + \sqrt{(13 - 8\varphi)/(17 - 2\varphi)}} > 3.48.$$

The numerator exceeds 3.7024, the inner square root is $< .0638$, and $3.48(1.0638) = 3.702024 < 3.7024$. Its TOTAL spatial norm is $17 - 2\varphi = 13 + d_C^2$. There is no artificial pure-D demand for this mixed point.

Let w be the ordinary reference ending eleven. Take the applicable head insertion/change above and centre insertion $(V_D - P_{11})/7$. The FINAL external coefficient is s_0. BEFORE using either mixed pressure, pay $d_C/7 + \mu_H d_H$ from that coefficient, and pay their P_{11}, P_5 prices from w. At mixed-head-five $\mu_H = .1$; the total external debit is $< .2125 < s_0$. Pure or fixed-sum head changes require no external debit. The P_{11} price remains positive, and its P_5 payment is independent. The head residual has zero sum on the first five coordinates; V_D is flat through twelve, so their FULL D-cross is zero. The single ambient completed square gives

$$\begin{aligned} \|x\|^2 + y^2 &> E_{11} + .1 + .0059 + \frac{2(G - d_C s_0 - f_{11})}{7} - \frac{2}{49} \\ &> .84816 + .1 + .0059 + \frac{2(3.48 - .875(.317) - 2.886)}{7} - \frac{2}{49} > 1.0037. \end{aligned}$$

The external component appears ONCE as the positive $s_0 e$ reference with its paid residual coefficient. No separate $y^2 > .1$ is added to a mixed ambient square that already spent it.

L.9 Ordinary D_{16}: every remaining finite fork and the final floor

For own zero and ANY finite fork label at least seven, put $u = \delta^{-2} \in (1, 100/81)$; $Q = 0$ was proved. The ACTUAL positive core

$$c + \delta^{-1} n_f + 2(\delta^{-1} - \delta) n_{14}$$

has cap/fork products zero, centre product $(3 - 4\delta^2)/(2\delta) < 0$, negative square and positive time. Its product with the OTHER fork is negative in each signed orientation; every other original wall is nonpositive. It gives pure pressure

$$V = (1^{13}, 3 - 2u, u - 1, \ \pm(1 - u)),$$
$$x \cdot V \geq G = \sqrt{16 - 2u} - \sqrt{-8 + 14u - 6u^2},$$
$$R = V^2 = 24 - 16u + 6u^2.$$

On the ENTIRE interval $14G - R > 32.51$. For a direct continuous proof, the chord $\sqrt{16 - 2u} \geq 4.011 - .27u$ has positive endpoint-square margins: at one $3.741^2 < 14$, and at $100/81$ its margin is $456199/81000000$. Set $z = u - 7/6$. Then the second radicand is $1/6 - 6z^2$, $R = 27/2 - 2z + 6z^2$, and $\sqrt{1/6 - 6z^2} \leq 1/\sqrt{6} - 3\sqrt{6}z^2$. Consequently

$$14G - R > 32.528 - 1.78z + 96.8748z^2 > 32.528 - .009 > 32.51.$$

Use $\sqrt{6} > 2.4494$, $14/\sqrt{6} < 5.716$, and $1.78^2 < .036(96.8748)$. Paying $1/7$ at P_{11} gives, WITHOUT head credit,

$$.84816 + .1 + \frac{32.51 - 14(2.886) + 11}{49} > 1.0115.$$

This excludes the entire finite-label range, without sampling native cosines.

For pure own-three/single-fork-four or five, the SAME actual cap closes that fork: retained A_{15} plus cap is finite D_{16}. For own-zero/pair-three/four, releasing the fork-three direction gives finite B_{16}. Keep old e throughout. Positive inverse finite projection gives an actual future endpoint, strict against all other original walls; the omitted fork is strict through its positive cap contact. The exact fork coshes are $7 + 8\delta > 12$ or $7 + 4\sqrt{2} > 12$. The genuine signed direction $V = P_{15} \pm e_{16}$, $V^2 = 16$, has demand $G > \sqrt{176/13} > 3.67$. Payment $1/7$ at P_{11} gives

$$.84816 + .1 + \frac{2(3.67 - 2.886)}{7} - \frac{5}{49} > 1.0701.$$

Own-zero/pair-three/five uses its ACTUAL finite H_3 projection, with positive correction $n_{f_3}/2 + (\varphi/2) n_{f_5}$. Its pure normalized displacement is EXACTLY P_{14}. For $q = 2\varphi^{-6} < 1/8$, its demand is

$$x \cdot P_{14} \geq \frac{\sqrt{14 - q}}{1 + \sqrt{q/14}} > 3.4.$$

Use $\sqrt{111/8} > 3.7249$, $\sqrt{1/112} < .095$, and $3.4(1.095) = 3.723$. The reference ending eleven with payment $1/7$ gives

$$.84816 + .1 + \frac{2(3.4 - 2.886)}{7} - \frac{3}{49} > 1.0337.$$

All remaining centre caps are therefore strong or impossible. The full actual curve through fourteen is now available, giving

$$\|x\|^2 + y^2 > E_{14} + .1 > .9084 + .1 > 1.$$

The ENTIRE ordinary/nonshort $D_{16} + A_1$ alternative is excluded. The short actual-I alternative has not received the ordinary coefficient $s_0 e$ or its nonshort bootstrap. It requires its own proof next.

M The full actual-I D_{16} case

M.1 The actual-I D_{16} foundation

The final classical consumer has ENTIRE forest $D_{16}+I$, with the ACTUAL isolated short opposite $c_I = \sqrt{h}\,p - \sqrt{1+h}\,e$, $0 < h < 1/2$. Its original parameters are $y^2 \geq q(h)$, $x_1 \geq t$, and $t + (8/5)\sqrt{q(h)} > .788$. No ordinary rank-one cost is assigned.

At the actual rank sixteen the generic bulk gives strong six through eleven. The D_4, D_3 sites twelve/thirteen have the SAME complete local inventories and actual left-H_4 data proved in the preceding ordinary section. Those geometric lists and points do not use its nonshort energy. All foreign old-I costs are retained; the only further weak rows are PURE golden own-five. Their actual pressures $G > 3.34, 3.47$ are flat through nine/ten, with full norms below twelve/thirteen.

If five is strong, its ordinary positive reference and the actual-I coordinate have one joint initial-block gain $> .05426$. Here is its precise payment. The first-five mean is $b < .361$; graft it to $\min(.6, \max(b, t))$. The other coordinates remain above the next slope. At $t \geq .6$, the block gain alone is $> (5/4)(.6 - .361)^2 > .05426$. Otherwise set $z = (t-b)_+$. Then $z + (8/5)\sqrt{q(h)} > .427$, so weighted Cauchy gives joint gain

$$> \frac{.427^2}{4/5 + 64/25} > .05426.$$

The head is not used. The separately paid actual left-H_4 packets give totals

$$.79805 + .05426 + (3.34 - 2.569)/3 - 3/36 > 1.0259,$$
$$.82436 + .05426 + .32(3.47 - 2.732) - .0256(3) > 1.0379.$$

Their residuals vanish on the initial block, so neither overlaps its change. If five is weak, its PURE actual point gives $x_1 > .5965, S_6 > 2.01815$. Positive interpolation gives $L_9 > .8610, L_{10} > .8870$; their terminal prices remain $> 1/6, .16$. The SAME actual packets then have pure D-energies $> 1.0346, 1.0463$, without adding any I energy. Thus six through thirteen are strong in BOTH branches before five, head or centre is concluded. At weak five specifically,

$$L_{13} > .7600299945 + \frac{(\sqrt{13} - \sqrt{6})^2}{7} > .95095.$$

We must exclude weak five arithmetically at its actual local rank. It is the literal PURE far helper, with integral $t_5 = 2g$, $202/125 < t_5 < 12/5$, and every other conjugate satisfies

$$5(t_5')^2 + 2t_5' - 1 < 0, \qquad -.69 < t_5' < .29.$$

Degree at least four has norm magnitude $< (12/5)(.69)^3 < 1$. In degree three a positive small conjugate would make it $< (12/5)(.69)(.29) < 1$. Both are negative, with product $1/t_5$, and integral trace one. Indeed the trace is positive and by AM–GM is < 2. Reality of the two small roots requires $t_5(t_5-1)^2 \geq 4$, forcing $t_5 > 2.3$. The integer second coefficient $1/t_5 - t_5^2 + t_5$ then lies in $(-883/300, -5877/2300) \subset (-3, -2)$, impossible. In degree two the norm is -1; positive norm would have magnitude $< (12/5)(.29) < 1$. Its trace $t_5 - 1/t_5 \in (0, 2)$ is one, hence $t_5 = \varphi$. Rational two is the actual parallel null. The PURE connected Lorentzian principal therefore forces ACTUAL $K = \mathbb{Q}(\varphi)$.

The weak-five time is $A_5^2 = (4 + 7\varphi)/2$. The actual I own invariant has $J = 4(1+h) \in \mathcal{O}_K$, $4 < J < 6, 0 < J' < 4$. The actual two-cycle makes $A_5^2(J-4)$ integral. Since $\mathrm{N}(4+7\varphi) = -5$, this factor is invertible modulo two, so $z = (J-4)/2 \in \mathbb{Z}[\varphi]$. Its full strip is $0 < z < 1, -2 < z' < 0$.

Its norm is -1, trace -1 or zero; zero would give the forbidden positive root one. Thus $z=\varphi-1$, $h=(\varphi-1)/2>.3$. The actual I energy then exceeds .06, from $(53/47)^2<13/10$. Adding it ONCE to $L_{13}>.95095$ contradicts the original norm. Five is therefore strong. Every individual floor five through thirteen is available before the head.

The head's COMPLETE finite-white list is isolation or simple-three at long leaf two, with all old-I costs kept. For the nonisolated row, $A_H^2=2g^2+2g+Q$, $C_H=(2g+1)/\sqrt{1-2Q}$. Own zero at positive Q has $c_H\cdot c_I=\beta(\sqrt{1+h}-\sqrt{h})>0$, contrary to acuteness; at zero cost time is zero. Own three has its affine triangle. The first-six-linear reference through thirteen has energy $>.87175$, mean $<.3364$, and any first-coordinate lift $a<1$ is legal. Positive Q gives $x_1>.73$; pure own at least five gives $x_1>2/3$. Either lift alone makes its energy >1. The remaining nonisolated own-four, $Q=0$, gives $x_1>.643$ and $A_H^2=1+\sqrt{2}<5/2$. Its actual mutual with c_I forces $A_H^2h\ge 1/4$, hence $h>.1$ and $y^2>.023$. The single graft and this disjoint energy total exceed 1.0075. Thus the head is isolated within D.

Write it as

$$c_H=A_Hp-g\omega_1-\beta e,\quad a=2g^2-1,\quad A_H^2=a+Q,\quad ah\ge Q.$$

Own zero/three/four has nonpositive time or violates the actual acuteness condition. Own six/parallel has its all-wall affine/null principal. Native five is impossible: at positive cost $ah<Q$, while pure time gives the forbidden positive actual mutual $\sqrt{ah}<1/2$. A remaining finite head has $a<1$; at positive cost $h>Q\ge 1/4$, and at zero cost the actual native gap gives $ah\ge 1/4$. Thus EVERY finite head forces $h>1/4$.

For a mixed head, or pure dotted head, $u=(1-Q)/g^2\in(0,1)$. The ACTUAL positive clearing $c_H+\beta e+gun_1$ is future, negative norm, and feasible against EVERY original wall including c_I. It gives PURE-D pressure

$$Z=(1-u/2)e_1+(u/2)e_2,\quad x\cdot Z\ge H=\frac{\sqrt{2-u}-\sqrt{u(1-u)}}{\sqrt{2}},\quad R=1-u+u^2/2.$$

At $\mu=.36$, its individual P_5 debit is $.072<.14$. Its full-reference gain

$$G_H=.72(H-b)-.1296(R-1/5)>.04 \tag{D16IH}$$

holds on the ENTIRE interval. For clarity the three chord seals are

u interval	$\sqrt{2-u}$ lower chord	R upper chord
$[0,1/2]$	$1.4142-.379u$	$1-.75u$
$[1/2,3/4]$	$1.4381-.4268u$	$.8125-.375u$
$[3/4,1]$	$1.4721-.4721u$	$.625-.125u$.

The endpoint-square comparisons are positive. Using $.5091<9\sqrt{2}/25<.5092$, substitution gives lower expressions

$$.35636922-.0958u-.5092\sqrt{u(1-u)},$$
$$.39283671-.16873u-.5092\sqrt{u(1-u)},$$
$$.43444611-.2242u-.5092\sqrt{u(1-u)}.$$

The circle inequality gives the following bounds:

square-root upper bound	gain lower bound
.5182	.04936922
.5365	.04022171
.5564	.04414611.

This proves the full scalar without sampling own parameters or spending I.

For a PURE finite head use the actual first-three-root finite pressure already proved with (R1S). Its native-seven/eight seals give gains $> .0325, .0345$. For every finite native at least nine, $\gamma > .939$, $1 < u < 1.135$; the finite scalar decreases with u on this range. At its upper endpoint $\sqrt{2-u} > .93$, the other radical is $< .284$, $R < .393$, so gain is $> .0456$. The actual mutual with c_I gives the compatible complete floors

finite own	$h >$	$y^2 > q_0$	$\eta_H >$
7	2/5	.083	.0325
8	7/20	.074	.0345
all ≥ 9	1/4	.055	.0456.

These heads and the cleared head each use only a zero-sum first-five residual, paid at P_5, and have no I component. They are orthogonal to every centre residual flat through eleven.

M.2 Actual $D_{16} + I$: the height reduction and unconditional centre

First suppose $h \geq .23$. Then $y^2 > .0516$, with $\sqrt{.0516} < .228$. Use either that fixed external reference and the cleared-head gain $> .04$, or a pure-finite-head row of the last table. In every case its total $q_0 + \eta_H$ is at least .0916, and its external coefficient $u_0 = \sqrt{q_0} > .22$. The entire SAME-rank centre inventory, actual points and all original-wall proofs are already supplied in Sections L.7–L.9. Those geometric proofs did not use nonshortness until their coefficient accounting. They apply here with every actual old-I contact retained, including the finite mixed isolated $u > 1$ and the unsealed left/fork-three arm.

Their complete nondirect point data are

centre role	actual pressure	D norm
own-five/fork-three, $Q = 0, 1/4$	$xV > 3.38$	< 14
own-zero/fork-five, $Q = 0$	$xV > 3.36$	13
own-zero/fork-five, $Q = 1/4$	$xV + dy > 3.48,\ d < .875$	13
own-zero/finite fork $\geq 7, Q = 0$	$14G - R > 32.51$	R
reciprocal own-three/fork-four or five; pair $(3,4)$	$xV > 3.67$	16
pair $(3,5)$, own zero	$xP_{14} > 3.4$	14.

Every V is flat through eleven. The mixed point has TOTAL spatial norm $13 + d^2$; it has not become a fictitious pure-D point. Use w_{11}, the applicable pure-D head foot and the centre residual $(V - P_{11})/7$. The individual P_5, P_{11} prices are paid first; the residual cross is zero. In the mixed row spend $d/7 < .125$ from $u_0 > .22$, keeping FINAL external coefficient u_0. The single ambient square is

$$E_{11} + q_0 + \eta_H + \frac{2(G - du_0 - f_{11})}{7} - \frac{R - 11}{49}.$$

Pure rows use $q_0 + \eta_H \geq .0916$; their totals exceed $1.0196, 1.0343, 1.0031, 1.0617, 1.0253$, respectively, with the continuous row using its proved $14G - R$ estimate. For the mixed row the cleared, native-seven, native-eight and native-at-least-nine head choices give totals $> 1.0116, 1.0203, 1.0173, 1.0189$. They follow by substituting the COMPLETE last table and $G > 3.48, d < .875$. Thus all centre exceptions are impossible under this contrary height. Only THEN is the actual E_{14} reference available, giving

$$E_{14} + q_0 + \eta_H > .9084 + .0916 = 1,$$

a contradiction. Therefore $h < .23$. Every finite head would require $h > 1/4$; it is excluded. The head is isolated within D, DOTTED, possibly contacting old I. Its cleared pure-D gain (D16IH) is now uniformly available.

We still must establish centre fourteen WITHOUT the contrary height. The only point rows left by its complete inventory are precisely those in the table above. For own-zero/single-fork finite label $m \geq 7$, $Q = 0$ is already proved. Set $t_f = 4\cos^2(\pi/m)$. The actual fork-product normalization makes the restricted old-D/cap Schur complement $4/t_f - 8 < 0$. Strict nonphysical positivity gives

$$3 < t_f < 4, \qquad 0 < (t_f)_\sigma < 1/2.$$

Degree at least three has nonzero integer norm $< 4(1/2)^2 = 1$. Degree one has no integer in $(3, 4)$. In degree two its norm is one and trace four, so $t_f = 2 + \sqrt{3}$: only native twelve remains. The connected Lorentzian principal forces ACTUAL $K = \mathbb{Q}(\sqrt{3})$.

The golden point rows instead force ACTUAL $K = \mathbb{Q}(\varphi)$. For own-five/fork-three, normalize by that ACTUAL three; the old-D spatial norm is $14\varphi+9 > 1$, independent of its optional old-I contact. For own-zero/single-fork-five, normalization to fork product minus one has Schur complement $4/(\varphi+1)-8 < 0$. For own-zero/pair-three/five, normalize by the three; its D-spatial norm is $4+(11/2)\varphi > 1$. Each restricted Gram is over the golden field and physically Lorentzian. The already proved original-root descent therefore identifies the ACTUAL field, even in the mixed old-I branch.

The actual invariant now has the sharper COMPLETE strip

$$J = 4(1 + h) \in \mathcal{O}_K, \qquad 4 < J < 123/25, \quad 0 < J_\sigma < 4. \tag{D16IJ}$$

In the golden full ring it leaves only $J = 3 + \varphi$. Indeed $J = a + b\varphi$ has $b = 1, 2$; at one $a = 3$, at two the physical interval gives $a = 1$ but its conjugate is negative. Thus $h = (\varphi - 1)/4$, $y^2 > .0359$. The exact rational seal is $309(9641)^2 - 2000(14360000) = 1204229 > 0$. In $\mathbb{Z}[\sqrt{3}]$, the same strip gives only $J = 3 + \sqrt{3}$, hence $h = (\sqrt{3} - 1)/4$, $y^2 > .04$. No rational J exists in the strip.

Use again ONE w_{11}, pure-D head gain $> .04$, actual centre point and FINAL external coefficient $u_0 = \sqrt{q_0}$. The head debit is $.072 < .14$; the centre debit is $1/7$. Their D-cross is zero. For golden rows choose $q_0 = .0359$, $.189 < u_0 < .19$; the mixed debit $d/7 < .125$ leaves a positive coefficient. For reciprocal rows use $q_0 = u_0 = 0$ and no I energy at all. The COMPLETE resulting totals are

own-five/fork-three	$.84816 + .0359 + .04 + .988/7 - 3/49 > 1.0039$
own-zero/fork-five, $Q = 0$	$.84816 + .0359 + .04 + .948/7 - 2/49 > 1.0186$
own-zero/fork-five, $Q = 1/4$	$.84816 + .0359 + .04 + .8555/7 - 2/49 > 1.0054$
reciprocal/pair $(3, 4)$	$.84816 + .04 + 1.568/7 - 5/49 > 1.0101$
pair $(3, 5)$	$.84816 + .0359 + .04 + 1.028/7 - 3/49 > 1.0096.$

The mixed numerator is $2(3.48 - .875(.19) - 2.886) = .8555$; its actual total point norm was retained before the coefficient debit.

For native twelve put $z = 8 - 4\sqrt{3}$. The already proved actual finite-fork point has

$$G = \sqrt{16 - 2z} - \sqrt{-8 + 14z - 6z^2}, \quad R = 24 - 16z + 6z^2.$$

The positive squares give $1.0716 < z < 1.072$, first radical > 3.722, second $< .337$, so $G > 3.385 > 3.38$, and $R < 13.75$. Choose $q_0 = .04, u_0 = .2$. The total exceeds $.84816 + .04 + .04 + .988/7 - (11/4)/49 > 1.0131$. Every nondirect centre role is impossible. Hence ALL individual five-through-fourteen floors now hold UNCONDITIONALLY.

M.3 Joint head height and both genuine D_{16} fork edges

For the actual dotted head let $Q = \beta_H^2$, $u = (1-Q)/g_H^2 \in (0,1)$. Acuteness with c_I gives $h(2g_H^2-1) \geq Q$ at $Q > 0$. At zero the positive actual mutual has native magnitude at least $1/2$, giving the same stronger inequality $h(2g_H^2 - 1) \geq 1/4$. Thus in ALL cases

$$h \geq k(u) = \frac{u}{4(2-u)}.$$

Use the actual w_{14} reference, head insertion $.36(Z - P_5/5)$, and the disjoint I energy ONCE. The total exceeds

$$.9084 + \eta_0(u) + q(h), \quad \eta_0(u) = .72(H(u) - .361) - .1296(R(u) - 1/5).$$

No initial-five/I gain is added to this overlapping head gain.

For $0 < u \leq 3/8$, $H'(u) \leq -1/4 - 1/(4\sqrt{2}) < -.425$, so $\eta_0' < -.1764$. Also $q'(h) \leq 1/4$, $k'(u) \leq 32/169$. Hence $\eta_0(u) + q(k(u))$ decreases on the ENTIRE interval. At $u = 3/8$,

$$H = (4\sqrt{13} - \sqrt{30})/16 > .55906, \quad R = .6953125, \quad k = 3/52, \quad q(k) > .014.$$

Use $\sqrt{13} > 3.60555$, $\sqrt{30} < 5.47723$, and $55(493)^2 - 52(507)^2 = 1147 > 0$. The total would exceed $.9084 + .0784107 + .014 = 1.0008107$. Thus

$$u > 3/8, \qquad h > 3/52, \qquad q(h) > .014, \qquad g_H^2 < (8/3)(1-Q). \tag{D16Iu}$$

From $h < .23$ and actual acuteness, $Q < 299/668 < 1/2$. The COMPLETE finite foreign alphabet therefore leaves only $\beta_H = 0, 1/2$. At positive contact, $g_H^2 < 2$ and $h > 1/12$.

For EITHER genuine fork direction $V = P_{15} \pm e_{16}$, $V^2 = 16$, an ACTUAL edge cosine $C \geq 37/4$ would give $xV \geq \sqrt{528/41} > 3.588$. Pay $1/7$ at P_{11}; the residual has square five and is orthogonal to the reference and the head residual. The latter still pays only its separate P_5 coefficient. The ONE square, with I counted once, gives

$$.84816 + .04 + .014 + \frac{2(3.588 - 2.886)}{7} - \frac{5}{49} > 1.00069.$$

Apply the contradictions separately to the two actual edges. Thus BOTH fork cosines satisfy

$$C < 37/4. \tag{D16IC}$$

No fork-classification or field-degree result was used to prove this cut. We will use it first, eliminating a redundant weaker length stage.

M.4 The complete actual $D_{16} + I$ fork roles

Release fork sixteen and retain the path $n_1, \ldots, n_{15} = A_{15}$ and old e. Its COMPLETE finite one-node list is isolation, helper-three at $j = 1, 2, 14, 15$, or helper-four at $j = 1, 15$. A second path contact makes a cycle; rank sixteen excludes exceptional completions, and a multiple bond must be terminal. Only endpoint-three at one/fifteen permits an old-I arm, with $\beta = 0, 1/2, 1/\sqrt{2}$. At next-to-end three, appending that arm gives nonfinite arms $1, 2, 13$. At helper-four it makes the four internal in a long path. Thus those four special roles have $\beta = 0$. Reversing the genuine fork sign covers the other outgoing edge separately.

For next-to-end-three at two put $t = 2\gamma$. Its exact data are

$$A_c^2 = 2t^2 + t, \qquad \Delta = 1/8, \qquad C = 4t + 1.$$

Own zero has zero time. At positive own, actual helper-three normalization makes t integral, ACTUAL $K = \mathbb{Q}(t)$, and $-1/2 < t_\sigma < 0$. The connected Schur complement is $-2t(2t+1) < 0$. The stronger INDEPENDENT cut (D16IC) gives $t < 33/16 < 5/2$. Degree at least three has norm < 1. In degree two its norm is -1, and the exact integral trace condition leaves $t = 1 + \sqrt{2}$, whose actual $C = 5 + 4\sqrt{2} > 37/4$, impossible. Degree one gives the empty actual-J strip (D16IJ). For long-end helper-four at one set $t = 2\sqrt{2}\gamma$. Its data are $A_c^2 = t^2 + t/2, \Delta = 1/16, C = 4t + 1$. Here the actual cap normalizes to square one, not two; its cycle through the four makes t integral. Schur complement $-t(2t+1)/2 < 0$ gives the SAME field/conjugate/degree argument, hence the same contradiction. This avoids an unnecessary weaker silver-role energy estimate.

The far next-three at fourteen has $A_c^2 = 2t^2 + 7t + 6, C = 4t + 7$, $t = 2\gamma$. Positive own gives $C \geq 11 > 37/4$; at own zero the actual future null is $c + n_{13} + 2n_{14} + n_{15} + n_{16}$. The far helper-four at fifteen has $A_c^2 = t^2 + (7/2)t + 3, C = 4t + 7$, $t = 2\sqrt{2}\gamma$. Positive own again violates the cut; at zero use $\sqrt{2}c + n_{13} + 2n_{14} + 2n_{15} + n_{16}$. Both kernels have zero selected products and norm, positive cap time, and all omitted ORIGINAL wall signs by positivity/acuteness. The full fork list is reduced to isolation or endpoint-three at one or the other fork fifteen.

For those two nonisolated roles write

$$c = A_c p - (t/2)\omega_{16} - \omega_j/2 - \beta e, \qquad j = 1, 15.$$

The exact unit geometry is

j	A_c^2	C
1	$2t^2 + t/2 - 1/2 + \beta^2$	$(8t+1)/\sqrt{17 - 32\beta^2}$
15	$2t^2 + 7t/2 + 1 + \beta^2$	$(8t+7)/\sqrt{17 - 32\beta^2}$.

Own zero at one has nonpositive time. At fifteen its restricted connected principal is rational Lorentzian even at zero, giving actual rational field and the impossible integral-J strip. Parallel has its positive future null.

At positive own the ACTUAL helper-three normalization gives raw square two, own product $-t$, integral t. The restricted Schur complements are $1 - 4t^2 - t$, $-(4t^2 + 7t + 2)$; both are physically negative. Thus ACTUAL $K = \mathbb{Q}(t)$, and

j	EVERY other t_σ
1	$(-1 - \sqrt{17})/8 < t_\sigma < (-1 + \sqrt{17})/8$
15	$(-7 - \sqrt{17})/8 < t_\sigma < (-7 + \sqrt{17})/8$.

(D16Itstrip)

A helper-four OLD-I contact $\beta = 1/\sqrt{2}$ is impossible: at fifteen positive own has $C \geq 15$; at one the cut leaves only own-three, with its actual all-three cycle. Hence $\beta = 0, 1/2$.

The same cycles exclude own-three in both roles. For every even finite own label $m \geq 4$, the coprime exponent $m - 1$ gives conjugate $-2\cos(\pi/m) \leq -\sqrt{2} < -1.4$, outside both intervals. For odd $m \geq 7$, exponent $m - 2$ gives $-2\cos(2\pi/m)$. It is below minus one for role one; for role fifteen it is below -1.4 when $m \geq 9$. At its remaining seven, exponent three gives a positive conjugate, outside the entirely negative near-fork interval. These inequalities follow from $\pi < 22/7, \cos z > 1 - z^2/2$. Only finite own-five remains: $t = \varphi, K = \mathbb{Q}(\varphi)$.

Its actual-J strip leaves $J = 3 + \varphi, h = (\varphi - 1)/4$. At $\beta = 0$, the two exact A_0^2 values are $(3 + 5\varphi)/2, (6 + 11\varphi)/2$. The actual c, c_I two-cycles are

$$(2 + 3\varphi)/2, \qquad (5 + 6\varphi)/2,$$

nonintegral. At $\beta = 1/2$, their actual mutual magnitude is

$$F = \sqrt{h(A_0^2 + 1/4)} - \sqrt{1+h}/2.$$

The rational brackets $1.618 < \varphi < 1.619$ give $.4085 < F < .411$ at one, and $.8325 < F < .835$ at fifteen. The first lies in $(0, 1/2)$; the second between $\varphi/2$ and $\sqrt{3}/2$. Both are forbidden ACTUAL native gaps. Thus every nonisolated fork has DOTTED own $t > 2$.

M.5 An individually isolated finite fork also fails

An isolated finite-own fork has

$$c_F = A_F p - \gamma\omega_f - \beta_F e, \qquad s = \gamma^2, \quad A_F^2 = 8s + \beta_F^2 - 1.$$

Actual old-D root normalization to square one and fork product minus one gives Schur complement $1/s - 8 < 0$. The Lorentzian-principal descent gives ACTUAL $K = \mathbb{Q}(s)$, $0 < s_\sigma < 1/8$. For small finite three/four/six this is rational, contradicting the integral actual-J strip. Five gives $s = (\varphi + 1)/4$. For every finite label at least seven, $t_f = 4s$ has physical value in $(3, 4)$, others in $(0, 1/2)$. The norm/trace proof above gives only $t_f = 2 + \sqrt{3}$, native twelve. The ACTUAL I strips then fix

K	s	h
$\mathbb{Q}(\varphi)$	$(\varphi + 1)/4$	$(\varphi - 1)/4$
$\mathbb{Q}(\sqrt{3})$	$(2 + \sqrt{3})/4$	$(\sqrt{3} - 1)/4$.

Both have $q(h) > .0359$.

We also need the COMPLETE old-I contact restriction for this finite fork. Put $B_0^2 = 8s - 1$. Its actual mutual with c_I is

$$F(\beta) = \sqrt{h(B_0^2 + \beta^2)} - \beta\sqrt{1+h},$$

strictly decreasing. Its sign is the sign of $hB_0^2 - \beta^2$. For own-five $hB_0^2 = s$; acuteness requires $\beta \le \gamma$. At $\beta = 1/2$, $F < 1/2$, because the positive-square difference is $1/2 + \sqrt{1+h}/2 - s > 0$. Thus every $0 < \beta < \gamma$ has forbidden $0 < F < 1/2$. Only $\beta_F = 0, \gamma$ remains. For own-twelve $hB_0^2 = 1 + h$; at $\beta = 1/2$, $1/2 < F < 1/\sqrt{2}$. At $\beta = 1/\sqrt{2}$, $0 < F < 1/2$; monotonicity covers every larger finite contact. These comparisons use $h < .23$: $\sqrt{1 + 5h/4} < 1.135$ and $\sqrt{(1+h)/2} - 1/4 - h > .7 - .48 > 0$. Therefore $\beta_F = 0$ for twelve. All foreign finite labels were included, without a separate field-degree census.

Now use the actual dotted head. Its cleared-head parameter u has the pressure $H(u), R(u)$ already proved. For a proposed finite fork, $q(h) > .0359$. At $u \le .48$, the derivative of its effective head gain is negative: $H' \le -1/4$ and $\eta' \le -.18 + .1296(1 - u) < 0$. At .48, $H > .5184, R = .6352$, giving gain $> .05692608$. The single actual reference would total $.9084 + .0359 + .05692608 > 1$. Hence $u > .48$, and the head integer $T = 4g_H^2$ satisfies

$$4 < T < 25/3, \qquad 0 < T' < 2.$$

The COMPLETE golden full-order list is

$$T = 2 + 2\varphi,\ 3 + 2\varphi,\ 2 + 3\varphi,\ 3 + 3\varphi.$$

Writing $T = a + b\varphi$ gives $b \le 3$; $b = 1$ has physical value below four, and $b = 2, 3$ each has precisely the two displayed constants. For $H_0 = h(T/2 - 1)$ their exact values are $1/4, 1/4 + h/2, 3/8, 3/8 + h/2$. All are below $1/2$, so old-I head contact is zero or $1/2$. At zero contact, the last three actual mutuals

lie in $(1/2, 1/\sqrt{2})$; at $1/2$, they lie in $(0, 1/2)$. Only $T = 2 + 2\varphi$ remains, with $g_H^2 = (\varphi + 1)/2 = 2s$ and $\beta_H = 0, 1/2$.

For $\mathbb{Q}(r)$, $r = \sqrt{3}$, the full strip leaves $T = 3 + r, 4 + 2r$. Their H_0 values are $1/4, 1/2$. The complete contacts are zero or $1/2$ at the first; zero, $1/2$, $1/\sqrt{2}$ at the second. The middle contact gives its forbidden $(0, 1/2)$ mutual, and the last gives $u = 2 - r < .48$. Thus the remaining head rows are $g_H^2 = (3 + r)/4, \beta_H = 0, 1/2$, or $g_H^2 = (2 + r)/2, \beta_H = 0$.

Both original actual root lines may be normalized OVER K to their nonzero own product minus one. Their ACTUAL mutual product is K-valued, so

$$U = \frac{A_H A_F - \beta_H \beta_F}{g_H \gamma} \in K.$$

This retains the WHOLE combined expression; neither separate unit time nor foreign product is assumed K-valued. For the golden table, equal zero/nonzero contact patterns give $U = 2\sqrt{2}$; the two mixed patterns give $U = \sqrt{6 + 2\varphi}$. Neither belongs to $\mathbb{Q}(\varphi)$; the second has nonsquare rational norm 44. For the three silver head rows, $\beta_F = 0$ and the exact U^2 values are

$$8, \qquad 6 + 2r, \qquad 24 - 8r.$$

The first requires $\sqrt{2} \notin \mathbb{Q}(r)$; the others have nonsquare norms $24, 384$. Every remaining actual raw Gram row contradicts $U \in K$. Thus NO isolated fork has finite own. Own zero has negative time; parallel has its actual positive null. Both fork opposites now have DOTTED own, whether isolated or nonisolated.

M.6 Every actual $D_{16} + I$ field has degree two, three or four

For a NONISOLATED dotted fork set $v = t$ at helper one, $v = t + 1$ at helper fifteen. The actual cut and (D16Itstrip) give integral v, ACTUAL $K = \mathbb{Q}(v)$, $0 < v < 4.9$, and every other $|v_\sigma| < .641$. Degree at least five would have nonzero norm $< 4.9(.641)^4 < 1$. Thus its actual degree is at most four.

For an ISOLATED dotted fork set $T = 4\gamma^2$. Its actual edge is $C^2 = 2T/(1 - \beta^2)$, so

$$4 < T < 1369/32 < 121/2.$$

Its actual own two-cycle is integral. Normalize its original root to own product minus one against old raw D-roots of square two. Its square is $2/T$, old dual projection square four, and restricted Schur complement $2/T - 4 < 0$. The connected Lorentzian principal gives ACTUAL $K = \mathbb{Q}(T)$. Strict nonphysical diagonal and Schur positivity give $0 < T_\sigma < 1/2$.

We prove the degree bound on the ENTIRE strip, including degrees five and six. At degree at least seven, the norm is $< (121/2)(1/2)^6 = 121/128 < 1$. For degree five let P be the monic integer minimal polynomial. The four small roots have product $\mathrm{N}(T)/T > 1/61$. Since $61(7/20)^4 < 1$, their average exceeds $7/20$. The four positive deficits $1/2 - T_\sigma$ then have average less than $3/20$. AM–GM gives

$$0 < |P(1/2)| < 61(3/20)^4 = 4941/160000 < 1/32.$$

But a nonzero monic degree-five integer polynomial at $1/2$ has magnitude at least $1/32$. It is nonzero because $T > 4$ and the minimal polynomial is irreducible. At degree six, $61(2/5)^5 < 1$ makes the five small-root average $> 2/5$; the deficit average is $< 1/10$. Thus $0 < |P(1/2)| < 61/100000 < 1/64$, again impossible. This proves degree at most four for an isolated fork, without a coefficient census.

These two roles exhaust either ACTUAL fork, and in each branch the generating field is the ACTUAL defining field. Degree one would make the integral J a rational integer in the empty interval $(4, 123/25)$. Consequently the whole forest has

$$[K : \mathbb{Q}] \in \{2, 3, 4\}.$$

No old-I contact has been set to zero in this isolated case. The nonisolated roles still require their complete exclusions below.

M.7 Nonisolated $D_{16} + I$ forks: quartic and old-I contact

Keep $v = t$ in the long-end role, $v = t + 1$ in the other-fork role. The actual field is generated by this integral parameter. Its physical bounds are $v < 4.644$, $v < 4.894 < 4.9$, respectively. The other windows are $(-.641, .391)$, $(-.391, .641)$. In degree four, any small conjugate of the wrong sign would give nonzero norm magnitude $< 4.9(.391)(.641)^2 < 1$. Thus the three small roots are all negative for the long-end role, positive for the other-fork role. Their absolute product is $1/v$, since $4.9(.641)^3 < 2$. Write their sum $s < 1.923$. Then $v > 3.79$, and AM–GM gives $s > 30/17$.

In the other-fork role, the integer trace $v + s$ is six. Hence $v > 4.077$, and $v + 3/v^{1/3} \le 6$ gives $v < 4.2$. For the pairwise small-root sum p, the integer third coefficient is $E_3 = vp + 1/v$. AM–GM gives $E_3 > 4.77 + 5/21 > 5$. But $p < 3(.641)^2$, so $E_3 < 4.2(1.232643) + 1/4 < 6$. Impossible. In the long-end role, integer trace $v - s$ is two. Thus $v < 3.923$; AM–GM requires $v - 2 \ge 3/v^{1/3}$, which forces $v > 3.9$ since $3.9(1.9)^3 < 27$. Its integer second coefficient $E_2 = p - vs$ is $< -6.177357 < -6$. Since $v < 4$, AM–GM gives $p > 1$, so $E_2 > 1 - 3.923(1.923) > -7$. Again impossible. Every nonisolated fork therefore has ACTUAL degree at most three, with BOTH permitted old-I contacts retained.

We next exclude its remaining positive old-I contact $\beta = 1/2$. Then

$$j = 1 : \ t < 107/32, \qquad j = 15 : \ t < 83/32.$$

Equivalently $v < 107/32, 115/32$. Degree one is already excluded by the integral actual-J strip.

In degree three, a wrong-sign small root would give norm $< (115/32)(.391)(.641) < 1$. Thus the two small roots are negative at one, positive at fifteen, with absolute product $1/v$. Their magnitudes a, b satisfy $a + b < 1.282$, $v > 2.43$, $a + b > 1$. At fifteen $v > 3$, so integer trace $v + a + b \in (4, 5)$, impossible. At one the trace $v - a - b$ is two. Thus $v < 3.282$, and $v(v-2)^2 \ge 4$ forces $v > 3.1$. The integer second coefficient $1/v - v(v-2) \in (-5, -3)$ is minus four. Its polynomial $X^3 - 2X^2 - 4X - 1 = (X+1)(X^2 - 3X - 1)$ is reducible. Positive old-I contact is therefore quadratic if it exists.

The complete quadratic trace/norm windows give EXACTLY

j	t
15	$(1 + \sqrt{13})/2, \quad 1 + \sqrt{2}$
1	$1 + \sqrt{2}, \quad 1 + \varphi, \quad (3 + \sqrt{13})/2.$

For coverage, at fifteen the trace of $v = t + 1$ is three/four. Trace three has norm minus one, trace four norm two. At one trace two has norm minus one; trace three has norm one or minus one. These follow from the actual strict root windows and the physical upper bounds above; all absolute norms are strictly below two or three as needed. The ACTUAL full quadratic-J strips leave

$$J = 3 + \sqrt{2}, \qquad 3 + \varphi, \qquad (5 + \sqrt{13})/2$$

in their respective fields. Write $c \cdot c_I = -F$. For positive old contact,

$$F = A_c\sqrt{h} - \sqrt{1+h}/2, \qquad F^2 = hA_c^2 + (1+h)/4 - \sqrt{h(1+h)A_c^2}.$$

The ACTUAL two-cycle puts F^2 in K, so $h(1+h)A_c^2$ must be a FULL K-square. On the COMPLETE five rows it equals $Z/64$, with

j, t	Z	$\mathrm{N}(Z)$
$15, (1+\sqrt{13})/2$	$29t + 37$	-81
$15, 1+\sqrt{2}$	$77 + 56\sqrt{2}$	-343
$1, 1+\sqrt{2}$	$47 + 32\sqrt{2}$	161
$1, 1+\varphi$	$77t - 33$	-605
$1, (3+\sqrt{13})/2$	$33t + 12$	$243.$

Every norm is negative or a positive nonsquare rational integer. Thus no row is a field square. Every nonisolated fork is orthogonal to OLD I:

$$\beta = 0.$$

Its contact with the actual I OPPOSITE has been retained and used, rather than declared zero.

M.8 Nonisolated $D_{16} + I$ forks: the complete cubic domain

Put $y_0 = J - 4$. In actual degree two or three, $y_0/2$ is NOT an algebraic integer. In degree two its norm would have magnitude $< (23/50)(2) < 1$. In degree three suppose $x = y_0/2$ were integral. Its physical value lies in $(0, 23/50)$, other roots $-b, -c$ in $(-2, 0)$; its norm is one. AM–GM gives $b + c > 2.94$, so integer trace $x - b - c = -3$. Then $b + c = x + 3$, and reality forces $x(x+3)^2 \geq 4$, so $x > 1/3$. The integer second coefficient $1/x - x^2 - 3x$ lies in $(0, 2)$, hence is one. Its polynomial $X^3 + 3X^2 + X - 1 = (X+1)(X^2 + 2X - 1)$ is reducible, contradicting actual degree three. This proves the half-integrality obstruction in both stated degrees.

At $\beta = 0$, the two nonisolated time formulas are $A_c^2 = q(v)/2$,

$$q(v) = 4v^2 + v - 1 \quad \text{or} \quad 4v^2 - v - 1.$$

The ACTUAL fork/c_I two-cycle makes $q(v)y_0/2$ integral. If $\mathrm{N}(q(v))$ were odd, its integral cofactor and integer Bezout would force $y_0/2$ integral. Thus $\mathrm{N}(q(v))$ is even. Both $q(v) \equiv v - 1 \pmod{2\mathcal{O}_K}$. For the cubic minimal polynomial P, the determinant norm therefore gives

$$P(1) \equiv 0 \pmod 2. \tag{D16Iparity}$$

This works in the FULL integer ring, without a monogenic-order assumption.

We exhaust every cubic sign pattern. For the long-end role $2 < v < 4.65$. Two positive small roots give norm $< 4.65(.391)^2 < 1$. With two negative roots $-a, -b$, their product is $1/v$, integer trace is two or three. At trace two, $3.1 < v < 3.282$; integer second coefficient is minus four, giving the reducible cubic already printed. At trace three, $4 \leq v < 4.282$; the coefficient is minus five or minus four. Minus five gives $(X+1)(X^2 - 4X - 1)$. Minus four gives $P(1) = -7$, violating (D16Iparity). For mixed signs $a, -b$, with $a < .391, b < .641$, norm is minus one; $v > 3.98, b > .55 > a$. The integral trace is four, and $v > 4$. Its second coefficient $-1/v + v(4 - v) \in (-4, 0)$ is minus three, minus two or minus one. Parity forces minus two. But

$$P(-b) = 1 + 2b - 4b^2 - b^3$$

for $P = X^3 - 4X^2 - 2X + 1$ is strictly concave on $[0, .641]$ and positive at BOTH endpoints, the second value .375101279. It has no required negative root. All long-end cubic cases fail.

For the other-fork role $3 < v < 4.9$, two negative small roots give norm $< 4.9(.391)^2 < 1$. With two positive roots, norm two forces $v > 4.86$, small-root sum > 1.27, and integer trace strictly between six and seven. For norm one, the trace is five or six. At six, $a + b > 1.1, v > 4$, while $(.641 - a)(.641 - b) > 0$ gives $ab > .294219 > 1/v$, impossible. At five, $3.718 < v \leq 4$; the integer second coefficient lies in $(4, 6)$, hence is five. Its polynomial $X^3-5X^2+5X-1 = (X-1)(X^2-4X+1)$ is reducible. For mixed signs $a, -b$, $a < .641, b < .391$, norm is minus one, $v > 3.98, a > .52$. The trace is five, hence $v > 4.359$. Its positive integer second coefficient $v(5 - v) - 1/v$ is one or two. Parity forces one. Only

$$P(X) = X^3 - 5X^2 + X + 1$$

remains; no polynomial enumeration was assumed.

Its discriminant is $404 = 4 \cdot 101$. The index of $\mathbb{Z}[v]$ in the FULL order is one or two. Index two would supply an integral $(A + Bv + Cv^2)/2$ with binary coefficients not all zero. Since $\mathrm{Tr}(v) = 5$, $\mathrm{Tr}(v^2) = 23$, integral trace leaves the three nonzero even-weight numerators. Their norms are $3/4, 9/2, -3/4$, not integers. Thus $\mathcal{O}_K = \mathbb{Z}[v]$. Exact endpoint signs isolate roots

$$4.74 < v < 4.75, \quad .60 < a < .61, \quad -.35 < b < -.34.$$

Suppose the required integral $J = A + Bv + Cv^2$ exists. Lagrange interpolation and its full strip $4 < J_v < 4.92, 0 < J_a, J_b < 4$ give $-2 < C < 2$, hence $C = -1, 0, 1$. The small-value difference gives $|B + C(a + b)| < 4/.94 < 4.26$; since $.25 < a + b < .26$, $-4 \leq B \leq 4$. At $C = -1$, $J_v - J_b = (v-b)(B-v-b) < 0$, impossible. At zero, positivity forces $B \geq 1$, so $J_v - J_b > 5.08 > 4.92$. At one, $J_v - J_a > (4.13)(-4+4.74+.60) > 4.92$. Every full-order coefficient case fails. The actual-I strip is empty in the last cubic field. No nonisolated cubic fork exists.

M.9 Nonisolated $D_{16} + I$ forks: the quadratic K-basis obstruction

It remains to exclude actual degree two. Normalize the connected old D roots OVER K to square two. Their sixteen spatial coordinate vectors $e_1, \ldots, e_{16}$ are then K-linear combinations of these roots. The actual fork root normalizes to $r_F = \sqrt{2}c_F$, square two and helper product minus one. Together with ANY nonzero original K-root r_I on old I, the eighteen vectors

$$e_1, \ldots, e_{16}, r_F, r_I$$

form an ACTUAL K-basis: r_F has nonzero time, r_I has its nonzero I-component. Normalize the head to own old-long product minus one, $r_H = c_H/(\sqrt{2}g_H)$, square $2/T$, $T = 4g_H^2$. Its coefficient on r_F is $A_H/(2g_H A_F) \in K$, by comparing the unique time coordinate. Consequently

$$R_{HF} = \frac{A_H^2}{TA_F^2} \in K^2. \tag{D16IHF}$$

Neither the unit e, p, nor separate unit time coordinates have been asserted K-vectors.

The actual head and old-I contact give the complete strips

Q_H	T	other T'
0	$4 < T < 32/3$	$0 < T' < 2$
1/4	$4 < T < 8$	$0 < T' < 3/2$.

The nonphysical Schur complement is $2(1-Q_H)/T'-1>0$, with the FULL old-I projection retained. At positive cost, actual acuteness also gives $T\geq 2+2/(J-4)$. At zero cost the actual head/c_I two-cycle gives $(T-2)(J-4)/2\in\mathcal{O}_K$.

The complete quadratic fork list comes from the strict physical/conjugate windows, not a field census. At long-end $j=1$, trace is two/three/four/five. Trace five would have noninteger norm strictly between one and two. Trace two has norm minus one; trace three and four each permit norm one, minus one or minus two. At the other-fork role, trace is one/two/three. Their negative norms are respectively three; one/two/three/four; two/three. Trace two/norm minus three is rational and excluded. Solving each exact integer pair and the full actual-J strip gives

j	t	J	$A_F^2(J-4)$
1	$1+\sqrt{2}$	$3+\sqrt{2}$	$3+(3/2)\sqrt{2}$, nonintegral
1	$1+\varphi$	$3+\varphi$	$4t-3/2$, nonintegral
1	$(3+\sqrt{13})/2$	$t+1$	$(3t+4)/2$, nonintegral
1	$(3+\sqrt{17})/2$	$t+1$	$(7t+5)/2$, nonintegral
1	$2+\sqrt{3}$	$3+\sqrt{3}$	$6t-1$
1	$2+\sqrt{5}$	$3+\varphi$	$5t+2$
1	$2+\sqrt{6}$	none	empty actual-J strip
15	$(1+\sqrt{13})/2$	$t+2$	$(3t+5)/2$, nonintegral
15	$1+\sqrt{2}$	$3+\sqrt{2}$	$3t+3/2$, nonintegral
15	$1+\sqrt{3}$	$3+\sqrt{3}$	$5t+5$
15	$1+\sqrt{5}$	$3+\varphi$	$9\varphi+6$
15	$(3+\sqrt{17})/2$	$t+1$	$5t+4$
15	$(3+\sqrt{21})/2$	$t+1$	$(14t+15)/2$, nonintegral.

The last field has a NONEMPTY actual J strip; it fails by integrality, not by an invented empty field. Every entry is an ACTUAL fork/c_I two-cycle. The full orders are $\mathbb{Z}[t]$ in the nonintegral rows. Only fields $\sqrt{3},\sqrt{5},\sqrt{17}$ and the five unmarked fork rows remain.

Their COMPLETE head lists are

K	Q_H	T
$\mathbb{Q}(s),\ s=\sqrt{3}$	0	$3+s,\ 4+2s$
	1/4	$3+s,\ 4+2s$
$\mathbb{Q}(\varphi)$	0	$2+2\varphi,\ 4+4\varphi$
	1/4	$2+2\varphi,\ 2+3\varphi,\ 3+3\varphi$
$\mathbb{Q}(\sqrt{17}),\ t=(3+\sqrt{17})/2$	0	$2+t,\ 2+2t$
	1/4	$2+t$.

For full coverage, in $\sqrt{3}$ the raw zero-cost list additionally has $5+2s$, excluded because its two-cycle is $(3+s)/2$. The positive-cost lower bound is $T\geq 3+s$. In the golden field the raw zero-cost list is $2+2\varphi,3+2\varphi,2+3\varphi,3+3\varphi,3+4\varphi,4+4\varphi$. Its two-cycle is $\{b-a+2+(a-2)\varphi\}/2$; both coefficients require a,b even. At positive cost the strict strips and $T\geq 2+2\varphi$ leave exactly the three displayed rows. In $\sqrt{17}$, raw zero-cost $1+t,2+t,2+2t,3+2t$ has two-cycle $\{(a-2)t+2b-3a+6\}/2$; a is even. At positive cost the raw $1+t,2+t$ list reduces to $2+t$ by acuteness. Writing $T=a+bw$ in each FULL order, the physical upper and conjugate width bound positive b, so these finite lists exhaust every element, including all dotted own and old-I contacts.

Compute the rational norms of the NECESSARY field square (D16IHF). For $\sqrt{3}$, the far/near

fork time norms are $-13/2, -25/2$:

Q_H	T	$\mathrm{N}(R_{HF})$, $j=1$	$j=15$
0	$3+s$	1/78	1/150
0	$4+2s$	1/13	$1/25^*$
1/4	$3+s$	1/208	$1/400^*$
1/4	$4+2s$	23/416	23/800.

For the golden field the time norms are $-19, -9$:

Q_H	T	$j=1$	$j=15$
0	$2+2\varphi$	1/76	$1/36^*$
0	$4+4\varphi$	1/304	$1/144^*$
1/4	$2+2\varphi$	11/1216	11/576
1/4	$2+3\varphi$	29/304	29/144
1/4	$3+3\varphi$	1/304	$1/144^*$.

For $\sqrt{17}$, only the near fork remains, time norm minus thirteen: the three displayed head rows have norms $1/208, 1/52, 1/1664$. All unstarred entries are positive rational nonsquares. They are direct applications of

$$\mathrm{N}(R_{HF}) = \frac{\mathrm{N}(T/2 - 1 + Q_H)}{\mathrm{N}(T)\mathrm{N}(A_F^2)}.$$

The starred square-norm cases REQUIRE a FULL field-square test. Their exact radicands and both possible rational square parts are

K, head row, near fork	R_{HF}	r^2 candidates
$\sqrt{3},\ Q_H = 0, T = 4+2\sqrt{3}$	$(7-4\sqrt{3})/5$	$4/5,\ 3/5$
$\sqrt{3},\ Q_H = 1/4, T = 3+\sqrt{3}$	$(2-\sqrt{3})/20$	$3/40,\ 1/40$
$\varphi,\ Q_H = 0, T = 2+2\varphi$	$(13-8\varphi)/6$	$5/6,\ 2/3$
$\varphi,\ Q_H = 0, T = 4+4\varphi;\ Q_H = 1/4, T = 3+3\varphi$	$(5-3\varphi)/12$	$3/16,\ 5/48.$

Each candidate fails rational squareness by an odd prime valuation. Thus no actual head/fork K-basis condition survives, including every norm-square exception and both head old-I contacts. All quadratic nonisolated forks are impossible.

Together with the complete quartic, cubic and old-I arguments, BOTH actual fork opposites are now isolated within D_{16} and DOTTED. Their independently varying own integers $T_i = 4\gamma_i^2$ generate the SAME ACTUAL field of degree two, three or four. No equality of parameters or vanishing contact with c_I has been imposed.

M.10 Isolated forks: full foreign principals and the shared actual cycles

Both ACTUAL fork caps are now isolated within D, dotted, with

$$c_i = A_i p - \gamma_i \omega_i - \beta_i e, \quad T_i = 4\gamma_i^2, \quad \omega_i^2 = 8, \quad \omega_1 \cdot \omega_2 = 7.$$

They are DISTINCT, since their positive own products are at different old forks. Each T_i generates the ACTUAL field. The old-I contact is finite or zero, and has not been equated to contact with c_I.

Normalize old D roots over K to square two and each cap to own product minus one. It has raw square $2/T$, old-D projection square four. Keep old I at ANY totally positive raw square $\ell \in K$.

The additional contact square is $2\beta^2\ell/T$, so its Schur cost is $2\beta^2/T$, independently of ℓ. Strict positivity of the FULL principal at every nonphysical place gives

$$0 < T_\sigma < (1-\beta_\sigma^2)/2, \qquad 4 < T < \frac{1369}{32}(1-\beta^2). \tag{D16Ifull}$$

All factors are positive; β^2 means its actual invariant embedding, not an assumed K-valued cosine. For degree d, the integral positive norm consequently satisfies

$$1 \le \mathrm{N}(T) < \frac{1369}{32\,2^{d-1}}\mathrm{N}(1-\beta^2).$$

For a positive finite label m, $E = \mathbb{Q}(\beta^2)$ has degree $\phi(m)/2$ dividing d. Pairing inverse cyclotomic roots gives $\mathrm{N}_E(4\sin^2(\pi/m)) = \Phi_m(1)$. These formulas retain all disconnected root scales until they cancel.

When BOTH $\beta_i = 0$, a second degree-INDEPENDENT lemma will be useful. Let r_i be the above ACTUAL raw caps, $P = r_1 \cdot r_2 \in K$, $Q = T_1T_2$. Subtracting their old-D projections leaves the SAME one-dimensional K-time line, so

$$(P - 7/2)^2 = \frac{4(2T_1-1)(2T_2-1)}{Q}.$$

The actual FIVE-cycle $r_1, n_{15}, n_{14}, n_{16}, r_2, r_1$ has Cartan factors $-T_1, -1, -1, -1, T_2P$; hence $X = QP$ is integral. The actual TWO-cycle gives $L = QP^2$ integral. At $P = 0$ both assertions are immediate. Expanding the ACTUAL residual determinant gives

$$L - 7X = \frac{15Q}{4} - 8T_1 - 8T_2 + 4, \qquad \frac{Q}{4} = 4Q - \frac{15Q}{4} \in \mathcal{O}_K. \tag{D16Iquarter}$$

No degree, ramification, equal-parameter, or individual Cartan-integrality premise entered this identity. It also implies $q(T_1)q(T_2) \in K^2$, $q(T) = T(2T-1)$, by clearing square denominators in the determinant.

M.11 The quartic isolated branch ends with the quarter-product

At $d = 4$, the complete positive old-I alphabet is

$$m = 3, 4, 5, 6, 8, 10, 12, 15, 16, 20, 24, 30.$$

Indeed $\phi(m)$ is two, four or eight. Any prime divisor has $p-1$ dividing one of those values, so $p = 2, 3, 5$; the allowed prime powers and products give exactly the list. The COMPLETE values of $256\mathrm{N}_K(1-\beta^2)$ are

m	3	4	5	6	8	10	12	15	16	20	24	30
$256\mathrm{N}(1-\beta^2)$	81	16	25	1	4	1	1	1	2	1	1	1.

They follow from the prime-power cyclotomic rule $\Phi_{p^a}(1) = p$, all other $\Phi_m(1) = 1$. For EVERY $m \ge 4$, the full norm bound gives $\mathrm{N}(T) < 34225/65536 < 1$. Only $\beta = 1/2$ or zero remains.

At $\beta = 1/2$, $4 < T < 4107/128 < 65/2$, other $T_\sigma < 3/8$. The positive norm is less than two, hence one. Its three reciprocal small conjugates $Y_i > 8/3$ have product T. For three positive numbers $> a$, their sum is $< 2a + \text{product}/a^2$; merge two at a time, since $(y-a)(z-a)/a > 0$. Thus

$$Y_1 + Y_2 + Y_3 < 16/3 + 9T/64.$$

The integral unit T^{-1} has trace strictly between eight and $16/3 + 585/128 + 27/512 < 10$, hence nine. Each $Y_i < 11/3$. The integral nonzero $T^{-1} - 3$ has physical absolute value < 3, others in $(-1/3, 2/3)$. Its nonzero integer norm has magnitude $< 3(2/3)^3 < 1$, impossible. Thus BOTH quartic forks have $\beta_i = 0$.

Their positive integer norms satisfy

$$1 \le \mathrm{N}(T_i) < 1369/256 < 6.$$

The shared actual TWO/FIVE-cycle identity (D16Iquarter) gives

$$256 \mid \mathrm{N}(T_1)\mathrm{N}(T_2).$$

But each norm is at most five, so their product is at most twenty-five. This contradiction excludes the ENTIRE quartic branch directly. The longer norm-two/unit/coefficient route is unnecessary once the shared quarter-product is available; its source is preserved rather than duplicated. This argument applies the already proved degree-independent actual-cycle lemma at degree four, and adds no geometry or finite numerical-output premise.

M.12 Cubic isolated forks: two reusable coefficient lemmas

We will use two COMPLETE cubic strip arguments several times. First, no cubic integer Z generating a totally real field can satisfy

$$0 < Z < 1369/64, \qquad 0 < Z_\sigma < 1/4.$$

Its positive norm is $< 1369/1024 < 2$, hence one, and $Z > 16$. The reciprocal small conjugates $r, s > 4$ have product Z, giving $r + s < 4 + Z/4$. The integral trace of the unit Z^{-1} lies in $(8, 2409/256) \subset (8, 10)$, so is nine. Writing $a = \mathrm{Tr}(Z)$, its polynomial is $X^3 - aX^2 + 9X - 1$. Its negative value at $1/4$ gives $a > 81/4$, while $a < Z + 1/2 < 1401/64 < 22$. Thus $a = 21$, but that cubic has discriminant -864. This contradicts total reality. In particular NO cubic fork parameter in the full isolated strip belongs to $2\mathcal{O}_K$.

Second suppose a cubic fork parameter V has norm eight, with physical $4 < V < 1369/32$, others in $(0, 1/2)$. Then $V > 32$, and its polynomial $X^3 - aX^2 + bX - 8$ obeys

$$34 \le a \le 43, \qquad b^2 > 32(a-1), \qquad 2b \le a + 31.$$

The first follows from $a \ge V + 2\sqrt{8/V} > 33$. The last follows by evaluating at $1/2$. The COMPLETE integer comparison is

a	34	35	36	37	38	39	40	41	42	43
b least	33	33	34	34	35	35	36	36	37	37
b greatest	32	33	33	34	34	35	35	36	36	37.

Only $(35, 33), (37, 34), (39, 35), (41, 36), (43, 37)$ remain. Their exact discriminants are

$$-17131, \quad -16124, \quad -11651,$$
$$-2704, \quad 11821.$$

Thus every such actual cubic parameter has the SAME polynomial

$$P_8(X) = X^3 - 43X^2 + 37X - 8. \tag{D16IP8}$$

No field or polynomial enumeration is being assumed.

M.13 Cubic isolated forks: every old-I contact vanishes

At degree three the full foreign alphabet is $m = 3, 4, 6, 7, 9, 14, 18$, since $\phi(m)/2$ divides three. The complete values of $64\mathrm{N}(1-\beta^2)$ are $27, 8, 1, 7, 3, 1, 1$. The shared full-principal norm bound immediately excludes six/nine/fourteen/ eighteen. At native four, T is a norm-one integer with physical $T < 1369/64$ and others $< 1/4$, impossible by the first cubic lemma. For native seven, put $s_0 = \sin^2(\pi/7)$. Its three conjugates satisfy $64s^3 - 112s^2 + 56s - 7 = 0$; the exact endpoint values $f(3/16) = -1/64$, $f(.189) = .015329216 > 0$, and $f' \geq 12$ on $[0, 1/4]$ give $3/16 < s_0 < .189$. The full principal then forces $\mathrm{N}(T) = 1$,

$$48/7 < T < 81/10, \quad x + y < 25/32, \quad Txy = 1.$$

AM–GM gives $x+y > 2/3$, so integer trace is eight. Its polynomial is X^3-8X^2+bX-1. At $1/2$, negativity gives $1 \leq b \leq 5$. The COMPLETE five discriminants are $-1871, -1563, -1175, -731, -255$. All are negative. Only $\beta = 0, 1/2$ remains at EACH fork.

For $\beta = 1/2$, let $n = \mathrm{N}(T)$. The full strip gives $n = 1, 2, 3, 4$. With cubic polynomial $X^3 - aX^2 + bX - n$, the exact root inequalities give

$$T > 64n/9, \quad a > T + 2\sqrt{n/T}, \quad a < 4203/128, \quad b^2 > n(4a-3), \quad 192b \leq 72a + 512n - 28.$$

The last uses its negative value at $3/8$. For $n = 4$, traces $30, 31, 32$ have respective least $b = 22, 23, 23$, greatest $b = 21, 22, 22$. No coefficient fits. Thus every positive-old-contact fork has norm one, two or three.

If BOTH forks had $\beta = 1/2$, let their actual own-normalized raw products with old I of square ℓ be $b_1, b_2 < 0$. Then

$$b_i^2 = \frac{\ell}{2T_i}, \qquad W = \frac{\ell}{2b_1 b_2} \in K, \quad W^2 = T_1 T_2.$$

Integral closure makes W integral. The ACTUAL triangle through old I makes $Z = WP$ integral, where $P = r_1 r_2$. The full old-D/old-I residual determinant gives

$$(2Z - 7W - 1)^2 = (8T_1 - 3)(8T_2 - 3).$$

Modulo $2\mathcal{O}_K$, this implies $W^2 = T_1 T_2 \in 2\mathcal{O}_K$. Taking cubic norms forces eight to divide a product of two members of $\{1, 2, 3\}$, impossible. Thus any positive-contact fork has a ZERO-contact partner.

Let its parameter be T, the zero-contact partner V, and $P = r_T r_V$. Their old-I cross projection is zero. The ACTUAL five-cycle gives $X = TVP \in \mathcal{O}_K$, and

$$(P - 7/2)^2 = \frac{(8T-3)(2V-1)}{TV}, \qquad (2X - 7TV)^2 = 4TV(8T-3)(2V-1).$$

Integral closure gives $TV \in 2\mathcal{O}_K$. At every prime above two, the two last factors are units, so the valuation of TV is EVEN. If $\mathrm{N}(T) = 1$ or three, T is a dyadic unit and $V \in 2\mathcal{O}_K$, contradicting the first cubic lemma. Hence $\mathrm{N}(T) = 2$.

Its principal ideal is a UNIQUE prime $\mathfrak{p}$ of norm two, valuation of T one. Let $e = v_{\mathfrak{p}}(2) \leq 3$. At every other dyadic prime, the product integrality already makes V divisible by two. At $\mathfrak{p}$, the even product valuation requires $v_{\mathfrak{p}}(V)$ odd. If $e = 1$ or three this again gives $V \in 2\mathcal{O}_K$. Therefore $e = 2$. Cubic degree leaves exactly one other dyadic prime of norm two. Avoiding even V forces valuations one at $\mathfrak{p}$, at least two at the other. Since $1 \leq \mathrm{N}(V) \leq 10$, its norm must be eight.

For the norm-two T, write $P_T = X^3 - aX^2 + bX - 2$. The identity $2 = T(T^2 - aT + b)$ proves, WITHOUT a monogenic-order assumption: odd b gives $e = 1$; even b, odd a gives $e = 2$; both even

gives $e = 3$. Thus a is odd, b even. The complete bounded coefficient comparison at $15 \le a \le 32$ is

a	15	17	19	21	23	25	27	29	31
b least even	12	12	14	14	14	14	16	16	16
b greatest even	10	10	12	12	12	14	14	16	16.

Only $(25, 14), (29, 16), (31, 16)$ remain, with discriminants $-984, 4(5099), 4(2263)$. The norm-eight partner has polynomial (D16IP8) and discriminant 11821. Generators of the SAME actual field must have polynomial discriminants differing by a rational square. But $5099, 2263, 11821$ are respectively three, seven, five modulo eight. Neither relevant odd rational ratio is a square. Every mixed-contact pair is impossible. Therefore BOTH cubic forks have $\beta_i = 0$.

M.14 The cubic zero-contact pair contradicts the same quarter-product

The positive cubic norms satisfy $1 \le n_i < 1369/128 < 11$. The degree-independent actual identity (D16Iquarter) forces $64 \mid n_1 n_2$. Their product is at most one hundred; the only factor pair with each at most ten is $n_1 = n_2 = 8$. The complete cubic norm-eight lemma then gives the SAME polynomial P_8 for both parameters. That polynomial has exactly ONE root above four and two below $1/2$, because either actual parameter has those conjugate bounds. Both physical T_i must be its unique large root. The physical embedding is injective, so $T_1 = T_2 = T$ is DERIVED.

Now $T^2/4$ is integral by the shared actual-cycle identity. Integral closure makes $T/2$ integral. It satisfies the first impossible cubic strip of Section M.12. This contradiction excludes the ENTIRE cubic branch, including all unequal parameters until equality was forced. No old contact with c_I was removed.

M.15 Quadratic isolated forks: the actual field list and square test

It remains to exclude actual degree two. We retain both independently varying fork parameters and the actual long head. The common full principal is (D16Ifull); no contact with the opposite c_I is removed. The old-I invariant already proved above satisfies

$$J = 4(1 + h) \in \mathcal{O}_K, \qquad 4 < J < 123/25, \quad 0 < J' < 4.$$

It generates the actual quadratic field, since a rational J could not have these two different strips.

Let D be the positive fundamental discriminant. In the full integer basis, $J - J' = b\sqrt{D}$, b a positive integer. Thus $D < (123/25)^2 < 25$. The complete fundamental list is $5, 8, 12, 13, 17, 21, 24$. At $D = 24$, the physical strip would give $J = 2 + \sqrt{6}$, whose conjugate is negative. At $D = 5$, coefficient two would give $J = 1 + 2\varphi$, also with negative conjugate. The other fields force coefficient one. The resulting complete table is

K	full integer generator w	J	$J(J-4)$
$\mathbb{Q}(\sqrt{2})$	$s = \sqrt{2}$	$3 + s$	$-1 + 2s$
$\mathbb{Q}(\sqrt{3})$	$r = \sqrt{3}$	$3 + r$	$2r$
$\mathbb{Q}(\sqrt{5})$	$\varphi = (1 + \sqrt{5})/2$	$3 + \varphi$	$-2 + 3\varphi$
$\mathbb{Q}(\sqrt{13})$	$\eta^2 = \eta + 3$	$2 + \eta$	$\eta - 1$
$\mathbb{Q}(\sqrt{17})$	$\eta^2 = \eta + 4$	$2 + \eta$	η
$\mathbb{Q}(\sqrt{21})$	$\eta^2 = \eta + 5$	$2 + \eta$	$\eta + 1$.

Here every $\eta = (1 + \sqrt{d})/2$ has the indicated radicand. This is a derived list for the actual field, not a field-list hypothesis.

We will repeatedly need a full field-square test. If $R \in \mathcal{O}_K$ is a K-square, its square root is integral. Its norm must be a nonnegative rational square d^2. Moreover

$$\left(\operatorname{Tr}\sqrt{R}\right)^2 = \operatorname{Tr} R + 2d \quad \text{or} \quad \operatorname{Tr} R - 2d \tag{D16IQsquare}$$

must be an integer square. Both signs must be tested. A square rational norm alone will never count as a field-square proof. When a square really survives, we display its root.

All interval tables below use the full bases in the first table. For a strip $L < T = a + bw < U$, $0 < T' = a + bw' < V$, the embedding difference bounds the positive integer b. For each such b, the short conjugate interval contains at most one integer a, except in the explicitly wider head strips. One then checks the physical strip. Thus the coefficient rules printed with each table prove completeness. For locating those integers, it is enough to use the strict bounds

$$1.41421 < \sqrt{2} < 1.41422, \quad 1.73205 < \sqrt{3} < 1.73206, \quad 2.23606 < \sqrt{5} < 2.23607,$$
$$3.60555 < \sqrt{13} < 3.60556, \quad 4.12310 < \sqrt{17} < 4.12311, \quad 4.58257 < \sqrt{21} < 4.58258.$$

Each follows by squaring positive rational endpoints. No polynomial or field enumeration is an input to these tables.

M.16 Quadratic isolated forks: every positive old-I contact is excluded

For a positive finite contact $\beta = \cos(\pi/m)$, the degree of $\mathbb{Q}(\beta^2)$ divides two. The complete alphabet is $m = 3, 4, 5, 6, 8, 10, 12$. For an irrational β^2, this subfield is the whole actual K. The full norm bound is

$$1 \le \mathrm{N}(T) < \frac{1369}{64}\mathrm{N}(1 - \beta^2).$$

Native twelve gives the physical upper bound $(1369/128)(2 - \sqrt{3}) < 4$. Native six has norm one, $8 < T < 1369/128$, $0 < T' < 1/8$; its integral trace is nine or ten, giving respectively $\mathbb{Q}(\sqrt{77})$ or $\mathbb{Q}(\sqrt{6})$, neither in our actual list. At native ten the field is golden, $T < 4.086$, $T' < .328$. Writing $T = a + b\varphi$ gives $1 < b = (T - T')/\sqrt{5} < 2$, impossible.

Further information comes from the actual two-cycle with c_I. Its unit mutual has magnitude

$$M = \sqrt{hA^2} - \beta\sqrt{1 + h},$$

and $4M^2$ is an integral invariant. All the squared terms belong to K, so expansion makes $h(1 + h)\beta^2A^2$ a K-square. We impose no prescribed label on the actual mutual with c_I.

At native four, clearing rational square denominators requires

$$R = J(J - 4)(4T - 1) \in K^2.$$

The full strips $4 < T < 1369/64$, $0 < T' < 1/4$ give exactly

K	T	R	$\mathrm{N}(R)$ or both trace tests
$\sqrt{2}$	$3 + 2s$	$21 + 14s$	$49;\ 56, 28$
$\sqrt{2}$	$10 + 7s$	$73 + 50s$	329
$\sqrt{3}$	$7 + 4r$	$96 + 54r$	468
φ	$2 + 3\varphi$	$22 + 33\varphi$	$121;\ 99, 55$
φ	$5 + 8\varphi$	$58 + 89\varphi$	605
$\sqrt{13}$	$4 + 3\eta$	$21 + 15\eta$	$81;\ 75, 39$
$\sqrt{17}$	$8 + 5\eta$	$80 + 51\eta$	76
$\sqrt{21}$	$2 + \eta$	$27 + 15\eta$	$9;\ 75, 63.$

The coefficient bounds $b \le 7, 6, 9, 5, 5, 4$ in field order, with the unique integer $0 < a + bw' < 1/4$, give every row after the physical bound. Every isolated norm is nonsquare. In the square-norm rows both integer trace tests fail. Thus no native-four contact survives.

At native five the golden strips give $T = 2+3\varphi, 4+6\varphi$. Here $b \le 6$ and $0 < a+b\varphi' < (\varphi+2)/8$. The necessary square

$$(1 + 4\varphi)(8T + \varphi - 3)$$

has values $113+177\varphi$ and $225+361\varphi$, of nonsquare norms $1441, 1529$. At native eight, $K = \mathbb{Q}(\sqrt{2})$, $b \le 2$, and only $T = 3 + 2s$ lies in the full strips. The necessary square is

$$(2 + 3s)(8T + s - 2) = 146 + 100s, \qquad \mathrm{N} = 1316,$$

again nonsquare. This removes every positive label at least four, including the genuine native-eight strip survivor.

For native three, $\beta = 1/2$, the necessary domain becomes

$$4 < T < 4107/128, \quad 0 < T' < 3/8, \qquad R = J(J-4)(8T-3) \in K^2.$$

The coefficient bounds are $b \le 11, 9, 14, 8, 7, 7$; the conjugate strip again supplies the unique a. The complete twenty rows and norms are

K	all T	corresponding $\mathrm{N}(R)$
$\sqrt{2}$	$3+2s;\ 6+4s;\ 10+7s;\ 13+9s$	497; 161; 2401; 1169
$\sqrt{3}$	$7+4r;\ 9+5r;\ 14+8r$	3156; 468; 4884
φ	$2+3\varphi;\ 4+6\varphi;\ 5+8\varphi;\ 7+11\varphi;\ 9+14\varphi$	1045; 781; 3949; 2981; 605
$\sqrt{13}$	$4+3\eta;\ 8+6\eta$	573; 789
$\sqrt{17}$	$5+3\eta;\ 8+5\eta;\ 11+7\eta$	188; 956; 2236
$\sqrt{21}$	$2+\eta;\ 9+5\eta;\ 11+6\eta$	141; 1437; 645.

All norms except 2401 are nonsquare. The remaining case really passes:

$$K = \mathbb{Q}(\sqrt{2}), \quad T = 10 + 7s, \qquad R = 147 + 98s = [7(1+s)]^2.$$

It must be retained. If it occurred at both forks, the two physical parameters would be equal and the actual unit mutual would be $1 - T/4$. Its two-cycle would be

$$(T-4)^2/4 = 67/2 + 21s,$$

which is not integral in the full ring $\mathbb{Z}[s]$. Thus a positive-contact fork must have a zero-contact partner V.

The complete zero-contact $\mathbb{Q}(\sqrt{2})$ strip gives

$$V = 3+2s,\ 6+4s,\ 10+7s,\ 13+9s,\ 16+11s,\ 17+12s,\ 20+14s.$$

For these actual caps let P be their raw product. The old-I cross projection is zero, so the actual residual identity and five-cycle give

$$(P - 7/2)^2 = \frac{(8T-3)(2V-1)}{TV}, \quad X = TVP \in \mathcal{O}_K, \quad (2X - 7TV)^2 = 4TV(8T-3)(2V-1).$$

Integral closure makes $X - 7TV/2$ integral, hence $TV/2$ integral. At the unique prime over two, normalized by $v(2) = 2$, the last two factors are units. Thus $v(TV)$ is even and at least two. Since

$\mathrm{N}(T) = 2$, $v(T) = 1$; the partner valuation must be odd and positive. The seven partner norms are $1, 4, 2, 7, 14, 1, 8$. Only

$$V = 10 + 7s, \quad 16 + 11s, \quad 20 + 14s$$

remains. But the determinant requires $TV(8T - 3)(2V - 1)$ to be a field-square. Its respective rational norms are

$$4 \cdot 7^3 \cdot 31, \quad 4 \cdot 7^5, \quad 16 \cdot 7^3 \cdot 47.$$

All are nonsquare. This excludes the last mixed pair. Both actual quadratic forks therefore have ZERO old-I contact. Their contacts with c_I have never been set to zero.

M.17 Quadratic isolated forks: the complete actual head list

We may now use the common quarter-product (D16Iquarter). It is not enough by itself in degree two, so we also retain the actual long head already proved isolated and dotted:

$$c_H = A_H p - g\omega_H - \epsilon e, \quad \omega_H^2 = 2, \quad g > 1, \quad \epsilon = 0 \text{ or } 1/2, \quad U = 4g^2.$$

The preceding original-vector argument gave $(1 - \epsilon^2)/g^2 > 3/8$. Actual raw own normalization gives head square $2/U$, raw old-D dual square one. The physical principal is Lorentzian over $\mathbb{Q}(U)$; actual-field descent gives $K = \mathbb{Q}(U)$. The FULL old-I Schur complement at the other embedding is $2(1 - \epsilon^2)/U' - 1 > 0$. Consequently

$$U \in \mathcal{O}_K, \qquad 4 < U < \frac{32}{3}(1 - \epsilon^2), \qquad 0 < U' < 2(1 - \epsilon^2). \tag{D16IQhead}$$

No raw scale of the disconnected old I has been prescribed.

At $\epsilon = 0$, the actual c_H, c_I two-cycle requires

$$(J - 4)(U - 2)/2 \in \mathcal{O}_K.$$

At $\epsilon = 1/2$, the same expansion as before requires $J(J-4)(2U-3) \in K^2$. Here is the COMPLETE pre-test list in (D16IQhead):

K	$\epsilon = 0$	$\epsilon = 1/2$
$\sqrt{2}$	$3 + s;\ 3 + 2s;\ 4 + 2s;\ 5 + 3s;\ 6 + 3s$	$3 + 2s;\ 4 + 2s$
$\sqrt{3}$	$3 + r;\ 4 + 2r;\ 5 + 2r$	$3 + r;\ 4 + 2r$
φ	$2 + 2\varphi;\ 3 + 2\varphi;\ 2 + 3\varphi;\ 3 + 3\varphi;\ 3 + 4\varphi;\ 4 + 4\varphi$	$2 + 2\varphi;\ 2 + 3\varphi;\ 3 + 3\varphi$
$\sqrt{13}$	$2 + \eta;\ 3 + \eta;\ 3 + 2\eta;\ 4 + 2\eta$	$2 + \eta;\ 3 + 2\eta$
$\sqrt{17}$	$2 + \eta;\ 3 + \eta;\ 4 + 2\eta;\ 5 + 2\eta$	$2 + \eta;\ 3 + \eta$
$\sqrt{21}$	$2 + \eta;\ 3 + \eta;\ 4 + 2\eta;\ 5 + 2\eta$	$2 + \eta;\ 3 + \eta.$

The coefficient bounds are $b \le 3, 3, 4, 2, 2, 2$. For each b, one or two integers satisfy $0 < a + bw' < 2$, respectively $3/2$, before the physical bound.

To show every deletion explicitly, the $\epsilon = 0$ cycle values, in the displayed order, are

$\sqrt{2}$	$1/2,\ (3 - s)/2,\ 1,\ 3/2,\ (2 + s)/2$
$\sqrt{3}$	$1,\ 2,\ (3 + r)/2$
φ	$1,\ (1 + \varphi)/2,\ 3/2,\ (2 + \varphi)/2,\ (3 + \varphi)/2,\ 1 + \varphi$
$\sqrt{13}$	$(3 - \eta)/2,\ 1/2,\ (4 - \eta)/2,\ 1$
$\sqrt{17}$	$(4 - \eta)/2,\ 1,\ 2,\ (2 + \eta)/2$
$\sqrt{21}$	$(5 - \eta)/2,\ 3/2,\ 3,\ (4 + \eta)/2.$

Every nonretained value is nonintegral in the FULL written integer basis. For $\epsilon = 1/2$, the corresponding square-test norms are

$\sqrt{2}$	161, 49
$\sqrt{3}$	36, 276
φ	121, 319, 99
$\sqrt{13}$	27, 81
$\sqrt{17}$	52, 4
$\sqrt{21}$	51, 15.

The square norm 81 in $\sqrt{13}$ is not sufficient: the radicand $9+3\eta$ has trace twenty-one and fails both tests 39, 3 in (D16IQsquare). The retained square roots in $\sqrt{2}, \sqrt{3}, \varphi, \sqrt{17}$ are $3+s, 3+r, 3+\varphi, 2+\eta$. The full head survivors are therefore

K	(U, ϵ)	$\mathrm{N}(H_U)$ in that order
$\sqrt{2}$	$(4+2s, 0)$, $(4+2s, 1/2)$	-128, -56
$\sqrt{3}$	$(3+r, 0)$, $(3+r, 1/2)$, $(4+2r, 0)$	-48, -18, -128
φ	$(2+2\varphi, 0)$, $(2+2\varphi, 1/2)$, $(4+4\varphi, 0)$	-64, -44, -256
$\sqrt{13}$	$(4+2\eta, 0)$	-192
$\sqrt{17}$	$(3+\eta, 0)$, $(3+\eta, 1/2)$, $(4+2\eta, 0)$	-64, -8, -256
$\sqrt{21}$	$(4+2\eta, 0)$	-192.

The constant used in the last column is

$$H_U = \begin{cases} 2U(U-2), & \epsilon = 0, \\ U(2U-3), & \epsilon = 1/2. \end{cases}$$

For each zero-old-contact fork, the SAME actual one-dimensional K-time line requires

$$H_U q(T) \in K^2, \qquad q(T) = T(2T-1). \tag{D16IQheadfork}$$

Indeed the squared raw residual-time norms are $(U-2)/U$, respectively $(2U-3)/(2U)$, for the head, and $2(2T-1)/T$ for a fork. Their product is a K-square: subtract the actual K-valued old-D and old-I projections from the actual raw mutual. Clearing SQUARE denominators gives exactly the displayed condition. It neither identifies disconnected raw root scales nor drops the head's old-I projection.

M.18 Quadratic isolated forks: every field and unequal pair closes

The full zero-old-contact strips are $4 < T < 1369/32$, $0 < T' < 1/2$. The coefficient bounds $b \le 15, 12, 19, 11, 10, 9$, with the unique a in the conjugate strip, give the COMPLETE lists

K	all T
$\sqrt{2}$	$3+2s$; $6+4s$; $10+7s$; $13+9s$; $16+11s$; $17+12s$; $20+14s$
$\sqrt{3}$	$7+4r$; $9+5r$; $14+8r$; $16+9r$; $21+12r$
φ	$2+3\varphi$; $4+6\varphi$; $5+8\varphi$; $6+9\varphi$; $7+11\varphi$; $9+14\varphi$; $10+16\varphi$; $11+17\varphi$; $12+19\varphi$
$\sqrt{13}$	$3+2\eta$; $4+3\eta$; $7+5\eta$; $8+6\eta$; $12+9\eta$
$\sqrt{17}$	$2+\eta$; $5+3\eta$; $8+5\eta$; $11+7\eta$; $16+10\eta$
$\sqrt{21}$	$2+\eta$; $4+2\eta$; $9+5\eta$; $11+6\eta$; $13+7\eta$.

We use the quarter-product at every prime over two, and the actual head condition (D16IQheadfork). All valuations give a uniformizer value one. The elementary quadratic splitting of two follows by

factoring the full basis polynomial modulo two; the ramified cases can equivalently be checked by the norms and division in their full bases.

For $K = \mathbb{Q}(\sqrt{2})$, the unique dyadic prime has $v(2) = 2$. The seven valuations in table order are $0, 2, 1, 0, 1, 0, 3$. The quarter-product requires a sum at least four, so either fork's valuation-zero rows are impossible. The remaining rows and q-norms are

$$(6 + 4s, -28), \quad (10 + 7s, -62), \quad (16 + 11s, -98), \quad (20 + 14s, -376).$$

Multiplication by the head norm -56 gives no rational square. For head norm -128, only $T = 16 + 11s$ passes this necessary norm test. It would occur at both forks, whose valuation sum would be two, contradicting the quarter-product.

For $K = \mathbb{Q}(\sqrt{3})$, the five valuations at the ramified prime, again $v(2) = 2$, are $0, 1, 2, 0, 0$. The quarter-product forces both $T = 14 + 8r$. Its q-norm is -156. None of its products with the three head norms $-48, -18, -128$ is a rational square. Thus the head condition fails.

For $K = \mathbb{Q}(\varphi)$, two is inert. Only $T = 4 + 6\varphi, 10 + 16\varphi$ has positive valuation, in each case one; every other row is a dyadic unit. Both forks must lie in this pair. Their q-norms are $-44, -220$. The head norms rule out $\epsilon = 0$; at $\epsilon = 1/2$ they force both $T = 4 + 6\varphi$. For its remaining head $U = 2 + 2\varphi$,

$$H_U q(T) = 3916 + 6336\varphi, \quad \mathrm{N} = 1936, \quad \mathrm{Tr} = 14168.$$

The two trace-square tests are $14256, 14080$, neither square. This contradicts the full field-square condition.

For $K = \mathbb{Q}(\sqrt{13})$, two is again inert. Only $T = 8 + 6\eta$ has positive dyadic valuation, equal to one. Both forks would have that parameter. The unique head gives

$$H_U q(T) = 38208 + 29328\eta, \quad \mathrm{N} = 20736, \quad \mathrm{Tr} = 105744.$$

Its two tests $106032, 105456$ are nonsquare, so this field also fails.

For $K = \mathbb{Q}(\sqrt{17})$, two splits. Label its primes by $\eta = 0, 1$ modulo two. The five valuation pairs are

$$(1, 0), \ (0, 2), \ (2, 0), \ (0, 1), \ (3, 1),$$

and the corresponding q-norms are $-2, -36, -100, -98, -304$. The quarter-product requires the two valuation pairs to sum componentwise to at least $(2, 2)$. At $\epsilon = 1/2$, head norm -8 permits only the first and fourth rows. No pair from $\{(1, 0), (0, 1)\}$ can meet that requirement. At $\epsilon = 0$, head norm -64 or -256 permits only the second and third rows. Their two primes force one of EACH:

$$T_1 = 5 + 3\eta, \qquad T_2 = 8 + 5\eta.$$

This genuinely unequal pair has not been discarded. But the shared actual fork time line requires $q(T_1)q(T_2)$ square, where

$$q(T_1)q(T_2) = 98940 + 63360\eta, \quad \mathrm{N} = 3600, \quad \mathrm{Tr} = 261240.$$

Its two trace tests are $261360, 261120$. The latter is $511^2 - 1$; the former lies strictly between 511^2 and 512^2. Neither is square.

Finally, in $K = \mathbb{Q}(\sqrt{21})$, two is inert. Only $T = 4 + 2\eta$ has positive valuation, equal to one, so both forks have that parameter. The unique actual head gives

$$H_U q(T) = 9888 + 5520\eta, \quad \mathrm{N} = 2304, \quad \mathrm{Tr} = 25296.$$

Its tests $25392, 25200$ are both nonsquare. For clarity the remaining larger square comparisons used above are

$$119^2 < 14256 < 120^2, \qquad 118^2 < 14080 < 119^2,$$
$$325^2 < 106032 < 326^2, \qquad 324^2 < 105456 < 325^2,$$
$$159^2 < 25392 < 160^2, \qquad 158^2 < 25200 < 159^2.$$

This exhausts all six fields derived from the actual old-I strip.

The isolated fork case is therefore impossible in degree two as well as degrees three and four. Together with the already proved degree bound, full actual fork inventory and nonisolated exclusions, this excludes the ENTIRE $D_{16} + I$ alternative at the original farthest vertex. Both forks, the long head and the actual I-opposite belong to that same original configuration. Every geometric height and price used earlier concerns its one original vector; the algebraic branches add no energy from a different point. No intrinsic face was assumed Coxeter or arithmetic, no prescribed lattice was introduced, and no two fork parameters were equated before equality was forced. The finite tables in this quadratic proof are complete hand arguments in explicitly bounded full integer bases, rather than new assumed script outputs.

N Complete forest dispatch and the final dimension deduction

The local arguments now meet the two necessary whole-forest covers. We make this last interface explicit: a list of possible forests is a necessary classification, while each row below is excluded by a theorem at its own actual rank and with its whole spectator forest retained.

N.1 The component inputs used before the partition count

Most older component floors in the partition proofs follow at once from stronger results already proved here:

whole component and hypothesis	available strict floor
A_{12}, arbitrary spectators	5/7
A_{14}, arbitrary spectators	4/5
A_{15}, arbitrary spectators	.8348
A_{16}, arbitrary spectators	.8703
A_{17}, arbitrary spectators	.9032.

Thus the older $A_{12} > .6725$, typed $A_{12} > .703101$, $A_{14} > .76$, typed $A_{14} > .7897732$, $A_{15} > .79$, $A_{16} > .83$ and $A_{17} > 6/7$ floors used in Propositions 7 and 12 require no separate field census or historical numerical chapter. The first two rows are (A12U) and Section J.1; the last three were proved by the one-reference simultaneous-foot calculation in Sections B.4 and B.5. These are whole-component estimates with arbitrary spectators, not higher-dimensional whole-forest exclusions.

The conditioned A_{11} floor needs its own short corollary; it cannot be deduced from the weaker uniform conclusion 13/20.

Corollary 21. *If an actual I, N_7 or O_7 occurs beside a whole A_{11}, then $e_{A_{11}} > .652761$, with arbitrary other finite spectators.*

Proof. The complete $N = 12$ inventory in Section J.6 is local and has no contrary component-energy hypothesis. It leaves the isolated feet and exactly the six pure nonisolated golden/silver time types printed in Section J.9. With an actual I, that section's two-cycle and denominator proof excludes all six independently of an assumed A_{11} energy.

With either actual typed-seven defect, the indefinite actual three-root principal gives $K = K_7$. Every one of the six nonisolated caps instead has an actual Lorentzian principal over $\mathbb{Q}(\varphi)$ or $\mathbb{Q}(\sqrt{2})$. The permitted original-root normalization and actual-field descent therefore force the whole actual field to be that quadratic field. This contradicts its cubic degree. The argument concerns the actual cap principal, not a unit-root normalizer adjoined to K. Thus all nonisolated exceptions are excluded in every stated branch.

Use the no-helper reference

$$H_5 = H_7 = 1.805, \quad H_6 = 1.839, \qquad E = .652761, \quad (\beta_5, \beta_6, \beta_7) = (.1635, .034, .1635).$$

Every retained site is strong or has its actual isolated foot. For any set of feet, use the SAME vector $Z = w - \sum \beta_i u_i n_i$. The continuous inequalities already proved in Section J.8 have no contrary-energy assumption: for $u \le 1/2$, the raw loss is $< .6085 < .61$; for $u \ge 1/2$, its loss after the smallest available coefficient .034 is $< .60775 < .61$. They apply to every pure or foreign foot. Every adjacent-foot cross has the nonpositive old-root sign. The single completed square therefore gives $e_A > E$. With no foot, the strict strong pressures give the same strictness. Neither necessary forest cover has been used in this proof. □

All ordinary geometric rates, defect compatibility and the rank-valid original-point/prefix interfaces still refer to their complete proofs in the general compact companion. The precise interfaces used are its inverse/projection/point-pressure lemmas, reciprocal-shadow proposition, ordinary-component appendix, actual typed-propagation appendix, uniform A-chain and B/D component floors, and actual long-prefix inventory. Our arithmetic helper seals and the explicit local-rank boundary checks then give the stronger prefix floors used in the two partition proofs. This distinguishes the shared geometric foundation from the arithmetic branch.

N.2 Every rank-seventeen row has a complete consumer

Apply Proposition 7 at the original farthest vertex. For groups one through fourteen of Table 1, the following is a complete dispatch. The cited sections contain the actual-wall, field and individual coefficient arguments; a final scalar is only their last step.

Group	Complete exclusion on the one original vector
1	The uniform $A_{11} > 13/20$ theorem in Section J.8 plus the ordinary rank-five cost $> 5/14$ gives $13/20+5/14 = 141/140 > 1$. The I energy is not needed.
2	The same uniform theorem, the actual defect counted once, and ordinary $T_4 > 2/7$ give $13/20 + .09 + 2/7 > 1$.
3	Section J.5 gives $5/7$. Every retained ordinary rank-five partition costs $> 5/14$. Their disjoint projections exceed one.
4	Use that same $A_{12} > 5/7$ component theorem. For $I + T_4$ the joint cost is $> 23/70 > 2/7$; for $I + T_3 + A_1$ it is $> 2/7 + .1$; for $I + 2T_2$ it is $> .21 + 1/7 > 2/7$. Each counts I exactly once.
5	Section J.1 gives $4/5$. Ordinary T_3 or $T_2 + A_1$ costs $> 17/70$; the actual $I + T_2$ costs $> .21$. Each total exceeds one.

Group	Complete exclusion on the one original vector
6	Section K.8 excludes the entire ordinary T_2 branch. Two ordinary A_1 or I+A_1 close with .8348+.2 >1 or .8348+.20204 >1, respectively. Section K.7 excludes both distinct actual typed branches, at their cubic field and with their complete foreign costs.
7	Section K.9 treats the SAME actual rank-one opposite, first its nonisolated alternative, then every isolated height. This includes ordinary A_1 and actual short I.
8	Sections C and C.5 prove actual endpoint isolation, the degree bound, the full two-dotted contradiction and finite-anchor field dispatch. Both quadratic and cubic domains of Proposition 10(i),(ii) are empty at the declared numerical output boundaries.
9	Section L.1 gives the actual typed head and complete rank-fifteen domain before its use: $.88989 + .0487 + .09 > 1$. Both N_7, O_7 are included.
10	Sections L.2 through L.5 give the complete original-rank short-cap inventory, actual degree two/three dispatch and full head/foreign/contact cases. Every remaining parameter model is excluded by its actual original head and I compatibility.
11	Sections I.2, I.3 and I.4 prove the B_{17}-specific reduction and actual inclusion in Proposition 10(iii). Its two surviving necessary pairs each fail the fully paid original-vector consumer.
12	Section L.1 gives $.86989 + .0487 + .09 > 1$ in the distinct D_{15} typed domain, with both actual defects.
13	Sections L.6 through L.9 exclude the ordinary rank-one branch. Sections M.1 through M.18 exclude the actual-I branch, including every actual degree two, three and four. The ordinary nonshort energy is never transferred to the short-I proof.
14	Sections I.5 through I.11 give the separate genuine-fork, head and field reduction for D_{17}. They independently enter the same algebraic domain (iii) and exclude its two survivors by their own original-wall parameter meanings.

All ordinary, J, actual-I and both actual typed alternatives were considered in the preceding cover, before this dispatch. There is no omitted A_{13} row. A bare seven bond has not been identified with a typed defect. Hence no rank-seventeen forest remains, proving Corollary 11 by the complete local arguments.

N.3 Every rank-sixteen row closes at its own rank

Use the fresh Proposition 12; no node is deleted from a rank-seventeen forest. For the complete nineteen groups of Table 3, the dispatch is

Group	Complete exclusion on the original vector
1,2	Section J.14 proves 404/625 under the actual I or either actual typed defect. The totals are $404/625+5/14 > 1$ and $404/625+ .09 + 2/7 > 1$.
3	Section J.8 gives $13/20 + 5/14 > 1$ for every retained ordinary rank-five forest.
4	Section J.10 proves the separate JOINT $A_{11} + I > 18/25$ with all remaining feet. Its disjoint $T_4 > 2/7$ gives $18/25 + 2/7 = 176/175 > 1$.
5	The uniform $A_{11} > 13/20$, the actual $O_7 > .09$, and Lemma 13 give $13/20 + .09 + 13/50 = 1$, strictly exceeded.
6,7,8	The universal Section J.5 gives 5/7. For group6 ordinary rank four costs $> 2/7$; for group7 the JOINT $I + T_3 > 2/7$. For group8 the actual defect propagates the last ordinary $T_2 > .2$, and $5/7 + .09 + .2 > 1$. The older weaker typed component rate needs no duplicate proof.
9	Section J.2 excludes ordinary A_3 and $T_2 + A_1$ with its 53/70 component rate. Section J.3 excludes actual $O_7 + A_1$ by the complete cubic full-order strip. Section J.4 excludes $I + T_2$ using both endpoints, each remaining foot and the ordinary 3/19 dihedral floor, with I counted once.
10	The universal $A_{14} > 4/5$ plus $2A_1 > .2$ or the JOINT $I + A_1 > .20204$ excludes those two reducible cases. Sections K.1 through K.3 exclude both actual typed defects. Sections K.4 through K.6 exclude the ordinary irreducible T_2 in its distinct field domain.
11	Section E takes the same actual rank-one opposite and exhausts its nonisolated/isolated dichotomy. The isolated proof covers every $h > 0$, including 1/2 and one.
12	Section D gives the full actual $N = 17$ endpoint, field and finite-anchor proof, including every one of the twenty-one time norms.
13	Sections F and F.1 give $.86989 + .0487 + .09 > 1$ with the genuine mixed factor and both actual typed defects.
14,18	Section E.8 and the common ordinary Section E.9 exclude nonisolated opposites and all isolated ordinary $h \geq 1/2$. Sections E.10 through E.14 and E.15 through E.18 give the separate short-I proofs for $0 < h < 1/2$. Equality $h = 1/2$ belongs to the ordinary theorem, so no boundary or actual rank-one opposite is lost.
15	Sections H.1 through H.5 give the actual whole-B_{16} reduction and cubic exclusion. The complete quadratic output Proposition 10(iv) leaves one pair; its actual edge consumer gives $1.00640742675 > 1$.

Group	Complete exclusion on the original vector
16	Section G proves the fresh short-layer exclusions before using $e_D > .82436$. Every rank-three remainder, including $I + T_2$ and $O_7 + A_1$, has the proved joint cost $> .21$; the total exceeds one.
17	The ordinary dihedral theorem and its actual D_{14} consumer in Section A exclude the $m \geq 4$ ordinary case. Section F separately proves the full actual D_{14} centre/head argument, then $.86989 + .0487 + .09 > 1$, for both typed branches.
19	The actual whole-D_{16} reduction Proposition 15 and manual dyadic proof Proposition 18 exclude the last forest. There is no whole-D_{16} census premise.

For the rank-one rows, taking the same actual opposite is essential. If it has a nonzero old-large-component contact, use the full nonisolated theorem. Otherwise its normalization has the isolated form and lies in exactly one of the stated height intervals. This dichotomy does not replace an actual wall by a formal Gram extension. All sums concern disjoint whole components of the SAME original x, after any overlapping head/end/foot pressures were jointly paid inside the indicated proof.

Thus groups one through eighteen prove Proposition 14; group nineteen proves that no rank-sixteen forest occurs.

N.4 The acyclic global implication and the verification boundary

The general compact companion is an internal geometric theorem of this project, cited as [4]. Its proof includes the exact dimension-eighteen exclusion and an independently proved tail for every dimension at least nineteen. That tail uses only the geometric parts of its mixed-source supplements. It uses no arithmetic exact-rank theorem, field census or conclusion of this paper. Hence Proposition 2 is a one-way input here.

If a compact arithmetic Coxeter polytope had dimension greater than fifteen, that companion would leave exact dimension seventeen or sixteen. The two complete covers and the preceding local dispatches exclude both. This proves Theorem 1. No induction down dimensions, face arithmeticity, hereditary Coxeter claim or transfer of a farthest vertex to a neighbour occurs.

The proof graph is therefore

shared original-vector geometry
$\longrightarrow$ general compact 17,

shared geometry + actual arithmetic structure
$\longrightarrow$ component and endpoint lemmas
$\longrightarrow$ fresh exact-rank covers and actual domain inclusions,

these inclusions + four declared finite outputs
$\longrightarrow$ exact 17 and exact 16 exclusions,

general compact 17 + both exact-rank exclusions
$\longrightarrow$ 15.

The actual-domain inclusions for B_{17} and D_{17} are distinct, although their quadratic algebraic domain is shared. The whole-D_{16} finish is manual. Neither should be conflated with the $D_{16} + I$

degree-two/three/four proof at ambient rank seventeen.

The algorithmic propositions retain their declared four numerical output premises: the quadratic A_{17} empty domain; the two cubic A_{17} empty finite-anchor domains; the two survivors in the shared quadratic B/D_{17} domain; and the one whole-B_{16} quadratic survivor. Section 3 proves domain finiteness, coverage, exact decision rules and termination. The actual geometric inclusion proofs are separate, and every nonempty output is followed by its explicit original-vector contradiction. The producer runs and same-code replays retain the execution scopes stated there.

The admissible quadratic-form model, actual K-root lines, cyclic integrality and positive definiteness at every nonphysical place are the named external arithmetic structure. The exact literal golden-lattice cutoff used in Section I.1 is a separate external premise, attributed to [3]; that section proves the passage to every compatible invariant lattice rather than infer it from similarity or commensurability. The external cutoff's precise scope and source-authentication limit belong in the assumption register. Elementary integer-ring, norm, trace, valuation and cyclotomic facts remain stated where used.

A conditional Lean verification may take these external facts, the named geometric companion interfaces and the four numerical outputs as explicit parameters, then verify the internal finite case dispatch, scalar and valuation consequences. This would verify exactly that implication, rather than assert an end-to-end formalization of the geometry or enumerations. The corresponding verification records state which conditional kernels were compiled and audited. They do not formalize the external geometry or the enumeration producers.

O Proof locations and source provenance

Each identifier denotes a theorem with its precise scope in the named supplement, relative to `results/`. The companion report `results/compact_arithmetic15_consolidation_source_ledger_20261024.md` gives full paths, literal hashes, acceptance/review bindings and restricted dependency edges. It is a source index, not an extra axiom.

The principal printed proof homes are:

Interface	Printed mathematical proof
A11/A10	Sections J.6–J.14; uniform 13/20, joint 18/25 and the distinct conditioned Corollary 21.
A12–A17	Sections J.5, J.2, J.1, K.1, K.4, K.7, K.8, K.9, D, C.
Classical typed and rank-one	Sections F, L.1, G, E.8, E.10, E.15.
B16/D16 plus I	Sections L.2–L.5 and M.1–M.18; ordinary D16 is separate in Section L.6.
Whole classical rows	Sections H.1, I.2, I.5 and Section 5; all four computed premises are in Section 3.
Covers and global implication	Propositions 7, 12 and Sections N.2–N.4; the general compact bound uses the independent companion interface `G17`.

The following source interfaces identify the companion, covers, endpoint reductions and four computed premises cited in the text. The full source ledger preserves the other local proof records. The mathematical arguments are printed above; these records retain their original execution and review scopes.

ID	Mathematical statement and primary source
G17	General compact bound of seventeen, final geometric composition and its complete dependency closure with 21 inputs. `compact18_whole_d18_uniform_dotted_head_all_initial_references_and_global17_composition_candidate_20261024.md`
F17	Complete cover in rank seventeen, Sections 2–5. `arithmetic_dimension17_complete_forest_frontier_candidate_20261021.md`
AE	Both actual A17 endpoints are isolated; complete field and strip argument with the golden exception. `arithmetic17_whole_a17_short_endpoints_isolation_and_golden_exclusion_candidate_20261022.md`
AD	One paid endpoint reference bounds the actual degree by five. `arithmetic17_whole_a17_joint_endpoint_graft_and_degree_five_reduction_candidate_20261022.md`
AF	Both dotted endpoints are excluded on their entire intervals; finite anchors and four actual fields. `arithmetic17_whole_a17_two_dotted_endpoints_exclusion_and_finite_field_reduction_candidate_20261022.md`
QA17	Complete quadratic domain and algorithm, contract Sections 1–4, followed by the recorded empty-output composition. `arithmetic17_a17_two_isolated_quadratic_endpoints_exact_census_contract_20261022.md` `arithmetic17_whole_a17_quadratic_empty_census_exclusion_composition_20261022.md`
CA17	Full cubic integer rings, coefficient boxes and exact square and sign decisions, Sections 2–4; recorded outcomes are stated in Proposition 10. `arithmetic17_a17_finite_cubic_endpoint_exact_census_contract_20261022.md`
QBD17	Complete census contract, Sections 1–5; two separate geometric inclusions and complete consumers of both surviving pairs. `arithmetic17_d17_one_isolated_quadratic_fork_exact_census_contract_20261022.md` `arithmetic17_whole_b17_fresh_long_head_short_cycle_and_coupled_exclusion_candidate_20261022.md` `arithmetic17_whole_d17_complete_quadratic_census_and_coupled_energy_exclusion_candidate_20261022.md`
F16	Complete cover in rank sixteen, Sections 2–7. `arithmetic_dimension16_fresh_complete_necessary_forest_frontier_candidate_20261022.md`
QB16	Full algorithm contract for quadratic rings and actual root scales, Sections 1–4. `arithmetic16_b16_isolated_quadratic_head_short_pair_exact_census_contract_20261023.md`

References

[1] I. Agol, M. Belolipetsky, P. Storm and K. Whyte, *Finiteness of arithmetic hyperbolic reflection groups*, Groups, Geometry, and Dynamics **2** (2008), 481–498, Lemma 5.2. `https://doi.org/10.4171/GGD/47`.

[2] M. Belolipetsky, *Arithmetic hyperbolic reflection groups*, `https://arxiv.org/abs/1506.03111`, Lemma 2.1 and Sections 1 and7, for the arithmetic and compactness background.

[3] V. O. Bugaenko, *Groups of automorphisms of unimodular hyperbolic quadratic forms over* $\mathbb{Z}[(\sqrt{5}+1)/2]$, Vestnik Moskov. Univ. Ser. 1 Mat. Mekh. (1984), no. 5, 6–12. `https://www.mathnet.ru/eng/vmumm3408`.

[4] Coxeter Polytopes Frontier Project, *Farthest vertices and dimension bounds for compact hyperbolic Coxeter polytopes*, general compact companion and its source-bound supporting proofs of the bound seventeen specified in Appendix O.